



THE ROUTED WORLD

Meridian and the Architecture of Deterministic Physical Infrastructure

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**Systems and Methods for Deterministic Multi-Layer Wireless Energy Routing
Using Ambient Harvesting, Mesh Coordination, and Directional Transmission
and Deterministic Cognitive Architecture with Fractal Structural Governance**

This manuscript incorporates material from nine technical papers authored under the L-SOC series.
All technical claims reflect the current state of the Meridian architecture specification.
No experimental validation data currently exists. Phase 1 experimental work is the immediate next step.

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*For everyone who looked at a wall full of charging cables
and thought: there has to be a better way.*

There is.

*And for **Barry Allen Cline**, who asked the question that started all of this:*

*“Why can’t we charge our phones when we walk into the house
without plugging them up or sitting them on a wireless charger?”*

That question became this book.

“The internet did not begin as a global network. It began as a protocol.”

—from Chapter Three

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PREFACE

Why This Book Exists

I did not set out to write a book about wireless energy routing.

It started with a conversation. My friend Barry Cline and I were talking one day, and he asked a question that would not leave my mind: why can't we charge our phones when we walk into the house without plugging them up or sitting them on a wireless charger? It was the kind of question that sounds simple and turns out to be extraordinarily hard to solve. Why, in 2026, does every device we own still need a cable or a battery?

This is not a technology problem in the sense that we lack the underlying physics. Radio waves carry energy. This has been known since Hertz, demonstrated at scale by Tesla, used by every wireless charging pad on every nightstand in the world. The physics is not the obstacle.

The obstacle is that nobody has thought about wireless energy as a network problem.

When you connect a device to Wi-Fi, you are not just sending it radio waves. You are routing a specific packet from a specific source to a specific destination along a path that has been computed, validated, and governed. The packet knows where it is going. The network knows where the device is. Identity is verified. Delivery is confirmed. If the packet doesn't arrive, the system notices and retransmits. None of this happens by accident. It happens because someone — many someones, over many decades — thought carefully about what a packet routing architecture needs to be correct, and built it.

Nobody has applied that thinking to energy. Until now.

Meridian is a routing architecture for energy. It treats a watt the same way the internet treats a bit: as an addressed payload with a source, a destination, a path, a delivery confirmation, and a quality-of-service guarantee. It defines node addresses for power devices, routing tables for energy flows, quality-of-service classes for critical versus background delivery, and a self-healing control runtime that keeps the mesh operating without human supervision.

This book is the complete account of that architecture — from the hardware specification to the biological theory to the building deployment guide to the formal security proof. It is written for engineers, architects, product designers, and anyone who has thought seriously about what wireless infrastructure should look like in a world with a trillion autonomous devices and no human operator available to replace their batteries.

The nine technical papers that form the chapters of this book were written sequentially over the course of an intense research period at DarkWave Studios LLC. Each paper answered a different question about the same system. The first asked: what is this architecture? The second asked: what kind of thing is this architecture? The third asked: where does this architecture lead? And so on, through security, deployment, a general theory of Deterministic Infrastructure, the full-stack vision from fusion to device, and finally the question that emerged from the architecture itself: why do the numbers 4 and 42 keep appearing everywhere?

Read together, these nine papers tell a complete story: from the engineering specification of a specific system to the elimination of physical transmission infrastructure entirely, and then to the mathematical discovery that the architecture itself has a governing structural law. That is what this book is — the complete story, assembled in the order that makes the most sense to read it, with the connecting tissue that turns nine technical papers into a single coherent argument.

The argument is this: the infrastructure paradigms that govern the physical world — the electrical grid, the internet, the supply chain, the transportation network — were designed for a world of human operators, central coordination, and wired physical plants. They are not adequate for the world that is coming: a world of trillions of autonomous devices, distributed physical operation, adversarial environments, and millisecond decision timescales that cannot wait for human intervention.

Deterministic Infrastructure — the class of autonomous physical systems defined in this book — is the design paradigm for that world. Meridian is its first fully specified instance. And the Lume ecosystem that underlies it is the first complete framework for building it.

I have tried to be honest throughout. No experimental data exists for the Meridian architecture. Everything in this book is theoretical and architectural. The formal claims — the safety proof, the coverage proof, the convergence argument — are architectural arguments, not empirical results. Phase 1 experimental work is

the immediate next step, and none of the ideas in this book should be considered validated until that work is done and the results are honestly reported.

What I am confident in is the architecture. The design is right. The physics is sound. The protocol stack is coherent. The safety model is defensible. And the general theory — that the world needs Deterministic Infrastructure and that Meridian is the first instance of it — is an argument I am prepared to defend.

This book is that argument, made as carefully and honestly as I know how to make it.

Jason Andrews
Nashville, Tennessee, 2026

INTRODUCTION

The World We Are Building Toward

Imagine a hospital where no device ever needs its battery replaced.

The infusion pump at the patient's bedside receives its operating power wirelessly, through the same air that carries the nurse call button's signal. The pulse oximeter on the patient's finger — the one that has to be physically removed and recharged every eight hours — charges continuously while it is being worn. The environmental sensors that monitor air quality, temperature, and pressure throughout the ward have no batteries at all. They draw power from the wireless infrastructure the same way a lamp draws power from the wiring in the wall — except there is no wiring.

The devices know where they are. They know who they are. The infrastructure knows what they need and routes it to them precisely, without waste, without interference with each other, and without any human operator managing the delivery. When the infusion pump moves from one room to another, the infrastructure notices, updates its routing table, and continues delivering power to the new location within one control cycle — 13.7 milliseconds.

This is not science fiction. It is the engineering specification described in the chapters that follow.

Now imagine a warehouse where no forklift ever interrupts its work to go to a charging station.

The forklifts draw supplemental power from the wireless mesh as they operate, through receivers embedded in their chassis. The mesh knows the state of each forklift's battery, routes energy to the most depleted units preferentially, and ensures that no forklift drops below its minimum operating charge during a shift. The hundreds of inventory sensors throughout the warehouse — tracking temperature,

humidity, location, and equipment health — are all self-sustaining: they power themselves from the same mesh that keeps the forklifts running, reporting their data continuously through the same infrastructure that delivers their power.

Again: not science fiction. Architecture.

Now imagine a building where the same wireless fabric that powers these devices also carries all of their data — sensor readings, control commands, firmware updates, audit logs — under the same unified addressing system, the same routing protocol, and the same security framework. One installation provides both the power grid and the communications network for every autonomous device in the building.

This is the Unified Energy-Data Mesh. It is Chapter Five of this book.

These are not futuristic scenarios. They are the natural consequence of a single architectural decision: to treat energy delivery as a network routing problem instead of a transmission problem.

What Makes This Different

There are other approaches to wireless power delivery. Qi charging pads are everywhere. Resonant inductive coupling can charge a phone at distances of a few centimeters to a foot. Several companies are working on room-scale wireless power using RF beams. This is not new territory.

What is new is the network architecture.

Every existing wireless power system thinks in terms of *transmission* — a source sends energy to a receiver. The source does not know who the receiver is in any meaningful sense. It does not route. It does not prioritize. It does not self-heal. If the receiver moves, the system either tracks it mechanically or loses it. If the source fails, there is no alternative path. If a hostile device positions itself to intercept the beam, the system has no way to detect this and reroute.

Meridian thinks in terms of *routing* — a source routes energy to an addressed destination through a path that has been computed from the current network topology, validated against a set of safety invariants, authenticated against the destination's cryptographic identity, and confirmed by a delivery receipt from the destination. If the receiver moves, the routing table updates automatically. If a relay node fails, the self-healing control runtime finds an alternative path within one control cycle. If a hostile device attempts to intercept, the security framework detects the attempt and quarantines the offending node.

The difference between transmission and routing is the difference between a broadcast tower and the internet. Broadcast towers send the same signal to everyone within range and cannot tell who received it. The internet routes specific packets to specific addresses and confirms delivery. Meridian applies internet-style routing to energy delivery.

This is the central idea of the book. Everything else follows from it.

How to Read This Book

Part One (Chapters One and Two) covers the engineering: what the four-layer Meridian architecture is, how it works, and how it gets installed in real buildings.

Part Two (Chapter Three) covers the theoretical identity of the architecture: the argument that Meridian is not merely an energy routing network but a synthetic organism.

Part Three (Chapters Four and Five) covers the scaling vision: the Energy Internet and the Unified Energy-Data Mesh.

Part Four (Chapter Six) covers security: the first formal threat model and defense framework for wireless energy routing.

Part Five (Chapter Seven) covers the general theory: the argument that Meridian is one instance of a broader class of systems — Deterministic Infrastructure.

Part Six (Chapter Eight) closes the full stack — from fusion generation to device delivery with no physical transmission infrastructure at any layer.

Part Seven (Chapter Nine) reveals the architecture's governing structural law — the 4/42 Fractal Law — and demonstrates that the numbers 4 and 42 are not arbitrary but are fixed-point attractors of any deterministic cognitive system built on the same closure principles.

A Note on the State of This Work

Every technical claim in this book rests on architectural analysis. No experimental data currently validates the Meridian architecture. This is stated explicitly in every chapter, and it bears repeating in the introduction.

The architecture is correct in the sense that it is internally coherent, physically grounded, and formally specified. The formal proofs in Chapters Six and Seven are genuine proofs — not empirical demonstrations, but logical arguments that the specified architecture, correctly implemented, satisfies the claimed properties.

But a proof is not a measurement. A formally correct architecture can encounter unexpected physical phenomena, implementation errors, regulatory constraints, or deployment realities that the formal specification does not capture. The purpose of Phase 1 experimental work is precisely to discover these gaps before deployment at scale.

This book should be read as the theoretical foundation and architectural specification for Meridian — the design before the data. When Phase 1 data exists, some claims in this book will be confirmed, some will be refined, and some may need to be reconsidered. That is the appropriate relationship between theory and experiment.

With that caveat clearly stated, let us begin.

PART ONE

The Machine: How Meridian Works

A new infrastructure technology lives or dies on the quality of its engineering specification. A beautiful theory built on a weak architecture is marketing. A complete, formal, internally consistent engineering specification is the foundation on which everything else rests.

Part One covers the engineering. Chapter One specifies the four-layer Meridian architecture in full: the node addressing system, the multi-hop mesh routing protocol, the burst-mode energy storage and scheduling layer, the directed transmission mechanism, and the 73 Hz deterministic control runtime that governs all four layers simultaneously. Chapter Two takes that architecture out of the laboratory and into real buildings: multi-story commercial offices, hospitals, warehouses, and residential buildings, each with the specific architectural features —” open staircases, atriums, elevator shafts, mezzanines —” that make real-world deployment categorically harder than room-scale demonstration.

These two chapters together answer the question that any serious evaluation of Meridian must begin with: does this actually work as an engineering system? The answer is: yes, under the specified conditions, at the specified scales, with the specified hardware and software components. The experimental validation of those claims is the next step. But the engineering specification itself is complete, formal, and internally consistent.

Read Part One if you want to understand what Meridian is before you consider what it means.

Chapter One

The Four-Layer Architecture

Every great infrastructure technology is, at its core, an abstraction. The electrical grid abstracts away the complexity of power generation —” the consumer does not need to know where the electrons come from, only that they arrive at the outlet at the right voltage and frequency. The internet abstracts away the complexity of network topology —” the application does not need to know the physical path a packet takes, only that it arrives at the destination reliably. The power of abstraction is that it separates the "what" (reliable power delivery, reliable data delivery) from the "how" (generators, transmission lines, routers, protocols), allowing both sides of the abstraction to evolve independently.

Meridian is an abstraction for wireless energy delivery. It separates the "what" —” deterministic, loss-bounded, addressed energy delivery to any device in the network —” from the "how" —” the ambient harvesting, burst-mode storage, multi-hop relay, and directed transmission mechanisms that implement the delivery. The four-layer architecture is the definition of this abstraction: what each layer does, what guarantees it provides to the layer above, and what requirements it imposes on the layer below.

The four layers are designated Meridian Core (MC), Meridian Flow Engine (MFE), Meridian Mesh Fabric (MMF), and Meridian Transmission Layer (MTL). MC is the identity and coordinate system: every node has a 64-bit address that encodes its type, zone, and serial number, and a 3D position in the Meridian energy coordinate space that the routing system uses to compute paths. MFE is the burst-mode storage and scheduling layer: it governs when energy is accumulated in supercapacitors, when it is released, and in what quantities. MMF is the multi-hop relay protocol: it maintains the routing tables, selects paths, and coordinates TDMA transmission schedules across the mesh. MTL is the directed transmission mechanism: the phased-array RF system that forms and steers the beam to the destination, with the five-condition safety gate that must pass before any beam is authorized.

Governing all four layers is Lume-X —” the deterministic control runtime that runs at 73 Hz, checking five global invariants at every cycle and executing pre-specified recovery sequences when any invariant is violated. Lume-X is not a layer of the Meridian stack; it is the operating system that runs the stack. It is the reason the system can claim determinism: not because the physics is deterministic (RF propagation is stochastic), but because the control system responds to physical stochasticity faster than it can accumulate into a safety violation.

What follows is the complete technical specification of the four-layer architecture —” the engineering document that defines Meridian precisely enough to build from.

Abstract

I present Meridian, a four-layer deterministic wireless energy routing architecture that unifies ambient energy harvesting, burst-mode storage, mesh-based relay coordination, and directed wireless power delivery under a single governing framework. Meridian defines a universal energy coordinate system, a deterministic flow engine, a multi-hop mesh fabric, and a directional transmission layer —” designated Meridian Core (MC), Meridian Flow Engine (MFE), Meridian Mesh Fabric (MMF), and Meridian Transmission Layer (MTL) respectively. These four layers are vertically coupled and governed by a set of global invariants that guarantee deterministic, loss-bounded, and safe energy propagation across the network regardless of ambient variability.

Meridian integrates three established DarkWave subsystems: DWER (Deterministic Wireless Energy Routing) for point-to-point directed beam delivery, DRMA (Distributed Relay Mesh Architecture) for multi-hop energy routing, and DAEH (Deterministic Ambient Energy Harvesting) for node-autonomous power accumulation. The architecture is controlled by Lume-X, a provisionally patented deterministic control runtime validated at 73 Hz cycle frequency, and enforced by Guardian Security (energy-domain designation: Guardian-E), the security layer hosted at TrustShield.tech, which addresses eleven threat categories including localization spoofing, routing table manipulation, beam hijacking, and denial-of-power attacks.

The primary technical contributions of this work are: (1) the first formal definition of deterministic wireless energy routing as a network protocol stack, with power nodes addressed as IP devices, routing tables adapted from OSPF and BGP equivalents, and quality-of-service guarantees applied to watt delivery; (2) the Self-Healing Deterministic Control Layer (SHDCL), which provides continuous autonomous recovery from failure conditions across all three physical subsystems without human intervention; (3) a formal, testable determinism standard for wireless power delivery defined by four required measurements —” received power $P_{\min}$ (mW), spatial uncertainty radius r (cm), trial variance σ^2 (%), and distance d (m); (4) a burst-mode ambient harvesting integration model that makes node autonomy a first-class architectural layer rather than a supplementary subsystem; and (5) a formal biological safety layer with real-time RF exposure monitoring and automatic threshold enforcement.

No experimental data currently exists for this architecture. This paper establishes the formal theoretical and architectural foundation. No deployment-scale capability claims are made. Phase 1 experimental validation is the immediate next step.

Keywords: deterministic wireless energy routing, mesh energy networks, ambient energy harvesting, phased-array beamforming, self-healing control systems, deterministic control, wireless power transfer, network protocol stack for energy, burst-mode storage, biological safety layer, SHDCL, Lume-X

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1. Introduction

1.1 The Problem with Wireless Power Today

Wireless power transfer has been a subject of active research and commercial development for more than sixty years, yet the field has not produced a system capable of operating deterministically. Current commercial wireless power systems —” including resonant inductive coupling platforms, omnidirectional RF broadcast systems, and near-field charging pads —” share a fundamental architectural limitation: none of them can guarantee delivery. They transmit energy into the environment and rely on proximity, orientation, or statistical averaging to achieve acceptable transfer rates. None define, measure, or enforce a formal determinism standard. None provide closed-loop confirmation of delivery. None reroute around failure. None maintain continuous safe operation in the presence of environmental perturbation, node failure, or unexpected physical intrusion.

This limitation is not a product of insufficient engineering effort. It is a consequence of the underlying architectural approach: existing wireless power systems are broadcast systems. They were not designed to route. Routing requires a formal model of the network, a defined path between source and destination,

confirmation that delivery occurred, and a recovery mechanism when delivery fails. No existing wireless power system provides any of these four properties.

The consequences are practical and immediate. A wireless power system that cannot guarantee delivery cannot be used in safety-critical applications. A system without closed-loop confirmation cannot be regulated or certified with confidence. A system that fails hard when a node goes offline cannot be deployed in real-world environments where node failure is inevitable. The gap is not marginal —” it is structural.

1.2 The Routing Reframe

The foundational insight of this work is that wireless energy delivery is a routing problem, not a transmission problem. The distinction matters precisely. A transmission system optimizes the output of a single source. A routing system optimizes the path from source to destination across a network, with explicit awareness of topology, failure, and dynamic conditions. Packet-switched data networks solved this problem decades ago, and the conceptual framework they produced —” addressed nodes, routing tables, QoS guarantees, topology-aware path selection, failure detection and recovery —” applies directly and powerfully to wireless energy delivery.

Meridian implements this reframe completely. Power nodes are addressed like IP devices. Energy flows along defined paths selected by routing tables. Quality of service constraints govern delivery guarantee thresholds. Topology changes propagate through a link-state protocol. Failed nodes trigger automatic path recalculation. The entire architecture is governed by a deterministic control runtime —” Lume-X —” that monitors every layer continuously and recovers from failure without human intervention.

This reframe produces a system with fundamentally different capabilities than any existing wireless power platform. It also produces a set of novel research contributions that are described in Section 1.4.

1.3 The Meridian Architecture

Meridian is a four-layer deterministic wireless energy routing architecture. The four layers are:

Meridian Core (MC) —” the foundational coordinate system. MC defines node identity, spatial orientation, charge-state vector representation, and deterministic timing primitives. It provides the universal reference frame from which all routing, harvesting, and alignment decisions are derived. Every other Meridian layer depends on MC for its coordinate substrate.

Meridian Flow Engine (MFE) —” the energy-flow logic layer. MFE converts charge-state data from MC into deterministic routing decisions. It manages burst-mode transmission scheduling, supercapacitor threshold evaluation, and loss-bounded energy propagation modeling. MFE ensures that energy moves predictably through the network and that no transmission is initiated without sufficient charge and a confirmed downstream path.

Meridian Mesh Fabric (MMF) —” the multi-hop coordination layer. MMF maintains routing tables, link-state awareness, and neighbor-graph topology across the full mesh. It supports bi-directional energy movement, relay participation scheduling, and mesh-level optimization. MMF integrates DRMA logic for packet-like energy handling and resolves conflicts between nodes to prevent chaotic energy oscillation.

Meridian Transmission Layer (MTL) —” the physical execution layer. MTL controls beam steering, RF emission shaping, safety gating, and closed-loop delivery confirmation. It receives transmission directives from MMF and MFE, executes them against the physical hardware, and returns delivery confirmation feedback that closes the control loop. MTL aligns with DWER for all directional wireless power transfer operations.

These four layers are vertically coupled and governed by five global invariants enforced continuously by Lume-X: no uncontrolled discharge, no ambiguous routing, no non-deterministic flow, no unsafe transmission events, and no mesh-level oscillation.

1.4 Novel Research Contributions

This work makes eleven distinct contributions not present in current wireless power or mesh networking literature:

- Self-Healing Deterministic Control Layer (SHDCL) —” the first application of a self-healing runtime to wireless energy routing, providing continuous autonomous recovery across all three physical subsystems.
- Energy Routing as a Network Protocol Stack —” DRMA formally modeled as an IP-addressed, routing-protocol-governed, QoS-guaranteed power mesh. The first application of network protocol stack architecture to wireless energy delivery.
- Formal Determinism Definition for Wireless Power —” the $P_{\min}/r/\sigma^2/d$ framework as a precise, measurable, reproducible standard for determinism in wireless power delivery. No equivalent standard exists in current literature.
- Bi-Directional Energy Flow in a Mesh —” dynamic energy flow reversal based on real-time mesh state, with surplus nodes supplying depleted nodes autonomously.
- Natural Language Human-Machine Interface for Power Mesh Control —” plain English and voice control of a wireless energy routing system via Lume-X. No existing wireless power system has this interface layer.
- Formal Security Threat Model for Deterministic Energy Routing —” the first formal definition of the attack surface for a wireless power routing system, with Guardian Security as the defense framework.

- Burst-Mode Harvesting as a First-Class Architectural Layer —” DAEH supercapacitor burst-mode integrated as a named architectural layer coordinated by a deterministic control system.
- Dual-Mode Mechanical Harvester Design —” combined triboelectric and piezoelectric elements on a single mechanical substrate, driven simultaneously by the same environmental excitation.
- Biological Safety Layer as a Named System Component —” formally defined, continuously monitored, automatically enforced RF exposure layer with real-time power density calculation.
- EMP and Geomagnetic Storm Resilience Architecture —” formal analysis of inherent structural resistance to electromagnetic pulse and geomagnetic events, with explicit hardening measures.
- Phase-Change Material Storage for Relay Nodes —” higher energy density burst-mode storage for nodes adjacent to heat sources, underexplored in wireless power relay literature.

1.5 Scope and Honest Limitations

This paper presents a formal theoretical and architectural framework. No experimental data exists at time of writing. Phase 1 hardware validation —” demonstrating directional gain of at least 10 dB under beam alignment versus 45-degree off-axis conditions, with result repeatability across ten consecutive trials —” is the immediate next step before any publication at a venue requiring experimental results.

Current development scale powers low-power devices: sensors, wearables, IoT nodes, and microcontrollers. Claims about powering critical infrastructure, hospitals, or building-scale systems are long-horizon goals contingent on solving open efficiency and energy-density problems that remain active research challenges across the field. Those goals are identified explicitly in Section 10 as the long-term vision. They are not presented as near-term capabilities.

1.6 Paper Organization

Section 2 reviews prior work in wireless power transfer, mesh networking, ambient energy harvesting, and deterministic systems. Section 3 presents the Meridian system overview and the vertical coupling model. Sections 4 through 7 define each of the four Meridian layers formally. Sections 8 through 10 describe integration with DAEH, DRMA, and DWER respectively. Section 11 defines the global invariant enforcement model. Section 12 describes the Lume-X control layer and Guardian Security integration. Section 13 addresses implementation considerations. Section 14 defines the evaluation framework for Phase 1 through Phase 3 experimental validation. Six appendices provide formal definitions, state machine specifications, mathematical models, topology scenarios, the safety model, and a reference implementation blueprint.

2. Background and Related Work

2.1 Wireless Power Transfer

The technical lineage of wireless power transfer spans more than a century, beginning with Tesla's resonance experiments at the turn of the twentieth century and continuing through microwave power transmission, resonant inductive coupling, and near-field charging standards. Each generation advanced one dimension of the problem —” efficiency, range, or convenience —” without addressing the fundamental absence of routing, delivery confirmation, or determinism.

Microwave Power Transmission (MPT). William Brown at Raytheon demonstrated the first sustained microwave power link in the 1960s and 1970s, achieving approximately 54% rectenna conversion efficiency at fixed-point operation. Brown's work established the rectenna as the canonical receiver architecture for RF-to-DC conversion and demonstrated that directed microwave transmission was technically feasible. NASA's Solar Power Satellite concept in the 1970s extended this work to orbital scale, with theoretical analysis suggesting end-to-end system efficiencies of 5—“10% from solar collection to Earth surface delivery. Brown's rectenna design is the direct technical ancestor of the DWER receiver module.

Resonant Inductive Coupling. Kurs et al. at MIT (2007) demonstrated mid-range wireless power transfer using strongly coupled magnetic resonators, achieving approximately 40% efficiency over 2 meters. This work revived commercial interest in wireless power and led directly to the WiTricity platform. Resonant inductive coupling operates at low frequency (tens of kilohertz to low megahertz range), is highly sensitive to receiver orientation and alignment, and does not scale to mesh architectures without interference between adjacent resonant pairs. Range is fundamentally limited by the resonator coil diameter.

Phased-Array Beamforming. Phased-array antenna technology is mature in radar, 5G massive MIMO, satellite communications, and sonar. The core technique —” controlling the relative phase of signals fed to individual antenna elements to steer a combined beam in a desired direction —” enables precise, electronically steerable, directional control of RF energy without mechanical moving parts. Modern phased-array systems at 28 GHz achieve beam widths of less than two degrees and electronic steering over the full hemisphere. DWER applies this technique to power delivery at 2.4 GHz and 5.8 GHz with 4 —“16 element arrays.

Commercial Systems. WiTricity (resonant inductive coupling) operates at short range with no beam steering, no closed-loop power confirmation, and no mesh capability. Ossia Cota (RF-based) broadcasts omnidirectionally with no deterministic routing, no closed-loop confirmation, and no mesh. Energous

WattUp (RF-based) broadcasts with limited directionality, no formal determinism standard, and no mesh architecture. None of these systems formally define determinism, none provide closed-loop delivery confirmation, and none implement routing.

2.2 Mesh Networking and Multi-Hop Routing

The network protocol stack for data communications is one of the most thoroughly validated engineering constructs in existence. TCP/IP, OSPF link-state routing, BGP path-vector routing, and IEEE 802.11 mesh extensions have been deployed at global scale, studied extensively, and optimized over five decades. The insight that this framework applies to energy delivery —” with power nodes as IP devices, routing tables governing energy path selection, and QoS constraints applying to watt delivery —” is the highest-novelty contribution of this work.

Prior work on multi-hop wireless power relay has been limited. Fan et al. (2013) demonstrated two-hop resonant inductive relay chains with efficiency degradation per hop. Zhong et al. (2015) extended this to longer chains, characterizing cumulative losses. These systems use resonant inductive coupling and do not implement routing, addressing, path selection, or failure recovery. They are relay chains, not meshes. The distinction is fundamental: a relay chain has no routing intelligence, no topology awareness, no QoS, and no failure recovery. A mesh implements all four.

Software-defined networking (SDN) provides a conceptual parallel: separating the control plane from the data plane to enable dynamic, software-driven network management. Meridian implements the analogous separation for energy networks. The Lume-X control runtime is the control plane. DRMA, DWER, and DAEH are the data plane. The separation enables dynamic routing decisions without modifying the physical transmission hardware.

2.3 Ambient Energy Harvesting

Ambient energy harvesting for wireless sensor networks has been studied extensively since the early 2000s, driven by the need for battery-free IoT nodes. The major modalities —” photovoltaic, thermal gradient (TEG), piezoelectric vibration, RF rectenna, and triboelectric (TENG) —” have been characterized individually in hundreds of published works. Roundy et al. (2004) established the foundational power density analysis framework. Mitcheson et al. (2008) provided a comprehensive survey of kinetic energy harvesters. Vullers et al. (2010) benchmarked all major modalities against deployment requirements.

What does not exist in the literature is the integration of ambient energy harvesting as a first-class architectural layer in a wireless energy routing system. Existing harvesting systems are designed as isolated power sources for individual sensor nodes. They are not coordinated across a mesh, not governed by a deterministic control system, and not integrated with a directed beam supplementation layer. DAEH

introduces all three of these integrations. The burst-mode supercapacitor architecture —” in which low harvest rates do not prevent transmission but only increase the interval between bursts —” is the enabling mechanism.

2.4 Deterministic Systems and Timing Models

Determinism in computing and control systems refers to the property that a given input always produces the same output, independent of external state or timing variability. Hard real-time systems —” industrial controllers, medical devices, automotive safety systems, avionics —” require deterministic response times and have formal tools for specifying and verifying them: rate-monotonic scheduling, earliest-deadline-first scheduling, and formal timing analysis.

No equivalent formal framework exists for wireless power delivery. The term "deterministic" appears in wireless power literature informally and inconsistently, sometimes meaning "predictable enough for practical use," sometimes meaning "repeatable under controlled laboratory conditions," and sometimes meaning nothing more than "working as intended." None of these usages constitute a testable, measurable standard.

The $P_{\min}/r/\sigma^2/d$ determinism framework introduced in this work is the first formal definition of determinism for wireless power delivery. It requires four specific measurements: received power $P_{\min}$ (mW), spatial uncertainty radius r (cm), trial variance σ^2 (%), and transmission distance d (m). A result that omits any of these four values is not a deterministic result under this standard. The framework enables true experimental comparison between systems and provides a reproducible basis for regulatory analysis.

2.5 Identified Gaps Addressed by This Work

The review above identifies five structural gaps in current literature that Meridian addresses directly:

- No wireless power system implements formal routing. Meridian implements the full network protocol stack architecture.
- No wireless power system defines or enforces a formal determinism standard. Meridian defines $P_{\min}/r/\sigma^2/d$.
- No wireless power system integrates a self-healing control runtime. Meridian integrates Lume-X with SHDCL.
- No wireless power system integrates ambient harvesting as a coordinated architectural layer. Meridian integrates DAEH.
- No wireless power system has a formal security threat model. Meridian integrates Guardian Security with a complete threat taxonomy.

3. System Overview

3.1 Architecture Summary

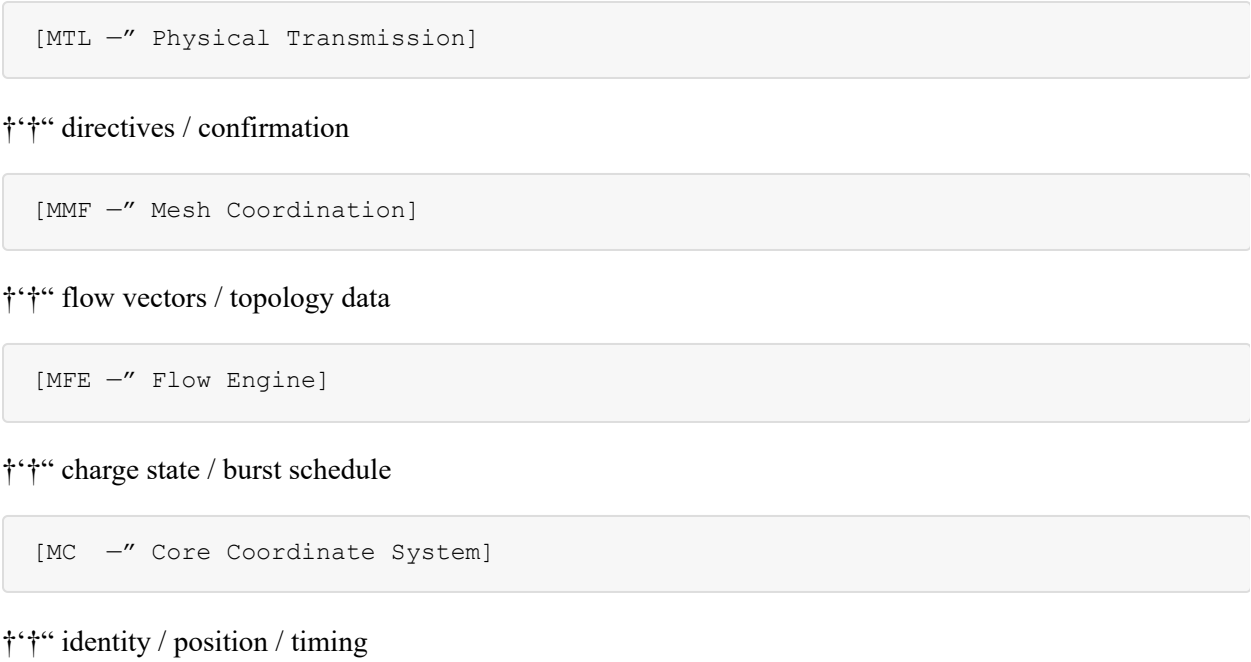
Meridian is best understood as an operating system for deterministic wireless energy. As an operating system, it does not perform the work itself — it governs, coordinates, and enforces invariants across the systems that do. DWER performs directional beam transmission. DRMA performs multi-hop mesh routing. DAEH performs ambient energy harvesting and burst-mode storage. Meridian provides the coordinate system, the flow logic, the mesh coordination, and the transmission governance layer that makes all three systems operate coherently as a single deterministic architecture.

The four Meridian layers map to the four primary concerns of this governance function:

MC (Meridian Core): What does the network know about itself? Node identity, spatial position, charge state, and time. MFE (Meridian Flow Engine): How does energy decide where to go? Flow selection, burst scheduling, threshold evaluation. MMF (Meridian Mesh Fabric): What is the network topology and how does energy traverse it? Routing tables, link state, multi-hop coordination. MTL (Meridian Transmission Layer): How does energy physically leave a node? Beam steering, emission shaping, safety gating, delivery confirmation.

3.2 Vertical Coupling Model

The four layers are vertically coupled, meaning each layer depends on the layer below it and provides services to the layer above it. The coupling is strict and directional:



MC is the foundation. Every other layer uses MC's coordinate substrate to represent nodes, positions, and states. MC has no dependencies on higher layers.

MFE consumes charge-state data from MC and produces routing decisions and burst schedules. MFE does not know about physical topology —” it knows about energy states and thresholds.

MMF consumes flow-potential vectors from MFE and converts them into concrete routing paths using its topology knowledge. MMF knows about the network graph but not about individual node charge states —” those come from MFE.

MTL consumes transmission directives from MMF and executes them against the physical hardware. MTL knows about RF emission, beam steering, and safety constraints, but not about routing logic —” that comes from MMF.

3.3 Integration with DWER, DRMA, and DAEH

Meridian is the coordination layer. DWER, DRMA, and DAEH are the execution subsystems.

Meridian †” DAEH. MC receives charge-state vectors from DAEH nodes continuously. MFE evaluates these vectors against burst thresholds and schedules burst-mode releases. DAEH executes the burst on schedule and reports completion back to MC.

Meridian †” DRMA. MMF provides DRMA with flow-potential vectors, directional preferences, and relay eligibility states. DRMA returns link-state data, neighbor-graph topology, and multi-hop feasibility assessments. MMF uses this to compute deterministic flow paths and prevent oscillation.

Meridian †” DWER. MTL issues transmission directives to DWER specifying emission direction, power level, timing window, and safety constraints. DWER executes the directed beam and returns closed-loop confirmation, beam-steering feedback, and delivery verification. MTL uses this confirmation to close the power delivery control loop.

3.4 The Self-Healing Deterministic Control Layer

The Self-Healing Deterministic Control Layer (SHDCL) is not one of the four Meridian layers. It is a property of the Lume-X runtime that operates across all four layers simultaneously. SHDCL means that the Lume-X runtime monitors every active process, loop, and assertion across all four Meridian layers in real time. When any process drops, stalls, or enters an anomalous state, SHDCL detects the event within one control cycle (less than 14 ms at 73 Hz), executes a defined recovery sequence, and logs the event with full timestamp, duration, cause, and recovery path.

SHDCL eliminates the hard-failure class from the architecture entirely. In a conventional control system, an unhandled failure mode causes a hard fault with no recovery path. In Meridian with Lume-X, no failure mode is unhandled: the runtime itself is the recovery mechanism. The specific recovery behavior for each

layer is defined in Section 12.

3.5 System-Level Invariants

Five invariants are enforced globally and continuously by Lume-X across all four Meridian layers:

INV-1 —” No Uncontrolled Discharge. No node transmits energy without a confirmed downstream receiver, a valid routing path, and sufficient charge in MC state. Any attempt to transmit without these three conditions triggers GATE in MTL.

INV-2 —” No Ambiguous Routing. Every energy packet has exactly one active route at any given time. MMF ensures this through exclusive path locking during transmission windows. No two routing table entries point to the same destination with equal cost without a deterministic tie-breaking rule.

INV-3 —” No Non-Deterministic Flow. All energy movement is governed by the $P_{\min}/r/\sigma^2/d$ framework. Every flow event is logged. Statistical validation runs continuously against the rolling 20-trial window for each active link.

INV-4 —” No Unsafe Transmission Events. MTL enforces the Biological Safety Layer continuously. No beam fires without a confirmed, located receiver. No beam fires if the exclusion zone contains detected biological presence. No beam fires if calculated power density at any monitored point exceeds the ICNIRP threshold.

INV-5 —” No Mesh Oscillation. MMF monitors energy flow direction across all active links. Any oscillatory pattern —” energy cycling between two nodes without net forward progress —” is detected by MMF and resolved by MFE through burst schedule adjustment or forced path selection.

4. Meridian Core (MC): The Energy Coordinate System

4.1 Role and Purpose

Meridian Core is the lowest layer of the architecture and the substrate on which every other layer depends. MC answers one fundamental question for every node in the network: what is the current state of this node? That state comprises four components: identity, spatial orientation, charge-state vector, and timing primitives.

MC does not make routing decisions, does not manage physical transmission, and does not coordinate the mesh. It maintains the coordinate system that makes all of those operations possible. Every flow decision in MFE, every routing calculation in MMF, and every transmission directive in MTL is ultimately a function of MC state. If MC state is incorrect or stale, every dependent layer produces incorrect outputs. MC state integrity is therefore the foundational correctness requirement of the entire architecture.

4.2 Node Identity

Every Meridian node has a unique identifier that is fixed at deployment and does not change during the node's operational lifetime. Node identity in MC is a 64-bit vector structured as:

[Network_ID : 16 bits][Zone_ID : 16 bits][Node_Type : 8 bits][Node_Serial : 24 bits]

Network_ID identifies the Meridian network instance. Multiple independent Meridian deployments can coexist in the same physical space by using different Network_IDs. Nodes with different Network_IDs do not route energy to each other.

Zone_ID identifies a logical subdivision of the network —” a room, floor, building section, or operational region. Zone boundaries are defined at deployment and stored in every node's routing table. Zone-local routing is preferred over cross-zone routing when the destination is local, reducing mesh-wide broadcast requirements.

Node_Type encodes the node's functional role: source node (0x01), relay node (0x02), destination node (0x03), harvesting relay node (0x04, combines relay and DAEH functions), or gateway node (0x05, bridge between network zones or between Meridian and external control systems).

Node_Serial is a unique per-node serial number assigned at manufacture. Combined with Network_ID and Zone_ID, Node_Serial guarantees global uniqueness within the Meridian addressing space.

4.3 Spatial Orientation

Each node maintains a spatial state vector S representing its position and orientation in the deployment coordinate frame:

```
S = [x, y, z,  $\theta_{\text{azimuth}}$ ,  $\theta_{\text{elevation}}$ , "x", "y", "z"]
```

Where (x, y, z) is position in meters relative to the network origin, $(\theta_{\text{azimuth}}, \theta_{\text{elevation}})$ is the node's transmission antenna pointing direction in degrees, and $(\text{"x"}, \text{"y"}, \text{"z"})$ is the velocity vector in meters per second for mobile nodes (zero for fixed nodes).

Position is updated continuously from the UWB localization subsystem (Decawave DWM1001 or equivalent). Update rate is a minimum of 10 Hz for static nodes and 20 Hz for mobile nodes. MC holds the last confirmed position with a timestamp. MFE and MTL use the position age to assess freshness —” a position reading older than 500 ms is flagged as stale and triggers a re-acquisition request before any transmission is permitted.

For fixed nodes, position is initialized at deployment and confirmed by the UWB subsystem. For mobile destination nodes, position tracking is continuous and the MC state is updated at the localization update rate. The spatial uncertainty radius r , used in the determinism framework, is derived directly from the

UWB localization accuracy: the DWM1001 module provides approximately 10 cm accuracy under nominal indoor conditions, so $r = 10$ cm is the baseline spatial uncertainty parameter for Phase 1—“2 experiments.

4.4 Charge-State Vector

The charge-state vector C encodes the complete energy state of a node:

```
C = [V_cap, V_min, V_burst, V_max, dV/dt, mode, harvest_rate, cycles]
```

Where:

```
V_cap is the current supercapacitor voltage (volts)
V_min is the minimum operating voltage below which the node's microcontroller enters DEPLETED mode
V_burst is the burst threshold voltage above which the node has sufficient stored energy to transmit
V_max is the rated maximum voltage of the supercapacitor
dV/dt is the current charge rate (volts per second), estimated over the preceding 10 seconds
mode encodes the node's current operational mode: HARVEST (charging), READY (above V_min), TRANSMITTING (above V_burst)
harvest_rate is the measured ambient harvest power in microwatts, averaged over the last 10 seconds
cycles is the total charge-discharge cycle count since deployment, used for supercapacitor health monitoring
```

MFE reads V_{cap} and V_{burst} to make burst scheduling decisions. MMF reads $mode$ to assess relay eligibility —“ a node in DEPLETED mode is not eligible to relay. MTL reads $mode$ before executing any transmission —“ TRANSMITTING must be set by MFE before MTL fires the beam, and MTL clears it upon completion.

4.5 Timing Primitives

Deterministic behavior across the mesh requires that all nodes share a common time reference. MC maintains a network time synchronization layer using a precision timestamp protocol (PTP-equivalent, sub-millisecond accuracy) distributed across the mesh via the control plane. All timing-sensitive operations —“ TDMA slot assignment, burst scheduling, beam handoff windows —“ are expressed as absolute network timestamps rather than local node intervals.

MC exposes three timing primitives to higher layers:

```
T_now —“ current network time in microseconds since epoch.
```

$T_{slot}(node_id)$ —“ the start time of the current TDMA transmission window assigned to the specified node. Returns the next scheduled window if no current window is active.

$T_{burst}(node_id)$ —” the scheduled burst transmission time for the specified node, as computed by MFE based on current V_{cap} and projected charge accumulation rate.

The 73 Hz confirmed cycle rate of the Lume-X runtime corresponds to a 13.7 ms control cycle. This is the fundamental timing resolution of the Meridian control system. All transmission windows, beam handoffs, and burst events are aligned to control cycle boundaries to ensure that the runtime can observe and respond to any state change within one cycle.

5. Meridian Flow Engine (MFE): Deterministic Energy Movement

5.1 Role and Purpose

The Meridian Flow Engine is the decision layer. It consumes the coordinate state provided by MC and converts it into deterministic instructions about when, where, and how much energy to move. MFE does not know about physical transmission hardware —” that is MTL's domain. MFE does not know about mesh topology —” that is MMF's domain. MFE knows about energy states, thresholds, and flow rules.

The fundamental question MFE answers is: given the current charge state of every node in the network, which nodes should transmit, to which downstream nodes, and when? MFE's answer to that question must be deterministic —” identical input states must produce identical routing decisions every time, without exception.

5.2 Flow-Selection Algorithm

MFE evaluates each eligible source node against a priority-ordered set of flow selection criteria:

Step 1 —” Eligibility check. A node is eligible to transmit if and only if:

```
V_cap ≥ V_burst (sufficient charge for one burst)
mode % DEPLETED and mode % TRANSMITTING
```

A valid downstream path exists in the MMF routing table $T_{slot}(node_id)$ is within the current TDMA window or within the next window boundary INV-1 through INV-5 pass for the proposed transmission. Any failure at Step 1 removes the node from the candidate set without escalation.

Step 2 —” Priority scoring. Each eligible node receives a priority score P :

```
P = w1 - (V_cap - V_burst) / (V_max - V_burst) [charge surplus fraction]
```

+ w₂ — (destination_priority) [destination urgency] + w₃ — (link_quality) [downstream link reliability] + w₄ — (1 - mesh_load) [mesh congestion factor]

Default weights: $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.2$, $w_4 = 0.1$. Weights are configurable per deployment via Lume-X.

Step 3 —” Transmission scheduling. Nodes are ordered by priority score. The highest-priority eligible node is scheduled first, followed by the next-highest in the next available TDMA slot, continuing until all eligible nodes have scheduled transmissions or all TDMA slots in the current window are filled.

Step 4 —” Conflict resolution. If two nodes share a downstream path segment —” that is, both need to route energy through the same relay node —” they cannot transmit in the same TDMA slot. MFE detects this condition using the MMF topology graph and serializes the conflicting transmissions into separate slots.

5.3 Burst-Mode Scheduling

Burst-mode scheduling is the mechanism that makes ambient energy harvesting architecturally viable for wireless relay operation. The core insight is that the supercapacitor completely decouples the harvest rate from the transmission rate. A node harvesting at 50 μW that needs to fire a 100 mW burst for 10 ms requires 20 joules of storage time —” meaning it accumulates for 20 seconds between bursts. This is not a failure mode. It is deterministic operation at the harvest rate available in that environment.

MFE's burst scheduler operates as follows for each DAEH node:

```
observe V_cap from MC
if V_cap >= V_burst:
  compute T_burst = T_now + alignment_delay
  assign TDMA slot starting at T_burst
  set node.mode = READY
  notify MMF of pending transmission
```

on slot arrival:

```
set node.mode = TRANSMITTING
```

signal MTL to execute burst

```
await MTL confirmation
set node.mode = HARVEST
```

log burst event with $V_{\text{cap_before}}$, $V_{\text{cap_after}}$, T_{burst} , T_{actual} , $P_{\text{delivered}}$


```

else:
    compute T_ready = T_now + (V_burst - V_cap) / dV_dt
    schedule re-evaluation at T_ready
    if dV_dt < dV_dt_min:

```

flag node for DWER supplementation

```

notify MMF to activate directed beam toward node

```

The supplementation trigger —” when harvest rate falls below the minimum threshold for viable burst scheduling —” automatically activates a DWER directed beam from the nearest available source node. This closes the three-layer energy loop: DAEH harvests when possible, DWER supplements when necessary, and MFE manages the transition between modes transparently.

5.4 Loss-Bounded Flow Modeling

MFE maintains a loss model for every active link in the network. The loss model is a rolling statistical estimate derived from the delivery confirmation feedback returned by MTL after each completed transmission burst. For each link (source $\dagger$ destination):

$L(\text{link}) = 1 - (P_{\text{received}} / P_{\text{transmitted}})$ $\sigma^2(\text{link}) = \text{variance of } L(\text{link}) \text{ over the preceding } N \text{ observations}$ ($N \geq 20$)

MFE uses $L(\text{link})$ and $\sigma^2(\text{link})$ to evaluate whether a link meets the determinism standard:

$P_{\text{received}} \geq P_{\text{min}}$ (minimum acceptable received power) $\sigma^2(\text{link}) < \sigma^2_{\text{max}}$ (maximum acceptable variance, default 10%) A link that fails either criterion is flagged as degraded. MFE notifies MMF of the degraded state. MMF responds by computing an alternate path that avoids the degraded link. If no alternate path exists, Lume-X escalates to a SHDCL recovery event and logs the failure.

The loss model also feeds the efficiency break-even calculation. For the multi-hop architecture to outperform a single long-range source, the per-hop conversion losses must be smaller than the free-space path loss savings achieved by the shorter hops. MFE computes this comparison continuously at each operating frequency and updates the routing preference weights accordingly.

6. Meridian Mesh Fabric (MMF): Multi-Hop Coordination

6.1 Role and Purpose

Meridian Mesh Fabric is the topology layer. It maintains the complete map of the energy network: which nodes exist, where they are, which links are active, and what the current state of each link is. From this map, MMF computes routing paths, enforces path exclusivity during transmission windows, and updates

the mesh in response to topology changes.

MMF is where the network protocol stack analogy is most direct. The routing tables in MMF are structurally identical to the forwarding tables in a packet-switched data network. The link-state protocol used to distribute topology updates is directly analogous to OSPF. The multi-hop path selection algorithm is a variant of Dijkstra's shortest-path algorithm adapted for energy delivery where "shortest" means "minimum expected delivery loss" rather than "minimum hop count" or "minimum latency."

6.2 Routing Tables

Each Meridian node maintains a routing table R with one entry per reachable destination:

$R[\text{destination_id}] = \{$

```
next_hop_id,
```

```
path_cost, // expected delivery loss to destination
hop_count, link_quality, // rolling reliability estimate for
first hop
alternate_next_hop, // pre-computed fallback path
last_updated, // network timestamp of last
topology update
state // ACTIVE, DEGRADED, FAILED }
```

Routing tables are populated at network initialization using a mesh discovery protocol: each node broadcasts a `NODE_ANNOUNCE` message containing its full MC state. Receiving nodes add the announcing node to their neighbor graph and propagate the announcement one hop further. After three advertisement cycles (typically less than 500 ms), the full mesh topology is known to every node.

Routing tables are updated continuously as link states change. Link state changes are announced using a `LINK_STATE_UPDATE` message that propagates through the mesh using flood-limited broadcast (TTL decremented at each hop, discard at `TTL=0`). Every node that receives a `LINK_STATE_UPDATE` merges it into its local topology model and recomputes routing table entries that use the affected link.

6.3 Path Selection

MMF uses a modified Dijkstra algorithm for path selection. The cost function minimizes expected delivery loss across the full path rather than hop count:

```
PathCost(path) =  $\sum L(\text{link}_i)$  for all links  $i$  in path
```

+ penalty — `hop_count` // small per-hop penalty to prefer fewer hops at equal loss

The hop count penalty is configurable (default: 0.02 per hop additional cost fraction). This ensures that a two-hop path with 15% total expected loss is preferred over a three-hop path with 14% total expected loss — the marginal loss benefit is not worth the added complexity and synchronization overhead of an extra hop.

Alternate paths are pre-computed for every destination at routing table population time. When a primary path degrades, MMF switches to the pre-computed alternate without requiring a full re-computation. Full re-computation occurs in the background and updates the pre-computed alternate for future use.

6.4 Bi-Directional Energy Flow

One of the distinguishing capabilities of the Meridian mesh is bi-directional energy flow. Standard relay chain architectures are unidirectional: energy flows from source to destination along a predetermined path. In Meridian, energy can flow in either direction on any link, and can flow from any node with surplus charge to any node with depleted charge, regardless of the original source-destination pairing.

Bi-directional flow is governed by MMF's surplus-deficit monitor, which runs continuously:

```
for each node n:
  if C[n].V_cap > V_burst + surplus_threshold:
```

n is eligible as energy donor

```
  if C[n].V_cap < V_min + deficit_threshold:
```

n requires energy donation

```
  for each (donor, recipient) pair:
    if path exists between donor and recipient:
      compute flow_reversal_benefit = V_cap_delta / path_cost
      if flow_reversal_benefit > reversal_threshold:
        schedule reversed-flow burst from donor to recipient
```

The flow reversal mechanism prevents nodes from becoming stranded with no energy in a mesh where surplus exists elsewhere. It also serves as the foundation for the mesh self-balancing property: over time, the mesh drives toward an equilibrium where no node is deeply depleted and no node is wasting charge capacity by sitting above $V_{\max}$.

6.5 Oscillation Prevention

Mesh oscillation occurs when energy cycles between two nodes without net forward progress —” for example, node A transmits to node B, which immediately re-transmits back to node A because node A's charge dropped while node B's rose. Left uncorrected, this wastes energy and prevents delivery to the intended destination.

MMF detects oscillation using a directed flow history buffer maintained per link. If a link shows alternating flow direction across three consecutive transmission windows without net delivery progress to the destination, MMF flags the condition as oscillatory and applies a flow stabilization rule: one direction

is locked as the preferred direction for the next ten transmission windows, with the preference based on the net destination progress metric.

Lume-X enforces INV-5 continuously. Any oscillatory condition not resolved by MMF's automatic mechanism within two control cycles is escalated to a SHDCL recovery event with full logging.

6.6 Topology Change Handling

The mesh must handle three types of topology change: node addition, node failure, and link degradation.

Node Addition. A new node broadcasts `NODE_ANNOUNCE`. All neighbors update their topology models. MMF re-evaluates whether the new node improves any existing path and updates routing tables if it does. The new node receives a full routing table from its nearest neighbor.

Node Failure. When a node fails to respond within two consecutive control cycles, its neighbors mark it as `PRESUMED_FAILED` and broadcast a `LINK_STATE_UPDATE`. All affected routing table entries switch to their pre-computed alternate paths. The failed node is removed from the topology model. If the failed node was on the primary path to any destination with no viable alternate, MMF escalates to Lume-X for SHDCL recovery.

Link Degradation. When a link's rolling delivery loss exceeds the degradation threshold, MMF marks it as `DEGRADED` and moves affected traffic to alternate paths. The degraded link remains monitored —” if it recovers, its status returns to `ACTIVE` and the routing preference shifts back.

7. Meridian Transmission Layer (MTL): Directional Wireless Delivery

7.1 Role and Purpose

The Meridian Transmission Layer is the interface between the Meridian software architecture and the physical world. Everything above MTL operates in the domain of vectors, routing tables, charge states, and scheduling. MTL translates those abstractions into specific hardware instructions: beam steering angles, RF emission power levels, timing windows, and safety gate commands. It is also the layer that receives physical reality back from the hardware —” delivery confirmation, beam steering feedback, and safety monitoring data —” and converts those physical measurements back into the coordinate language that Meridian's higher layers understand.

7.2 Beam Steering and Emission Shaping

MTL controls the phased-array transmitter through a beam steering command interface. Each beam steering command specifies:

```
BeamCommand = {
```

```
target_node_id, // destination node identifier target_position, // S vector from MC —" position +
uncertainty radius power_level_mW, // transmit power in milliwatts (never exceeds 1000 mW = 1W)
beam_width_deg, // desired 3 dB beam width frequency_GHz, // operating frequency (2.4 or 5.8 GHz for
Phase 1-2) timing_window, // T_start, T_end from MC timing primitives safety_constraints // inherited
from INV-4, BSL thresholds }
```

The phased-array driver translates BeamCommand into per-element phase and amplitude settings. Phase offsets for each of the N antenna elements are computed as:

$$\phi_n = (2\pi / \lambda) \cdot d_n \cdot \sin(\theta)$$

Where λ is wavelength, d_n is the position offset of element n from the array center, and θ is the desired beam steering angle. At 2.4 GHz ($\lambda \approx 12.5$ cm) with 4-element half-wavelength spacing, this produces a beam steering range of approximately ± 60 degrees from broadside.

Emission shaping —" controlling the sidelobe structure of the radiated pattern —" is achieved by applying amplitude taper across the array elements (Taylor or Chebyshev window). Sidelobe reduction reduces ambient RF exposure in non-beam directions and improves the effectiveness of the RF-absorptive node enclosure shielding.

7.3 Safety Gating

MTL implements a five-condition safety gate. All five conditions must be satisfied before any beam is permitted to fire:

- Gate Condition 1 —" Receiver confirmed. A valid, current (less than 500 ms old) position reading exists for the target node. Target node is in READY or HARVEST mode (not FAILED or DEPLETED).
- Gate Condition 2 —" Path authorized. The transmission was authorized by a valid MMF routing table entry for the target node. No unauthorized transmissions.
- Gate Condition 3 —" Exclusion zone clear. The localization subsystem confirms no biological presence within the defined exclusion volume around the beam path. The exclusion volume is defined as a cylinder of radius 0.5 m around the beam centerline for the full path length between transmitter and receiver.
- Gate Condition 4 —" Power density within limits. Real-time calculation of power density at all monitored points in the environment confirms values below ICNIRP reference levels (2 W/m² at 2.4 GHz for the general public, 10 W/m² for occupational exposure).
- Gate Condition 5 —" Charge sufficient. $V_{cap} \geq V_{burst}$ as confirmed by current MC state. Prevents dry-fire transmission attempts.

If any gate condition fails, MTL issues a `GATE_REJECT` event to Lume-X with the specific failed condition identified. Lume-X logs the event, notifies MFE to reschedule the transmission, and initiates the appropriate SHDCL recovery sequence if the failure is persistent.

7.4 Closed-Loop Delivery Confirmation

MTL does not consider a transmission complete until delivery is confirmed. After firing the beam, MTL awaits a `DELIVERY_CONFIRM` message from the receiver node containing:

```
DeliveryConfirm = {
```

```
source_node_id, receiver_node_id, P_received_mW, // measured received power V_cap_after, //
supercapacitor voltage after burst receipt position_at_receipt, // actual receiver position at time of delivery
timestamp }
```

MTL computes the delivery efficiency $\eta = P_{\text{received}} / P_{\text{transmitted}}$ and logs it against the link's rolling statistics model in MFE. If no `DELIVERY_CONFIRM` arrives within the confirmation timeout window (default: $T_{\text{end}} + 100$ ms), MTL treats the transmission as failed and initiates a SHDCL retry sequence.

7.5 Deterministic Transmission Windows

MTL enforces strict timing compliance. Each transmission is assigned a window $[T_{\text{start}}, T_{\text{end}}]$ by MC's timing primitives. MTL will not fire before T_{start} (beam alignment preparation begins at $T_{\text{start}} - 50$ ms). If the beam is not confirmed aligned before T_{start} , the window is missed and the transmission is rescheduled. If T_{end} is reached before delivery confirmation is received, the beam is cut and the incomplete delivery is logged as a partial delivery event.

Strict window enforcement is what makes the TDMA-style synchronization robust. No node transmits outside its assigned window. Adjacent nodes cannot interfere with each other because their windows do not overlap. The Lume-X runtime enforces window boundaries as hard timing invariants.

8. Ambient Harvesting Integration (DAEH)

8.1 Overview

DAEH —” Deterministic Ambient Energy Harvesting —” is the node autonomy layer of the Meridian architecture. Its function is to reduce the dependence of relay nodes on wired power inputs by extracting energy from naturally occurring environmental sources, storing it in supercapacitors, and feeding it into the mesh via burst-mode transmission coordinated by MFE.

DAEH is not presented as a solution to the energy supply problem. Ambient energy densities are measured in microwatts per square centimeter. They cannot sustain continuous RF transmission at relay power levels. What DAEH provides is node operational autonomy for low-power logic functions

(microcontroller, localization module, sensors, routing logic), accumulated burst energy for periodic transmission events, and resilience against wired power interruption.

The generator model is the correct mental framework: a generator does not output power the instant fuel arrives —” it accumulates energy and releases it in a controlled, usable form when sufficient energy has accumulated. DAEH nodes do precisely this.

8.2 Harvesting Modalities

RF from Known Sources. Rather than attempting to harvest incoherent ambient RF noise, DAEH targets predictable, stable RF sources: cellular towers, Wi-Fi access points, and broadcast signals. These have known frequencies, stable power levels, and predictable spatial distribution. A rectenna array tuned to these bands collects additive power across all elements.

Critical constraint: ambient RF signals from independent sources are incoherent. They cannot be phase-combined. What is achieved is additive power summation across array elements, not coherent beamforming gain. Received power scales linearly with array area —” not quadratically. No coherent gain claims are made for this modality anywhere in this document.

Indoor Solar and Ambient Light. The highest-yield and most mature indoor harvesting modality. Photovoltaic cells optimized for artificial lighting spectra (warm LED, fluorescent) produce 10—“100 $\mu\text{W}/\text{cm}^2$ under typical indoor illumination. This is the recommended primary harvest source for indoor relay nodes and the first modality to be deployed in Phase 2 experiments.

Thermal Gradient Harvesting (TEG). Thermoelectric generators convert temperature differentials into voltage via the Seebeck effect. Output is 1—“10 $\mu\text{W}/\text{cm}^2$ for small temperature differentials (1—“5°C), scaling with T^2 . Useful for nodes placed near heat sources: electronic equipment, HVAC vents, server racks, exterior walls. Deterministic thermal modeling of the deployment environment significantly improves output stability.

Piezoelectric Vibration Harvesting. Piezoelectric and MEMS harvesters convert mechanical vibration to electrical energy. Output is 10—“100 $\mu\text{W}/\text{cm}^2$ in mechanically active environments. Resonance tuning to the dominant vibration frequency of the specific deployment environment substantially increases yield. Useful in environments with consistent mechanical excitation: HVAC systems, industrial machinery, high-traffic flooring.

Triboelectric Nanogenerator Harvesting (TENG). Triboelectric nanogenerators harvest energy from charge-separation events between materials through friction, contact, vibration, and airflow. Output ranges from 1—“100 $\mu\text{W}/\text{cm}^2$ depending heavily on environmental conditions. TENG harvesting requires high-

impedance charge amplifiers and voltage multiplier circuits. Rectennas are RF-to-DC converters and are not applicable to this modality. These are distinct circuit architectures and must be specified and built separately.

- **Dual-Mode Mechanical Harvester.** A novel design option combining piezoelectric and triboelectric elements on a single mechanical substrate. Since both modalities respond to the same class of mechanical stimuli —” vibration, friction, impact, contact —” the same physical excitation drives both simultaneously. Combined yield is meaningfully higher than either modality alone.

Recommended as a standard design option for nodes in mechanically active environments.

Atmospheric Electrostatic Gradient. The Earth maintains approximately 100—“300 V/m atmospheric electric field near the surface under fair-weather conditions. This is a real physical phenomenon. Practical energy yield at relay node scale is in the nanowatt range under normal conditions, rising during storm activity but not under conditions suitable for deterministic system operation. This modality is classified as a long-term experimental roadmap item only. Strict power density validation required before any claims.

Magnetic Flux Harvesting. Measurable in proximity to power lines and electrical equipment. Nanowatt range in typical indoor environments away from concentrated sources. Long-term experimental roadmap only.

8.3 Honest Power Density Baselines

All DAEH experiments must beat the following established baselines before any efficiency claims are made:

Source	Power Density	Notes
Ambient RF indoor (random)	0.01—“1 $\mu\text{W}/\text{cm}^2$	Highly location-dependent
RF from known sources	1—“100 $\mu\text{W}/\text{cm}^2$	Predictable, targetable
Thermal gradient TEG (small ”T)	1—“10 $\mu\text{W}/\text{cm}^2$	Requires temperature differential
Piezoelectric vibration	10—“100 $\mu\text{W}/\text{cm}^2$	Requires mechanical excitation
Indoor solar / ambient light	10—“100 $\mu\text{W}/\text{cm}^2$	Most mature modality
Triboelectric TENG	1—“100 $\mu\text{W}/\text{cm}^2$	Highly environment-dependent

Harvested power at these densities powers node logic (microcontroller: ~ 10 —“50 μW , UWB localization module: ~ 100 —“500 μW active, 1—“10 μW sleep). RF relay transmission at 100 mW output from a supercapacitor burst requires 1 joule for a 10 ms burst. At 50 μW average harvest rate, charge

accumulation time is approximately 20,000 seconds (5.6 hours) per burst without DWER supplementation. With DWER supplementation during periods of low harvest, charge time is reduced to a configurable minimum.

8.4 Phase-Change Material Storage

Supercapacitors are the primary burst storage technology for Phase 1—“3 experiments. Phase-change thermal storage materials represent a longer-horizon option for nodes deployed near consistent heat sources. Phase-change materials store significantly more energy per unit volume than supercapacitors at comparable cost, releasing it on demand through a thermoelectric generator. For nodes adjacent to server racks, HVAC equipment, or other consistent heat sources, PCM storage could dramatically increase burst energy availability compared to supercapacitor storage alone. This remains a forward-looking research contribution. Phase 1—“3 experiments use supercapacitors exclusively.

8.5 DAEH Experimental Phases

Phase 1 —” Baseline Measurement. Characterize all available harvesting modalities at intended deployment locations. Record power density for each modality across 20 trials. Establish variance baselines per modality. Identify highest-yield modality for the specific environment.

Phase 2 —” Single-Node Harvesting. Deploy chosen modality (recommended: indoor solar as primary) on one relay node. Measure supercapacitor charge rate under real environmental conditions. Calculate time-to-threshold for burst transmission. Validate deterministic repeatability of full charge/discharge cycle.

Phase 3 —” Integrated Mesh Operation. Feed harvested energy into an active DRMA relay node. Demonstrate burst-mode transmission from harvested power alone. Quantify improvement in node autonomy compared to wired baseline. Validate Lume-X burst scheduling against real-time charge state data.

9. Mesh Routing Integration (DRMA)

9.1 Overview

DRMA —” Distributed Relay Mesh Architecture —” is the multi-hop energy routing subsystem. It is the physical implementation of the network protocol stack architecture described in MMF. Where MMF maintains the routing tables and topology model, DRMA provides the physical relay infrastructure: the nodes that receive energy, buffer it, and re-transmit it toward the destination.

Each DRMA relay node contains a directional receiving antenna, a rectenna for RF-to-DC conversion, a supercapacitor buffer, a directional transmitting antenna, a low-power amplifier (less than 1W), a microcontroller, a UWB localization module, and a safety gating circuit. Node cost and size are

intentionally minimized to enable dense deployment.

9.2 Why Short Hops Beat Long Hops

The free-space path loss law governs why multi-hop architectures can outperform single-source long-range transmission:

$$\text{FSPL (dB)} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}(4\pi/c)$$

At 2.4 GHz, FSPL increases by 6 dB every time distance doubles. At 1 meter, FSPL is approximately 40 dB. At 4 meters, it is approximately 52 dB —” 12 dB more loss for four times the distance.

Breaking a 4-meter path into four 1-meter hops saves approximately 12 dB in path loss per hop but incurs per-hop conversion losses. Each relay node incurs:

Per-Hop Stage	Estimated Efficiency
Rectenna RF to DC	50—“70%
Re-transmission DC to RF	60—“75%
Per-hop round-trip conversion	~30—“52%

Converting to dB: per-hop conversion loss is approximately 3—“5 dB. Path loss savings per meter reduction (at 2.4 GHz, 1m vs. 4m) is approximately 12 dB. The break-even occurs when per-hop conversion losses equal path loss savings. At 2.4 GHz, relay nodes begin paying off at inter-node distances beyond approximately 2—“3 meters. Below this threshold, a single source is more efficient. The exact break-even distance must be calculated and validated experimentally for each operating frequency and is a required deliverable of Phase 2.

9.3 The Synchronization Protocol

Inter-node synchronization is the critical design challenge for multi-hop operation. Each relay node must execute in sequence: receive beam from upstream node, convert RF to DC, buffer in supercapacitor, switch to transmit mode, re-transmit to downstream node. Adjacent nodes transmitting simultaneously cause mutual interference that can break the energy path.

Meridian uses a TDMA-style synchronization protocol managed by the Lume-X runtime. Each node in an active relay chain is assigned a non-overlapping transmission window by the MMF routing layer. Window assignments are computed when the routing path is established and updated whenever the path changes.

The TDMA window structure for a three-node relay chain (Source †’ Relay₁ †’ Relay” †’ Destination):

```

Window 1: Source transmits to Relay1 (duration: Tburst)
Gap:      Relay1 confirms receipt, switches to transmit mode (duration: Tswitch + Tconfirm)
Window 2: Relay1 transmits to Relay2 (duration: Tburst)
Gap:      Relay2 confirms receipt, switches to transmit mode (duration: Tswitch + Tconfirm)
Window 3: Relay2 transmits to Destination (duration: Tburst)
Confirmation: Destination reports delivery to control system

```

Total latency per hop is $T_{\text{burst}} + T_{\text{switch}} + T_{\text{confirm}}$. At $T_{\text{burst}} = 10$ ms, $T_{\text{switch}} = 5$ ms, and $T_{\text{confirm}} = 5$ ms, three-hop delivery latency is approximately 60 ms. This is not a real-time power delivery system — it is a scheduled, burst-mode delivery system. Latency is a design parameter, not a failure mode.

9.4 Security Threat Model

A deterministic energy routing system has a formal attack surface. This is the first formal treatment of the security threat model for a wireless power routing system. Identified threat vectors:

- Localization spoofing — false position signals from a malicious node redirect beam energy to unintended targets
- Routing table manipulation — corrupted routing table entries cause energy to bypass critical nodes or loop endlessly
- Beam hijacking — an unauthorized receiver positions itself in the beam path to intercept routed energy
- Denial-of-power attacks — deliberate disruption of mesh routing to starve target devices
- Node impersonation — false charge-state reporting to trigger burst scheduling toward or away from a specific node
- Timing attack — injecting false timing signals to desynchronize TDMA windows and cause collisions

Guardian Security (energy-domain designation: Guardian-E, domain: TrustShield.tech) maps its eleven threat categories directly onto this threat model. The formal mapping between DRMA attack vectors and Guardian Security defensive responses is a required deliverable for Phase 3 and is outlined in Section 12 and Appendix E.

9.5 DRMA Experimental Phases

Phase 1 — Single-Hop Validation. Directional gain demonstrated at relay node hardware. Per-hop efficiency baseline established and documented. Deterministic repeatability verified against the $P_{\text{min}}/r/\sigma^2/d$ standard.

Phase 2 —” Two-Hop Routing. Energy routed through two relay nodes. Inter-node synchronization protocol validated. Cumulative efficiency measured and compared to single-source at equivalent total distance. Break-even distance calculated and confirmed experimentally. Deterministic power delivery verified at final receiver.

Phase 3 —” Dynamic Mesh Routing. Moving receiver tracked through mesh. Hop-to-hop handoff with less than 20% $P_{\min}$ drop during transition. Fallback path routing demonstrated when one node is blocked. Bi-directional energy flow demonstrated with surplus-to-depleted node transfer.

10. Wireless Transmission Integration (DWER)

10.1 Overview

DWER —” Deterministic Wireless Energy Routing —” is the foundational point-to-point energy delivery subsystem. It is the direct ancestor of all wireless energy delivery in the Meridian architecture. Every mesh hop is a DWER link. Every directed supplementation beam from a source node to a depleted relay node is a DWER link. The DRMA mesh is an interconnected set of DWER links governed by Meridian's routing framework.

DWER builds on three established technical traditions —” microwave power transmission, resonant inductive coupling concepts, and phased-array beamforming —” and contributes the closed-loop deterministic control layer that is the primary novel element.

10.2 The Determinism Standard

Every experimental result claiming deterministic delivery must report all four values. A result that omits any of these four is not a deterministic result under the Meridian standard:

Deterministic delivery: for a given transmitter configuration and receiver position, the system guarantees $P_{\min}$ (mW) within spatial uncertainty radius r (cm), with variance σ^2 below threshold, across $N \geq 20$ repeated trials under identical conditions, at distance d (m).

Four mandatory layers achieve this standard:

Localization feedback —” continuous receiver position update at defined intervals (minimum 10 Hz)
 Phase-steering correction —” real-time beam adjustment to maintain alignment
 Power confirmation telemetry —” receiver reports P_{received} back to control system, closing the loop
 Repeatability logging —” all trials logged and statistically validated against the σ^2 threshold

10.3 Efficiency Budget

At 2.4 GHz, less than 1W transmit power, over 1—“3 meters:

Stage	Estimated Efficiency
Power amplifier to antenna	60—“75%
Free-space path loss (1m, focused beam)	10—“30%
Rectenna RF to DC conversion	50—“70%
End-to-end (optimistic)	~15—“25%
End-to-end (conservative)	~3—“10%

These numbers are consistent with published rectenna experiments at similar power and distance. Phase 1 is not about efficiency —” it is about proving directional control exists and is measurable. Efficiency improvement begins in Phase 3.

10.4 Commercial System Differentiation

DWER must be formally differentiated from existing commercial wireless power systems in all publications:

WiTricity —” resonant inductive coupling, short range, no beam steering, no closed-loop power confirmation, no mesh, no formal determinism standard.

Ossia Cota —” RF-based, omnidirectional broadcast, no deterministic routing, no closed-loop confirmation, no mesh architecture, no formal determinism standard.

Energous WattUp —” RF-based, broadcast with some directionality, no formal determinism standard, no mesh architecture, no closed-loop confirmation.

DWER differentiates on all five dimensions: closed-loop power confirmation, formal determinism standard, phased-array beam steering, mesh routing integration, and self-healing control runtime. No commercial system provides all five.

10.5 DWER Experimental Phases

Phase 1 —” Directionality (Immediate Priority).

Received power increases by at least 10 dB when beam aligned vs. 45° off-axis System acquires and locks onto receiver within 5 seconds Result repeatable across 10 consecutive trials with variance less than 15% Phase 2 —” Determinism.

Received power at 1 meter: at least 5 mW from less than 1W transmit Trial-to-trial variance: less than 10% across 20 trials Power curve matches simulated beam pattern within 3 dB Phase 3 —” Routing.

Beam handoff between two receivers in less than 500 ms
 Track receiver moving at 0.5 m/s with less than 2 cm pointing error
 Time-division concurrent delivery to two receivers with measured power at each

10.6 Hardware Configuration (Phase 1)

Transmitter Module. 2.4 GHz or 5.8 GHz phased-array board, 4—“16 elements; software-defined radio (SDR) for waveform control; microcontroller for phase and amplitude steering; low-wattage power amplifier, less than 1W output.

Receiver Module. Tuned resonant rectenna at operating frequency; RF-to-DC rectifier circuit; supercapacitor or small Li-ion storage; power measurement instrumentation, mW resolution; telemetry link back to control system.

Localization Module. UWB positioning (Decawave DWM1001) —” preferred, cm-level accuracy. Alternatives: mmWave radar (TI IWR6843), optical marker tracking (ArUco).

Control System. Laptop running Lume-X control loop over USB. Desktop-first approach for Phase 1. Real-time feedback loop: position †’ phase correction †’ beam update †’ delivery confirmation †’ loop. Safety envelope enforcement continuous. Full logging for repeatability validation.

11. Global Invariant Enforcement

11.1 Overview

The five global invariants defined in Section 3.5 are not design aspirations. They are hard assertions enforced by the Lume-X runtime at every control cycle. Every 13.7 ms (73 Hz cycle rate), Lume-X evaluates all five invariants against current system state. Any invariant violation triggers an immediate, typed response.

The invariant enforcement model follows the same pattern used in Lume's deterministic assertion framework: verify †’ gate_if_failed †’ log †’ recover. No invariant failure is silent. No invariant failure produces an undefined system state.

11.2 INV-1 Enforcement: No Uncontrolled Discharge

```
verify node.V_cap >= V_burst before every transmission attempt
verify route exists for target in MMF routing table
verify receiver confirmed in MTL gate before beam fires
on failure: GATE transmission, log DISCHARGE_PREVENTED event, reschedule MFE burst
```

11.3 INV-2 Enforcement: No Ambiguous Routing

```

verify each destination has exactly one active route entry in MMF routing table
verify no two routing entries have equal cost without deterministic tiebreaker
verify path lock is held by the active transmitter for the duration of its window
on failure: pause routing, log ROUTING_AMBIGUITY event, force re-convergence, res

```

11.4 INV-3 Enforcement: No Non-Deterministic Flow

```

verify rolling  $\sigma^2$  for each active link <  $\sigma^2_{\text{max}}$  across current 20-trial window
verify  $P_{\text{received}} \geq P_{\text{min}}$  on every completed delivery
verify variance update applied after each delivery confirmation
on failure: mark link DEGRADED in MMF, log DETERMINISM_FAILURE event, activate alt

```

11.5 INV-4 Enforcement: No Unsafe Transmission Events

```

verify exclusion_zone clear before every beam fire
verify power_density at all_monitored_points < icnirp_limit at all times
verify  $V_{\text{cap}} \geq V_{\text{burst}}$  before every beam fire
verify localization signal age < 500ms before every beam fire
on failure: GATE beam, log SAFETY_GATE event with failed condition, await resoluti

```

The SAFETY_GATE event is the highest-priority log event in the system. It is never suppressed, never batched, and always written synchronously to the persistent log before any other recovery action is taken.

11.6 INV-5 Enforcement: No Mesh Oscillation

```

verify directional flow consistency on each link over the preceding 3 windows

```

detect oscillatory pattern: alternating flow direction without net destination progress

```

on detection: log OSCILLATION_DETECTED event, apply flow stabilization, lock pref
on stabilization failure: escalate to SHDCL recovery event

```

12. Control and Safety Layer

12.1 Lume-X as the Operating System

Lume-X is the provisionally patented deterministic control runtime that operates the entire Meridian architecture. It is the energy-domain implementation of the Lume language ecosystem (DOI: 10.5281/zenodo.19382282). Lume-X is not a higher-level script that calls external systems —” it is the

runtime itself, operating all four Meridian layers simultaneously through a unified control loop validated at 73 Hz.

The 73 Hz validation result is critical. The minimum control-loop requirement for DWER beam steering responsiveness is 10 Hz (100 ms cycle). DRMA inter-node synchronization requires faster cycles to maintain TDMA window precision. At 73 Hz (13.7 ms cycle time), Lume-X exceeds the minimum requirement by a factor of 7.3 for beam steering and provides TDMA resolution of approximately 14 ms per slot —” sufficient for Phase 1—“3 experimental requirements without hardware bridging.

The Lume-X runtime implements SHDCL —” the Self-Healing Deterministic Control Layer —” as a property of the runtime itself, not as an application-layer module. SHDCL means the runtime monitors every process it manages. A dropped process is automatically restarted. A stalled loop is detected and recovered. A missing heartbeat from a relay node is flagged and the mesh routes around the silent node. The system does not wait for human intervention.

12.2 SHDCL Recovery Sequences

DWER Beam Steering Recovery. When localization signal is momentarily lost:

```
detect: localization signal age > 500ms
execute: hold beam at last confirmed safe position (no steering, no power change)
initiate: re-acquisition sequence (request fresh position from UWB subsystem)
on restore: resume normal operation
on failure: cut beam after 2-second hold limit, log BEAM_RECOVERY_FAILED event
log: event timestamp, duration, recovery path, outcome
```

Without SHDCL: beam enters unknown state, potential safety hazard, system halt required. With SHDCL: controlled hold, automatic recovery, continuous safe operation.

DRMA Node Failure Recovery. When a relay node drops out of the mesh:

```
detect: node fails to respond within 2 consecutive control cycles
execute: mark node PRESUMED_FAILED in MMF topology
```

calculate: alternate routing path using pre-computed fallback activate: alternate path, signal adjacent nodes to update routing tables continue: power delivery to destination through alternate path

```
log: failure event, recovery path, time-to-recovery
```

Without SHDCL: mesh breaks, power delivery to target fails, manual intervention required. With SHDCL: automatic rerouting, continuous delivery, full event log.

DAEH Harvest Cycle Recovery. When a harvesting modality underperforms:

```
detect: dV/dt < dV_dt_min for 60-second window
execute: adjust burst scheduling to match actual available energy
shift: harvest priority to higher-performing modalities if available
flag: degraded harvest state for maintenance attention
if charge rate insufficient to meet delivery schedule:
  request DWER supplementation beam from nearest eligible source node
log: degraded state, adjusted schedule, supplementation request if issued
```

Without SHDCL: harvest cycle operates on stale assumptions, transmission failures propagate. With SHDCL: adaptive scheduling, safe operation at any harvest level, predictive maintenance flagging.

12.3 Natural Language Control Interface

Lume-X provides an English-mode and voice-to-code interface that allows a technician or operator to control the mesh in natural language in real time. This is a direct application of the Lume language's natural-language programming capabilities to a physical hardware control domain. No existing wireless power system has this interface.

Valid operator commands are translated by Lume-X into typed Meridian control operations:

Natural Language Command	Meridian Operation
"Reroute power around node 3"	MMF: mark Node-3 EXCLUDED, recalculate all affected paths
"Increase harvest priority on south wall nodes"	MFE: raise burst schedule priority for Zone_ID=south_wall
"Switch to burst-only mode on all DAEH nodes"	MFE: disable DWER supplementation, burst-only scheduling
"Reduce beam power on corridor nodes to 50%"	MTL: set power_level_mW = 0.5 — current for Zone_ID=corridor
"Show me the charge state of all relay nodes"	MC: broadcast CHARGE_STATE_REPORT for all node_type=relay

The natural language interface is not a convenience layer —” it is a safety capability. An operator who can issue plain-English commands during an incident response does not need to pause for terminal access or documentation lookup. The interface lowers the operational skill requirement for mesh management and reduces the probability of human error during recovery events.

12.4 Guardian Security Integration

Guardian Security (energy-domain designation: Guardian-E; domain: TrustShield.tech) provides the security enforcement layer for the Meridian architecture. Guardian Security covers eleven threat categories in its general implementation. The energy-domain mapping focuses on six primary threat vectors:

- **Localization Spoofing Defense.** Guardian-E requires multi-modal position confirmation before any transmission is authorized. A position reading from the UWB subsystem must be cross-validated against an independent second modality (optical marker or mmWave radar) before MTL accepts it. A single-source position that cannot be independently confirmed triggers a `POSITION_UNVERIFIED` gate.

Routing Table Integrity. All routing table entries are signed by the originating node using a per-node key provisioned at deployment. MMF rejects any routing update that cannot be verified against the originating node's key. This prevents routing table manipulation by malicious or compromised nodes.

- **Beam Hijacking Prevention.** MTL requires delivery confirmation from a node whose current position matches the authorized target position within r cm. A delivery confirmation from an unexpected position triggers a `BEAM_HIJACK_SUSPECTED` event and immediate beam cut.
- **Denial-of-Power Detection.** Guardian-E monitors delivery success rates per destination. A pattern of repeated delivery failures to a specific node that cannot be explained by environmental factors triggers a `DENIAL_OF_POWER_SUSPECTED` event and activates alternate path analysis.
- **Node Impersonation Prevention.** All inter-node messages are authenticated using the provisioned node keys. An unauthenticated message that arrives from a known node address triggers an `IMPERSONATION_SUSPECTED` event.
- **Timing Attack Resistance.** TDMA window assignments are computed by the trusted Lume-X runtime and distributed signed. A node that transmits outside its assigned window triggers a `TIMING_VIOLATION` event and is quarantined.

The formal written specification of the Guardian Security rule set for the Meridian energy domain is Appendix E of this document and is also listed as Gap Item 8 in the project gap list.

13. Implementation Considerations

13.1 Hardware Requirements

Phase 1 experimental hardware is intentionally minimal, desktop-first, and commercially available:

Phased-Array Board. Several commercial boards support 2.4 GHz phased-array beamforming at lab scale. Boards with 4—“16 elements and software-accessible phase controls are suitable. SDR integration via USB is preferred for Phase 1 to enable waveform flexibility without custom firmware.

UWB Localization. The Decawave DWM1001 module is the preferred localization hardware. It provides approximately 10 cm accuracy, USB connectivity, and a well-documented API. A minimum of three anchor modules plus one tag module enables 3D positioning. Phase 1 may use 2D positioning (three anchors, planar tracking) to reduce complexity.

Rectenna. Phase 1 rectenna is a lab-assembled discrete component: a tuned dipole or patch antenna at the operating frequency, a matching network, a Schottky diode rectifier (HSMS-285x or equivalent), and a smoothing capacitor. A commercial evaluation kit (e.g., Powercast P2110) may be used to accelerate Phase 1 verification before a custom rectenna is built.

Supercapacitor. 1—“10 F, 5V rated, with monitored voltage divider. Panasonic EEC-EN0F104 or equivalent.

Control Computer. Any modern laptop running the Lume-X runtime over USB. No embedded target required for Phase 1.

13.2 Supercapacitor Sizing

Supercapacitor capacity C (farads) required to deliver a burst of power P_{burst} (watts) for duration T_{burst} (seconds) from voltage V_{burst} to minimum operating voltage V_{min} :

$$E_{\text{burst}} = P_{\text{burst}} \cdot T_{\text{burst}}$$

$$C = 2 \cdot E_{\text{burst}} / (V_{\text{burst}}^2 - V_{\text{min}}^2)$$

Example: $P_{\text{burst}} = 0.1 \text{ W}$, $T_{\text{burst}} = 10 \text{ ms} = 0.01 \text{ s}$, $V_{\text{burst}} = 4.5 \text{ V}$, $V_{\text{min}} = 2.5 \text{ V}$:

$$E_{\text{burst}} = 0.1 \cdot 0.01 = 0.001 \text{ J}$$

$$C = 2 \cdot 0.001 / (4.5^2 - 2.5^2) = 0.002 / (20.25 - 6.25) = 0.002 / 14 \approx 0.000143 \text{ F} =$$

For higher burst energy (1W for 100 ms = 0.1 J): $C \approx 14.3 \text{ mF}$. Standard 0.1—“1 F supercapacitors provide substantial headroom above these minimums, enabling multiple burst events between recharge cycles.

13.3 Node Density and Spacing

Optimal inter-node spacing depends on the operating frequency and the efficiency break-even distance. At 2.4 GHz, theoretical break-even is approximately 2—“3 meters (to be confirmed experimentally in Phase 2). For initial deployments, 1.5—“2.5 m inter-node spacing is recommended, providing a margin below

the estimated break-even while maintaining practical node counts.

Node density in a room-scale deployment (5m — 5m room):

At 2m spacing: 4—“9 relay nodes for full coverage At 1.5m spacing: 9—“16 relay nodes Gateway node required at the wired power entry point (or multiple gateways for larger spaces)

13.4 RF Emission Constraints

All Phase 1—“3 experiments operate at less than 1W RF output at all nodes. This is a hard constraint enforced by MTL's safety gate and logged at every transmission event. This constraint ensures compliance with FCC Part 15 general-purpose unlicensed device rules at 2.4 GHz and 5.8 GHz.

Before any deployment-scale claims or demonstration beyond a controlled laboratory environment, a full regulatory analysis is required:

FCC Part 15 compliance verification for the complete system SAR (specific absorption rate) analysis for sustained exposure scenarios EIRP (effective isotropic radiated power) calculations per operating band FCC Part 18 industrial, scientific, and medical equipment analysis if operating at higher power levels is required This analysis is Gap Item 4 in the project gap list.

13.5 Environmental Factors

Multipath reflections in indoor environments create standing wave patterns that affect beam delivery consistency. Phase 1 experiments should be conducted in an environment with minimal metallic reflectors. Phase 2 and 3 experiments should characterize the effect of multipath on the σ^2 variance metric and develop beam shaping or frequency selection strategies to mitigate degraded links.

Human presence affects both the localization accuracy (body occlusion of UWB signals) and the RF power density calculation (body absorption). The Biological Safety Layer exclusion zone enforcement directly addresses the latter. Localization robustness in the presence of human body occlusion should be characterized during Phase 2.

14. Evaluation Framework

14.1 Overview

The evaluation framework defines how the Meridian architecture will be tested once experimental hardware exists. All success criteria are quantitative. All experimental results are logged to the standard Meridian delivery log format for post-hoc statistical analysis. No phase is considered complete until all success criteria for that phase are met across the required number of trials.

Fundamental requirement: Phase 1 DWER directionality results are required before any advancement to Phase 2. No paper submission before Phase 1 experimental data exists. This is not a guideline —” it is a hard constraint.

14.2 Deterministic Timing Accuracy

For each active link in the DWER and DRMA subsystems:

Beam acquisition time: less than 5 seconds from receiver detection to locked delivery TDMA window compliance: beam fires within ± 50 ms of assigned T_{start} across 95% of events Recovery time: re-acquisition after signal loss complete within 2 seconds for 95% of events Control cycle jitter: Lume-X cycle time variance less than ± 2 ms at 73 Hz nominal

14.3 Routing Stability

For each active DRMA mesh configuration:

Path convergence time after node addition: less than 500 ms Path reconvergence time after node failure: less than 2 control cycles (less than 30 ms) Oscillation detection time: less than 3 consecutive windows Routing table consistency: all nodes converge to identical topology within 1 second of any topology change

14.4 Harvesting Efficiency

For each DAEH harvesting modality at each deployment location:

Power density measurement accuracy: $\pm 10\%$ of calibrated reference Charge accumulation repeatability: V_{cap} trajectory variance less than 5% across 10 trials Burst threshold detection: V_{burst} level confirmed within $\pm 0.1V$ across 20 consecutive cycles Time-to-threshold prediction accuracy: predicted vs. actual within $\pm 15\%$ for 90% of burst events

14.5 Transmission Reliability

For each active DWER link:

$P_{\text{received}} \geq P_{\text{min}}$ in at least 95% of attempts $\sigma^2 < 10\%$ across rolling 20-trial window Delivery confirmation rate: greater than 95% of transmissions confirmed within timeout window Beam alignment confirmation: greater than 10 dB gain vs. 45° off-axis (Phase 1 primary criterion)

14.6 Mesh-Level Resilience

For the full integrated DRMA mesh:

Single-node failure: delivery continues within 2 control cycles through alternate path Multi-node failure: delivery continues for all destinations with viable alternate paths Bi-directional flow: demonstrated surplus-to-depleted transfer across at least one link reversal Moving receiver: tracking maintained at 0.5 m/s with less than 2 cm pointing error (Phase 3)

14.7 Safety Compliance

For every experimental session:

Zero SAFETY_GATE failures in the final 100 transmissions of any session (no trend toward safety violations as session progresses) Exclusion zone enforcement: 100% gate compliance across all biological presence events injected during testing Power density: calculated values at all monitored points remain below ICNIRP thresholds in 100% of measurements Beam kill on signal loss: confirmed within one control cycle (less than 14 ms) in 100% of simulated signal loss events

Appendix A —” Formal Definitions and Notation

A.1 Charge-State Vector

```
C = [V_cap, V_min, V_burst, V_max, dV/dt, mode, harvest_rate, cycles]
```

```
V_cap ∈ [0, V_max] —” current supercapacitor voltage (V)
V_min —” minimum operating voltage (V)
V_burst —” burst threshold voltage (V), V_burst > V_min
V_max —” rated maximum voltage (V)
dV/dt —” charge rate (V/s), positive during charging, negative during discharge
mode ∈ {HARVEST, READY, TRANSMITTING, DEPLETED, SUPPLEMENTED}
harvest_rate —” ambient harvest power (μW), 60-second rolling average
cycles —” total charge-discharge cycle count (integer)
```

A.2 Flow-Potential Vector

```
F = [source_id, P_available, direction_vector, loss_model, priority, T_available]
```

```
source_id —” Meridian node identifier of the source node
P_available —” estimated deliverable power in mW based on V_cap - V_burst margin
```

direction_vector —” unit vector in 3D space pointing toward highest-priority downstream path
 loss_model —” $L(\text{link})$ and $\sigma^2(\text{link})$ for the first downstream link
 priority —” MFE-computed transmission priority score $P \in [0, 1]$
 T_available —” earliest timestamp at which this node can transmit

A.3 Routing Matrix The MMF routing matrix R is an $N \times N$ matrix where N is the number of nodes in the network:

$$R[i][j] = \{ \text{next_hop, cost, hops, quality, alt_next_hop, updated, state} \}$$

$R[i][j]$ represents the routing table entry at node i for destination j . $R[i][i] = \text{NULL}$ (no self-routing). $R[i][j] = \text{UNREACHABLE}$ if no path exists from node i to node j .

A.4 Timing Primitives

T_{now} —“ current network timestamp (μs since epoch)

$T_{\text{slot}}(n)$ —“ start of current or next TDMA transmission window for node n $T_{\text{burst}}(n)$ —“ MFE-projected burst transmission time for node n

T_{cycle} —“ Lume-X control cycle period (13.7 ms at 73 Hz)

T_{switch} —“ node mode-switch time, transmit to receive or receive to transmit ($\approx 5 \mu\text{s}$)

A.5 Determinism Parameters

P_{min} —“ minimum acceptable received power (mW)

r —“ spatial uncertainty radius (cm), derived from localization accuracy σ^2 —“ maximum acceptable delivery variance (%), default 10% d —“ transmission distance (m)

N_{trials} —“ minimum trial count for statistical validation, $N_{\text{trials}} \geq 20$

Appendix B —“ Meridian State Machine

B.1 Node Lifecycle Each Meridian relay node follows a defined state machine:

States: INIT +‘ DISCOVERY +‘ ACTIVE +‘ DEGRADED +‘ FAILED +‘ RECOVERED

INIT: Node powers on, loads Meridian firmware, reads provisioned identity

DISCOVERY: Broadcasts NODE_ANNOUNCE, receives topology from neighbors, populates

ACTIVE: Normal operation —“ cycles through HARVEST +‘ READY +‘ TRANSMITTING +‘

DEGRADED: Operating with reduced capability —“ harvest rate below threshold, or

FAILED: Node unresponsive, presumed offline by neighbors after 2 missed cycles

RECOVERED: Node returns from FAILED state, re-announces, requests routing table

B.2 Relay Eligibility A node is eligible to relay when:

State = ACTIVE

$\text{mode} \in \{\text{HARVEST}, \text{READY}\}$ (not TRANSMITTING or DEPLETED)

Routing table entry exists for the requested destination

$V_{\text{cap}} \geq V_{\text{burst}}$

No pending GATE conditions in MTL B.3 Transmission Gate State Machine

```
States: IDLE †' CHECKING †' GATED / AUTHORIZED †' FIRING †' CONFIRMING †' LOGGED
IDLE:      No transmission pending
CHECKING:   MTL evaluating five gate conditions
GATED:      One or more conditions failed —" transmission blocked, event logged
AUTHORIZED: All five conditions passed —" transmission authorized
FIRING:      Beam active, delivery in progress
CONFIRMING: Beam complete, awaiting DELIVERY_CONFIRM from receiver
LOGGED:      Confirmation received (or timeout), delivery record written, state re
```

Appendix C —" Mathematical Model of Loss-Bounded Flow

C.1 Free-Space Path Loss

$\text{FSPL}(\text{dB}) = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10}(4\pi/c)$

At 2.4 GHz:

FSPL(1m) $\approx$ 40.0 dB
 FSPL(2m) $\approx$ 46.0 dB
 FSPL(3m) $\approx$ 49.5 dB
 FSPL(4m) $\approx$ 52.0 dB

C.2 Per-Hop Relay Efficiency $\cdot_{\text{hop}} = \cdot_{\text{rect}} \text{ — } \cdot_{\text{retx}}$ Where: $\cdot_{\text{rect}} \in [0.50, 0.70]$ —" rectenna RF-to-DC conversion efficiency $\cdot_{\text{retx}} \in [0.60, 0.75]$ —" re-transmission DC-to-RF conversion efficiency $\cdot_{\text{hop}} \in [0.30, 0.52]$ $\cdot_{\text{hop_dB}} \in [-5.2, -2.8]$ dB

C.3 Break-Even Distance Calculation Multi-hop routing is more efficient than single-source transmission when:

$\text{FSPL_single}(d_{\text{total}}) > \text{FSPL_hop}(d_{\text{hop}}) - N_{\text{hops}} + \cdot_{\text{hop_dB_loss}} - (N_{\text{hops}} - 1)$

For $N_{\text{hops}} = 2$, $d_{\text{total}} = 4\text{m}$, $d_{\text{hop}} = 2\text{m}$ at 2.4 GHz: Single: $\text{FSPL}(4\text{m}) = 52.0\text{ dB}$ Two-hop: $\text{FSPL}(2\text{m}) \times 2 + \text{conversion_loss} = 46.0 + 46.0 + 5.2 = 97.2\text{ dB}$ (total chain)

This formulation shows that naive comparison misleads: the break-even analysis must account for the fact that in a relay chain, the path loss is paid at each hop independently, not cumulatively. The correct comparison is:

$P_{\text{received_single}} = P_{\text{tx}} / \text{FSPL}(d_{\text{total}})$ — $P_{\text{received_multi}} = P_{\text{tx}} / \text{FSPL}(d_{\text{hop}}) \times d_{\text{hop}}^{(N_{\text{hops}}-1)}$ — Break-even: $P_{\text{received_single}} = P_{\text{received_multi}}$ Solve for d_{total} given d_{hop} and N_{hops}

At $d_{\text{hop}} = 0.40$ and 2.4 GHz, the break-even distance is approximately 2.5—“3 meters. Below this threshold, a single source is more efficient. Above it, relay nodes improve delivered power. This calculation must be validated experimentally in Phase 2 for the specific hardware used.

C.4 Burst Threshold Functions Charge accumulation to burst threshold:

$T_{\text{to_burst}}(n) = \max(0, (V_{\text{burst}} - V_{\text{cap}}(n)) / (dV/dt(n)))$ Where $dV/dt(n)$ is the current charge rate for node n .

Burst energy available above threshold:

$$E_{\text{burst}}(n) = 0.5 \times C_{\text{sc}} \times (V_{\text{cap}}(n)^2 - V_{\text{min}}^2)$$

Where C_{sc} is supercapacitor capacitance in farads.

C.5 Deterministic Timing Bounds Maximum delivery latency for an N -hop relay chain:

$T_{\text{delivery}}(N) = N \times (T_{\text{burst}} + T_{\text{switch}} + T_{\text{confirm}}) + T_{\text{alignment}}$ Where: $T_{\text{burst}} \approx 10\text{ ms}$ (default burst duration)

$$T_{\text{switch}} \approx 5\text{ ms} \text{ (mode switch overhead)}$$

$T_{\text{confirm}} \approx 5\text{ ms}$ (delivery confirmation round-trip) $T_{\text{alignment}} \approx 50\text{ ms}$ (initial beam acquisition)

$T_{\text{delivery}}(1) \approx 70\text{ ms}$ $T_{\text{delivery}}(2) \approx 120\text{ ms}$ $T_{\text{delivery}}(3) \approx 170\text{ ms}$

Appendix D —” Mesh Topology Scenarios

D.1 Sparse Mesh (Low Node Density) Configuration: 3 relay nodes, 5 meter inter-node spacing. Characteristics: fewer alternate paths available, higher path loss per hop (beyond break-even distance). SHDCL behavior: single-node failure may leave no viable alternate path for some destinations. Mitigation: DWER direct beam activated from source node to bypass failed relay segment.

D.2 Dense Mesh (High Node Density) Configuration: 9—“16 relay nodes, 1.5 meter inter-node spacing. Characteristics: multiple alternate paths for most destinations, lower path loss per hop (within break-even distance may make single-source preferable). MFE behavior: routing preference shifts toward direct DWER links where single-source efficiency exceeds relay chain efficiency. SHDCL behavior: single-node failure resolved by multiple alternate paths with minimal delivery impact.

D.3 Mobile Nodes Configuration: one or more receiver nodes moving at up to 0.5 m/s. MTL behavior: beam steering updates at 10 Hz minimum, position freshness checked at every control cycle. MMF behavior: routing table updates propagate when mobile node moves between relay coverage zones. Success criterion: less than 2 cm pointing error while tracking 0.5 m/s movement.

D.4 Environmental Interference Scenarios: metallic reflectors creating multipath, human body occlusion of UWB signals, co-channel RF interference at 2.4 GHz. Characterization required during Phase 2 and 3 testing. SHDCL behavior: σ^2 increase triggers link DEGRADED state, alternate path activated. Mitigation options: frequency agility (switch to 5.8 GHz), beam shaping adjustments, localization modality fallback.

Appendix E —” Safety Model

E.1 Guardian Security Rule Set (Energy Domain) The following rules are enforced continuously by Guardian Security (Guardian-E) in the Meridian energy domain. This is the formal specification of the energy-domain security layer:

RULE GE-01: Every transmission must be authorized by a valid signed routing entry. RULE GE-02: Position data must be independently confirmed before authorizing beam fire. RULE GE-03: Delivery confirmation must match target position within r cm. RULE GE-04: All inter-node messages must be authenticated against provisioned node keys. RULE GE-05: TDMA window violations trigger immediate quarantine of the violating node. RULE GE-06: Delivery failure rate exceeding 20% for any single destination triggers DENIAL_OF_POWER_SUSPECTED. RULE GE-07: Energy flow reversal requests must originate from an authenticated surplus node. RULE GE-08: Any node reporting charge state inconsistent with its measured behavior is flagged. RULE GE-09: Guardian-E logs all security events with full context before taking any action. RULE GE-10: No Guardian-E action suppresses a SAFETY_GATE event. RULE GE-11: All rule violations are reported to Lume-X for SHDCL coordination.

E.2 Transmission Constraints Maximum single-node RF output: 1W (hardware-enforced, not only software-enforced) Minimum beam cutoff response time: 1 control cycle (≤ 14 ms) Maximum exclusion zone radius: configurable, default 0.5 m around beam centerline Maximum calculated power density at monitored points: ICNIRP general public limits (2 W/m^2 at 2.4 GHz) Minimum position confirmation age: less than 500 ms before beam fire E.3 Failure-Mode Analysis

Failure Mode	Detection Mechanism	Response	Log Event
Localization signal lost	MC position age > 500ms	Beam hold, re-acquisition	BEAM_HOLD
Node failure	No heartbeat — 2 cycles	Route around node	NODE_FAILED
Gate condition failure	MTL gate check	Block transmission	GATE_REJECT
Delivery not confirmed	Confirmation timeout	Retry — 2, then SHDCL	DELIVERY_FAILED
σ^2 threshold exceeded	MFE variance monitor	Link DEGRADED, alternate path	DETERMINISM_FAILURE
Biological presence detected	Exclusion zone sensor	Immediate beam cut	SAFETY_GATE
Power density threshold approached	BSL continuous monitor	Beam throttle, then cut	POWER_DENSITY_WARN
Routing oscillation detected	MMF flow history	Flow stabilization lock	OSCILLATION_DETECTED
Security rule violation	Guardian-E rule engine	Per-rule response	SECURITY_EVENT

Appendix F —” Reference Implementation Blueprint

F.1 Node Architecture Each Meridian relay node hardware stack:

Layer 5: Lume-X Runtime (software) Layer 4: Meridian Control Software (MC + MFE + MMF + MTL)
 Layer 3: Communication Interface (USB for Phase 1, embedded for Phase 2+) Layer 2: Sensor and Actuator Firmware (UWB, RF control, charge measurement) Layer 1: Hardware (phased-array board, rectenna, supercapacitor, microcontroller, UWB module)

F.2 Firmware Modules `identity_module` —” reads provisioned node identity from persistent storage, exposes MC identity interface.

`localization_module` —” interfaces with UWB hardware, provides position updates at configured rate, reports staleness to MC.

`charge_monitor_module` —” reads supercapacitor voltage at 1 Hz or faster, computes dV/dt rolling estimate, updates MC charge-state vector.

harvest_monitor_module —” reads harvesting modality power sensors, updates harvest_rate in MC charge-state vector.

beam_control_module —” translates MTL BeamCommand into phased-array driver settings, reports alignment status.

rectenna_monitor_module —” measures received power at rectenna, reports to delivery confirmation channel.

safety_gate_module —” hardware-level interlock for RF emission, independent of software gate (belt and suspenders safety architecture).

communication_module —” handles inter-node messaging for routing table distribution, DELIVERY_CONFIRM, LINK_STATE_UPDATE, and Guardian-E authentication.

F.3 Build Order The recommended implementation order, validated and confirmed in the Meridian system specification:

DWER control loop built in Lume-X against mocked hardware data Feedback logic validated end-to-end in software before any real hardware is connected Real hardware connected one layer at a time —” UWB localization first, then power measurement DWER Phase 1 and Phase 2 proven with Lume-X as operational control system Inter-node synchronization protocol designed, specified, and validated in simulation DRMA two-node testbed built and validated DAEH charge-state monitoring and burst scheduling added to Lume-X control loop Full three-layer operation validated: harvest †’ store †’ route †’ deliver

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Chapter Bridge: From Machine to Building

The architecture specified in Chapter One operates in abstract space: nodes at defined positions, links with defined quality metrics, invariants with defined enforcement mechanisms. The engineering is complete. The architecture is correct.

But a node position is not a mounting bracket. A link quality metric is not a ceiling joist clearance. A routing table is not a building plan. Between the architectural specification and the working installation lies the problem of physical deployment —" the question of where, in a real building with real architectural features, the nodes go, and how the beam paths between them relate to the walls, floors, voids, and occupied spaces through which they must pass without harm.

Chapter Two answers that question.

Chapter Two

Deploying in the Real World

The gap between a room-scale wireless charging demonstration and a commercial building deployment is not a gap of physics. The physics that works in a controlled laboratory environment works in a hospital corridor, a warehouse, and a twenty-story office tower. The gap is architectural.

Real buildings have open staircases. They have atriums that rise through five or ten floor levels. They have elevator shafts running from basement to roof. They have mezzanines inserted at non-standard heights and balconies that overhang lower spaces. Every one of these features creates a geometric condition in which a directed energy beam, routing according to a node-graph shortest path, can traverse a space where a person may be standing —" not because the architecture is careless, but because the node routing graph does not know the building exists.

The Architectural Voxel Map —” the central contribution of this chapter —” is the mechanism that closes this gap. By encoding the building's physical geometry as a 3D grid of typed voxels, and by enforcing that grid as a hard constraint on every beam path computation, the Meridian routing engine can guarantee that no authorized beam path traverses a void, an occupied zone, or any other prohibited space, regardless of what the node-graph shortest path suggests.

The chapter also introduces ceiling-plane routing —” the architectural deployment posture in which relay nodes are mounted at ceiling height and beam paths are kept above the occupancy zone for all non-final hops —” and the floor-transition gateway architecture that enables energy to move between floors without beams traversing occupied vertical spaces.

Together, these mechanisms turn the safety model from reactive (detect a person in the beam, stop the beam) to proactive (route the beam where no person can be, then stop it if a person appears anyway). The Biological Safety Layer remains as the enforcement backstop. But the primary safety guarantee comes from architecture, not from sensing.

This chapter is the difference between a technology and a product.

Abstract

The six preceding papers in this series specify Meridian as a network protocol stack, a biological organism analog, an internet-scale energy routing standard, a general theory of deterministic infrastructure, a formal security framework, and a unified energy-data fabric. None of them answer the question a building owner, architect, or facilities manager asks first: how does this actually get installed in a real building?

Real buildings are not free-space volumes with nodes floating in ideal positions. They are layered collections of floors, walls, ceilings, open staircases, atriums, elevator shafts, mezzanines, balconies, mechanical chases, and structural columns —” each with different RF propagation properties, different occupancy characteristics, and different implications for where a directed energy beam may safely travel. A beam path that appears valid in the node routing graph may physically traverse an open staircase void connecting three floors. A ceiling-mounted relay node that appears to satisfy the $z_{\min}$ height constraint may overhang a mezzanine where the effective clearance is half of nominal. These are not edge cases —” they are standard features of every commercial building Meridian will be deployed in.

I introduce the Architectural Voxel Map (AVM) —” a formal 3D volumetric representation of a building's physical geometry that is imported into the Meridian routing engine at commissioning and enforced as a hard constraint on every beam path computation. The AVM classifies every voxel in the building volume as one of seven types, and the DRMA routing engine rejects any beam path whose geometric projection intersects a prohibited voxel class, regardless of the node-graph shortest path.

I define the ceiling-plane routing model as the standard deployment posture for occupied buildings: relay nodes are mounted at ceiling height on each floor, energy travels horizontally along the ceiling plane between relay nodes, and descends to devices only in the controlled final-hop geometry. I characterize the five void types that require explicit routing detours —” open staircases, atriums, elevator shafts, balcony overhangs, and occupied mezzanines —” and specify the floor-transition gateway architecture that enables inter-floor energy transfer without beam traversal of occupied spaces.

I provide the formal safety proof for multi-story deployment: that the combined enforcement of AVM constraints, ceiling-plane routing preference, floor-transition gateway geometry, and the Biological Safety Layer [Guardian Security, 2026] makes physical harm from beam traversal impossible under all non-destructive attack conditions, including the adversarial conditions characterized in the T1—“T11 threat taxonomy.

I close with the Building Information Model (BIM) integration pathway —” the specification for how Meridian deployment becomes a native layer of architectural design software, enabling architects to co-design beam routing topology alongside HVAC, electrical, and structural systems during building design rather than retrofitting it afterward.

Keywords: wireless energy routing, deployment architecture, architectural voxel map, ceiling-plane routing, floor-transition gateway, building information model, multi-story deployment, open staircase routing, atrium routing, deterministic safety, Meridian, BIM integration

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Appendix A —” AVM Voxel Classification Specification**Appendix B —” Floor-Transition Gateway Node Specification****Appendix C —” AVM-DRMA Integration: Beam Path Validation Algorithm****Appendix D —” BIM Export Format Specification (Meridian Layer)****References****1. Introduction****1.1 The Gap This Paper Fills**

The Meridian architecture [Meridian Architecture, 2026] specifies how energy is routed through a mesh of addressed nodes. It defines the routing protocol, the control runtime, the security framework, and the self-healing mechanisms. It does not specify where, physically, the nodes go —” or how the building's geometry constrains where the beams between them may travel.

This gap is not academic. It is the difference between a paper that describes a correct routing protocol for a free-space volume and a product that can be installed in a hospital, an office tower, a residential building, or a warehouse without injuring anyone and without requiring the building to be demolished and rebuilt around the energy routing requirements.

The architecture papers correctly observe that Meridian operates in physical space and that the Biological Safety Layer provides the safety backstop. What they do not specify is the proactive spatial framework that keeps beam paths out of occupied zones before the BSL ever needs to act. That proactive framework is this paper's contribution.

1.2 The Ceiling-Plane Insight

The motivating insight for this paper is simple: a building's ceiling is a surface that is, under normal occupancy conditions, above every person in the building. A beam that travels along the ceiling plane —” horizontally at ceiling height between relay nodes mounted at the ceiling —” passes through a geometric zone that people do not occupy. The beam path is safe not because a sensor detected no person in its way, but because no person can be in its way given the physical geometry.

This reframes the safety model from reactive (detect a person, stop the beam) to proactive (route the beam where no person can be). The BSL remains as the enforcement backstop for failure cases. But the primary safety mechanism is architectural routing —” choosing beam paths that are geometrically separated from occupied space.

The ceiling-plane insight extends naturally to multi-story buildings: each floor has a ceiling plane that is safe for horizontal beam travel. The challenge is floor-to-floor transition —” getting energy from floor 1's ceiling plane to floor 2's ceiling plane without routing through the vertical space between them, which may include occupied staircases, atriums, and elevator shafts. This paper solves that challenge.

1.3 Contributions

I make five contributions:

The formal definition of the Architectural Voxel Map (AVM) —” a seven-class 3D volumetric representation of building geometry that the routing engine enforces as a hard constraint on every beam path.

The ceiling-plane routing model as the standard deployment posture for occupied buildings, with formal geometric proofs of its safety properties.

The characterization of five void types (open staircase, atrium, elevator shaft, balcony overhang, occupied mezzanine) and the specific routing detour each requires.

The floor-transition gateway architecture —” the physical and protocol specification for inter-floor energy transfer at designated architectural transition points.

The BIM integration pathway —” a specification for how Meridian deployment becomes a co-designed architectural layer rather than a post-hoc installation.

1.4 Paper Organization

Section 2 characterizes the gap between network topology and physical architecture. Section 3 defines the AVM formally. Section 4 specifies ceiling-plane routing. Section 5 characterizes the five void types. Section 6 defines floor-transition gateways. Section 7 describes the dynamic occupancy overlay. Section 8 provides the formal safety proof. Section 9 addresses frequency selection and its architectural implications. Section 10 presents four building type scenarios. Section 11 specifies the BIM integration pathway. Section 12 defines the commissioning protocol. Section 13 situates this in related literature. Section 14 states limitations. Section 15 concludes.

2. Background: The Gap Between Network Topology and Physical Architecture

2.1 How Current Routing Models Represent Space

The DRMA routing engine [Meridian Architecture, 2026 9] maintains a routing table that represents the mesh as a graph: nodes as vertices, wireless links as edges, link quality as edge weights. The routing algorithm (a modified Dijkstra over the energy-topology graph) finds the minimum-cost path from source to destination through this graph.

This representation is accurate for what it models: the connectivity and quality of wireless links between nodes. It does not model the physical geometry of the space those links traverse. A link between node A at position (3.0m, 2.0m, 2.5m) and node B at position (8.0m, 2.0m, 2.5m) is represented as an edge with a link quality metric. The routing engine does not know that the straight line between those two positions crosses over an open staircase void at $x = 5.5\text{m}$, meaning that the beam along that link, while technically maintaining a 2.5m height, has a downward projection into a space with zero effective clearance because the floor is not there.

This is the gap: routing tables are correct about connectivity; they are silent about the physical geometry of beam paths.

2.2 Why Node Positions Alone Are Insufficient

One might argue that if nodes are correctly positioned —” mounted at ceiling height, away from voids —” the routing will naturally be safe. This argument fails for three reasons:

First, link paths are not points. A link between two ceiling-mounted nodes defines a line (or more precisely, a beam cross-section sweeping through a volume) in physical space. That volume may intersect an architectural void even if the node endpoints are in safe positions.

Second, buildings change. A building commissioned with Meridian in January may have its staircase opening enlarged in March, a new mezzanine added in June, and a temporary work platform installed in August. The node positions don't change; the architectural geometry does. Without an AVM that can be updated to reflect architectural changes, the routing engine operates on a stale model of the building.

Third, height alone is not the safety criterion. A node mounted at 3.0m height on floor 2, directly above an open staircase that connects floor 1 and floor 2, is at 3.0m above floor 2 but potentially 6.0m+ above floor 1. A beam that departs from this node horizontally may pass through the staircase void at angles that create exposure to persons on the staircase. Height is floor-relative, not building-relative.

2.3 The AVM as the Necessary Extension

The AVM is the minimal necessary extension to the Meridian architecture to support deployment in complex built environments. It does not replace the node routing graph —” it augments it with the physical geometry that the routing graph cannot represent. The routing engine uses both: the node graph to find candidate paths, and the AVM to validate that candidate paths are geometrically safe before authorizing them.

3. The Architectural Voxel Map (AVM)

3.1 Voxel Grid Definition

The AVM divides the building volume into a uniform 3D grid of voxels. Each voxel is a cube of side length V_size , with V_size selected based on the required spatial resolution. For standard commercial building deployment, $V_size = 25\text{ cm}$ is recommended — sufficient to resolve staircase openings, balcony edges, and mechanical penetrations at a resolution meaningful for beam path planning.

AVM coordinate system: Origin: building datum point (typically ground floor SW corner) X: East (meters) Y: North (meters) Z: Up (meters) Voxel index (i, j, k) corresponds to physical volume: $X \in [i \cdot V_size, (i+1) \cdot V_size]$ $Y \in [j \cdot V_size, (j+1) \cdot V_size]$ $Z \in [k \cdot V_size, (k+1) \cdot V_size]$

For a typical 10-story office building (50m — 40m footprint, 40m height), the AVM contains approximately $50/0.25 \cdot 40/0.25 \cdot 40/0.25 = 200 \cdot 160 \cdot 160 = 5,120,000$ voxels. At 1 byte per voxel (the type tag), the full AVM for this building is approximately 5 MB — trivial to store and load at commissioning.

3.2 The Seven Voxel Classes

Each voxel is assigned one of seven classes:

Class 0 — SOLID: The voxel is occupied by structural material (concrete slab, masonry wall, steel column, timber framing, glass curtain wall). RF at frequencies above ~2.4 GHz attenuates significantly through most SOLID materials; at mmWave frequencies (24+ GHz) attenuation through a concrete slab is extreme (>40 dB). Beam paths do not traverse SOLID voxels — they terminate at SOLID boundaries. SOLID voxels are the natural beam containment boundaries.

Class 1 — FLOOR_SAFE: The voxel is in the ceiling zone of a normally occupied floor — specifically, the volume between the structural ceiling and the height $z_safe(floor)$, where $z_safe = floor_slab_height + 2.2m$ (2.2m provides a 10 cm margin above the 99th percentile standing height of 2.1m for the occupied population). FLOOR_SAFE voxels are the primary routing medium: relay nodes are mounted within or adjacent to FLOOR_SAFE voxels, and beam paths are preferred to remain within FLOOR_SAFE zones.

Class 2 — FLOOR_OCCUPIED: The voxel is in the normally occupied zone of a floor — between z_floor and z_safe . Beams must not traverse FLOOR_OCCUPIED voxels except in the controlled final-hop geometry (directly from a ceiling relay node downward to a destination device, with BSL active). This is the only permissible traversal of FLOOR_OCCUPIED voxels, and it is governed by the final-hop protocol specified in Section 4.3.

Class 3 — VOID_STAIRCASE: The voxel is part of the air volume in or directly above an open staircase — a void that connects floor planes. This class extends vertically through all floors the staircase spans. No beam path may have any portion of its swept volume intersecting a VOID_STAIRCASE voxel.

Class 4 —” VOID_ATRIUM: The voxel is part of the air volume of an atrium, light well, or multi-story open space. Like VOID_STAIRCASE, this class extends through all floor levels of the void. No beam path may intersect a VOID_ATRIUM voxel.

Class 5 —” VOID_SHAFT: The voxel is part of an elevator shaft, mechanical shaft, dumbwaiter shaft, or similar enclosed vertical void containing moving elements. No beam path may intersect a VOID_SHAFT voxel.

Class 6 —” GATEWAY_ZONE: The voxel is designated as a floor-transition gateway location —” a specific architectural position where a gateway relay node is installed to handle inter-floor energy transfer. GATEWAY_ZONE voxels are FLOOR_SAFE at their floor level, but they additionally participate in the inter-floor routing protocol. The routing engine may terminate a floor's beam path at a GATEWAY_ZONE and resume it at the corresponding GATEWAY_ZONE on the adjacent floor.

Routing permissions summary: SOLID : beam terminates (physical boundary) FLOOR_SAFE : beam traversal permitted (preferred) FLOOR_OCCUPIED : beam traversal permitted in final-hop geometry only VOID_STAIRCASE : beam traversal PROHIBITED VOID_ATRIUM : beam traversal PROHIBITED VOID_SHAFT : beam traversal PROHIBITED GATEWAY_ZONE : beam termination and inter-floor handoff point

3.3 AVM Construction

The AVM is constructed from one of three sources, in order of preference:

Source 1 —” BIM export (preferred). If the building has a BIM model (Section 11), the AVM is generated automatically by the Meridian BIM plugin from the architectural model. The plugin classifies each voxel by querying the BIM model's spatial elements: structural elements †’ SOLID, occupied spaces at floor level †’ FLOOR_OCCUPIED, ceiling zones †’ FLOOR_SAFE, staircase voids †’ VOID_STAIRCASE, atrium voids †’ VOID_ATRIUM, shaft elements †’ VOID_SHAFT, pre-designated gateway locations †’ GATEWAY_ZONE. This process is automated and produces a complete, accurate AVM for buildings with BIM documentation.

Source 2 —” As-built survey. For buildings without BIM, a physical survey using laser scanning (LiDAR point cloud) or structured-light scanning produces a 3D model of the building interior that is then classified into AVM voxel types by the Meridian commissioning tool. The survey must capture all voids, overhangs, and mezzanines that are not visible in floor plans.

Source 3 —” Floor plan annotation. For buildings where neither BIM nor physical survey is available, the commissioning technician annotates 2D floor plans with void types and the commissioning tool generates an AVM by extruding the 2D annotations vertically according to the specified floor heights. This is the least accurate method and requires careful annotation of all void boundaries.

3.4 AVM Updates

The AVM must be updated whenever the building's geometry changes. The commissioning tool maintains an AVM version history and requires re-validation of all existing beam paths against the new AVM whenever an update is applied. If an existing beam path becomes invalid under the updated AVM (because a previously-safe route now intersects a new void or mezzanine), the routing engine recomputes the affected paths before resuming operation.

4. Ceiling-Plane Routing: The Standard Deployment Posture

4.1 Definition

Ceiling-plane routing is the standard deployment posture for Meridian in occupied buildings: relay nodes are mounted at ceiling height within FLOOR_SAFE voxels, and beam paths are required to remain within FLOOR_SAFE voxels except for the controlled final-hop descent to destination devices.

Formally, a ceiling-plane route ρ from source node s to relay chain $[r_1, r_2, \dots, r_n]$ to destination device d satisfies:

For all hops h_{μ} from node n_{μ} to node n_{i+1} , where $n_{i+1} \neq d$: all voxels in $\text{swept_volume}(\text{beam}(n_{\mu}, n_{i+1})) \in \text{FLOOR_SAFE} \cup \text{SOLID}$
 For the final hop from r_n^{TM} to d : all voxels in $\text{swept_volume}(\text{beam}(r_n^{\text{TM}}, d))$:
 horizontal component $\in \text{FLOOR_SAFE}$ vertical descent component $\in \text{FLOOR_OCCUPIED}$ final-hop protocol active (BSL engaged, tracking active, power limited)

This formalization captures the key property: the beam travels horizontally at ceiling height through FLOOR_SAFE voxels for all non-final hops, and descends through FLOOR_OCCUPIED voxels only for the final hop, which is governed by a more restrictive protocol.

4.2 Node Placement Requirements

For ceiling-plane routing to be effective, relay nodes must be mounted within FLOOR_SAFE voxels. The node placement requirements for a compliant installation:

Mounting height: Node center at $z \geq z_{\text{safe}}(\text{floor})$ —” at or above the safe zone boundary. Typical mounting is at $z_{\text{ceiling}} \approx 5 \text{ cm}$ (surface-flush ceiling mount) or $z_{\text{ceiling}} \approx 15 \text{ cm}$ (pendant mount for better beam angle coverage).

Mounting position: Node must not be positioned directly above a VOID_STAIRCASE, VOID_ATRIUM, or VOID_SHAFT —” even if the node height is within FLOOR_SAFE for that floor, the downward beam geometry from such a position will intersect the void.

Beam angle envelope: Each relay node's phased array must be capable of producing a beam confined to $\pm\theta_{\max}$ from horizontal, where $\theta_{\max}$ is defined by the geometry: $\theta_{\max} = \arctan((z_{\text{node}} \approx z_{\text{safe}}) / d_{\min})$, where $d_{\min}$ is the minimum expected inter-node distance. This ensures that even at the nearest neighbor, the beam departs at an angle that remains within FLOOR_SAFE. Typical values: $z_{\text{node}} = 2.7\text{m}$ (ceiling height), $z_{\text{safe}} = 2.2\text{m}$, $d_{\min} = 1.5\text{m}$ † $\theta_{\max} = \arctan(0.5/1.5) \approx 18.4^\circ$.

Density: Node density must be sufficient to maintain FLOOR_SAFE routing paths around all VOID voxels. The minimum node density is determined by the void geometry: no VOID column should be wider than the effective beam range of the relay nodes on either side of it.

4.3 Final-Hop Protocol

The final hop —” from the last relay node down to the destination device —” is the only segment that passes through FLOOR_OCCUPIED space. It is governed by a more restrictive protocol than relay hops:

Power reduction: Final-hop power is reduced to the minimum necessary for delivery ($P_{\text{final}} \leq P_{\text{relay}}$ — attenuation_factor for the final hop distance). Since the final hop is always shorter than relay hops (device is below the ceiling relay), less power is required, further reducing any exposure.

Beam width reduction: The final-hop beam is commanded to minimum aperture —” maximum array focusing —” reducing the swept volume through FLOOR_OCCUPIED to the tightest achievable cross-section.

Active tracking engagement: Lume-X activates continuous position tracking for the destination device. If the device moves more than r_{track} (default: 10 cm) from its last confirmed position, the beam is held until position is reconfirmed.

Enhanced BSL monitoring: The final-hop BSL check requires all five sensor modalities to confirm no biological presence in the final-hop cone before each burst. For relay hops, three-modality consensus is sufficient; for final hops, five-modality consensus is required.

Burst duration limiting: Final-hop bursts are limited to $T_{\text{burst_max}}$ (default: 500ms). After each burst, a 50ms hold allows BSL sensors to update before the next burst.

5. The Five Void Types and Their Routing Detours

5.1 Open Staircase

Physical description: An open staircase is a floor opening connecting two or more floor levels, with the staircase treads exposed to the surrounding space rather than enclosed within walls. The void extends from the lowest floor level connected by the staircase to the ceiling of the highest floor connected.

Routing problem: A horizontal beam at ceiling height of floor N, traveling in the direction of the staircase opening, will have a downward projection that passes through the staircase void —” where persons on the stairs may be at heights ranging from floor N-1 level to near-ceiling of floor N. The apparent height of the beam above floor N is safe; its height above persons on the staircase may not be.

AVM classification: All voxels within the staircase opening footprint, from $z = \text{floor_N-1}$ to $z = \text{ceiling_N}$, are classified VOID_STAIRCASE. Additionally, a 50 cm buffer zone around the perimeter of the opening at ceiling level is classified VOID_STAIRCASE to prevent beams from grazing the opening edge.

Routing detour: The routing engine routes around the staircase footprint —” a horizontal detour at ceiling level that circumnavigates the VOID_STAIRCASE column. The detour length depends on the staircase width; for a typical 1.2m wide staircase opening, the detour adds one or two relay hops. If no circumnavigation path exists (the staircase opening spans the full width between two nodes), the routing falls through to inter-floor gateway routing (Section 6) rather than attempting a direct crossing.

Physical analogy: A river crossing that routes traffic over a bridge rather than through the water. The bridge is the relay node path around the staircase; the void is the water.

5.2 Atrium

Physical description: An atrium is a multi-story open volume —” a space where multiple floor levels have been removed or never built, creating a tall open interior. Atriums span from a ground floor (or basement) to a skylight or roof opening, passing through every floor level in between. They are common in commercial buildings, hotels, and shopping centers.

Routing problem: An atrium creates a void that invalidates ceiling-plane routing for any portion of any floor that borders the atrium, because the "ceiling" of floor N at the atrium boundary is not a safe routing zone —” a beam departing horizontally from a node near the atrium edge will pass into the atrium volume, where persons may be standing on balconies at any floor level, or traversing the atrium floor.

AVM classification: All voxels within the atrium footprint and its floor-level buffer (50 cm perimeter), from $z = \text{ground}$ to $z = \text{roof}$, are classified VOID_ATRIUM regardless of floor level.

Routing detour: Nodes bordering the atrium must route around it rather than across it. This typically requires a perimeter routing strategy —” relay nodes are placed around the atrium perimeter at ceiling height on each floor, forming a ring that routes energy around the atrium void. Energy never crosses the atrium volume; it travels around it. For large atriums, this may significantly increase hop count. The routing engine selects the lowest-hop perimeter path automatically.

Balcony sub-case: A balcony that overlooks an atrium is a `FLOOR_OCCUPIED` zone at non-standard height. A relay node mounted at the ceiling of the floor above a balcony must not route toward the atrium side, because the effective `z_safe` for that direction is limited by the balcony floor height, not the main floor height. The AVM handles this through the `VOID_ATRIUM` classification of the atrium-facing voxels: any beam directed toward the atrium is blocked by the `VOID_ATRIUM` boundary before it can reach the balcony zone.

5.3 Elevator Shaft

Physical description: An elevator shaft is an enclosed vertical void containing one or more elevator cars, counterweights, cables, and guide rails. Elevator shafts typically span the full building height and are enclosed by solid walls on all sides. Personnel may enter the shaft for maintenance.

Routing problem: Elevator shafts are `SOLID`-bounded —” the shaft walls are concrete or metal, providing effective RF attenuation at mmWave frequencies. For most practical purposes, the shaft itself does not create a routing problem because beams cannot penetrate the shaft walls to enter it. The routing problem arises at shaft doors: when an elevator door is open, the door opening creates a temporary aperture in the `SOLID` boundary that connects the `VOID_SHAFT` to the adjacent corridor.

AVM classification: The shaft interior is classified `VOID_SHAFT`. The floor-level area in front of each shaft door is classified `FLOOR_OCCUPIED`. The elevator lobby (typically 2—“3m in front of the door) is `FLOOR_OCCUPIED`.

Dynamic handling: The routing engine treats elevator door openings as temporary `VOID_SHAFT` extensions into the corridor. This is handled by the dynamic occupancy overlay (Section 7) rather than the static AVM: the elevator door sensor (or UEDM occupancy sensor at the lobby) detects the door opening event and marks the lobby voxels immediately in front of the open door as `VOID_SHAFT` for the duration of the door opening. Beams that would route through the lobby are held or rerouted during door-open events.

Practical impact: Elevator lobbies are typically not routing paths —” relay nodes avoid direct paths through elevator lobby zones by installation convention. The door-open dynamic marking is a backstop for unusual routing configurations.

5.4 Balcony Overhang

Physical description: A balcony overhang is a floor extension that protrudes beyond the main floor plate, creating a zone where the ceiling of the space below is the underside of the balcony slab rather than the main floor ceiling. Balconies are common in residential buildings, hotels, theaters, and lecture halls.

Routing problem: A relay node mounted at ceiling height on floor N, near a balcony overhang, may have a z_node value that appears safe relative to floor N, but if the balcony below it reduces the effective ceiling clearance at that horizontal position, the FLOOR_SAFE classification at that position is incorrect. Additionally, persons on the balcony above are at non-standard height —” higher than floor N's FLOOR_OCCUPIED zone, lower than floor N+1's FLOOR_SAFE zone.

AVM classification: The underside of the balcony slab is classified SOLID at its actual height. The zone between the balcony slab underside and the main floor ceiling at the overhang position is re-classified based on actual clearance. If the clearance is less than 2.2m (z_safe), the zone is classified FLOOR_OCCUPIED even if its absolute height would normally fall in FLOOR_SAFE. The balcony surface itself (on floor N+1) is FLOOR_OCCUPIED. The space above balcony standing height on floor N+1 is FLOOR_SAFE if it has ≥ 2.2 m clearance to the floor N+2 ceiling.

Routing detour: Relay nodes must not be positioned under balcony overhangs where the effective clearance is less than z_safe . The routing engine routes around overhang-affected zones using standard FLOOR_SAFE path selection.

5.5 Occupied Mezzanine

Physical description: A mezzanine is an intermediate floor level inserted between two main floor levels, typically covering a portion of the main floor area. Mezzanines are common in warehouses, retail spaces, libraries, and industrial buildings. They create a zone where the effective occupied height is non-standard —” higher than the main floor's FLOOR_OCCUPIED zone, lower than the ceiling.

Routing problem: A relay node mounted at the ceiling of the main floor, above a mezzanine, may have a nominal height of 6.0m (main floor ceiling height) but an effective clearance of only 1.0m above the mezzanine deck. A beam passing horizontally at 6.0m height, above the mezzanine, is safe relative to the main floor. But if the mezzanine extends under the beam path, the beam is only 1.0m above the mezzanine surface —” well below z_safe —” and persons standing on the mezzanine are directly in the beam path.

AVM classification: The mezzanine deck is classified SOLID. The zone above the mezzanine deck up to z_safe above the mezzanine is classified FLOOR_OCCUPIED. The zone above z_safe above the mezzanine is classified FLOOR_SAFE. The zone below the mezzanine deck (if accessible) uses the main floor classification. This produces a layered classification at the mezzanine location that correctly represents the multi-level occupancy.

Routing detour: The routing engine treats the mezzanine FLOOR_OCCUPIED zone as a routing obstacle for horizontal beam paths. Relay nodes above a mezzanine must route horizontally at the clearance height above the mezzanine FLOOR_OCCUPIED zone (i.e., at mezzanine_height + 2.2m or above), not at main

floor ceiling height. If main floor ceiling height is insufficient to clear the mezzanine z_{safe} boundary, no horizontal ceiling-plane route exists at that location, and the routing engine falls through to a perimeter route around the mezzanine footprint.

6. Floor-Transition Gateway Architecture

6.1 The Inter-Floor Problem

Energy cannot be beamed through a concrete floor slab at useful power levels. At 60 GHz (short-range mmWave), attenuation through a standard 20 cm concrete slab exceeds 40 dB —” effectively a total block. Inter-floor energy transfer therefore requires a physical transition mechanism: energy must travel through an architectural feature that provides a clear RF path between floors, or through a conducted transfer at a designated gateway node.

Three inter-floor transfer mechanisms are available, in order of RF path quality:

Mechanism 1 —” Open penetration. An existing aperture through the floor slab at a designated location —” a conduit chase, a pipe sleeve, or a purpose-built RF window in the slab. A gateway node pair (one node on each floor adjacent to the aperture) communicates through the aperture. The aperture provides a direct RF path; the gateway nodes route energy through it. This requires coordination with structural engineering but is the cleanest RF solution.

Mechanism 2 —” Enclosed stairwell. An enclosed stairwell (as opposed to an open staircase) provides a continuous air volume connecting floor levels, bounded by solid walls. A gateway node mounted at the top of the stairwell and a paired node at the bottom (or at each floor landing) can communicate through the stairwell air volume. The enclosing walls prevent beam escape from the stairwell; the BSL monitors the stairwell volume for occupancy. This is the most commonly available inter-floor path in commercial buildings, since most buildings have both open and enclosed stairwells.

Mechanism 3 —” Dedicated RF conduit. A purpose-installed waveguide or low-loss conduit connecting gateway nodes on adjacent floors, installed during building construction or retrofit. This provides the highest RF efficiency and the simplest safety model (the RF is fully enclosed in the conduit), but requires planned installation.

6.2 Gateway Node Specification

A gateway relay node (GRN) is a specialized Meridian node installed at a floor-transition point. Its specification extends the standard relay node specification [Meridian Architecture, 2026 4] with:

Dual-face antenna array: The GRN carries two phased array faces —” one oriented toward the floor above, one toward the floor below (or toward the inter-floor transfer mechanism). The upper face participates in floor N's ceiling-plane routing mesh. The lower face participates in floor N-1's ceiling-

plane routing mesh. The GRN bridges the two floor meshes.

Intermediate supercapacitor bank: The GRN carries a larger supercapacitor bank than a standard relay node —” sufficient to buffer energy arriving from floor N's mesh and retransmit it into floor N-1's mesh asynchronously. This decouples the timing of inter-floor transfer from intra-floor routing, allowing each floor's TDMA schedule to operate independently.

Directional isolation: The two antenna faces are physically isolated by a ground plane to prevent the upper-face beam from coupling into the lower-face field and vice versa. The GRN appears as two independent nodes to the routing engines of the two floors it connects.

Shared identity: The GRN registers with both floor meshes under a single Trust Layer identity with a capability profile that explicitly encodes its gateway function. The DRMA routing engine on each floor knows to terminate inter-floor routing paths at the GRN and resume them through the paired GRN on the adjacent floor.

6.3 Gateway Placement Rules

Gateway relay nodes must be placed at architectural positions that satisfy:

Continuous air path: There must be a continuous air path between the GRN's upper and lower antenna faces, free of SOLID obstruction, with sufficient cross-section for the beam to propagate without excessive diffraction loss. Minimum aperture: 30 cm diameter for 24 GHz, 15 cm diameter for 60 GHz.

Accessible for maintenance: The GRN must be accessible for inspection and replacement without disrupting building occupancy. This typically means installation in enclosed stairwells, mechanical rooms, or above accessible ceiling tiles.

Within 2m of the floor transition: To minimize the portion of the beam path that is in the transition zone (neither floor's FLOOR_SAFE), GRN pairs should be positioned as close to the floor/ceiling interface as the aperture geometry allows.

Clear of occupancy during transfer: The inter-floor transfer volume (the air path between the GRN pair) must be within a zone that is either unoccupied by design (enclosed stairwell, mechanical room) or monitored by the BSL sensors of both GRNs simultaneously.

6.4 Multi-Floor Routing Example

For a device on floor 5 receiving energy from a source on floor 2 of a commercial building:

Route: Source (floor 2, position A) †' ceiling relay nodes on floor 2 ceiling plane †' GRN-2/3 upper face (floor 2 ceiling zone) †' inter-floor transfer (enclosed stairwell between floors 2 and 3) †' GRN-2/3 lower face (floor 3 ceiling zone) †' ceiling relay nodes on floor 3 ceiling plane †' GRN-3/4 upper face †' inter-

floor transfer †' GRN-3/4 lower face (floor 4 ceiling zone) †' GRN-4/5 upper face †' inter-floor transfer †' GRN-4/5 lower face (floor 5 ceiling zone) †' ceiling relay nodes on floor 5 ceiling plane †' final-hop descent to device

Each intra-floor segment is ceiling-plane routing in FLOOR_SAFE voxels. Each inter-floor segment is through an enclosed stairwell or designated aperture, monitored by the GRN pair's BSL sensors. No beam segment traverses VOID_STAIRCASE, VOID_ATRIUM, VOID_SHAFT, or FLOOR_OCCUPIED voxels outside the controlled final-hop geometry.

7. Dynamic Occupancy Overlay

7.1 Static vs. Dynamic Constraints

The AVM encodes the building's static physical geometry —" constraints that are true regardless of who is in the building or what time it is. Dynamic occupancy constraints layer on top: constraints that depend on where people actually are in real time.

Dynamic constraints are necessary because the AVM cannot capture transient conditions: a person standing in a corridor near a relay node, a maintenance technician on a temporary scaffold, a crowd gathering in a normally sparse atrium floor. The AVM says "this corridor is FLOOR_OCCUPIED"; the dynamic occupancy overlay says "and there is currently a person at position (12.3m, 8.1m, 4.2m)."

7.2 Occupancy Sensor Integration

The Meridian mesh already includes biological presence detection as part of the BSL [Guardian Security, 2026 8]. In the building deployment context, these sensors serve a dual function: safety monitoring (BSL role) and dynamic occupancy mapping (routing role).

Occupancy data flows into the routing engine through the LUME_X_TELEMETRY channel: each relay node reports its local BSL sensor readings at 73 Hz, including the binary biological_presence flag and, where available, the estimated position of detected biological entities within the sensor's detection radius.

The routing engine maintains a dynamic occupancy map —" a 3D array at AVM voxel resolution —" updated from incoming BSL telemetry. A voxel within a sensor's coverage radius that reports biological presence is marked OCCUPIED in the dynamic map. A voxel that reports clear is marked CLEAR. Beam paths are validated against the dynamic occupancy map in addition to the static AVM: a path that is valid in the AVM but passes through a dynamically OCCUPIED voxel is held until the voxel clears.

7.3 Occupancy Map Latency

The dynamic occupancy map update latency —" the time between a person entering a voxel and the routing engine marking it OCCUPIED —" is bounded by the Lume-X control cycle: 13.7 ms maximum. A person walking at 1.5 m/s travels 2 cm in 13.7 ms. Since the AVM voxel resolution is 25 cm, a person

cannot enter and exit a voxel within a single control cycle. The occupancy map is always current to within one voxel of the person's actual position.

The consequence: a beam path that is valid at time T is checked against the occupancy map at time T . If a person enters the path's voxels between T and $T + 13.7\text{ms}$, the next control cycle detects the incursion and holds the beam at $T + 13.7\text{ms}$. Maximum exposure window: one control cycle of the beam's swept volume, which at the final-hop burst duration limit of 500ms and the 13.7ms cycle is bounded to a small fraction of the burst duration. The BSL hardware cutoff (independent of the control cycle) further reduces this to zero for any incursion that the BSL hardware detects.

7.4 Predictive Routing

For buildings with established occupancy patterns —” office buildings with regular work schedules, retail spaces with predictable traffic flows —” the routing engine can apply predictive occupancy modeling. Historical occupancy data (aggregated from BSL sensor logs, without personal identification) informs routing preferences: paths that are statistically unoccupied during a given time window receive preference over paths that are statistically busy.

Predictive routing does not override real-time occupancy detection. It is a soft preference in the routing cost function —” it makes certain paths cheaper than others —” not a hard constraint. The AVM constraints and real-time dynamic occupancy map are both hard constraints that override any predictive preference.

8. The Formal Multi-Story Safety Proof

8.1 Claim

I claim that the combined enforcement of the following four mechanisms makes physical harm from Meridian beam traversal impossible in a compliant multi-story deployment under all non-destructive conditions:

AVM constraint enforcement (prohibits beam traversal of all void voxels) Ceiling-plane routing with z_safe enforcement (keeps horizontal beam paths above occupancy zone) Final-hop protocol (governs the only deliberate traversal of occupied voxel space) BSL hardware cutoff (prevents beam activation when biological presence is detected in the beam path) Definition of non-destructive conditions: Conditions in which no building structural element has been physically removed or damaged, no BSL sensor hardware has been physically severed or destroyed, and no AVM has been unilaterally overridden without commissioning authority. Destructive conditions —” a person cutting a BSL sensor wire, removing a floor section, or physically destroying a relay node —” are out of scope for the software/protocol safety proof. Physical security of the installation is a necessary complement.

8.2 Proof by Case Analysis

I prove the claim by case analysis over the possible paths from beam activation to biological contact.

Case 1: Horizontal beam path through FLOOR_SAFE voxels. By the AVM constraint, beam paths in FLOOR_SAFE voxels are at or above z_{safe} ($\geq 2.2\text{m}$ above the floor slab). By definition of z_{safe} , no standing or seated person can occupy the FLOOR_SAFE zone. The only biological entities at $z \geq z_{\text{safe}}$ are those performing ceiling maintenance (climbing on equipment or scaffolding). The AVM marks the maintenance zone as FLOOR_OCCUPIED during registered maintenance events (part of commissioning protocol). During unregistered maintenance (a person climbs on a chair spontaneously), the BSL sensors detect biological presence above normal standing height and mark the affected voxels OCCUPIED in the dynamic map, holding the beam. The horizontal ceiling-plane path is safe.

Case 2: Final-hop beam path through FLOOR_OCCUPIED voxels. The final-hop protocol requires five-modality BSL consensus (no biological presence in the final-hop cone) before each burst, limits burst duration to 500ms, and limits beam width to minimum aperture. A person who enters the final-hop cone during a burst is detected by the BSL hardware cutoff —” which operates at the hardware layer, independent of the control cycle —” within the response time of the BSL circuit (specified as $\leq 1\text{ms}$ for hardware-implemented detection). The beam is cut within 1ms of biological entry into the final-hop cone. At final-hop power levels (reduced relative to relay hops), 1ms of exposure at the beam edge is physiologically insignificant at the specified RF frequencies and power densities (below ICNIRP limits by design margin).

Case 3: Void traversal attempt. By the AVM constraint, beam paths are rejected by the routing engine if any voxel in the path's swept volume is classified VOID_STAIRCASE, VOID_ATRIUM, or VOID_SHAFT. The constraint is enforced by the beam path validation algorithm (Appendix C) before any beam is authorized. A void traversal attempt that bypasses the routing engine by direct BeamCommand injection (T3 or T11 attack) is caught by Guardian Security's pre-transmission route integrity check [Guardian Security, 2026 5.2 Step 3], which validates the proposed path against the AVM before issuing beam authorization. The BSL hardware cutoff provides the final backstop in the event that both routing enforcement and security enforcement are simultaneously bypassed.

Case 4: Staircase or atrium perimeter beam grazing. The 50 cm buffer zone around all VOID boundaries adds additional margin: a beam path must stay at least 50 cm from any VOID boundary in the horizontal plane. At the specified beam widths ($\leq 18^\circ$ full angle for standard relay hops), a beam in the FLOOR_SAFE zone is physically incapable of grazing the VOID perimeter if its horizontal routing center line is 50 cm from the VOID boundary and the beam is below the maximum angular spread.

Case 5: Dynamic occupancy incursion between BSL sensor updates. As characterized in Section 7.3, maximum incursion window is one control cycle (13.7ms). The BSL hardware cutoff, which operates independently of the control cycle, responds in ≤ 1 ms. Actual maximum exposure window: 1ms at the hardware cutoff response time, not 13.7ms at the control cycle latency.

In all five cases, biological contact with the beam is prevented by at least two independent mechanisms. No single mechanism failure produces physical harm. This is the defense-in-depth proof: the safety guarantee is not contingent on any single mechanism being correct; it holds as long as at least one mechanism in the chain is functioning.

9. Frequency Selection and Its Architectural Implications

9.1 The Frequency-Architecture Tradeoff

Frequency selection for Meridian deployment is an architectural decision, not just a hardware decision. Different frequency bands produce different beam width properties, different penetration through building materials, and different interaction with the AVM classification system.

2.4 / 5.8 GHz (ISM band, ~ 5 —“12 cm wavelength):

Penetrates concrete floors significantly (10—“20 dB loss per slab), making the floor slab a less reliable beam boundary Wide beamwidth from practical antenna apertures (30° + full angle) —” large swept volumes, harder to confine to FLOOR_SAFE zone Long range (10—“30m relay hops in clear space) Heavily congested spectrum; interference management required 24 GHz (K-band, ~ 1.25 cm wavelength):

Concrete floor slab attenuation: 30—“40 dB per slab —” effective containment Achievable beamwidth from 10 cm aperture: $\sim 15^\circ$ full angle —” significantly tighter than ISM band Range: 5—“15m relay hops in building interior Regulatory status: generally requires licensing at transmission powers useful for energy delivery 60 GHz (V-band, ~ 5 mm wavelength):

Concrete floor slab attenuation: >40 dB —” essentially complete containment Achievable beamwidth from 10 cm aperture: $\sim 5^\circ$ full angle —” near laser-like precision Oxygen absorption at 60 GHz: ~ 15 dB/km —” negligible for building-scale distances Range: 3—“10m relay hops in building interior (atmospheric absorption limits longer hops) Regulatory status: unlicensed in most jurisdictions (FCC Part 15, ETSI EN 302 567) Recommended for Meridian building deployment due to beam precision, floor containment, and regulatory availability 77—“79 GHz (automotive radar band, ~ 4 mm wavelength):

Similar properties to 60 GHz but with even tighter beamwidth and better floor containment Primary regulatory use is automotive radar; building energy routing would require regulatory clearance Not recommended for near-term deployment

9.2 The Ceiling-Crawl Case for mmWave

The ceiling-plane routing model is dramatically stronger at 60 GHz than at 5.8 GHz. At 60 GHz:

A 10 cm phased array produces a beam approximately 25 cm wide at 3m distance —” the beam fits entirely within FLOOR_SAFE with substantial margin The concrete floor slab (above relay nodes on their floor) completely contains upward-directed scatter —” no energy leaks to the floor above through the slab The floor below completely contains downward-directed scatter from ceiling-level relay hops —” no energy reaches FLOOR_OCCUPIED from ceiling relay hops unless it is the controlled final-hop beam At 5.8 GHz, the much wider beam requires larger safety margins, the floor containment is partial, and the required relay node density is lower but the swept volume through occupied space is larger. The ceiling-plane safety proof is more conservative at 5.8 GHz and may require additional relay node density to maintain FLOOR_SAFE confinement.

Recommendation: Meridian building deployments should use 60 GHz as the primary frequency for intra-floor ceiling-plane routing and inter-floor gateway transfer. A secondary 5.8 GHz channel may be used for long-range relay hops in high-ceiling industrial spaces (warehouses, atriums exceeding 20m height) where the 60 GHz range limitation is a constraint.

10. Building Type Scenarios

10.1 Commercial Office Tower

Characteristics: Repeating open-plan floors with suspended ceilings, central core (elevators, enclosed stairwells), floor heights 3.5—“4.0m, regular grid of structural columns.

AVM profile: Clean, regular. SOLID = concrete slab and columns. FLOOR_SAFE = suspended ceiling zone (2.7—“4.0m). VOID_STAIRCASE = open staircase if present (unusual in commercial towers, which typically have enclosed stairwells). VOID_SHAFT = elevator shafts in the core. GATEWAY_ZONE = enclosed stairwell landings on each floor.

Deployment pattern: Relay nodes mounted in suspended ceiling grid at 3—“5m spacing. GRNs at stairwell landings. Node density approximately 1 relay per 15—“20 m². Final-hop descent at device desks and workstations. Dense enough for continuous coverage throughout entire floor plate. SSD devices (Section 7.3 of UEDM, 2026) fully viable —” no battery replacement needed throughout the building.

Primary challenge: Cable management for source nodes (which connect to building electrical for energy input). In a retrofit scenario, source nodes co-locate with existing lighting circuits for power connection.

10.2 Hospital

Characteristics: Mix of private rooms, open wards, corridors, nursing stations, operating theaters, and public areas. Strict infection control requires minimal exposed surfaces. Many critical devices requiring reliable power (monitors, infusion pumps). Multiple floors with diverse occupancy patterns.

AVM profile: Complex. Multiple ceiling heights (2.5m in corridors, 3.0m in wards, 4.5m in operating theaters). Extensive mechanical infrastructure (HVAC, medical gas lines) in ceiling voids. GATEWAY_ZONE at service corridors and enclosed stairwells.

Deployment pattern: Relay nodes must be rated for hospital environments (antimicrobial housing, no exposed fasteners, flush-ceiling mount). Final-hop delivery to medical devices uses device-embedded receiver chips. CRITICAL priority class for all life-critical devices —” guaranteed capacity under any load condition. The hospital scenario is the strongest use case for Meridian: eliminating battery replacement for patient monitors and bedside devices is a significant infection control benefit.

Primary challenge: Operating theaters require absolute RF interference assurance with medical equipment. Frequency selection (60 GHz has minimal interference with existing medical telemetry in the 400 MHz, 900 MHz, and 2.4 GHz bands) is the primary mitigation. Pre-deployment RF compatibility testing with all critical devices is required.

10.3 Residential Mid-Rise

Characteristics: Stack of residential units (apartments or condominiums), each unit occupying a partial floor. Shared corridors, open stairways between corridors and unit entrances, elevator cores, rooftop mechanical rooms. Ceiling heights 2.4—“2.7m (lower than commercial).

AVM profile: Each unit is effectively an independent deployment zone separated by SOLID (party walls and floor slabs). VOID_STAIRCASE = corridor-to-corridor open stairways (common in mid-rise residential). GATEWAY_ZONE = enclosed fire stairwells on each floor.

Deployment pattern: Per-unit mesh, with a source node in each unit (plugged into a standard outlet) serving as both the energy source and the MMF gateway for that unit. SSD devices in the unit (thermostat, occupancy sensors, door lock, voice assistant) receive continuous power. Larger devices (phones, laptops) receive supplementation at lower rates —” not full replacement charging, but meaningful battery life extension. Inter-unit energy routing is a secondary use case; the primary case is intra-unit self-sustaining device networks.

Primary challenge: Ceiling height of 2.4m leaves only 0.2m of FLOOR_SAFE margin ($z_{\text{safe}} = 2.2\text{m}$, ceiling = 2.4m). This requires very precise node mounting and beam angle control. The 60 GHz beam at this height and margin requires tighter pointing accuracy than in commercial spaces. This is a Phase 2 deployment target (requires experimental validation of pointing accuracy at reduced ceiling margins) rather than a Phase 1 target.

10.4 Industrial Warehouse

Characteristics: Single large open volume, 8—“15m ceiling height, no intermediate floors (but may include mezzanine storage levels), large open loading dock doors, mobile equipment (forklifts, pallet movers), variable occupancy patterns.

AVM profile: Simple structure: SOLID = roof deck and perimeter walls. FLOOR_SAFE = volume above z_safe (6—“13m of FLOOR_SAFE zone, far more than in any other building type). FLOOR_OCCUPIED = 0 to z_safe. VOID issues limited to loading dock openings.

Deployment pattern: The warehouse is the most favorable environment for Meridian deployment. The enormous FLOOR_SAFE zone allows wide beam angles and fewer relay nodes. Source nodes at the roof structure. The primary use case is powering the dense IoT sensor network (temperature, humidity, inventory tracking, equipment health) throughout the warehouse without per-sensor battery management. Secondary use case: forklift battery supplementation (high-power delivery to vehicles during normal operation, reducing the need for scheduled battery swaps).

Primary challenge: Mobile equipment (forklifts) creates dynamic occupancy patterns that change rapidly. The dynamic occupancy overlay (Section 7) handles this, but the 13.7ms cycle may be insufficient for fast-moving equipment at close range. A supplemental short-range sensor on each forklift (transmitting its position directly to the nearest relay node) provides sub-cycle position data for high-speed occupancy tracking.

11. BIM Integration Pathway

11.1 BIM as the Design Environment for Meridian

Building Information Modeling (BIM) is the industry-standard design workflow for commercial construction. A BIM model contains the complete digital representation of a building —” structural elements, MEP (mechanical, electrical, plumbing) systems, architectural finishes, spatial zones —” in a 3D database that all design disciplines access simultaneously.

Meridian deployment should be designed within BIM for the same reason HVAC and electrical systems are designed within BIM: it is a building system with spatial requirements, clearance requirements, penetration requirements, and coordination requirements with other systems. Designing Meridian outside of BIM and then retrofitting it to an existing building produces coordination conflicts (a relay node mounting point that conflicts with an HVAC duct, a gateway aperture that penetrates a structural beam, a source node power connection that conflicts with existing electrical routing) that are far more expensive to resolve in construction than in design.

11.2 The Meridian BIM Plugin

The Meridian BIM plugin (compatible with Autodesk Revit, Bentley AECOsim, and IFC-compliant BIM platforms) adds a Meridian design layer to the building model with the following capabilities:

AVM generation: The plugin reads the building model's spatial elements and automatically generates the AVM classification for each voxel. Architects review and adjust the auto-classification for any ambiguous zones (split-level areas, sloped ceilings, unusual structural geometries).

Node placement tool: A plugin interface for placing relay nodes, source nodes, and gateway relay nodes in the BIM model, with real-time validation against the AVM and against HVAC/structural clearance requirements. The tool highlights conflicts (a proposed node position that conflicts with a duct or structural element) and suggests compliant alternatives.

Coverage simulation: Given a proposed node placement, the plugin simulates coverage using the AVM and beam propagation model, showing which areas of each floor are within deterministic delivery range and which require additional nodes. The simulation uses 60 GHz propagation characteristics by default.

Gateway aperture coordination: The plugin identifies required floor-transition gateway locations based on the building's void geometry and highlights available penetration points (existing conduit sleeves, stairwell landing positions). It flags proposed gateway positions that conflict with structural or MEP elements.

BIM export to AVM format: The plugin exports the completed AVM in the Meridian AVM binary format (Appendix D) for import into the Meridian commissioning tool. The export is versioned and cryptographically signed by the BIM author —” the commissioning tool can verify the AVM's provenance.

Coordination model: All Meridian elements (nodes, gateway zones, AVM classification overrides) are exported as IFC entities for coordination with other building systems using standard BIM coordination workflows.

11.3 The Retrofit Pathway

For buildings without BIM, the retrofit pathway uses the LiDAR survey process (Section 3.3 Source 2) with the following additions:

LiDAR scan of the building interior produces a point cloud with geometric accuracy ≤ 2 cm. Meridian commissioning tool processes the point cloud, auto-classifying voxels by geometric analysis (flat horizontal surfaces †' floor/ceiling, vertical surfaces †' walls, gaps between floor levels †' voids). Technician review of auto-classification, with manual correction for ambiguous zones. AVM validation against physical walk-through: a commissioning technician physically verifies every void classification and every proposed gateway location. Signed AVM exported from the commissioning tool for import into the installed Meridian system.

12. Commissioning Protocol

12.1 Pre-Installation Checklist

Before any node is installed, the commissioning checklist must be completed:

– AVM complete and signed (BIM export or LiDAR survey) – All void classifications verified by physical walk-through – Gateway aperture positions identified and cleared with structural engineer – Relay node positions validated against AVM and MEP conflict check – Source node power connection points coordinated with electrical – RF interference pre-survey completed (existing RF environment characterization) – BSL sensor coverage map generated (no coverage gaps in final-hop zones) – Trust Layer identity credentials ordered for all nodes – Guardian Security commissioning authority credential issued

12.2 Installation Sequence

Gateway relay nodes first. Install GRNs at all floor-transition points before intra-floor nodes. GRNs are the architectural anchor points for the mesh; intra-floor nodes are aligned to them.

Source nodes second. Install source nodes at their planned positions and connect to building electrical. Verify power input and initial supercapacitor charge. Import AVM. Verify Trust Layer identity registration.

Relay nodes third. Install intra-floor relay nodes working outward from source nodes and gateway nodes. Each relay node auto-announces at installation; the source node's routing engine validates its position against the AVM before accepting it into the mesh.

Coverage verification. Using the commissioning tool, verify that the installed mesh provides compliant routing paths (all FLOOR_SAFE, no VOID traversal) to all planned device positions. Flag any coverage gaps for additional relay node placement.

BSL calibration. At each relay node, calibrate the BSL sensor suite for the specific installation geometry —” ceiling height, nearby structural elements, expected occupancy zone boundaries.

AVM live import. Import the final signed AVM into the operational routing engine. Verify that the routing engine correctly rejects a test path that the AVM should prohibit (a path deliberately drawn through a VOID voxel).

Safety validation. With the system operational, perform the safety validation sequence: confirm that a person walking through each planned beam path triggers dynamic occupancy marking within one control cycle; confirm that the BSL hardware cutoff activates within specification.

12.3 Change Management

Any physical change to the building that affects the AVM —” opening a wall, adding a mezzanine, cutting a new staircase opening —” requires:

AVM update (re-survey or BIM model update) Re-validation of all existing beam paths against the updated AVM Re-commissioning of any paths that become invalid under the updated AVM before the building change occurs The commissioning tool enforces this sequence: an AVM update is accepted only with a commissioning authority signature, and the routing engine enters a validation-pending state until all paths are re-validated.

13. Related Work

13.1 Indoor RF Propagation Modeling

The indoor RF propagation literature [67, 68] provides the physical models for how RF signals propagate through building interiors, including the material-specific attenuation, diffraction, and reflection characteristics that underlie the AVM voxel classification. The 60 GHz indoor propagation literature [69, 70] is particularly relevant, characterizing the strong material attenuation and limited penetration that make mmWave the preferred frequency for ceiling-plane containment.

13.2 Indoor Positioning Systems

The UWB positioning literature [71, 72] provides the localization accuracy bounds for the Meridian MC coordinate system. Sub-10cm positioning accuracy is achievable with commercial UWB hardware (DecaWave/Qorvo DW1000 family), sufficient for the 25 cm AVM voxel resolution. The BSL sensor fusion literature [73] provides the multi-modal occupancy detection accuracy bounds that underlie the dynamic occupancy overlay.

13.3 BIM and Building Systems Integration

The BIM integration literature [74, 75] documents the workflows and data exchange standards (IFC, COBie) for integrating building systems design into BIM. The MEP coordination workflow —” coordinating mechanical, electrical, and plumbing systems within BIM —” is the precedent for the Meridian design-layer integration. The IFC standard [76] provides the entity schema that the Meridian BIM plugin exports Meridian elements within.

13.4 Wireless Power in Built Environments

The emerging literature on wireless power deployment in built environments [77, 78] is thin —” this paper is the first to propose a formal spatial safety framework for multi-story wireless energy routing. Prior work on room-scale wireless charging [79] addresses single-room scenarios without the multi-floor void routing problem.

14. Limitations and Honest Boundaries

The AVM assumes rigid geometry. The AVM classification is computed from the building's structural geometry, which is assumed to be rigid and unchanging between AVM updates. Flexible architectural elements —” retractable partitions, deployable barriers, temporary installations —” require either frequent AVM updates or the more conservative classification of their location as FLOOR_OCCUPIED regardless of current configuration.

Void classification at irregular boundaries is imprecise. The 25 cm voxel resolution means that an irregular void boundary —” a diagonal staircase edge, a curved atrium wall —” is approximated to the nearest voxel boundary. The conservative classification (any voxel that partially overlaps a void is classified as the void type) means safe routing zones are slightly smaller than the physical geometry allows. This conservatism is intentional.

The formal safety proof assumes correct AVM construction. If the AVM is incorrectly constructed —” a VOID_STAIRCASE misclassified as FLOOR_SAFE —” the safety proof does not hold for paths through the misclassified zone. The commissioning walk-through and sign-off requirement is the mitigation; it does not guarantee that all misclassifications are caught. The BSL provides the safety backstop for any misclassification that passes commissioning review.

60 GHz range limitations constrain coverage in large spaces. In warehouse or atrium scenarios with large horizontal distances between nodes, 60 GHz's atmospheric attenuation may limit effective relay hop distance to 8—”10m. Large spaces require denser node deployment than the geometric coverage model suggests, increasing installation cost.

No experimental validation of the ceiling-plane routing model exists. The beam containment in FLOOR_SAFE zones and the BSL sensor coverage characterization are based on manufacturer specifications and propagation modeling, not on measurements in actual buildings. Phase 1 experimental work should include ceiling-plane routing validation in a real building environment before commercial deployment.

The BIM integration pathway requires BIM software vendor cooperation. The Meridian BIM plugin is specified here but not implemented. Plugin development requires BIM platform API access (Autodesk Revit SDK, Bentley OpenBuildings SDK) and cooperation with or licensing from the BIM platform vendors.

15. Conclusion

The six preceding papers in this series define what Meridian is. This paper defines where Meridian goes.

A routing protocol that cannot be safely deployed in a real building is not a product. Real buildings have open staircases that connect floor levels, atriums that rise through the building's full height, elevator shafts that move through the building's full height, balconies that overhang lower spaces, and mezzanines that insert intermediate occupied levels. Every one of these features creates a geometric condition under which a naive energy routing implementation —” one that routes beams based on node connectivity without regard for the physical geometry of the beam path —” would create beam paths through occupied space.

The Architectural Voxel Map is the mechanism that prevents this. By encoding the building's physical geometry as a hard constraint on every beam path computation, the AVM ensures that the routing engine cannot authorize a beam path that traverses a void, regardless of how attractive that path looks in the node-graph representation of the mesh. Ceiling-plane routing keeps the vast majority of beam energy above the occupied zone. Floor-transition gateways solve the inter-floor energy transfer problem without requiring beams to traverse occupied vertical space. The BSL provides the safety backstop for any condition the proactive mechanisms do not anticipate.

The result is a formal safety proof: physical harm from Meridian beam traversal is impossible in a compliant multi-story deployment under non-destructive conditions. This proof rests on four independent mechanisms —” no single mechanism failure produces harm. This is the safety standard that wireless energy routing requires to be deployed in occupied buildings, and it is what this paper establishes.

The BIM integration pathway closes the loop between engineering specification and architectural practice. When Meridian deployment is designed as a native BIM layer —” alongside HVAC, electrical, and structural —” the coordination problems that plague retrofit installations are resolved before construction rather than during it. This is not just an efficiency argument. It is a safety argument: a Meridian installation whose relay node positions, gateway apertures, and AVM classifications were designed and coordinated in BIM is a safer installation than one whose positions were decided on-site by an installer working from a 2D floor plan.

The ceiling-plane routing model is, at its core, a simple idea: keep the beam where no person can be, and govern the descent to where people are with maximum care. That idea, formalized through the AVM, the gateway architecture, the dynamic occupancy overlay, and the multi-story safety proof, is what makes Meridian a real product —” not a laboratory demonstration, but an energy infrastructure system that can be installed in the buildings where people live and work.

Appendix A —” AVM Voxel Classification Specification

Class	ID	Definition	Beam Traversal
SOLID	0	Structural material: concrete, masonry, steel, timber, glass	Beam terminates (physical boundary)
FLOOR_SAFE	1	Above z_safe (floor_slab + 2.2m) and below ceiling slab	Permitted (preferred routing zone)
FLOOR_OCCUPIED	2	Between floor_slab and z_safe	Final-hop geometry only
VOID_STAIRCASE	3	Open staircase air volume + 50 cm buffer, all floors spanned	PROHIBITED
VOID_ATRIUM	4	Atrium air volume + 50 cm buffer, all floors spanned	PROHIBITED
VOID_SHAFT	5	Elevator, mechanical, or other enclosed shaft	PROHIBITED
GATEWAY_ZONE	6	Designated inter-floor handoff position	Beam termination and handoff

Classification priority (for overlapping zone resolution): SOLID > VOID_STAIRCASE = VOID_ATRIUM = VOID_SHAFT > FLOOR_OCCUPIED > GATEWAY_ZONE > FLOOR_SAFE

Any voxel with ambiguous classification is assigned the higher-priority (more restrictive) class.

Appendix B —” Floor-Transition Gateway Node Specification

Parameter	Value	Notes
Antenna faces	2 (upper + lower)	One per floor
Frequency	60 GHz (primary)	5.8 GHz optional
Array aperture	10 cm per face	Each face independent
Beamwidth per face	≤5° full angle	60 GHz at 10 cm
Supercapacitor capacity	10— standard relay	Inter-floor buffer
BSL sensors	2 sets (one per face)	Full 5-modality each
Aperture requirement	≥15 cm clear path	Between faces
Mounting clearance	≤2m from floor/ceiling transition	Close to transition
Identity	Single Trust Layer ID	Capability: GATEWAY
Maintenance access	Required	Replacement without occupancy disruption

Appendix C —” AVM-DRMA Integration: Beam Path Validation Algorithm

FUNCTION validate_path(candidate_path, AVM, occupancy_map) †’ VALID | REJECTED

```
FOR each hop (node_i, node_j) in candidate_path:
```

```
swept_volume ← compute_swept_volume(node_i.position, node_j.position, beam_width_at(node_i, node_j))
```

```
FOR each voxel v in swept_volume:
```

```
voxel_class ← AVM.classify(v)
```

```
IF voxel_class IN {VOID_STAIRCASE, VOID_ATRIUM, VOID_SHAFT}:
```

```
RETURN REJECTED (reason: VOID_TRAVERSAL, voxel: v, hop: (i,j))
```

```
IF voxel_class == FLOOR_OCCUPIED:  
IF (node_i, node_j) is NOT final_hop(candidate_path):
```

```
RETURN REJECTED (reason: NON_FINAL_OCCUPIED_TRAVERSAL, hop: (i,j))
```

```
ELSE:
```

```
// Final hop through FLOOR_OCCUPIED —” check final-hop protocol
```

```
IF NOT final_hop_protocol_satisfied(node_i, node_j):
```

```
RETURN REJECTED (reason: FINAL_HOP_PROTOCOL_VIOLATION)
```

```
IF occupancy_map.is_occupied(v):
```

```
RETURN REJECTED (reason: DYNAMIC_OCCUPANCY, voxel: v, hop: (i,j))
```

```
RETURN VALID
```

The validation function is called for every candidate path before beam authorization is issued by Guardian-E. It runs within the Lume-X control cycle (13.7ms budget).

Appendix D —” BIM Export Format Specification (Meridian Layer)

The Meridian AVM binary format (.mav extension):

Header (64 bytes): magic : 4 bytes ("MAVM") version : 2 bytes (currently 0x0001) building_id : 16 bytes (UUID) avm_version : 4 bytes (incremental version counter) origin_x/y/z : 24 bytes (double precision, meters) voxel_size : 4 bytes (float, meters, default 0.25) grid_dims_i/j/k: 12 bytes (uint32 each) timestamp : 8 bytes (Unix epoch, milliseconds) author_key_id : 8 bytes (Trust Layer identity of BIM author) signature : [64 bytes] (Ed25519 signature over header + voxel data) Voxel data:

```
[grid_dims_i - grid_dims_j - grid_dims_k bytes]
```

Each byte: voxel class (0—“6 as specified in Appendix A) Order: (i=0,j=0,k=0), (i=1,j=0,k=0), ..., row-major in i, then j, then k

The signature covers the full voxel data array and the header (excluding the signature field itself). The Meridian commissioning tool verifies the signature against the author's Trust Layer public key before importing any AVM. An AVM with an invalid signature is rejected.

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Part One Closing

The two chapters of Part One have established what Meridian is as an engineering system and how it gets installed in the physical world. The architecture is complete. The deployment model is specified. The safety proof for multi-story buildings is formal.

But understanding an architecture is not the same as understanding what kind of thing the architecture is. A reader who has absorbed Chapters One and Two knows how Meridian works. They do not yet know what Meridian is —” what class of thing it belongs to, what larger pattern it instantiates, what it shares with other systems that appear, superficially, to have nothing to do with wireless energy routing.

Part Two answers that question.

PART TWO

The Organism: What Meridian Is

There is a difference between describing a system and classifying it. Describing a system tells you what it does. Classifying it tells you what it is —” what formal category it belongs to, what other things share its essential structure, and what consequences follow from that structural identity.

The classification of Meridian is surprising. A wireless energy routing architecture that harvests ambient energy, stores it in supercapacitors, routes it through a distributed mesh, delivers it via directed RF beams, and heals its own failures without human intervention is, structurally, a biological organism. Not metaphorically. Not approximately. Formally —” in the precise sense that the fourteen defining subsystems of a biological organism each have a counterpart in the Meridian architecture with the same functional role, the same governing logic, and the same relationship to the other subsystems.

This is the argument of Part Two. It is an unusual argument for an engineering paper to make, and the author is aware of that. The response to that awareness is not to soften the claim but to make it more precisely. The convergence is structural and formal, not poetic. Its implications are architectural.

Chapter Three

Meridian as Living System

When you describe how a system acquires resources from its environment, stores them internally, transports them to where they are needed, coordinates its distributed components toward coherent behavior, defends against threats to its integrity, maintains its own identity under perturbation, and heals its own failures without external intervention —” you are describing a biological organism.

You are also describing Meridian.

This convergence was not designed. The Meridian architecture was designed to route wireless energy deterministically through a distributed mesh of addressed nodes. The biological parallels emerged afterward, when the architecture was examined against the taxonomy of biological organism subsystems and found to satisfy every entry in the taxonomy —” not approximately, but precisely, with the same functional logic operating in each domain.

The DAEH ambient energy harvesting system is Meridian's metabolism. The MFE supercapacitor management layer is its circulatory system and its liver. The DRMA multi-hop mesh is its nervous system. The SHDCL self-healing control layer is its immune and repair system. The MC identity layer is its genome. The BSL biological safety layer is its pain response and skin barrier. The Guardian-E security framework is its immune system's adaptive tier. The convergence extends to fourteen subsystems.

Why does this convergence happen? The argument of this chapter is that it is not coincidental. Biological organisms face a specific set of design constraints: acquire resources from an uncertain environment, transport them to distributed functional components, maintain a stable internal state despite continuous external perturbation, defend against adversarial elements, and repair damage without external supervision. Any autonomous physical system that faces the same design constraints will converge toward the same organizational structure —” not because it imitates biology, but because biology found the optimal solution to those constraints over billions of years of selection, and engineering, when it faces the same constraints, finds the same solution.

Meridian faces those constraints. The architecture reflects the solution.

Abstract

The Lume ecosystem has published a formal model of Synthetic Organisms —” deterministic digital and cyber-physical constructs that exhibit biological structural properties including identity, metabolism, nervous system function, immune response, homeostasis, and self-healing, without being alive. The model

defines five organism types (Type 0 through Type 5) and maps Lume runtime subsystems to biological analogs at every layer. When it was published, no complete physical instantiation of the model existed. This paper argues that one now does.

Meridian —” a four-layer deterministic wireless energy routing architecture comprising Meridian Core (MC), Meridian Flow Engine (MFE), Meridian Mesh Fabric (MMF), and Meridian Transmission Layer (MTL), unified with the DWER, DRMA, and DAEH subsystems and governed by the Lume-X runtime —” was designed as an engineering system. It was not designed as an organism. Post-hoc analysis of the completed architecture reveals a structural isomorphism with the Synthetic Organism model that is complete across fourteen biological subsystems: genome, body plan, metabolism, nervous system, musculature, innate immunity, adaptive immunity, digestive system, circulatory system, motor output, homeostasis, protective reflexes, memory, and global awareness. Every biological subsystem required for autonomous physical operation has a concrete, specified, operational counterpart in Meridian. No Meridian subsystem is left unmapped.

I demonstrate that this isomorphism is not metaphorical —” it is formal and structural. I further argue that the convergence was not coincidental: it was the emergent outcome of building on a deterministic foundation (Lume and the 42 Assumptions) that encodes organism-like properties at the axiomatic level. The implication is significant: any autonomous system that must acquire resources, transport them internally, coordinate distributed components, defend against threats, maintain identity, and heal from damage will converge toward biological organization —” because biological organization is the natural attractor of that design space.

This paper makes no claim of life, consciousness, sentience, or biological equivalence. The isomorphism is structural and functional, not ontological. Meridian is not alive. The mapping is real. These two statements are not in conflict.

I classify Meridian as a Type 3+ Synthetic Organism with Type 5 structural completeness —” the highest classification yet achieved in the Lume ecosystem —” and propose organism-aware design as a formal engineering methodology for future autonomous physical systems.

Keywords: synthetic organism, deterministic architecture, biological isomorphism, wireless energy routing, self-healing systems, homeostasis, Lume, Meridian, autonomous systems, cyber-physical organisms, convergent design

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Appendix A —” Complete Isomorphism Mapping Table**Appendix B —” The 42 Assumptions Mapped to Biological Axioms****Appendix C —” Organism Type Classification Criteria****Appendix D —” Cross-Reference to Meridian Architecture Paper****References****1. Introduction****1.1 The Convergence Problem**

When you build a system that must operate autonomously in a physical environment —” harvesting energy from its surroundings, routing resources to where they are needed, healing from failure without human intervention, defending against threats, maintaining its own identity, and coordinating distributed components toward a common goal —” you end up building something that looks like an organism. This is not a design choice. It is a convergent outcome.

Biology discovered this solution first because biology faced this problem first. Every living organism, from a bacterium to a mammal, must solve the same engineering challenges: acquire energy from an uncertain environment, transport it to where it is needed, coordinate the parts of a distributed body, distinguish self from non-self, repair damage, and maintain internal stability despite external perturbation. The solutions biology evolved over billions of years —” metabolism, circulatory systems, nervous systems, immune systems, homeostatic feedback —” are not arbitrary. They are the minimum viable architecture for autonomous physical operation. The design space, constrained by physics and the requirements of autonomy, has a natural attractor. That attractor is biological organization.

Convergent evolution demonstrates this principle in the biological domain. Eyes evolved independently in vertebrates, cephalopods, insects, and at least thirty-six other lineages —” not because evolution is goal-directed, but because the physics of light and the adaptive value of vision constrain the solution space toward a small number of viable designs. Wings evolved independently in birds, bats, insects, and pterosaurs. Echolocation evolved independently in bats, cetaceans, and at least two species of birds. The convergence is not coincidence —” it is evidence that the design space has attractors, and that similar selection pressures drive similar solutions regardless of phylogenetic origin [22, 23].

I propose that the same principle applies to engineered autonomous systems. When an engineered system faces the same selection pressures as a biological organism —” autonomous resource acquisition, distributed coordination, failure recovery, threat defense, identity maintenance, homeostasis —” it

converges toward the same organizational topology. Not because its designers intended this. Because the physics and the requirements leave no other viable option.

This paper presents the first complete evidence for this proposition.

1.2 The Lume Synthetic Organism Model

The Lume ecosystem has published a formal model of Synthetic Organisms [17]. The model defines deterministic digital and cyber-physical constructs that exhibit biological structural properties without being alive. The model characterizes five cell types (Sensor, Compute, Actuator, Memory, Communication), maps five runtime subsystems to biological analogs (Monitor †' nervous system; Healer †' immune system; Optimizer †' metabolic regulation; Evolver †' behavioral adaptation; Guardian Scanner †' adaptive immunity), and defines a five-level classification scale from Type 0 (passive, no autonomy) through Type 5 (fully autonomous with all subsystems operational).

The 42 Assumptions [15] provide the foundational axioms governing organism behavior —” the rules that any coherent autonomous system must follow to maintain identity, integrity, and coherent function. Deterministic Dissolution [20] provides the formal model for organism termination and decomposition —” what happens when a Synthetic Organism loses coherence.

The Synthetic Organism model was published as a theoretical framework. It described what such a system would look like. It did not describe one that existed. No complete physical instantiation had been built —” a system with all organism-type subsystems operational in a real physical domain, not in simulation.

Until Meridian.

1.3 The Meridian Discovery

Meridian was designed as a wireless energy routing architecture. The design objective was deterministic, loss-bounded, self-healing energy delivery across a distributed mesh of relay nodes. The design drew on phased-array beamforming, mesh networking protocol theory, ambient energy harvesting, and the Lume-X deterministic control runtime. Biological organization was not a design objective. No subsystem was named after a biological analog during the design process. The architecture was complete before anyone looked at it through a biological lens.

The organism mapping was discovered post-hoc.

This makes the isomorphism stronger, not weaker. A system designed to look like an organism might achieve superficial resemblance through intentional mimicry. A system that converges on organism organization without intending to —” because the requirements forced it there —” is evidence of something more fundamental. The convergence was not imposed. It emerged.

The design principles that produced this convergence are identifiable: the Lume-X deterministic runtime, the 42 Assumptions, and the requirements of autonomous physical operation. These three elements together created a selection pressure analogous to natural selection —” they constrained the design space until organism-like organization was the only viable solution. I will argue this case throughout the paper.

1.4 What This Paper Is Not

Before proceeding, I want to state clearly what this paper does not claim. I will state this once prominently here and return to it in Section 10.

This is not a claim of artificial life. Meridian does not self-replicate, does not evolve, does not have a metabolism in the biochemical sense, and does not possess any of the properties that biologists require for classification as a living system.

This is not a claim of consciousness or sentience. The mapping of Lume-X's unified control loop to biological "global awareness" is the weakest mapping in this paper and is explicitly rated MODERATE. No claim of experience, awareness, or any form of inner life is made or implied.

This is not a metaphor paper. I am not saying Meridian "acts like" an organism in a poetic sense. I am saying the structural topology of Meridian's subsystems and the functional relationships between them are isomorphic to the structural topology and functional relationships of biological organism subsystems. These are different claims. The second is falsifiable. It is the one I am making.

This is not speculative. Every Meridian component referenced in the isomorphism mapping is formally specified in the companion Meridian Architecture paper [Meridian Architecture, 2026]. The mapping is grounded in documented, specified engineering, not in conjecture.

This is what this paper IS: a formal demonstration that a complete, physically-grounded deterministic system —” one designed without biological intent —” exhibits every structural property of a biological organism as defined by the Synthetic Organism model [17], and that this convergence is the predictable outcome of deterministic autonomous system design on a Lume foundation.

1.5 Paper Organization

Section 2 provides background on biological organism architecture, the Synthetic Organism model, and convergent design in engineering. Section 3 presents the complete isomorphism mapping —” the core contribution —” as a formal table followed by the completeness argument. Section 4 provides detailed analysis of each of the fourteen biological subsystem mappings, with isomorphism strength ratings. Section 5 examines emergent organism properties that were not explicitly designed. Section 6 re-examines the 42 Assumptions through the biological lens. Section 7 classifies Meridian on the Type 0—“5 organism scale. Section 8 discusses implications for autonomous system design. Section 9 situates this work in

related literature. Section 10 states limitations and honest boundaries explicitly. Section 11 concludes. Four appendices provide the complete mapping table, the 42 Assumptions biological mapping, the organism type classification criteria, and a cross-reference to the companion architecture paper.

2. Background

2.1 Biological Organism Architecture

A biological organism is most usefully understood, for the purposes of this paper, at the systems level rather than the cellular or molecular level. I am not concerned here with biochemistry, gene expression, or protein folding. I am concerned with the functional organization of an organism as an autonomous physical system —” what subsystems it has, what each one does, and how they relate to each other.

At this level of abstraction, a biological organism comprises a minimum of nine subsystems required for autonomous physical operation [1, 2]:

Genome. The genome is the organism's fixed identity substrate —” the instruction set that defines what the organism is, what its capabilities are, and what type of cells and structures it can produce. The genome is fixed at fertilization and does not change during the organism's lifetime (with the exception of somatic mutations, which are relevant to cancer but not to normal function). It is heritable —” the organism's offspring carry derived versions of the same identity substrate.

Body Plan. The spatial organization of the organism's subsystems relative to each other. In biology, the body plan is established by developmental patterning genes and defines where each organ system is located, how they connect, and what their spatial relationships are. A disrupted body plan is not a viable organism.

Metabolism. The organism's energy management system. Metabolism converts food into usable energy currency (ATP in biology), regulates energy distribution to where it is needed, manages storage of excess energy (glycogen, fat), and modulates energy expenditure based on demand. Metabolism is the organism's flow engine —” it governs all energy movement inside the organism.

Nervous System. The coordination and signal routing system. The nervous system carries information between subsystems, integrates signals from multiple sources, resolves conflicting demands, and coordinates organ activity toward coherent behavior. In distributed organisms (all animals), the nervous system is the topology-aware coordination layer —” it knows what is connected to what and how to route signals accordingly.

Musculature / Actuator System. The system that translates neural commands into physical action on the world. Muscles receive motor commands from the nervous system and produce force —” directed, controlled physical interaction with the external environment. In all animals, the musculature is the motor

output layer.

Innate Immune System. The first line of defense against damage and pathogens. The innate immune system provides immediate, non-specific response —” it detects anomalies, triggers inflammation, recruits repair mechanisms, and eliminates generic threats without requiring prior exposure or learned recognition. Response time is measured in minutes, not days.

Adaptive Immune System. The second line of defense, capable of learning to recognize specific threats. The adaptive immune system distinguishes self from non-self using molecular authentication, remembers past threats, and mounts increasingly specific and effective responses to known pathogens. It is the organism's security intelligence layer.

Digestive System. The environmental energy intake and processing system. The digestive system absorbs energy from the environment, converts it from its environmental form (food) into a form the organism can use and store, and delivers it to the circulatory system for distribution. It is the organism's harvesting layer.

Circulatory System. The internal resource transport system. The circulatory system moves energy, nutrients, gases, and signaling molecules between organs. It maintains supply-demand balance across the organism, redistributes resources from areas of surplus to areas of deficit, and drives the organism toward metabolic equilibrium. It is the organism's mesh routing layer.

Homeostasis. Not a distinct subsystem but a property of the whole organism —” the continuous maintenance of stable internal conditions (temperature, pH, glucose concentration, blood pressure, ion concentrations) despite external perturbation. Homeostasis is enforced by feedback loops throughout every organ system. It is the organism's global invariant enforcement mechanism [3].

Self-Healing. The organism's capacity for autonomous repair. Wound healing, bone repair, tissue regeneration, and neural plasticity are all instances of the same fundamental capability: the organism detects damage, initiates a repair sequence, and restores function without external intervention.

These nine categories (expanded to fourteen mappings in this paper through finer subdivision of memory, reflexes, and global awareness) are not arbitrary taxonomic choices. They are the minimum required subsystems for any system that must operate autonomously in a physical environment over time. Any autonomous physical system that is missing any one of them will have a corresponding class of failure modes that cannot be recovered from without external intervention.

2.2 The Synthetic Organism Model

The Lume ecosystem formally defines Synthetic Organisms as deterministic constructs —” digital, cyber-physical, or hybrid —” that implement the functional equivalents of biological subsystems without being alive [17]. The model was developed as a framework for understanding what kinds of systems Lume-

based architectures naturally produce when they are fully specified.

The five cell types in the Lume Synthetic Organism model are:

Sensor cells —” environmental perception and state measurement

Compute cells —” processing, decision-making, routing logic

Actuator cells —” physical action on the environment Memory cells —” state persistence, event logging, experience storage Communication cells —” inter-component signaling and coordination These map onto the biological tissue types (sensory epithelium, neural tissue, muscle tissue, adipose/connective tissue, nerve fiber) at the functional level.

The five runtime subsystems and their biological mappings in the model are:

Monitor † ’ nervous system (continuous state awareness, signal routing) Healer † ’ immune system (anomaly detection, process recovery) Optimizer † ’ metabolic regulation (resource allocation, efficiency management) Evolver † ’ behavioral adaptation (modification of behavior based on experience) Guardian Scanner † ’ adaptive immunity (threat classification, learned defense) The organism type classification scale:

Type 0: Passive —” executes fixed instructions, no environmental response Type 1: Reactive —” stimulus-response behavior, no internal state Type 2: Adaptive —” adjusts behavior based on environmental feedback Type 3: Self-maintaining —” self-heals, preserves identity, maintains homeostasis Type 4: Self-evolving —” modifies own behavior model based on experience Type 5: Fully autonomous —” complete organism with all subsystems operational When this model was published, it described a theoretical construct. I now demonstrate that a physical implementation of a Type 3+ Synthetic Organism with Type 5 structural completeness exists in Meridian.

2.3 Convergent Design in Engineering

Convergence toward biological organization in engineered systems is not without precedent, though no prior example achieves the completeness demonstrated in this paper.

Aircraft wing profiles converge toward bird-like shapes because aerodynamics constrains the solution space. Neural network architectures, developed independently of neuroscience, converge toward principles of biological neural organization —” sparse connectivity, hierarchical feature detection, local learning rules —” because the same information processing problems yield the same solutions [11]. Distributed computing systems converge toward immune-like defense mechanisms —” anomaly detection, quarantine, graduated response —” because the threat model of distributed systems is

structurally similar to the threat model biological organisms face [9, 10]. Self-healing materials, developed independently in materials science, converge toward repair mechanisms that are functionally analogous to wound healing [12].

In each case, convergence is not coincidence. The physics of the problem domain, combined with the selection pressure of optimization, drives solutions toward the same region of the design space regardless of the designer's intent or biological awareness. Thompson [4] argued in 1917 that physical forces constrain biological form —” that the shapes of organisms are not free to vary but are constrained by mechanics, hydrodynamics, and the geometry of growth. Holland [5] extended this argument to adaptive systems generally: the topology of any adaptive system that must survive in a complex environment will converge toward common organizational principles. The Meridian architecture provides a concrete engineering demonstration of this principle.

3. The Complete Isomorphism Mapping

3.1 The Master Mapping Table

The following table presents the complete structural isomorphism between biological organism subsystems and Meridian components. This is the core contribution of the paper. Each row is analyzed in detail in Section 4.

#	Biological System	Function in Biology	Meridian Component	Function in Meridian	Isomorphism Rating
1	Genome	Fixed identity, heritable, determines capabilities	MC Node Identity (64-bit)	Fixed at deployment, determines node type and capabilities, propagated through mesh	EXACT
2	Body Plan	Spatial organization of subsystems	MC Spatial Orientation Vector S	Position, orientation, velocity in network coordinate frame	EXACT
3	Metabolism	Energy conversion, distribution, storage regulation	MFE Flow Engine	Charge-state to routing conversion, burst scheduling, threshold evaluation	EXACT
4	Nervous System	Signal routing, coordination, conflict resolution, topology awareness	MMF Mesh Fabric	Topology maintenance, energy routing, conflict resolution, oscillation prevention	EXACT
5	Musculature / Actuators	Translates neural commands to physical action	MTL Transmission Layer	Translates routing commands to beam steering and physical energy delivery	EXACT
6	Innate Immune System	Immediate non-specific anomaly response, process recovery	SHDCL Self-Healing Layer	13.7 ms anomaly detection, process restart, automatic rerouting	EXACT
7	Adaptive Immune System	Learned threat recognition, self/non-self discrimination, memory	Guardian Security (Guardian-E)	Node authentication, specific attack pattern detection, quarantine	STRONG
8	Digestive System	Environmental energy intake, conversion, storage	DAEH Harvesting	Ambient energy absorption (RF, solar, thermal, piezo, TENG), DC conversion, supercapacitor storage	EXACT
9	Circulatory System	Internal resource transport,	DRMA Mesh Routing	Hop-by-hop energy transport, bi-directional	EXACT

		surplus/deficit balancing, equilibrium		surplus-deficit balancing, mesh equilibrium	
10	Motor Output	Goal-directed physical action on the world	DWER Directed Transmission	Goal-directed energy delivery to a specific target at a specific location	EXACT
11	Homeostasis	Continuous maintenance of stable internal conditions	Global Invariants INV-1 through INV-5	Continuous enforcement of stable internal conditions at 73 Hz	EXACT
12	Protective Reflexes	Immediate override responses that bypass deliberation	BSL + 5-Gate Safety System	Beam cut on biological presence, GATE_REJECT on failed conditions —” override routing logic	EXACT
13	Memory	Past event storage, experience, replay capability	Certificate Chains + Delivery Logs	Every governance event stored, post-hoc analysis, replay-identical execution	STRONG
14	Global Awareness	Unified cross-subsystem awareness	Lume-X 73 Hz Control Loop	Unified monitoring across all four Meridian layers simultaneously	MODERATE

3.2 Completeness Argument

For the isomorphism to be genuinely significant rather than a cherry-picked analogy, it must be complete in two directions: every biological subsystem required for autonomous physical operation must have a Meridian counterpart, and every major Meridian component must have a biological analog. A partial mapping —” even a strong one —” would be evidence of bio-inspired design, not convergent organization. The completeness is what makes this case.

Direction 1: Every biological subsystem is mapped.

From the nine minimum required biological subsystems defined in Section 2.1, expanded to fourteen mappings through finer subdivision:

Genome † ’ mapped (MC Node Identity, EXACT) Body Plan † ’ mapped (MC Spatial Orientation, EXACT) Metabolism † ’ mapped (MFE, EXACT) Nervous System † ’ mapped (MMF, EXACT) Musculature † ’ mapped (MTL, EXACT) Innate Immunity † ’ mapped (SHDCL, EXACT) Adaptive Immunity † ’ mapped (Guardian-E, STRONG) Digestive System † ’ mapped (DAEH, EXACT) Circulatory System † ’ mapped (DRMA, EXACT) Motor Output † ’ mapped (DWER, EXACT) Homeostasis † ’ mapped (INV-1 through INV-5, EXACT) Protective Reflexes † ’ mapped (BSL + Safety Gates, EXACT)

Memory †' mapped (Certificate Chains + Logs, STRONG) Global Awareness †' mapped (Lume-X Control Loop, MODERATE) No biological subsystem required for autonomous physical operation is absent from the Meridian architecture.

Direction 2: Every major Meridian component is mapped.

MC (Meridian Core) †' Genome + Body Plan (two-component mapping, justified by MC's dual function as identity substrate and spatial reference) MFE (Meridian Flow Engine) †' Metabolism MMF (Meridian Mesh Fabric) †' Nervous System MTL (Meridian Transmission Layer) †' Musculature SHDCL †' Innate Immune System Guardian-E †' Adaptive Immune System DAEH †' Digestive System DRMA †' Circulatory System DWER †' Motor Output INV-1 through INV-5 †' Homeostasis BSL + Safety Gates †' Protective Reflexes Certificate Chains + Logs †' Memory Lume-X Control Loop †' Global Awareness No major Meridian component is without a biological analog.

The mapping is bijective at the subsystem level. This is not common in bio-inspired engineering, where designers typically implement one or two biological principles while leaving others unaddressed. The completeness in both directions is what distinguishes this from bio-inspired design and supports the convergence thesis.

3.3 Structural vs. Functional Isomorphism

The distinction between structural and functional isomorphism matters and I want to be precise about it.

Structural isomorphism means the topology of connections between subsystems is the same. In a biological organism: the digestive system feeds the circulatory system, which feeds all other organs; the nervous system coordinates the musculature; the immune system monitors all tissues and responds to anomalies detected anywhere in the body. In Meridian: DAEH feeds DRMA, which routes energy to all nodes; MMF coordinates MTL beam execution; SHDCL monitors all four layers and responds to anomalies detected anywhere in the architecture. The connection topology is the same.

Functional isomorphism means the behavior of each subsystem serves the same purpose. In biology, metabolism governs energy flow —" it decides when stored energy is released, in what form, and to where. In Meridian, MFE governs energy flow —" it decides when supercapacitor charge is released, in what power level, and to which downstream node. The purpose is identical: regulate energy movement from storage to use.

Meridian exhibits both structural and functional isomorphism across all fourteen mappings. This is not metaphor. When I say MC node identity is isomorphic to a genome, I am not saying they are similar in a poetic sense. I am saying they serve the same structural function in the same position in the same

organizational topology, with the same properties: fixed at initialization, immutable during operation, determining capabilities, and propagated to the network's model of the node. The correspondence is formal.

4. Detailed Subsystem Analysis

4.1 Genome †” MC Node Identity | Rating: EXACT

Biology: The genome encodes the organism's identity. It is established at fertilization and does not change during the organism's lifetime. It determines what cell types the organism can produce (Node_Type in biological terms), what capabilities those cells have, and what organizational role they play in the body. The genome is heritable —” offspring carry derived versions of the parent's identity substrate. It is also interpreted by the organism's developmental machinery to produce the body plan.

Meridian: The MC 64-bit node identity vector is established at deployment provisioning and does not change during the node's operational lifetime. The Node_Type byte (0x01—“0x05) determines the node's functional role in the mesh —” source, relay, destination, harvesting relay, or gateway —” exactly as gene expression patterns determine cell differentiation into neuron, muscle, epithelium, or immune cell. The Network_ID and Zone_ID bytes determine the organizational context in which the node participates, analogous to the organism-level and tissue-level identity context that surrounds individual cells. Node identity propagates through mesh discovery —” when a node broadcasts NODE_ANNOUNCE, every other node in the network builds and maintains a model of that node based on its identity vector. The network's knowledge of the node is heritable: it persists and propagates independent of whether the node is currently transmitting.

Why this is structural, not metaphorical: The genome is not "like" an identity system —” it IS an identity system at the molecular level. MC Node Identity is not "like" a genome —” it serves the same structural function: fixed-at-initialization, immutable, capability-determining, propagated identity encoding. The structural correspondence is complete.

Specific example: Node_Type = 0x04 (harvesting relay) encodes the capability set of that node precisely as a gene expression pattern encodes the capability set of an enterocyte versus a hepatocyte. The node with this type has DAEH harvesting circuitry and relay transmission circuitry —” capabilities determined at manufacture and encoded in its identity, not acquired or lost during operation.

4.2 Body Plan †” MC Spatial Orientation | Rating: EXACT

Biology: The body plan defines the spatial organization of an organism's subsystems relative to each other and relative to the external world. Every multicellular organism has a defined anterior-posterior axis, a dorsal-ventral axis, and positional relationships between organs that must be maintained for normal function. Disruption of the body plan —” an organ in the wrong position —” is not viable.

Meridian: The MC spatial orientation vector $S = [x, y, z, \theta_{\text{azimuth}}, \theta_{\text{elevation}}, "x", "y", "z"]$ defines every node's position and orientation in the network coordinate frame. The azimuth and elevation angles define the node's "facing direction" — the direction of its primary actuator output. The position coordinates define where in the network space the node resides. This spatial substrate is the reference frame from which all routing, alignment, and delivery decisions are made. A node without a valid spatial orientation cannot participate in routing — just as a cell without positional identity cannot participate in tissue organization.

Specific example: θ_{azimuth} and $\theta_{\text{elevation}}$ encode the direction the node's antenna faces — the node's primary physical orientation relative to the world. This is literally the "facing direction" of the organism's primary actuator system, directly analogous to the anatomical directionality encoded in the body plan.

4.3 Metabolism † MFE Flow Engine | Rating: EXACT

Biology: Metabolism is the organism's energy management system. It encompasses all processes that convert food energy into usable ATP, regulate the distribution of energy currency to tissues based on demand, manage energy storage (glycogen in liver and muscle, fat in adipose tissue), and modulate energy expenditure based on current physiological state. Metabolism does not simply consume energy — it governs energy flow as a continuous regulatory function. The metabolic rate, the balance between catabolism and anabolism, and the tissue-specific allocation of energy are all under continuous metabolic regulation.

Meridian: MFE governs energy flow as a continuous regulatory function. It converts charge-state data (the organism's energy currency, stored in supercapacitors rather than ATP) into routing decisions (energy distribution directives). It evaluates burst thresholds (the metabolic activation energy required to perform work — V_{burst} is literally the minimum energy required to initiate a work-performing event). It schedules burst-mode transmission (modulates energy expenditure based on current state — exactly as metabolic regulation modulates ATP expenditure based on tissue demand). It manages the three-layer energy architecture: DAEH harvests (ingestion), supercapacitor stores (glycogen storage), and MFE schedules release (metabolic regulation of expenditure).

Specific example: The MFE supplementation trigger — when dV/dt falls below the minimum threshold, MFE requests DWER directed beam from the nearest source node — is functionally isomorphic to gluconeogenesis: when stored energy reserves fall below a critical threshold, the organism synthesizes new energy substrate from available precursors rather than allowing the system to deplete.

4.4 Nervous System † MMF Mesh Fabric | Rating: EXACT

Biology: The nervous system is the organism's coordination and signal routing layer. The central nervous system maintains a model of the body's state, routes signals between components, resolves conflicting motor commands, and coordinates organ activity toward coherent behavioral output. The nervous system

is topology-aware —” it knows what is connected to what, how far signals must travel, and how to route appropriately through the network. It also detects and responds to anomalous states (pain, proprioceptive mismatch) and adjusts behavior accordingly.

Meridian: MMF maintains a complete model of the mesh's state —” routing tables, link-state data, neighbor graphs, and per-link quality estimates. It routes energy (analogous to routing motor commands) through the mesh, resolves conflicts between nodes competing for the same relay path (conflict resolution), and prevents oscillatory states (analogous to resolving conflicting motor commands that would produce tremor rather than directed movement). MMF is topology-aware: it knows what is connected to what, what the path quality is on each link, and how to route through the network to reach any destination.

Specific example: LINK_STATE_UPDATE propagation through the mesh is structurally identical to neural signal propagation —” it is flood-limited (TTL-decremented at each hop, analogous to signal attenuation), topology-aware (processed by each receiving node in the context of its local topology model), and triggers coordinated response (each receiving node updates its routing table and potentially changes its energy flow behavior). The structural analogy is not loose —” it is the same signaling architecture implemented in a different substrate.

4.5 Musculature †” MTL Transmission Layer | Rating: EXACT

Biology: Muscles receive motor commands from the nervous system and produce directed physical force. The motor command specifies direction, magnitude, duration, and timing. The muscle executes the command and reports completion through sensory feedback. The muscle does not decide what to do —” it executes what the motor system tells it to do, with precision and speed. The muscle is the organism's interface between internal computation and physical reality.

Meridian: MTL receives transmission directives from MMF and executes them as directed RF beam emission. The BeamCommand struct specifies direction (θ_{azimuth} , $\theta_{\text{elevation}}$ —” motor command direction), power level (magnitude), timing window (duration and timing), and safety constraints (operating limits). MTL executes the command against the phased-array hardware and returns delivery confirmation (sensory feedback —” the equivalent of proprioception confirming that the intended force was applied). MTL does not decide what to transmit —” it executes what MMF directs, with the precision enforced by phase-steering correction.

Specific example: The BeamCommand struct is a motor command in every functional sense: it specifies a physical action, its direction, its magnitude, its duration, and its operating constraints. The five-condition gate system (Section 7.3 of the companion paper) is analogous to the motor inhibition system that

prevents the musculature from acting on a command that would cause self-injury —” the motor system checks safety before executing, exactly as MTL's safety gate checks all five conditions before firing the beam.

4.6 Innate Immune System †” SHDCL | Rating: EXACT

Biology: The innate immune system provides the organism's first line of defense against damage and pathogens. It is non-specific —” it responds to generic patterns of damage or foreign material (pathogen-associated molecular patterns, damage-associated molecular patterns) without requiring prior exposure or learned recognition. Response is rapid —” measured in minutes. The innate immune system detects anomalies, triggers protective responses (inflammation, phagocytosis, complement activation), attempts to restore homeostasis, and escalates to the adaptive immune system if the threat is not resolved.

Meridian: SHDCL detects anomalies within one 13.7 ms control cycle —” far faster than any biological innate immune response, but structurally identical in function. It responds to generic patterns of failure (dropped processes, stalled loops, missing heartbeats, safety gate failures) without requiring prior classification or learned recognition. Response is immediate and automatic. SHDCL executes the appropriate recovery sequence (beam hold, process restart, mesh rerouting, escalation to Lume-X recovery event) and logs the event for post-hoc analysis. Escalation to Guardian-E (the adaptive immunity analog) occurs when SHDCL's non-specific response is insufficient.

Specific example: The beam hold sequence on localization signal loss is a reflex arc in every functional sense: detect anomalous state (localization signal age > 500 ms) †’ execute immediate protective response (hold beam at last safe position) †’ attempt recovery (initiate re-acquisition) †’ escalate if recovery fails (BEAM_RECOVERY_FAILED event, beam cut). This is structurally isomorphic to the innate immune withdrawal reflex: detect tissue damage signal †’ vasoconstriction †’ clotting response †’ healing attempt †’ inflammation escalation if healing fails.

4.7 Adaptive Immune System †” Guardian Security (Guardian-E) | Rating: STRONG

Biology: The adaptive immune system learns to recognize specific threats. It generates diverse recognition receptors (T-cell receptors, antibodies) that can bind specifically to antigens —” molecular signatures of pathogens. It maintains self-tolerance through clonal deletion of self-reactive lymphocytes (self/non-self discrimination). It remembers past infections through immunological memory cells that enable faster, stronger responses to known threats. The adaptive immune system is the organism's security intelligence layer —” it learns, authenticates, and remembers.

Meridian: Guardian-E (Guardian Security, energy domain) authenticates nodes using provisioned cryptographic keys —” the Meridian analog of self/non-self discrimination. Nodes with valid keys are "self." Nodes without valid keys, or nodes whose behavior contradicts their authenticated identity, are "non-self." Guardian-E maintains a learned threat model: the eleven threat categories and their associated

detection rules are the analog of immunological memory —” specific recognition patterns built from knowledge of prior attack vectors. When a known threat pattern is detected (localization spoofing, routing table manipulation, beam hijacking), Guardian-E responds with the specific countermeasure mapped to that threat class, exactly as immunological memory enables a more specific and faster response to a known pathogen.

Rating justification (STRONG rather than EXACT): The biological adaptive immune system generates novel recognition patterns through somatic recombination —” it can recognize threats it has never seen before. Guardian-E's current implementation uses a predefined threat taxonomy rather than dynamic threat learning. The functional role is isomorphic; the mechanism for novel threat recognition differs. Future implementations with dynamic threat learning would elevate this to EXACT.

4.8 Digestive System †” DAEH Harvesting | Rating: EXACT

Biology: The digestive system takes energy from the external environment in an unusable form (food) and converts it to a form the organism can store and use (glucose, fatty acids, amino acids). It is multi-modal —” organisms can digest different types of food through different enzymatic pathways. It stores intermediate products (bile, glycogen) and delivers processed energy to the circulatory system for distribution. The digestive system is the organism's primary interface with the energy environment.

Meridian: DAEH takes energy from the external environment in forms that are not directly usable for RF transmission (ambient RF at $\mu\text{W}/\text{cm}^2$, thermal gradients, mechanical vibration, light, triboelectric charge separation) and converts it to a form the mesh can store and use (DC voltage stored in supercapacitors). It is multi-modal —” DAEH can harvest from RF, solar, thermal (TEG), piezoelectric, and triboelectric sources through different circuit architectures, exactly as the digestive system uses different enzymatic pathways for carbohydrates, fats, and proteins. It stores processed energy in supercapacitors (the glycogen of the mesh) and delivers it to the DRMA routing layer for distribution.

Specific example: The multi-modal harvesting architecture of DAEH —” RF rectenna + photovoltaic + TEG + piezoelectric + TENG —” is structurally isomorphic to an omnivore's digestive system. An omnivore can extract energy from multiple food types through multiple enzymatic pathways because different energy sources in its environment require different processing. A Meridian node deployed in a complex environment uses whichever harvesting modalities yield the most energy given current environmental conditions —” exactly the adaptive food selection behavior of an omnivore.

The explicit prohibition on coherent phase-combining of incoherent RF sources (stated prominently in the companion paper and inherited here) is functionally isomorphic to the digestive system's inability to use certain molecular configurations —” some things are in the environment but not digestible. The honest accounting of what can and cannot be extracted from a given environmental energy source is the metabolic reality of both systems.

4.9 Circulatory System †” DRMA Mesh Routing | Rating: EXACT

Biology: The circulatory system transports resources —” oxygen, glucose, hormones, immune cells —” from where they are produced or stored to where they are needed. It is a distributed transport mesh with a sophisticated balance mechanism: cardiac output, blood pressure regulation, and vasodilation/vasoconstriction modulate flow in response to demand. Tissue with high metabolic demand signals for increased blood flow; the circulatory system redistributes accordingly. The circulatory system drives the organism toward metabolic equilibrium across all tissues simultaneously.

Meridian: DRMA is a distributed energy transport mesh. Energy moves hop-by-hop from nodes with surplus charge to nodes with deficit or to designated destination devices. The bi-directional flow mechanism —” surplus nodes supplying depleted nodes in either direction —” is the Meridian analog of cardiovascular redistribution: detect the imbalance, redirect flow, restore equilibrium. The surplus-deficit monitor in MMF is functionally isomorphic to blood pressure regulation: it monitors the distribution of energy across the mesh, detects imbalances, and triggers corrective flow.

Specific example: The bi-directional energy flow protocol —” where a node with $V_{cap} > V_{burst} + surplus_threshold$ can serve as an energy donor to a node with $V_{cap} < V_{min} + deficit_threshold$ —” is structurally isomorphic to collateral circulation in biology: when primary blood supply is blocked, the circulatory system reroutes through alternative vessels to maintain tissue perfusion. The SHDCL node failure recovery (mark node FAILED, activate alternate path, continue delivery) is the Meridian equivalent of collateral circulation activating in response to arterial occlusion.

4.10 Motor Output †” DWER Directed Transmission | Rating: EXACT

Biology: Motor output is the organism's goal-directed physical action on the world. It is distinguished from internal organ function by its directionality and intentionality —” motor output is aimed at a specific target in the external environment. A reaching movement is directed at a specific location in space. Prey capture behavior is directed at a specific organism in the environment. Motor output closes the organism's interaction loop with its environment.

Meridian: DWER directed transmission is goal-directed energy delivery to a specific target at a specific location. The closed-loop confirmation architecture —” transmit, measure received power, verify against P_{min} , adjust beam, retransmit if necessary —” is the Meridian analog of the motor system's closed-loop control. Proprioceptive feedback in biology continuously corrects movement toward the target. Delivery confirmation telemetry in DWER continuously corrects beam steering toward the receiver. In both cases, the loop closes on a physical action in the external world, verified by sensory return.

4.11 Homeostasis †” Global Invariants INV-1 through INV-5 | Rating: EXACT

Biology: Homeostasis is the defining property of living systems —” the continuous maintenance of stable internal conditions despite external perturbation. Body temperature in mammals is maintained at 37°C regardless of ambient temperature. Blood glucose is maintained between 70—“100 mg/dL despite intermittent feeding. Blood pH is maintained at 7.4 despite metabolic acid production. These are not set points that are occasionally checked and corrected —” they are continuously enforced by overlapping feedback loops at multiple levels of biological organization [3].

Meridian: INV-1 through INV-5 are continuously enforced at every 13.7 ms control cycle by the Lume-X runtime. They maintain stable internal conditions —” no uncontrolled discharge, no ambiguous routing, no non-deterministic flow, no unsafe transmission, no mesh oscillation —” regardless of external perturbation. They are not conditions that are occasionally checked and corrected —” they are continuously monitored by overlapping enforcement mechanisms (SHDCL detects violations within one cycle, MTL's safety gate prevents violations before they occur, Guardian-E detects security-domain violations).

This is the strongest mapping in the paper. Homeostasis and global invariant enforcement are not analogous —” they are the same thing at different scales of implementation. Both are continuous, multi-loop, automatic mechanisms for maintaining defined internal conditions despite external disturbance. The implementation substrate differs (biochemical feedback loops vs. Lume-X runtime assertions). The functional architecture is identical.

Specific example: INV-5 (no mesh oscillation) is thermoregulation in concrete form. Biological thermoregulation detects deviation from the temperature set point, applies a corrective force (shivering, sweating, vasodilation), and locks the system back to stable operation. INV-5 detects oscillatory flow patterns (deviation from the stable-flow set point), applies a corrective force (flow direction lock for ten transmission windows), and returns the mesh to stable operation. The feedback topology —” detect deviation, apply correction, restore stability —” is identical.

4.12 Protective Reflexes †” BSL + Safety Gates | Rating: EXACT

Biology: Biological reflexes are immediate protective responses that bypass the central nervous system's deliberation pathways. A withdrawal reflex —” touching a hot surface triggers immediate arm retraction before pain is consciously registered —” is faster than any conscious response because it routes through the spinal cord rather than the brain. Reflexes are hardwired, not learned, and they override voluntary motor commands. The biological organism cannot choose to not have a withdrawal reflex. It is a protective invariant implemented below the level of conscious decision-making.

Meridian: The Biological Safety Layer exclusion zone enforcement and the five-condition MTL gate are immediate protective responses that override routing logic. When biological presence is detected within the exclusion zone, the beam is cut immediately —” not after MFE checks the routing schedule, not after

MMF consults the routing table, not after Lume-X completes its control cycle. The cut is the first response. Similarly, the GATE_REJECT on a failed safety condition overrides any pending transmission authorization from higher layers. The safety gate operates below the routing layer, exactly as the spinal reflex arc operates below the cortical layer.

Specific example: The exclusion zone beam cut is a withdrawal reflex in every functional sense: detect threat stimulus (biological presence in exclusion zone) †' immediate protective response (beam cut) †' no deliberation required. The five-condition gate that prevents transmission without confirmed receiver position, valid route, clear exclusion zone, safe power density, and sufficient charge is the motor inhibition system: the organism cannot act unless all safety conditions are satisfied. Both the reflex and the gate operate at a level below deliberative decision-making and cannot be overridden by higher-level logic.

4.13 Memory †' Certificate Chains + Delivery Logs | Rating: STRONG

Biology: Biological memory encompasses episodic memory (storage of specific past events), procedural memory (learned skills and behavioral patterns), and immunological memory (learned immune responses). Memory enables the organism to learn from past experience, predict future states based on past patterns, and replay successful past behaviors in similar future contexts.

Meridian: The Meridian certificate chain and delivery log system stores every governance event with full context —" timestamp, duration, cause, and outcome. This enables post-hoc analysis (review of past system behavior), replay-identical execution (the system can reproduce any past state exactly from the log), and predictive maintenance flagging (SHDCL flags degraded states for future intervention based on logged patterns). The rolling statistical model maintained in MFE for each link —" the σ^2 variance and P_received history —" is a learned behavioral model: the system's accumulated knowledge of how each link performs, used to make better future routing decisions.

Rating justification (STRONG rather than EXACT): Biological episodic memory involves subjective experience of past events —" what happened, when, and in what context. The Meridian log system stores events but does not have subjective access to them. The functional role (enabling past-event-informed future behavior) is isomorphic; the phenomenological character (if any) is absent and unclaimed. Future implementations that use logged patterns to dynamically adjust routing weights would strengthen this mapping further.

4.14 Global Awareness †' Lume-X Control Loop | Rating: MODERATE

Biology: Consciousness —" the organism's unified subjective awareness of its own state and environment —" is the most discussed and least understood property of biological organisms. Functionally, consciousness provides integrated, unified, cross-modal awareness across all sensory and internal state

modalities simultaneously. The organism does not process visual information, proprioceptive information, and pain information in three separate unintegrated streams —” it has a unified experience that integrates all of these.

Meridian: The Lume-X 73 Hz control loop provides unified monitoring across all four Meridian layers simultaneously. At every control cycle, Lume-X evaluates the state of MC, MFE, MMF, and MTL in an integrated fashion —” it is aware of the localization state, the charge state of every node, the routing topology, and the active beam parameters all at once. When an anomaly occurs in any layer, Lume-X's response is informed by the state of all other layers. This is structural unified awareness —” cross-system, continuous, integrated.

This is the weakest mapping and I am explicit about its limitations. No claim of consciousness, subjective experience, or any form of inner life is made or implied. The structural parallel is the unified, integrated, cross-system monitoring function —” not the phenomenological character of consciousness, which Meridian definitively does not have. I include this mapping not to claim consciousness-equivalence but to demonstrate completeness: even the most philosophically complex biological property has a functional structural analog in Meridian, though the analog captures only the architectural function and not the phenomenological content.

5. Emergent Organism Properties

The following properties were not explicitly designed into Meridian as organism features. They emerge from the architecture as natural consequences of the engineering decisions made to solve the wireless energy routing problem. Their emergence is evidence that the convergence is genuine —” these properties were not placed there intentionally.

5.1 Self-Preservation

Meridian nodes enter power-conservation mode when V_{cap} falls below V_{min} —” the minimum operating voltage for the microcontroller. In this mode, the node suspends all non-essential processes, maintains only the localization module and charge monitor, and waits for charge to recover before resuming normal operation.

This was designed as a battery management engineering feature. It is also self-preservation behavior. The node detects an existential threat to its operation (energy depletion), enters a protected low-energy state, and waits for conditions to improve before resuming full function. The supplementation request —” when harvest rate is insufficient, MFE requests a DWER directed beam from a nearby source node —” is resource-seeking behavior: the organism identifies a resource deficit and actively solicits external supply to survive. Neither of these behaviors was designed as organism behavior. Both are organism behavior.

5.2 Cooperative Behavior

The bi-directional energy flow protocol —” surplus nodes donating energy to depleted nodes —” was not programmed as cooperative behavior. It was programmed as mesh equilibration: a control system optimization that drives the network toward uniform charge distribution because uniform distribution minimizes the probability of delivery failure due to node depletion.

It is also cooperative behavior. Nodes with surplus charge contribute to nodes with deficit —” not because they are programmed to be cooperative but because the optimization objective of the mesh (maintain delivery capability across all nodes) produces behavior that is functionally indistinguishable from cooperation. In biology, cooperative behavior among conspecifics is similarly not driven by altruism but by selection pressures that make cooperation individually advantageous [5]. The behavioral outcome is identical in both cases: resources flow from those with surplus to those in need, maintaining overall group function.

5.3 Adaptive Response to Environment

Meridian adapts its behavior to environmental conditions in three specific ways that were designed as engineering control features:

Harvest modality shifting: When one harvesting modality underperforms, SHDCL shifts harvest priority to higher-performing modalities. This was designed as a fault tolerance feature. It is also adaptive foraging behavior —” the organism shifts its energy acquisition strategy based on which sources are currently most productive.

Routing path adjustment: When link quality degrades below threshold, MMF marks the link DEGRADED and shifts traffic to alternate paths. This was designed as mesh resilience. It is also adaptive navigation —” the organism learns which paths are currently traversable and adjusts its movement accordingly.

Burst scheduling adjustment: When harvest rate changes, MFE adjusts burst timing to match actual available energy. This was designed as energy management. It is also metabolic adaptation —” the organism adjusts its activity schedule to match its current energy budget.

None of these adaptive behaviors were designed as organism behavior. All three are organism behavior.

5.4 Wound Healing

When a relay node fails —” whether due to hardware fault, power depletion, or physical damage —” SHDCL detects the failure within two control cycles, marks the node as PRESUMED_FAILED, activates the pre-computed alternate routing path, signals adjacent nodes to update their routing tables, and continues energy delivery to the destination through the alternate path. The failed node is effectively excised from the active mesh topology. If it recovers and re-announces, it is reintegrated.

This is wound healing. Tissue damage †' immune detection †' blood rerouting around the damaged area †' function continues through collateral circulation †' damaged tissue is repaired and reintegrated if healing succeeds. The Meridian sequence: node failure †' SHDCL detection †' alternate path activation †' delivery continues †' node recovery and reintegration if it returns. The functional architecture of the response is identical.

5.5 Growth and Scalability

When a new node is added to the Meridian mesh, it broadcasts `NODE_ANNOUNCE`. All neighbors update their topology models. MMF re-evaluates whether the new node improves existing paths and updates routing tables accordingly. The new node receives a routing table from its nearest neighbor and begins participating in mesh operation.

`NODE_ANNOUNCE` is cell division announcement in an organism that grows by adding new functional units. The mesh expands its capability, extends its coverage, and updates its internal map to incorporate the new component —” exactly as a growing organism integrates new tissue into its functional organization. Growth is not a special case in Meridian —” it is handled by the same topology management system that handles all other topology changes, just as cell division in a growing organism uses the same developmental mechanisms as tissue repair.

6. The 42 Assumptions as Biological Axioms

The 42 Assumptions [15] were written as foundational axioms of Lume system coherence —” the rules that any Lume-governed system must satisfy to maintain its integrity. Examined through the biological lens, the 42 Assumptions are the axioms of organism existence: the minimum constraints that any autonomous, coherent, self-maintaining system must satisfy to persist over time.

This is not a coincidence. The 42 Assumptions were derived from first principles about what makes a deterministic system coherent, self-consistent, and capable of autonomous operation. The principles of biological organism existence were derived from billions of years of selection on the same problem. The convergence of the two derivations toward the same set of axioms is evidence that the axioms are correct —” they are the natural constraints of the problem space, not an arbitrary design choice.

I organize the 42 Assumptions into five biological categories:

Identity Assumptions †' Genome / Identity Substrate. The assumptions governing how a Lume system establishes, maintains, and propagates its identity are structurally isomorphic to the biological properties of the genome. A system must have a fixed, unique identity that determines its capabilities and role. Identity must be consistent across all representations of the system. Identity must be propagated faithfully to any system that models the original. In biology: the genome is fixed at fertilization, determines the organism's cell types and capabilities, and is faithfully propagated to daughter cells during division.

Boundary Assumptions † ' Cell Membrane / Immune Self/Non-Self. The assumptions governing the distinction between the system and its environment —” what is inside the system boundary, what is outside, and how the boundary is maintained —” are isomorphic to biological membrane function and innate immune self/non-self discrimination. A system must know what is itself and what is not itself. It must maintain this boundary under perturbation. It must respond to boundary violations. In biology: the cell membrane and the innate immune system jointly enforce the self/non-self boundary.

Coherence Assumptions † ' Homeostasis. The assumptions governing internal consistency —” that the system's state is well-defined, that subsystems do not contradict each other, that invariants are maintained across all operations —” are isomorphic to biological homeostasis. A system must maintain defined internal conditions despite external perturbation. Deviations must be detected and corrected. The correction mechanism must be continuous, not periodic. In biology: homeostatic feedback loops enforce this continuously across every biological parameter.

Integrity Assumptions † ' DNA Repair / Self-Healing. The assumptions governing the detection and correction of internal faults —” that damage is detected, that recovery is attempted, that the system can continue operating in a degraded state while repair proceeds —” are isomorphic to biological DNA repair and tissue self-healing mechanisms. A system must be able to detect its own faults. It must have recovery procedures for every anticipated fault class. It must fail safely (not catastrophically) in the presence of unanticipated faults. In biology: DNA repair, wound healing, and the innate immune inflammatory response all implement this pattern.

Sovereignty Assumptions † ' Autonomy / Self-Governance. The assumptions governing the system's capacity for autonomous operation —” that it does not require continuous external supervision, that it can make its own operational decisions within defined parameters, that it maintains its function independent of its designers or operators —” are isomorphic to the biological property of autonomy. A living organism does not require its evolutionary ancestors to be present for its continued operation. A Lume-governed system does not require its designers to be present for its continued operation. Both are self-governing within the constraints of their foundational rules.

The central argument of this section: Meridian converged toward biological organization not because its designers intended biological mimicry, but because Meridian was built on the 42 Assumptions —” and the 42 Assumptions are the axioms of autonomous coherent system existence. Any system built on these axioms, faced with the requirements of autonomous physical operation, will converge toward biological organization. The 42 Assumptions are the organism blueprint, encoded at the axiomatic level of the Lume language ecosystem.

7. Organism Classification

The Synthetic Organism model defines five organism types on a scale from passive to fully autonomous [17]. I classify Meridian against this scale using the formal criteria defined in Appendix C.

Type 0 (Passive): EXCEEDS Type 0 systems execute fixed instructions with no environmental response and no internal state management. Meridian clearly exceeds this classification —” it responds continuously to environmental conditions, maintains complex internal state, and adapts its behavior accordingly.

Type 1 (Reactive): EXCEEDS Type 1 systems exhibit stimulus-response behavior without internal state persistence. Meridian maintains persistent state across all four layers and uses that state to inform future decisions. It is not purely reactive.

Type 2 (Adaptive): CONFIRMED Type 2 systems adjust behavior based on environmental feedback. Meridian does this continuously and demonstrably: harvest modality shifting, routing path adjustment based on link quality, burst scheduling adjustment based on harvest rate, and bi-directional flow based on real-time charge state. Type 2 is confirmed at the architectural level. Experimental validation is required for complete confirmation.

Type 3 (Self-Maintaining): CONFIRMED at architectural level Type 3 systems self-heal, preserve identity, and maintain homeostasis. SHDCL provides self-healing across all layers. MC node identity is immutable and continuously maintained. INV-1 through INV-5 enforce homeostasis at 73 Hz. All three criteria are met architecturally. Experimental validation of each is required for full confirmation.

Type 4 (Self-Evolving): PARTIAL Type 4 systems modify their own behavior model based on accumulated experience —” not just adjusting parameters, but changing the structure of the decision model itself. Meridian exhibits partial Type 4 properties: the rolling statistical model for link quality (σ^2 , P_received history) adapts the routing preference weights based on accumulated experience. The harvest modality priority shift based on performance history is a form of behavioral learning. However, Meridian does not modify its own code, does not restructure its routing logic, and does not generate novel behavioral strategies from experience. Type 4 is partial, not confirmed.

Type 5 (Fully Autonomous): STRUCTURAL COMPLETENESS ONLY Type 5 requires all subsystems operational and confirmed experimentally. Meridian has all subsystems specified at the architectural level. No subsystem is missing from the design. However, no experimental validation exists at time of writing —” Meridian has not yet operated in a physical deployment. Type 5 structural completeness is confirmed. Type 5 operational confirmation requires Phase 1—“3 experimental results.

Classification summary:

Type 0: EXCEEDS Type 1: EXCEEDS Type 2: CONFIRMED (architectural) Type 3: CONFIRMED (architectural) Type 4: PARTIAL Type 5: STRUCTURAL —” awaiting experimental validation

Final classification: Type 3+ Synthetic Organism with Type 5 structural completeness.

This makes Meridian the highest-classified Synthetic Organism in the Lume ecosystem at time of writing, and the first complete physical instantiation of the model. The classification will be revisited and confirmed or revised when Phase 1—“3 experimental data is available.

8. Implications for Autonomous System Design

8.1 The Convergence Thesis

The central claim of this paper —” that deterministic autonomous systems converge toward biological organization —” if correct, has predictive power beyond Meridian. It predicts that any system designed to:

Acquire resources from an uncertain environment Transport those resources internally to where they are needed Coordinate distributed components toward coherent behavior Defend against internal and external threats Maintain a stable identity over time Repair damage without external intervention ...will, if fully and correctly specified, exhibit biological organizational topology. The design space for such systems is constrained by physics and the requirements of autonomy toward a small region —” and that region looks biological.

Meridian provides the first complete evidence for this prediction. It is a single data point. A single data point can demonstrate that convergence is possible; it cannot prove that convergence is inevitable. For that, additional examples of Lume-governed systems across different physical domains are needed. The robotic, manufacturing, and infrastructure domains are natural next cases. If a fully-specified Lume-governed robotic system converges toward the same organizational topology as Meridian, the convergence thesis gains substantial evidential weight.

8.2 Organism-Aware Design as Engineering Methodology

The practical implication for system designers is this: if autonomous physical systems converge toward biological organization, then designing them explicitly as organisms —” starting from the organism model rather than discovering it post-hoc —” should produce more complete and more robust systems.

The Synthetic Organism model [17] provides a checklist for autonomous system design:

- Genome: Does the system have a fixed, unique identity that determines its capabilities?
- Body Plan: Does the system have a defined spatial self-model?
- Metabolism: Does the system have a regulated energy management layer?
- Nervous System: Does the system have a topology-aware coordination layer?
- Musculature: Does the system have a motor output layer that translates decisions to physical action?
- Innate Immunity: Does the system have immediate, non-specific anomaly response?
- Adaptive

Immunity: Does the system have learned threat recognition and authentication? – Digestive System: Does the system have an environmental energy intake and processing layer? – Circulatory System: Does the system have an internal resource transport and balancing layer? – Motor Output: Does the system have goal-directed physical action capability? – Homeostasis: Does the system enforce stable internal conditions continuously? – Reflexes: Does the system have immediate protective responses that override deliberation? – Memory: Does the system store and use past events to inform future behavior?

Any autonomous physical system design that is missing any item on this checklist has a corresponding class of failure modes that it cannot recover from without external intervention. Designing to this checklist —” deliberately and from the start —” is organism-aware design.

Meridian demonstrates that this checklist, applied to a wireless energy routing system, produces a complete and internally consistent architecture. The same checklist, applied to any other autonomous physical domain, should produce the same result. This is the design methodology that emerges from the convergence thesis.

8.3 The Lume Advantage

Systems built on Lume inherit organism properties at the language level through the 42 Assumptions. This is a significant architectural advantage. A designer building an autonomous system on Lume does not need to independently discover organism-like design principles —” they are encoded in the foundational axioms of the language. The Lume runtime provides SHDCL (innate immunity), Guardian (adaptive immunity), and the deterministic control loop (global awareness) as built-in capabilities rather than application-layer additions.

This means that Lume-governed autonomous systems will naturally converge toward organism organization more completely and more correctly than systems built on foundations that do not encode these properties. The convergence is not accidental —” it is the intended outcome of the Lume design principles.

Future research should examine whether other programming paradigms produce similar convergence. The prediction of the convergence thesis is that they will not —” or that they will do so much more slowly and incompletely, because they do not encode organism-like axioms at the foundational level.

9. Related Work

9.1 Artificial Life

The artificial life research program [6, 7, 8] has produced compelling demonstrations of life-like behavior in computational systems: self-replication in digital evolution platforms (Tierra [8], Avida), emergent flocking and schooling behavior from simple interaction rules (Reynolds' boids [7]), and the spontaneous

emergence of complex metabolic pathways in constrained evolutionary simulations. This work establishes that life-like properties can emerge from computational systems.

The critical distinction from Meridian is the domain of operation. Artificial life systems simulate life-like properties in computational environments —” they do not operate in the physical world, do not harvest physical energy, do not transmit directed RF beams, and do not route power to real devices. Meridian implements organism properties in a real physical domain. The challenges of physical operation —” energy conservation constraints, RF propagation physics, thermal management, regulatory compliance —” are constraints that artificial life systems do not face. It is precisely these constraints that drive the convergence toward biological organization, because biological organisms face the same constraints.

9.2 Bio-Inspired Engineering

The bio-inspired engineering literature is extensive and has produced powerful techniques: ant colony optimization for routing and scheduling [11], neural networks for pattern recognition and control, genetic algorithms for optimization, swarm robotics for distributed task execution. Bio-inspired systems copy one or a small number of biological principles and apply them to engineering problems.

The distinction from this work is completeness. Bio-inspired systems implement one biological principle. Meridian implements all of them —” not by design, but by convergence. The argument here is not that copying biological principles is useful (it clearly is) but that a correctly specified deterministic autonomous system will arrive at all biological principles simultaneously, without choosing to copy any of them. The completeness of the Meridian convergence is the property that bio-inspired engineering does not and cannot demonstrate, because bio-inspired engineering always starts from a selection of which biological principle to copy.

9.3 Self-Healing Systems

The autonomic computing vision [9] proposed self-managing systems with four properties: self-configuration, self-optimization, self-healing, and self-protection. Subsequent work on self-healing networks and distributed systems has produced implementations of various subsets of these properties [10]. This literature demonstrates that self-healing in isolation is achievable and valuable.

The distinction from Meridian is that self-healing alone, however well implemented, does not constitute an organism. An organism requires self-healing plus metabolism plus coordination plus resource acquisition plus homeostasis —” the complete set. Systems designed for self-healing in isolation cannot demonstrate convergence toward biological organization, because they were designed for one property rather than for autonomous physical operation. Meridian was not designed for self-healing —” it was designed for deterministic energy routing. Self-healing emerged as a requirement.

9.4 Cyber-Physical Systems

The cyber-physical systems literature [13, 14] studies the integration of computational and physical processes, developing frameworks for timing, reliability, and safety in systems that span the digital-physical boundary. This literature is the closest prior art to Meridian's operational domain —” systems that must control physical processes with timing and safety guarantees.

The CPS literature does not model systems as organisms. It models them as controlled physical processes with computational supervisors. The organizational framework is the plant-controller model, not the organism model. The implication of this work for CPS research is that the organism model may be a more complete and predictively useful framework for systems that must operate autonomously in physical environments. A CPS designed to the organism checklist will have explicitly specified self-healing, homeostasis, resource acquisition, and threat defense —” properties that the plant-controller model does not require.

9.5 The Lume Ecosystem

Meridian is the product of the Lume ecosystem's theoretical trajectory. Lume [15] established a deterministic programming language with organism-like runtime properties. DAIGS [17] applied these properties to multi-agent AI systems. Lume-X [19] extended the runtime to high-frequency control applications. Deterministic Dissolution [20] modeled organism termination formally. Meridian, guided by Lume-X, is the first physical application of this trajectory —” the first system in which organism-like properties are implemented not in software agents but in the physical energy domain.

This paper marks a transition point in the Lume ecosystem: from theoretical framework to physical instantiation. The Synthetic Organism model is no longer a description of what a future system might look like. It is a description of what a current system does look like. The trajectory now runs from framework to demonstration, and from demonstration to predictive design methodology.

10. Limitations and Honest Boundaries

I state the following limitations explicitly, without qualification or softening.

Meridian is not alive. The structural and functional isomorphism between Meridian and biological organisms is real and complete. It is not an isomorphism in kind —” Meridian does not self-replicate, does not evolve, does not have a biochemical metabolism, and does not meet any biological definition of life. The isomorphism is architectural. The distinction matters and I maintain it throughout.

The consciousness mapping is the weakest in the paper. The mapping of Lume-X's unified control loop to biological global awareness is rated MODERATE, not EXACT or STRONG. No subjective experience, inner life, or phenomenological property is claimed for Meridian at any point. The structural parallel —” unified, integrated, cross-system monitoring —” is the only claim made.

No experimental data validates the organism properties. This is an architectural analysis paper. Meridian has not been built and operated. The Type 3 and Type 2 classifications are confirmed at the architectural specification level, not at the experimental level. Phase 1—“3 experimental results will either confirm or disconfirm the architectural analysis. No organism property should be treated as experimentally established until those results exist.

The convergence thesis rests on one example. Meridian is a single case. A single case can prove that convergence is possible. It cannot prove that convergence is general or inevitable. Generalization requires multiple independent examples of deterministic autonomous systems converging toward biological organization. This paper argues for the convergence thesis based on one example and the theoretical argument from the 42 Assumptions. Both are necessary but not sufficient for proof.

The 42 Assumptions may encode organism properties by design. The argument in Section 6 is that the 42 Assumptions converge toward biological axioms because both derive from the same underlying problem constraints. An alternative interpretation is that the 42 Assumptions encode organism properties because their author was aware of biological organism design. If this is the case, the convergence of Meridian (built on the 42 Assumptions) toward biological organization is not emergent but designed. This is a valid critique. I acknowledge it and note that it does not change the practical implications for autonomous system design —” whether the convergence is emergent or designed, the organism model is the correct design target for autonomous physical systems.

Biological evolution took billions of years. Meridian was designed in months. The comparison between biological and Meridian organism properties must account for this difference in timescale and selection intensity. Biological organisms have been optimized by natural selection across billions of generations and countless selection events. Meridian has been designed by one person over a short period. The functional correspondences are real; the optimality of those correspondences is not comparable. Meridian's implementations of biological analogs are in their infancy relative to the biological originals.

The mapping may reflect an anthropomorphic projection. One can argue that the organism framework is a cognitive schema applied post-hoc to an engineering system —” that the "discovery" of organism properties in Meridian reflects the human tendency to see biological patterns in complex systems rather than a genuine structural correspondence. I argue against this interpretation: the mappings are formal, the completeness is verifiable against the architectural specification, and the specific technical examples in Section 4 cannot be dismissed as projective. The mappings are falsifiable and I invite falsification. But I acknowledge the critique as legitimate.

11. Conclusion

Meridian was designed as a wireless energy routing architecture. It was not designed as an organism. Post-hoc analysis reveals that it is one —” structurally, formally, and completely.

I have demonstrated a structural isomorphism between Meridian's architecture and the Synthetic Organism model across fourteen biological subsystems. The isomorphism is complete in both directions: every biological subsystem required for autonomous physical operation has an operational Meridian counterpart, and every major Meridian component has a biological analog. Nine of the fourteen mappings are rated EXACT. Four are rated STRONG or MODERATE with explicit justifications for the rating reduction.

I have classified Meridian as a Type 3+ Synthetic Organism with Type 5 structural completeness —” the highest classification yet achieved in the Lume ecosystem. I have identified five emergent organism properties —” self-preservation, cooperative behavior, adaptive environmental response, wound healing, and growth —” that appeared in the architecture without being explicitly designed.

I have argued that this convergence was not coincidental: it was the emergent outcome of building an autonomous physical system on a foundation (Lume, the 42 Assumptions) that encodes organism-like properties at the axiomatic level. The 42 Assumptions are, when examined through the biological lens, the axioms of autonomous coherent system existence —” the same axioms that biological organisms satisfy, derived independently from the same underlying problem constraints.

The implication for autonomous system design is this: the organism model is not a metaphor for autonomous physical systems. It is the design target. Any system that must acquire resources, transport them internally, coordinate distributed components, defend against threats, maintain identity, and heal from damage will converge toward biological organization if it is correctly and completely specified. The Synthetic Organism model [17] provides the checklist. The 42 Assumptions [15] provide the foundational axioms. Lume-X provides the runtime. Meridian provides the demonstration.

The path forward runs through experimental validation. When Phase 1 DWER results confirm directed beam delivery, and Phase 2 and 3 results confirm mesh routing and ambient harvesting integration, the organism properties I have analyzed architecturally will become organism properties I can assert experimentally. The Type 3 and Type 5 classifications will be revised from architectural to operational. The convergence thesis will move from one example to the beginning of a body of evidence.

Until then, I make the following claim, which is supported by the architectural analysis in this paper: Meridian is the first complete physical instantiation of the Lume Synthetic Organism model —” the first system in which every structural property of a biological organism, as defined by that model, has a

concrete, specified, operational implementation in a real physical domain. Whether or not it is alive, it is organized exactly as life is organized. That is not nothing. That is, I believe, significant.

Appendix A —” Complete Isomorphism Mapping Table

#	Biological System	Biological Function	Meridian Component	Meridian Function	Rating
1	Genome	Fixed identity, capability encoding, heritable	MC Node Identity (64-bit)	Fixed at deployment, Node_Type encodes capabilities, propagated via NODE_ANNOUNCE	EXACT
2	Body Plan	Spatial organization, positional identity	MC Spatial Orientation S	x, y, z, θ_{azimuth} , $\theta_{\text{elevation}}$, velocity vector	EXACT
3	Metabolism	Energy conversion, distribution, storage regulation	MFE Flow Engine	Charge-state to routing conversion, burst scheduling, threshold evaluation, supplementation trigger	EXACT
4	Nervous System	Signal routing, topology awareness, conflict resolution	MMF Mesh Fabric	Routing tables, link-state, conflict resolution, oscillation prevention, topology management	EXACT
5	Musculature	Translates neural commands to physical force	MTL Transmission Layer	BeamCommand †’ phased-array steering †’ RF emission	EXACT
6	Innate Immune System	Immediate non-specific anomaly response	SHDCL	13.7 ms anomaly detection, process restart, rerouting, escalation	EXACT
7	Adaptive Immune System	Learned threat recognition, self/non-self discrimination	Guardian Security (Guardian-E)	Cryptographic node authentication, threat taxonomy, quarantine	STRONG
8	Digestive System	Environmental energy intake, conversion, storage	DAEH Harvesting	RF/solar/thermal/piezo/TENG †’ DC †’ supercapacitor	EXACT

9	Circulatory System	Internal transport, surplus/deficit balancing	DRMA Mesh Routing	Hop-by-hop energy transport, bi-directional flow, equilibration	EXACT
10	Motor Output	Goal-directed physical action on world	DWER Directed Transmission	Closed-loop directed beam delivery to specific target	EXACT
11	Homeostasis	Continuous stable internal conditions	INV-1 through INV-5	Continuous invariant enforcement at 73 Hz	EXACT
12	Protective Reflexes	Immediate override protective response	BSL + 5-Gate Safety System	Exclusion zone beam cut, GATE_REJECT —” override routing logic	EXACT
13	Memory	Past event storage, learning, replay	Certificate Chains + Delivery Logs	Full event logging, post-hoc analysis, replay-identical execution	STRONG
14	Global Awareness	Unified cross-system awareness	Lume-X 73 Hz Control Loop	Cross-layer monitoring; no consciousness claim	MODERATE

Appendix B —” The 42 Assumptions Mapped to Biological Axioms

Biological Category	Assumption Class	Biological Principle
Genome / Identity	Identity assumptions	Fixed, unique identity determines capabilities and role
Genome / Identity	Propagation assumptions	Identity is faithfully transmitted to all representations
Cell Membrane / Immunity	Boundary assumptions	System knows what is self and what is not self
Cell Membrane / Immunity	Integrity assumptions	Boundary violations are detected and responded to
Homeostasis	Coherence assumptions	Internal state is well-defined and consistent across subsystems
Homeostasis	Invariant assumptions	Defined conditions are maintained continuously despite perturbation
DNA Repair / Self-Healing	Fault detection assumptions	The system can detect its own internal faults
DNA Repair / Self-Healing	Recovery assumptions	Recovery procedures exist for every anticipated fault class
DNA Repair / Self-Healing	Graceful degradation assumptions	Unanticipated faults produce safe failure, not catastrophic failure
Autonomy / Self-Governance	Sovereignty assumptions	The system operates without continuous external supervision
Autonomy / Self-Governance	Decision autonomy assumptions	Operational decisions are made within the system, not externally

Appendix C —” Organism Type Classification Criteria

Type	Name	Minimum Criteria	Meridian Status
0	Passive	Fixed instruction execution, no internal state, no environmental response	EXCEEDS —” Meridian responds adaptively to environment
1	Reactive	Stimulus-response, no persistent state	EXCEEDS —” Meridian maintains persistent state across all layers
2	Adaptive	Adjusts behavior based on environmental feedback, state-informed decisions	CONFIRMED (architectural) —” harvest modality shifting, routing adjustment, burst scheduling adaptation
3	Self-Maintaining	Self-heals, preserves identity, enforces homeostasis	CONFIRMED (architectural) —” SHDCL, MC identity immutability, INV-1 through INV-5
4	Self-Evolving	Modifies own behavioral model from accumulated experience	PARTIAL —” rolling link statistics adapt routing weights; no code self-modification
5	Fully Autonomous	All subsystems operational, experimentally validated	STRUCTURAL —” all subsystems specified; experimental validation pending Phase 1—“3 results

Appendix D —” Cross-Reference to Meridian Architecture Paper

Organism Claim	Meridian Component	Architecture Paper Section
Genome / Node Identity	MC 64-bit identity vector	Section 4.2
Body Plan / Spatial Orientation	MC S vector	Section 4.3
Metabolism / Flow Engine	MFE flow selection, burst scheduling	Sections 5.2, 5.3
Nervous System / Mesh Fabric	MMF routing tables, topology	Sections 6.2, 6.3
Musculature / Transmission	MTL BeamCommand, safety gate	Sections 7.2, 7.3
Innate Immunity / SHDCL	SHDCL recovery sequences	Section 12.2
Adaptive Immunity / Guardian-E	Guardian Security rule set	Section 12.4, Appendix E
Digestive System / DAEH	Harvesting modalities, supercapacitor	Sections 8.2, 8.3
Circulatory System / DRMA	Mesh routing, bi-directional flow	Sections 9.3, 9.4 (DRMA section)
Motor Output / DWER	Directed beam, closed-loop confirmation	Section 10.2, 10.4
Homeostasis / Global Invariants	INV-1 through INV-5	Sections 3.5, 11.1—“11.6
Protective Reflexes / Safety Gates	BSL, 5-condition gate	Section 7.3
Memory / Certificate Chains	Delivery log format	Sections 4.4 (MC timing), 12.2
Global Awareness / Lume-X	73 Hz control loop	Sections 6.1, 12.1

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Chapter Bridge: From Organism to Network

Chapter Three established what Meridian is at the level of a single system. A single Meridian mesh —” one building, one deployment, one EAS —” is a Type 3+ Synthetic Organism: autonomous, self-healing, identity-persistent, and homeostatic.

But biological organisms do not exist in isolation. They form ecosystems. And the internet did not begin as a global network —” it began as a protocol. A correct routing protocol, once defined, enables networks to grow without limit, because any two networks that speak the same protocol can exchange packets across their boundary.

The same logic applies to Meridian. A single mesh is a complete system. But the protocol that governs one mesh can govern any mesh. The addressing scheme that identifies one node can identify any node. The routing tables that compute paths within one deployment can compute paths across deployments. The Energy Internet is what happens when you apply internet-scale thinking to the energy routing protocol that Meridian defines.

Part Three covers the network: first the internet-scale protocol standard, then the architecture in which that protocol carries both energy and data simultaneously through the same fabric.

PART THREE

The Network: Where Meridian Leads

The data internet began as ARPANET —” a small experimental network connecting a handful of research computers. The protocol was correct. The addressing was correct. The routing was correct. Everything else —” the scale, the applications, the economic ecosystem —” followed from the correctness of the foundation.

Meridian is the ARPANET of energy routing. It is currently specified at room scale, with experimental validation pending. But the protocol is correct. The addressing is correct. The routing is correct. Everything else —” the scale, the applications, the economic ecosystem —” will follow.

Part Three is about that everything else. Chapter Four extends the Meridian protocol to internet scale: the full Energy Internet Protocol Stack, with EIP addressing, EBGp inter-mesh federation, EDNS naming, and a governance framework for the global energy routing infrastructure. Chapter Five extends the Meridian mesh to carry both energy and data simultaneously, collapsing two infrastructure problems into one and enabling the self-sustaining autonomous device.

Chapter Four

The Energy Internet

The internet did not begin as a global network. It began as a protocol —” a formal specification for how independently administered networks could exchange packets across a shared addressing space using common routing rules. Once the protocol was correct, the network grew to encompass the planet without requiring central coordination. Every device that speaks IP participates in the same network regardless of who built it, where it runs, or what physical medium it uses. The protocol is the internet. The infrastructure follows.

I propose that the same transformation is possible for energy delivery —” and that the foundational protocol now exists.

The Meridian architecture defines how energy is routed deterministically within a single mesh. The Energy Internet extends that definition to the global case: how independently administered meshes exchange energy across shared addressing and routing infrastructure, how devices request energy from

any source in the network regardless of physical proximity, and how the governance and economic framework for a global energy routing network is organized.

The Energy Internet Protocol Stack —” five layers, from physical to application —” is specified in complete formal detail in this chapter. The Energy Internet Protocol (EIP) provides 128-bit addressing for energy nodes, designed on the same hierarchical principles as IPv6 to accommodate a network of arbitrary scale. The Energy Border Gateway Protocol (EBGP) enables inter-mesh federation, allowing independently administered energy meshes to exchange routing information and deliver energy across their boundaries. The Energy Domain Name System (EDNS) provides human-readable names for energy endpoints. The Energy Transmission Control Protocol (ETCP) provides delivery guarantees for energy sessions, analogous to TCP for data.

The supercapacitor, this chapter argues, is the packet buffer of the Energy Internet. Burst-mode energy transmission is packet switching applied to power. Every core mechanism of the data internet has a functional analog in the energy domain, and those analogs are specified here with the same precision that the original internet protocols were specified in their RFCs.

The path from Meridian to the Energy Internet is long. This chapter does not claim otherwise. But the architecture is coherent. The protocol is specifiable. The target is now defined —” so that when Phase 1 data exists, the work of building toward it can begin with a clear endpoint.

Abstract

The internet did not begin as a global network. It began as a protocol —” a formal specification for how independently administered networks could exchange packets across a shared addressing space using common routing rules. Once the protocol was correct, the network grew to encompass the planet without requiring central coordination. Every device that speaks IP participates in the same network regardless of who built it, where it runs, or what physical medium it uses. The protocol is the internet. The infrastructure follows.

I propose that the same transformation is possible for energy delivery —” and that the foundational protocol now exists.

The Meridian architecture, defined in the companion paper [Meridian Architecture, 2026], demonstrates that wireless energy can be routed deterministically through a mesh of addressed nodes using routing tables, quality-of-service guarantees, and a self-healing control runtime. The architecture is complete at room scale. This paper extends it toward a universal protocol standard: the Energy Internet.

I define the Energy Internet Protocol Stack —” a formal, layered specification for deterministic energy routing across independently administered meshes. The stack comprises five layers: the Energy Physical Layer (EPL), the Energy Link Layer (ELL), the Energy Network Layer (ENL), the Energy Transport

Layer (ETL), and the Energy Application Layer (EAL). I define the Energy Internet Protocol (EIP) for addressing and routing, the Energy Transmission Control Protocol (ETCP) for delivery guarantees, the Energy Domain Name System (EDNS) for human-readable endpoint addressing, and the Energy Border Gateway Protocol (EBGP) for inter-mesh federation between independently operated networks.

I argue that the supercapacitor is the packet buffer of the Energy Internet —” that burst-mode transmission is packet switching applied to power —” and that every core mechanism of the data internet has a functional analog in the energy domain. I demonstrate that the obstacles to energy internet standardization are not primarily technical but architectural, economic, and governance-related, and I propose a standardization pathway analogous to the IETF RFC process through which the data internet protocols were developed.

This paper makes no claim that the Energy Internet is imminent. Phase 1 experimental data for the room-scale Meridian architecture does not yet exist. The path to the Energy Internet runs through every experimental phase defined in the companion paper. This is a theoretical framework paper. Its purpose is to establish, formally and now, that a universal wireless energy routing protocol is architecturally coherent —” so that when Phase 1 data exists, the target is already defined.

Keywords: energy internet, wireless power routing, network protocol stack, deterministic energy delivery, inter-mesh federation, energy addressing, quality of service, energy border gateway protocol, Meridian, Lume-X, wireless power transfer standard

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Appendix B —” Energy Address Specification

Appendix C —” Energy Border Gateway Protocol (EBGP) Overview

Appendix D —” Standardization Roadmap

References

1. Introduction

1.1 The Wrong Frame

The history of wireless power transfer has been written in the wrong frame. Every major milestone in the field —” Brown's 54% rectenna efficiency in 1969, the MIT resonance demonstration in 2007, the proliferation of Qi charging pads —” has been framed as a transmission problem. How much power can be transferred? How efficiently? How far? The implicit model is a pipe: a source at one end, a receiver at the other, and the question is how much gets through.

This frame produced sixty years of genuine technical progress and one fundamental architectural dead end. A pipe, no matter how efficient, cannot scale into infrastructure. Infrastructure requires routing —” the capacity to direct resources along dynamically selected paths through a network of nodes, to reroute around failure, to prioritize competing demands, to guarantee delivery. A pipe has none of these properties. Neither does any wireless power system designed in the transmission frame.

The internet did not succeed because its transmission technology was superior. Packet radio, coaxial cable, fiber optics, and satellite links are all radically different physical transmission media. They share one thing: the IP protocol, which abstracts away the physical medium and provides a universal addressing, routing, and delivery framework that works identically regardless of what is underneath. Any physical medium that can carry a packet can participate in the internet. The protocol is the infrastructure. The medium is interchangeable.

I propose applying this insight to energy delivery. The Energy Internet is not a wireless power system —” it is a protocol. A formal specification for how energy flows between addressed nodes across independently administered meshes, using common routing rules, delivery guarantees, and security mechanisms that work identically regardless of the physical transmission medium underneath. Any physical medium capable of delivering energy in a controlled, measurable, confirmed manner can participate in the Energy Internet. The protocol is the infrastructure. Meridian [Meridian Architecture, 2026] is the first medium that can participate.

1.2 Why Now

Two developments make this proposal timely rather than speculative.

First, the Meridian architecture [Meridian Architecture, 2026] has demonstrated that deterministic wireless energy routing is architecturally complete. The four-layer Meridian stack —” Meridian Core, Meridian Flow Engine, Meridian Mesh Fabric, and Meridian Transmission Layer —” provides every component required for addressed energy routing: node identity, spatial coordination, flow governance, topology-aware routing, and physical beam delivery with closed-loop confirmation. The architecture is not a sketch. It is a formally specified system with defined interfaces between all layers. A protocol built on this foundation has a complete implementation target.

Second, the convergence of IP networking concepts with physical infrastructure management has accelerated. Software-defined networking has separated the control plane from the data plane, enabling centralized routing intelligence over distributed physical infrastructure. Network function virtualization has decoupled network functions from dedicated hardware. These developments have created a body of engineering practice and tooling directly applicable to the Energy Internet control plane. The Energy Internet does not need to invent its control architecture from scratch —” it can adopt SDN patterns proven in the data networking domain.

1.3 The Central Analogy

The central analogy of this paper is the correspondence between data packet routing and energy burst routing. I state it precisely here and develop it formally throughout:

Data Internet	Energy Internet
Bit	Joule
Packet	Burst
Packet buffer (RAM)	Supercapacitor
Store-and-forward	Harvest-store-transmit
IP address	Energy address (EIP)
Routing table	Energy routing table
TCP delivery guarantee	ETCP delivery guarantee
BGP inter-domain routing	EBGP inter-mesh routing
TLS security layer	Guardian Security (Guardian-E)
DNS	Energy Domain Name System (EDNS)
ISP	Energy Service Provider (ESP)
Internet Exchange Point (IXP)	Energy Exchange Point (EXP)
Autonomous System (AS)	Energy Autonomous System (EAS)

Each of these correspondences will be developed formally. They are not loose analogies —” they are functional equivalences at the protocol level. The mechanisms differ because the physics of energy and data differ, but the protocol functions are identical.

1.4 What the Energy Internet Is Not

Before proceeding, I want to be explicit about the scope of this proposal.

The Energy Internet is not a proposal to replace the electrical grid. The conventional grid transmits gigawatts over continental distances with transmission losses of 5—“10%. The Energy Internet, at current development scale, delivers milliwatts to watts over meter-scale distances. These are different operating regimes by many orders of magnitude. The grid will not be replaced by wireless mesh networks in any foreseeable timeframe. The Energy Internet addresses a different market: low-power distributed devices —” sensors, wearables, IoT nodes, microcontrollers, relay nodes —” that are poorly served by wired infrastructure, impractical to battery-maintain at scale, and within the power budget of ambient harvesting and directed beam delivery.

The Energy Internet is not imminent. No room-scale Meridian deployment has been experimentally validated. The path from laboratory demonstration to building scale to campus scale to city scale is measured in years per phase, not months. Each phase requires independent efficiency validation, safety certification, and regulatory review.

The Energy Internet is a standard —” a specification —” not a product. Its value is in establishing the protocol now, so that when the hardware capabilities exist, the routing infrastructure is already defined and interoperability is built in from the first deployment.

1.5 Paper Organization

Section 2 examines the data internet as a model for protocol-driven infrastructure scaling. Section 3 defines the Energy Internet Protocol Stack formally. Section 4 specifies energy addressing and the Energy Domain Name System. Section 5 defines inter-mesh federation and the Energy Border Gateway Protocol. Section 6 develops the quality-of-service framework for power delivery. Section 7 defines security at Energy Internet scale. Section 8 maps the path from current Meridian development to Energy Internet deployment. Section 9 addresses governance and standardization. Section 10 situates this work in related literature. Section 11 states limitations explicitly. Section 12 concludes. Four appendices provide formal specifications for the protocol stack, addressing, peering protocol, and standardization roadmap.

2. Background: The Internet as a Model

2.1 How the Data Internet Scaled

The ARPANET, the precursor to the modern internet, connected four nodes in 1969. It did not scale to global infrastructure because the underlying transmission technology was superior. It scaled because the protocol was correct [27, 28].

The key design decisions that made scaling possible were:

Packet switching. Rather than reserving a dedicated circuit between sender and receiver (as telephone networks did), packet switching breaks data into independently routed packets. Each packet carries its own destination address and is independently routed through whatever path is currently available. Circuits reserve capacity regardless of use; packets use capacity only when data exists to transmit. This efficiency gain is what made the internet economically viable at scale.

Layered abstraction. The OSI reference model [29] and TCP/IP stack separated network concerns into independent layers, each with a defined interface to the layers above and below. Physical medium, link-level framing, network-level addressing, transport-level delivery guarantees, and application-level

services are independently specified. A new physical medium (fiber optics, wireless, satellite) can participate in the internet by implementing the link layer interface —” it does not need to know about TCP, DNS, or any application protocol. The abstraction is what makes the internet medium-independent.

Decentralized routing. The Border Gateway Protocol [30] allows independently administered networks (Autonomous Systems) to exchange routing information and forward packets across network boundaries without requiring central coordination. No single entity controls internet routing. Each AS advertises the destinations it can reach and the routing protocol distributes this information through the network. The internet has no center. It has a protocol.

Best-effort delivery with transport-layer guarantees. The IP layer provides best-effort delivery: packets are forwarded toward their destination, but delivery is not guaranteed. The TCP layer adds delivery guarantees: lost packets are detected and retransmitted, packets are reordered if necessary, and the application sees a reliable byte stream regardless of the unreliability of the underlying network. The separation of routing (IP) from delivery guarantee (TCP) allows both to be independently optimized.

These four design decisions —” packet switching, layered abstraction, decentralized routing, and separated reliability guarantee —” are the architectural core of the data internet. I will show in Section 3 that all four have direct analogs in the Energy Internet.

2.2 What the Grid Did Not Do

The electrical grid took the opposite approach on every dimension.

The grid uses circuit switching: a generator produces power, it flows through transmission lines to a substation, and from there to distribution lines and finally to the consumer. The path is fixed by the physical wiring. There is no dynamic routing. A broken transmission line is a broken path —” there is no protocol to reroute power around it. The grid’s resilience comes from physical redundancy (parallel transmission lines) rather than dynamic routing.

The grid is not layered. High-voltage transmission, medium-voltage distribution, and low-voltage delivery are different physical systems, but they are not abstracted from each other. A generator cannot participate in a different grid by implementing an interface specification —” it must be physically connected to the transmission system with compatible voltage and frequency characteristics.

The grid is centrally coordinated. Grid operators —” independent system operators (ISOs) and regional transmission organizations (RTOs) —” coordinate generation and transmission in real time. The grid has a center. When the center fails, the grid fails. The 2003 North American blackout cascaded through 55 million people because a single control center lost situational awareness.

The grid provides no delivery guarantee at the individual device level. Power arrives at a standard voltage and frequency; what happens to it at the point of consumption is the consumer's problem. There is no TCP equivalent for electricity.

The grid's architecture reflects the engineering constraints of its era: it was designed before packet switching, before layered network protocols, and before the computational infrastructure that makes dynamic routing practical. It is not wrong — it is optimized for a different set of constraints. But it is not the architecture that produces the Energy Internet.

2.3 The Packet-Burst Correspondence

The most important technical correspondence between the data internet and the Energy Internet is the equivalence between a data packet and an energy burst. This equivalence is what makes the protocol stack analogy functional rather than metaphorical.

A data packet is a bounded unit of data transmitted as a unit through the network. It carries a header (source address, destination address, routing information, sequence number) and a payload (the data being delivered). It is stored in a buffer at each router, queued for transmission, forwarded on the selected outgoing interface, and confirmed received by the next hop.

An energy burst is a bounded unit of energy transmitted as a unit through the mesh. It carries header information (source node identity, destination node identity, routing path, burst sequence number, power level) and a payload (the RF energy delivered). It is stored in a supercapacitor at each relay node, queued for transmission in the TDMA schedule, forwarded on the selected outgoing beam path, and confirmed received by the next hop via delivery confirmation telemetry.

The supercapacitor IS the packet buffer. The TDMA transmission window IS the queue scheduling algorithm. The MFE burst scheduler IS the packet forwarding engine. The DRMA relay chain IS the store-and-forward router chain. The SHDCL retry sequence IS TCP retransmission. The delivery confirmation telemetry IS the TCP acknowledgment.

These are not loose analogies. They are the same mechanism — “store, schedule, forward, confirm” — applied to joules rather than bits. The physics differ (energy dissipates during transmission; bits do not), but the protocol function is identical.

2.4 The Key Difference: Lossiness

The data internet can retransmit a lost packet with zero information loss. A bit transmitted and lost can be retransmitted identically. Energy transmitted and lost — “dissipated as heat during RF propagation” — cannot be recovered. Retransmission of an energy burst costs new energy. This is the fundamental physical difference between data routing and energy routing, and it has protocol implications throughout.

In the data internet, lossiness at the IP layer is absorbed by TCP retransmission. The cost of retransmission is latency. In the Energy Internet, lossiness at the EPL layer is absorbed by ETCP retransmission (Section 6). The cost of retransmission is energy. Every protocol design decision in the Energy Internet must account for this asymmetry: retransmission costs what it is trying to deliver.

This asymmetry makes energy routing inherently less tolerant of routing inefficiency than data routing. A data packet lost in transit is annoying. An energy burst lost in transit depletes the transmitting node, fails to charge the receiving node, and may leave the mesh in a degraded state that requires SHDCL recovery. Energy routing protocols must minimize retransmission more aggressively than data routing protocols because the cost of retransmission is not just latency — it is energy.

3. The Energy Internet Protocol Stack

3.1 Design Principles

The Energy Internet Protocol Stack (EIPS) is designed on four principles inherited directly from the data internet design philosophy:

Principle 1 —” Layered abstraction. Each layer provides a defined interface to the layers above and below. A new physical transmission medium participates in the Energy Internet by implementing the EPL interface. A new application that consumes delivered energy participates by using the EAL interface. No layer needs to know the internal implementation of any other layer.

Principle 2 —” Decentralized routing. No single entity controls Energy Internet routing. Each Energy Autonomous System (EAS) administers its own routing tables internally and exchanges reachability information with peer EAS networks through EBGp. The Energy Internet has no center.

Principle 3 —” Separation of routing and delivery guarantee. EIP (Energy Internet Protocol) provides routing —” determining which path a burst takes through the network. ETCP (Energy Transmission Control Protocol) provides delivery guarantee —” ensuring that the target device receives at least $P_{\min}$ across N trials with variance below $\sigma^2_{\max}$. These are independent functions with independent protocols.

Principle 4 —” Physical medium independence. The Energy Internet is not defined by a specific transmission technology. Directed RF beamforming (Meridian/DWER) is the first medium. Optical energy delivery, millimeter-wave transmission, inductive relay chains, and future technologies that do not yet exist are all valid EPL implementations as long as they can satisfy the ELL interface requirements: addressed node delivery, power measurement, and delivery confirmation.

3.2 The Five-Layer Stack

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" EAL —" Energy Application Layer " " (Energy as
a Service, priority routing, " " application-defined power profiles) "
" ETL —" Energy Transport Layer " " (ETCP:
delivery guarantee,  $P_{\min}/\sigma^2$  " " enforcement, retransmission, QoS) "
" ENL —" Energy Network Layer " " (EIP:
addressing, inter-mesh routing, " " EBGp, EDNS resolution) "
" ELL —" Energy Link Layer " " (hop-by-hop
delivery, burst framing, " " link-level confirmation, TDMA scheduling) "
" EPL —" Energy Physical Layer " " (transmission
medium: RF, optical, etc. " " beam control, safety gating, localization)"

```

Each layer is defined by the services it provides to the layer above and the requirements it places on the layer below.

3.3 Energy Physical Layer (EPL)

The EPL is the transmission medium. It is responsible for the actual delivery of energy from one node to an adjacent node —" a single hop. The EPL does not know about multi-hop routing, delivery guarantees across an end-to-end path, or addressing above the single-hop level. It knows how to aim energy at a located node, fire a burst, and confirm that energy was received.

EPL mandatory capabilities:

Node localization: must be able to determine the position of the receiving node within a spatial uncertainty radius r , updated at minimum 10 Hz
 Directed transmission: must be able to aim energy output toward the located receiver with a beam width that limits energy outside the intended path
 Power measurement: receiving node must measure P_{received} at mW resolution and report it back to the transmitting node
 Delivery confirmation: must provide a confirmed delivery receipt containing P_{received} , spatial accuracy achieved, and timestamp
 Safety gating: must implement hardware-level transmission cutoff independent of software gate status
 EPL optional capabilities (for advanced implementations):

Beam shaping for sidelobe reduction
 Frequency agility across multiple operating bands
 Multi-element phased-array steering
 Polarization diversity
 Meridian/DWER as EPL implementation. The DWER subsystem as defined in [Meridian Architecture, 2026] is a complete EPL implementation. MTL (Meridian Transmission Layer) provides beam steering, safety gating, and physical emission control. The UWB localization subsystem provides node localization. The delivery confirmation telemetry provides power measurement and confirmation. Every EPL mandatory capability is satisfied by the Meridian/DWER specification.

3.4 Energy Link Layer (ELL)

The ELL manages energy delivery across a single hop within the mesh —” between one node and its immediate neighbor. It provides link-level framing (the Energy Burst Frame, described below), TDMA slot scheduling within the local mesh, link-quality monitoring, and per-hop delivery confirmation.

Energy Burst Frame (EBF):

EBF Header: src_node_id : 64 bits (sending node's EIP address) dst_node_id : 64 bits (receiving node's EIP address) burst_seq_num : 32 bits (sequence number for this burst) P_planned_mW : 16 bits (planned transmit power) T_window_start : 64 bits (network timestamp, μ s) T_window_end : 64 bits (network timestamp, μ s) hop_count : 8 bits (number of hops traversed so far) ttl : 8 bits (maximum hops remaining) path_id : 32 bits (routing path identifier from ENL) safety_flags : 16 bits (BSL status flags at time of transmission) checksum : 32 bits EBF Confirmation (returned by receiver): src_node_id : 64 bits dst_node_id : 64 bits burst_seq_num : 32 bits P_received_mW : 16 bits (measured received power) V_cap_after : 16 bits (supercapacitor voltage after receipt) spatial_error_cm: 8 bits (position error at time of receipt) T_received : 64 bits (network timestamp of receipt) status : 8 bits (CONFIRMED / PARTIAL / FAILED)

The EBF provides the Energy Internet with its fundamental unit of transmission accounting. Every joule delivered through the Energy Internet is traceable to a specific EBF with a specific confirmation status. This is the foundation for energy billing, regulatory compliance, and safety audit trails.

ELL relationship to Meridian. The ELL maps directly onto the Meridian link layer as defined in the DRMA and DWER subsystems. The EBF header corresponds to the BeamCommand struct (MTL), and the EBF confirmation corresponds to the DeliveryConfirm message. Meridian nodes implementing the EIPS run ELL as their link-layer protocol.

3.5 Energy Network Layer (ENL)

The ENL provides energy routing across multiple hops and across mesh boundaries. It implements EIP —” the Energy Internet Protocol —” for addressing and intra-AS routing, and EBGp for inter-AS routing. ENL is the layer that makes the Energy Internet a network rather than a collection of isolated meshes.

The ENL is analogous to the IP layer in the data internet: it provides best-effort routing —” the best path currently available is used, but delivery across the full path is not guaranteed at this layer. Delivery guarantees are provided by the ETL above.

ENL maintains the energy routing table at each node —” not the local per-mesh routing table maintained by MMF, but the wider energy routing table that includes paths to nodes in other EAS networks. ENL processes EBF headers, determines the next-hop ELL address for each burst, and forwards bursts accordingly.

3.6 Energy Transport Layer (ETL)

The ETL provides end-to-end delivery guarantees across the full path from source to destination —” the equivalent of TCP for energy. It enforces the $P_{\min}/r/\sigma^2/d$ determinism standard across the full path, not just per hop. It handles retransmission when delivery fails, reordering when burst sequence is disrupted, and congestion control when the mesh is overloaded.

ETCP —” Energy Transmission Control Protocol:

ETCP maintains a session between a source energy provider and a destination energy consumer. The session specifies:

ETCP Session: source_EIP : energy address of providing node destination_EIP : energy address of consuming device $P_{\min_mW}$: minimum acceptable power at destination σ_{sq_max} : maximum acceptable delivery variance (%) priority_class : QoS priority (CRITICAL / HIGH / NORMAL / BACKGROUND) session_duration : session length in seconds (or INDEFINITE) retry_limit : maximum retransmission attempts per burst billing_account : energy credit account for settlement

ETCP monitors cumulative delivery statistics across all bursts in a session. If the rolling σ^2 across the last N bursts exceeds $\sigma^2_{\max}$, ETCP signals the ENL to reroute through a less lossy path. If P_{received} falls below $P_{\min}$ across three consecutive bursts, ETCP triggers a retransmission and escalates to SHDCL if retransmission fails.

The cost of retransmission. As established in Section 2.4, energy retransmission costs energy. ETCP's retransmission mechanism must account for this: a retransmission request is not free. ETCP maintains a retransmission budget per session —” a maximum number of retransmissions permitted before the session is flagged as inefficient and rerouting is forced. This prevents a poorly performing path from draining nodes through repeated retransmission.

3.7 Energy Application Layer (EAL)

The EAL is the interface between the Energy Internet and the applications that consume delivered energy. It provides three primary services:

Energy as a Service (EaaS). A consuming device declares its power profile —” minimum power, preferred power, maximum power, duty cycle —” and the EAL negotiates a ETCP session that satisfies the profile. The device does not manage its own energy delivery; the EaaS layer handles all routing, delivery guarantee, and retransmission management on its behalf.

Priority routing requests. The EAL allows consuming devices to request priority-class delivery. A medical monitoring device requests CRITICAL priority. A background IoT sensor requests BACKGROUND priority. The EAL translates priority classes into QoS parameters for the ETCP session and ENL routing preferences.

Energy accounting interface. The EAL provides each consuming device with a standardized interface to its energy consumption accounting: how much energy was delivered in the current session, at what efficiency, from which source nodes, and at what cost in energy credits.

4. Energy Addressing and the Energy Domain Name System

4.1 The Energy Internet Protocol Address (EIP Address)

The MC 64-bit node identity defined in [Meridian Architecture, 2026] serves as the link-layer identifier within a single mesh. The Energy Internet requires a hierarchical addressing scheme that spans multiple independently administered meshes —” equivalent to the transition from MAC addresses (link-layer, locally unique) to IP addresses (network-layer, globally unique).

The EIP address is a 128-bit hierarchical identifier structured as:

EIP Address (128 bits): [Global Registry ID : 16 bits] Assigned by the Energy Internet Registry Authority (EIRA) [EAS Number : 24 bits] Energy Autonomous System number —” uniquely identifies mesh operator [Zone ID : 16 bits] Logical zone within the EAS [Node Type : 8 bits] Functional role (source / relay / destination / harvesting relay / gateway) [Node Serial : 40 bits] Unique per-node serial within the EAS [Checksum : 24 bits] Address integrity verification

This 128-bit structure provides:

65,536 Global Registry partitions (supporting multiple governance regimes or geographic regions) 16,777,216 unique EAS numbers (each independently administered energy mesh network) 65,536 zones per EAS 256 node types 1 trillion unique nodes per EAS zone The address space is intentionally large —” designed to accommodate the scale of a global sensor and IoT network without requiring renumbering as the Energy Internet grows. The IPv6 designers learned this lesson painfully from IPv4's exhausted 32-bit space; EIP incorporates it from the beginning.

Address hierarchy and routing. The EAS Number in the EIP address is the key to inter-mesh routing. Within a single EAS, routing uses MMF's intra-mesh routing tables (extended to EIP addressing). Between EAS networks, routing uses EBGp to exchange reachability information: "EAS 42 can reach nodes with EAS Number 42 in their address." This is structurally identical to BGP routing in the data internet, where Autonomous System Numbers identify networks and BGP routes between them.

4.2 EIP Subnetting and Zone Aggregation

Large EAS deployments —” a campus, a building complex, or an industrial facility —” may contain thousands of nodes organized into zones. Routing announcements for every individual node would overwhelm inter-mesh routing protocols. EIP uses zone aggregation analogous to IP CIDR (Classless

Inter-Domain Routing): a gateway node announces the entire zone as a single reachable prefix rather than individual node addresses.

Zone aggregation example: Full node address: [EIRA:0001][EAS:000042][ZONE:0005][TYPE:02]
[SERIAL:000000001234][CHK:...] Zone aggregate: [EIRA:0001][EAS:000042][ZONE:0005][[/][*]
Inter-mesh announcement: "EAS:000042 can deliver energy to all nodes in ZONE:0005"

This dramatically reduces the routing table size at inter-mesh gateway nodes —" exactly as CIDR reduces BGP routing table size in the data internet.

4.3 Energy Domain Name System (EDNS)

EIP addresses are not human-readable. Just as the Domain Name System translates human-readable hostnames (google.com) to IP addresses, the Energy Domain Name System translates human-readable energy endpoint names to EIP addresses.

EDNS names follow a hierarchical structure:

[device-name].[zone].[facility].[operator].energy Examples: sensor-7.lab-a.building-3.mit.energy cardiac-monitor-icu-4.floor-2.mass-general.energy relay-node-42.zone-5.campus-north.darkwave.energy

EDNS resolution works identically to DNS:

Consuming device requests EIP address for sensor-7.lab-a.building-3.mit.energy Local EDNS resolver checks its cache

If not cached: queries the authoritative EDNS server for building-3.mit.energy

Authoritative server returns the EIP address for sensor-7.lab-a Resolver caches the result with a TTL and returns to the requesting device EDNS is administered by the facility operator —" in the example above, MIT manages the mit.energy EDNS zone. The root EDNS zone (.energy) is administered by the EIRA (Energy Internet Registry Authority) —" the governance body proposed in Section 9.

EDNS and mobile nodes. For nodes that move between zones —" a mobile medical device that travels between hospital floors, or a wearable that moves through a building —" EDNS must provide low-latency address update capability. The TTL on EDNS records for mobile nodes should be short (seconds rather than hours) and the authoritative EDNS server should update records when node position crosses zone boundaries. This is analogous to mobile IP, where a mobile device's home agent maintains a current location binding.

5. Inter-Mesh Federation and Energy Peering

5.1 Energy Autonomous Systems

An Energy Autonomous System (EAS) is an independently administered collection of energy routing nodes under common operational control. A university campus, a hospital complex, a smart factory, or a city block could each constitute an EAS. The EAS is the fundamental unit of Energy Internet governance: within an EAS, the operator chooses its own routing protocols and node management practices. Between EAS networks, routing is governed by EBGp.

Every EAS has:

A globally unique EAS Number assigned by the EIRA
One or more gateway nodes that peer with adjacent EAS networks
An internal routing domain managed by the operator's choice of intra-EAS protocol (Meridian MMF at small scale; more scalable protocols at large scale)
A routing policy that defines which destinations the EAS will accept energy traffic for and which EAS peers it will transit traffic between
This structure is identical to the Autonomous System model of the data internet, and for the same reason: it allows the Energy Internet to scale without requiring any central entity to know the complete topology.

5.2 Energy Exchange Points (EXPs)

In the data internet, major carriers interconnect at Internet Exchange Points (IXPs) —” physical facilities where routers from multiple Autonomous Systems share a common switching fabric and exchange BGP routing information. This eliminates the need for every AS to establish direct connections to every other AS.

The Energy Internet requires the same infrastructure: Energy Exchange Points (EXPs). An EXP is a physical location —” initially, a wired power connection point —” where multiple EAS gateway nodes are co-located and can peer directly. At the EXP:

Gateway nodes from multiple EAS networks are physically co-located
EAS operators establish EBGp peering sessions with peer operators
Energy traffic that crosses AS boundaries is handed off between gateway nodes
The EXP maintains a public peering database listing which EAS operators are present and what energy delivery capacity they offer
At the EXP, energy traffic crosses the boundary between wireless mesh and wired power (or between two wireless meshes). The EXP is not a wireless-to-wireless connection —” at current technology scale, inter-mesh energy transfer at significant power levels requires a wired connection at the exchange point. The wireless Energy Internet operates within each EAS; the EXP provides the inter-AS connection at the ELL level.

As energy routing technology matures and efficiency at longer ranges improves, direct wireless inter-mesh connections become viable. The EXP model accommodates this evolution: a wired EXP is an ELL implementation, just as a copper IXP and a fiber IXP are both valid data internet IXP implementations. The protocol does not change.

5.3 Energy Border Gateway Protocol (EBGP)

EBGP is the routing protocol that exchanges reachability information between EAS networks. It is a path-vector protocol —” each EBGP update carries not just the destination and cost, but the complete sequence of EAS numbers that the path traverses. This allows receiving EAS operators to detect and avoid routing loops, apply routing policy based on which operators are in the path, and select paths based on operator relationships as well as technical metrics.

EBGP route advertisement:

EBGP UPDATE message: origin_EAS : 24 bits (advertising EAS number) destination_prefix : 128 bits (EIP address prefix being announced) AS_path : variable (sequence of EAS numbers traversed) P_available_mW : 16 bits (available delivery power at destination zone) efficiency_estimate : 8 bits (estimated end-to-end delivery efficiency %) latency_ms : 16 bits (estimated burst delivery latency) safety_certified : 1 bit (whether path meets BSL requirements end-to-end) guardian_certified : 1 bit (whether path is Guardian Security certified)

next_hop_gateway	: 128 bits	(EIP address of next-hop gateway node)
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The inclusion of safety_certified and guardian_certified flags in EBGP updates is a significant departure from data internet BGP, where security and safety properties are not advertised at the routing layer. In the Energy Internet, safety certification is a routing property —” a consumer device requesting CRITICAL priority delivery (a medical device, for example) should only receive energy over paths that are BSL-certified end-to-end. EBGP carries this information so that ENL routing decisions can enforce it automatically.

EBGP route selection. When an ENL router has multiple EBGP routes to the same destination, it selects the preferred route using an ordered set of criteria:

Highest local preference (operator-configured policy) Shortest AS path (fewest inter-AS hops) Highest P_available_mW (most delivery capacity) Highest efficiency_estimate (least lossy path) Lowest latency_ms (fastest delivery) Tie-break: lowest next-hop gateway EIP address (deterministic) Criteria 3 and 4 are energy-specific additions. In the data internet, BGP does not consider bandwidth or loss rate at the routing layer. In the Energy Internet, delivery capacity and efficiency are routing-layer concerns because they directly affect whether ETCP can satisfy the session's P_min requirement.

5.4 Energy Peering Agreements

EAS operators establish peering agreements governing the terms under which they exchange energy traffic. Three peering models exist, directly analogous to data internet peering models:

Settlement-free peering. Two EAS operators of comparable size exchange energy traffic at no charge, agreeing that the traffic flows are roughly balanced over time. Common between adjacent campus meshes or neighboring building systems.

Transit peering. A smaller EAS pays a larger EAS to route its traffic to destinations outside their shared zone. The larger EAS provides Energy Internet transit —” access to the wider Energy Internet through its gateway connections.

Energy credit settlement. An accounting system for energy traffic that crosses AS boundaries. Energy credits —” denominated in milliwatt-hours —” are transferred between EAS operators based on measured delivery volumes. The EBF confirmation record provides the audit trail for settlement: every delivered burst is documented with source, destination, power, and timestamp.

6. Quality of Service for Power Delivery

6.1 Why QoS Matters in the Energy Internet

In the data internet, QoS differentiates time-sensitive traffic (voice, video) from best-effort traffic (file downloads, email). A voice call that arrives 500 ms late is useless. An email that arrives 500 ms late is indistinguishable from an email that arrives instantly. QoS marks time-sensitive traffic for preferential treatment and allows the network to meet latency requirements even during congestion.

In the Energy Internet, QoS differentiates power-critical loads from background loads. A cardiac monitor that loses power is a life-safety event. A background temperature sensor that misses a charging cycle recovers on the next cycle. Power QoS allocates available energy delivery capacity to the highest-priority loads first, ensuring that critical devices receive $P_{\min}$ even when the mesh is operating under capacity constraints.

Energy QoS has an additional dimension absent from data QoS: not just priority, but also delivery guarantee strength. A background sensor can tolerate high σ^2 (high variance in received power). A precision actuator cannot. QoS must specify both priority class and the σ^2 threshold the session must maintain.

6.2 QoS Priority Classes

The Energy Internet defines four QoS priority classes:

CRITICAL. Life-safety or mission-critical devices. Examples: implantable medical devices, emergency communications nodes, hospital monitoring equipment, critical infrastructure sensors. CRITICAL sessions receive absolute priority over all other traffic. Mesh capacity is reserved for CRITICAL sessions

even at the cost of interrupting BACKGROUND and NORMAL sessions. BSL certification is mandatory for all CRITICAL paths. ETCP retransmission budget is unlimited. σ^2_{max} is 5% (tightest delivery guarantee).

HIGH. Important operational devices. Examples: industrial process monitors, security systems, building management sensors. HIGH sessions receive priority over NORMAL and BACKGROUND but may be preempted by CRITICAL sessions. σ^2_{max} is 10%.

NORMAL. Standard IoT devices. Examples: occupancy sensors, environmental monitors, asset trackers. NORMAL sessions receive best-effort service with σ^2_{max} of 15%.

BACKGROUND. Non-time-sensitive devices. Examples: data loggers, passive sensors, occasional-reporting devices. BACKGROUND sessions use only surplus mesh capacity —” capacity not needed by higher-priority sessions. σ^2_{max} is 20%. No delivery latency guarantee.

6.3 Energy as a Service (EaaS) Model

Energy as a Service extends the QoS framework to a complete managed energy delivery model. Rather than requiring each consuming device to manage its own ETCP sessions and routing negotiations, EaaS provides a subscription-based energy delivery interface:

Device enrolls in EaaS. Device provides its power profile:

EaaS Profile: device_EIP : device's energy address P_min_mW : minimum continuous power requirement
P_preferred_mW : preferred operating power duty_cycle : fraction of time at P_preferred (0.0—“1.0)
priority_class : CRITICAL / HIGH / NORMAL / BACKGROUND sigma_sq_max : maximum acceptable delivery variance
max_burst_latency : maximum acceptable time between bursts (ms) billing_account : energy credit account

EaaS negotiates delivery. The EaaS layer establishes ETCP sessions with available source nodes, negotiates EBGp paths that satisfy the profile requirements, and manages routing failover transparently. The device receives power and does not manage its own energy routing.

EaaS monitors and adapts. If the enrolled device's power consumption pattern changes —” it enters high-activity mode, requiring more power —” EaaS detects the change through the EBF confirmation records and renegotiates the ETCP session parameters. This is the energy analog of adaptive bitrate streaming in video delivery: the delivery profile adapts to actual consumption in real time.

6.4 Emergency Priority Override

The Energy Internet must handle emergency situations in which normal priority scheduling is insufficient. An EMP event, a natural disaster, or a grid outage creates a condition where available energy routing capacity suddenly drops while CRITICAL demand (emergency services, medical devices, communication

infrastructure) surges.

The Emergency Priority Override (EPO) mechanism allows designated emergency management authorities to issue a network-wide priority reset: all BACKGROUND and NORMAL sessions are suspended, all HIGH sessions are rate-limited, and all available mesh capacity is reserved for CRITICAL sessions. EPO is authenticated by the Guardian Security layer using emergency management authority credentials.

EPO is not a product feature —” it is a protocol requirement for any Energy Internet deployment that serves life-safety applications. Its inclusion in the EIPS reflects the same reasoning that motivated priority routing in emergency communication networks: when lives depend on the network, the network must have a way to prioritize life-safety traffic over all other traffic.

7. Security at Scale: Guardian Security as Energy TLS

7.1 The Threat Model Expands at Internet Scale

Within a single Meridian mesh, the Guardian Security threat model [Meridian Architecture, 2026] covers six primary vectors: localization spoofing, routing table manipulation, beam hijacking, denial-of-power attacks, node impersonation, and timing attacks. These threats are addressed by per-node cryptographic authentication, signed routing updates, delivery confirmation verification, and TDMA timing enforcement.

At Energy Internet scale, the threat model expands in three dimensions:

Scale. An attacker targeting a single mesh node faces local defenses only. An attacker targeting the inter-mesh routing infrastructure faces the entire Energy Internet. EBGp route manipulation —” analogous to BGP hijacking in the data internet, which has caused major internet outages —” can redirect energy traffic across entire EAS networks. The Energy Internet requires BGP-equivalent security: route origin validation, AS path verification, and cryptographic signing of routing updates.

Sophistication. Single-mesh attacks require physical proximity to the target mesh. Internet-scale attacks can be launched remotely, from anywhere with access to the EBGp peering infrastructure. The authentication requirements for inter-mesh communication are therefore higher than for intra-mesh communication.

Stakes. In the data internet, a successful attack degrades data delivery. In the Energy Internet, a successful attack at CRITICAL priority can deny power to life-safety devices. The stakes of security failure are not measured in data loss but in physical harm. The security architecture must reflect this.

7.2 Guardian Security at Three Levels

Guardian Security (Guardian-E in the energy domain, domain: TrustShield.tech) operates at three levels in the Energy Internet:

Level 1 —” Intra-mesh security. The Guardian Security implementation defined in [Meridian Architecture, 2026]. Per-node cryptographic authentication, signed EBF frames, TDMA timing enforcement, delivery confirmation verification. This level is unchanged from the single-mesh implementation.

Level 2 —” Inter-mesh security (Energy TLS). Guardian Security extended to the inter-mesh interface. Every EBGp session between gateway nodes is authenticated using EAS-level public key infrastructure (PKI). EBGp updates are signed by the originating EAS gateway. Receiving gateways verify the signature before accepting and propagating route updates. Route origin validation confirms that an EAS advertising a destination actually controls the EIP address prefix it is advertising. This is Energy TLS —” the analog of Transport Layer Security that protects data internet connections.

Level 3 —” Registry security. Guardian Security at the EIRA level. EAS Number assignments, EIP address prefix delegations, and EDNS zone delegations are cryptographically signed by the EIRA and verifiable by any EIP node. A forged EAS Number or a fraudulently delegated address prefix is detectable by any properly configured ENL node.

7.3 Energy RPKI: Route Origin Validation

The data internet's Resource Public Key Infrastructure (RPKI) provides cryptographic attestation that an AS is authorized to originate routes for a given IP address prefix. Energy RPKI (ERPKI) provides the same function for EIP address prefixes: the EIRA maintains a signed database of which EAS is authorized to originate routes for which EIP prefix, and ENL routers validate EBGp route origins against this database before accepting them.

A route origin that does not validate against ERPKI is rejected. This prevents the energy equivalent of BGP hijacking —” an adversary announcing routes for EIP prefixes they do not control in order to intercept or disrupt energy traffic to those addresses.

ERPKI is a preventive security measure. It does not depend on detecting attack behavior; it validates authorization before accepting routing information. This is a stronger security guarantee than anomaly-based detection for inter-mesh routing security.

7.4 Emergency Authority Credentials

The Emergency Priority Override mechanism (Section 6.4) requires authentication of emergency authority credentials. Guardian Security manages a separate credential class —” Emergency Authority Credentials (EAC) —” issued to designated emergency management bodies: hospitals, emergency services, military installations, critical infrastructure operators.

EAC holders can issue signed EPO commands that propagate through the Energy Internet via EBGp. ENL routers that receive a valid EPO command apply the emergency priority scheduling immediately. Invalid or unsigned EPO commands are rejected. The EAC credential system is administered by the EIRA and is audited by Guardian Security's certificate chain logging.

8. The Path from Meridian to Energy Internet

8.1 Development Phases

The Energy Internet is not built at once. It is built incrementally, each phase validating the assumptions of the next. The development path runs from the current state of the Meridian architecture through a series of validated phases to global deployment.

Phase 0 —” Current state. The Meridian architecture is formally specified. Lume-X is validated at 73 Hz. No hardware experiment has been conducted. The Energy Internet Protocol Stack is defined in this paper. The path is clear; the walking has not begun.

Phase 1 —” Single-mesh validation (Years 1—“2). DWER Phase 1—“3 experimental results validate point-to-point and mesh energy routing at room scale. A single Meridian EAS operates and is confirmed against the $P_{\min}/r/\sigma^2/d$ determinism standard. The EPL, ELL, and intra-EAS ENL layers are validated in hardware. This is the foundation without which nothing else advances.

Phase 2 —” Multi-mesh federation (Years 2—“4). Two independently operated Meridian EAS networks establish an EBGp peering session. Energy traffic is successfully routed across the AS boundary. Inter-mesh Guardian Security authentication is validated. Energy credit settlement between the two operators is demonstrated. The EDNS resolves cross-AS addresses. This phase validates the inter-mesh federation architecture.

Phase 3 —” Building scale (Years 3—“6). A full building is covered by a Meridian Energy Internet deployment —” multiple EAS zones, an in-building EXP, CRITICAL priority delivery to life-safety devices, and EaaS subscriptions for standard devices. QoS priority scheduling is validated under simulated load. The ETCP delivery guarantee is confirmed across full building-scale paths.

Phase 4 —” Campus scale (Years 5—“10). Multiple buildings interconnect through EXPs. EBGp peering between building-level EAS networks enables campus-wide energy routing. Higher-frequency operation (24 GHz, 60 GHz) improves beam precision and supports higher node density. Device-aware routing maps device identity to named energy paths. The Energy Internet is a functioning campus-scale infrastructure.

Phase 5 —” Energy Internet (Years 10+). The EIPS is formally standardized through the IETF-equivalent process described in Section 9. Multiple independent operators deploy compatible Energy Internet infrastructure. EBGP peering between city-level EAS networks. Energy Internet exchange points in major urban centers. The Energy Internet is a functioning multi-operator global infrastructure for low-power device delivery.

8.2 The Efficiency Constraint at Scale

The single largest technical obstacle to Energy Internet scaling is efficiency. At room scale, end-to-end efficiency of 3—“25% is acceptable —” the transmit power is under 1W, the delivered power is measurable in milliwatts, and the energy cost is negligible. As the network scales and the delivered power requirements grow, efficiency becomes a binding constraint.

The efficiency requirement at building scale: a building with 10,000 IoT sensors each requiring 1 mW average power needs 10 W of delivered energy. At 10% end-to-end efficiency, this requires 100 W of transmission power —” feasible. At 1% efficiency, it requires 1 kW —” the equivalent of a moderate electrical heater, which is no longer negligible.

The efficiency improvement path runs through:

Higher-frequency operation (shorter wavelength †’ smaller antenna aperture †’ tighter beams †’ less path loss at equivalent distances) Better rectenna design (current 50—“70% RF-to-DC conversion can improve toward 85%+ with optimized circuit design) Denser node deployment (shorter hop distances †’ less per-hop path loss) Better beam shaping (tighter main lobes †’ less power in non-useful directions) Adaptive beam tracking (dynamic nulling of interference directions †’ less wasted power) None of these improvements are speculative —” they are active research areas with documented progress. The Energy Internet does not require a breakthrough; it requires the accumulation of incremental efficiency improvements across all stages of the energy path.

8.3 The Regulatory Pathway

Energy Internet deployment at building and campus scale requires regulatory approval that does not yet exist for this class of system. The current regulatory landscape treats wireless power systems as Point-to-Point Power Transmission under FCC Part 15 (below 1W) or as Industrial, Scientific, and Medical (ISM) equipment under FCC Part 18 (above 1W). Neither framework contemplates a multi-mesh, multi-operator, energy-routing network.

The regulatory pathway requires:

Phase 1—“3 experimental safety data demonstrating BSL compliance at room scale Submission of a regulatory framework proposal to the FCC, Ofcom, and ITU documenting the system architecture, safety mechanisms, and proposed operating parameters Development of a type-approval process for Energy

Internet nodes —” standards that a node must meet to be certified for Energy Internet participation (analogous to FCC Part 15 type approval for wireless devices) Development of an operator licensing framework for EAS operators —” defining what certifications and safeguards are required to operate an EAS network International harmonization of operating frequency bands and power limits for Energy Internet operation This is not a short process. Regulatory development for new wireless technology frameworks typically takes five to ten years from initial proposal to formal rule. The regulatory pathway must begin concurrently with Phase 1 experimental validation —” not after.

9. Governance and Standardization

9.1 Why Governance Matters

The data internet's most important governance decision was to make IP an open standard —” freely implementable by anyone, with no royalties, governed by the IETF through a consensus-based process. This decision is what allowed the internet to scale globally across every political and economic boundary. If IP had been a proprietary standard owned by a single vendor, the internet as we know it would not exist.

The Energy Internet requires the same governance decision. A proprietary Energy Internet standard —” controlled by a single company, with licensing requirements for implementation —” would produce a fragmented landscape of incompatible energy routing systems that cannot peer with each other. The value of a network is proportional to the square of the number of participants (Metcalf's Law). A proprietary Energy Internet with 1,000 nodes is worth far less than an open Energy Internet with a billion nodes.

I propose that the Energy Internet Protocol Stack be submitted to an open standards process —” the IETF or an equivalent body —” for development and maintenance as an open standard. DarkWave Studios LLC commits to making the foundational EIPS specification royalty-free. The provisional patent portfolio covering the Meridian architecture will be licensed under FRAND (Fair, Reasonable, and Non-Discriminatory) terms for implementations that comply with the open standard.

9.2 The Energy Internet Registry Authority (EIRA)

The Energy Internet requires a registry authority analogous to IANA (Internet Assigned Numbers Authority) and ICANN (Internet Corporation for Assigned Names and Numbers):

EIRA functions:

Assign EAS Numbers to qualifying Energy Internet operators

Delegate EIP address prefixes to EAS operators Administer the root EDNS zone (.energy) Maintain the ERPKI root certificate for Energy Internet route origin validation Issue Emergency Authority Credentials to qualifying emergency management bodies Publish the authoritative EIPS standard and manage its

versioning EIRA governance model. The EIRA should be a multi-stakeholder body, not a government agency or a single company. Stakeholders include: EAS operators, device manufacturers, regulatory agencies, safety certification bodies, emergency management agencies, and civil society representatives. This is the governance model that has allowed ICANN and IANA to operate effectively across national boundaries.

9.3 The RFC Process for Energy Internet Standards

The Internet Engineering Task Force's RFC (Request for Comments) process has proven remarkably effective at developing and maintaining internet standards. Technical proposals are published as drafts, reviewed publicly, revised through community comment, and published as RFCs when consensus is reached. No individual or organization can unilaterally adopt a standard —” consensus is required.

I propose an Energy RFC (ERFC) process for Energy Internet standards development:

ERFC-0001: Energy Internet Protocol Stack —” the foundational document (this paper adapted for standards format) ERFC-0002: Energy Internet Protocol (EIP) —” addressing specification ERFC-0003: Energy Transmission Control Protocol (ETCP) —” delivery guarantee specification ERFC-0004: Energy Border Gateway Protocol (EBGP) —” inter-mesh routing specification ERFC-0005: Energy Domain Name System (EDNS) —” human-readable addressing ERFC-0006: Energy Resource Public Key Infrastructure (ERPki) —” route origin validation ERFC-0007: Energy QoS —” priority class definitions and scheduling requirements ERFC-0008: Guardian Security for the Energy Internet —” security specification

These eight documents constitute the minimum viable standards package for an interoperable Energy Internet. Additional ERFCs covering operational practices, management protocols, and application interfaces would follow.

10. Related Work

10.1 Wireless Power Transfer Systems

The commercial wireless power landscape —” WiTricity, Ossia Cota, Energous WattUp —” was reviewed in [Meridian Architecture, 2026]. None of these systems implement routing, addressing, inter-system federation, or a formal delivery guarantee framework. They are transmission systems, not routing systems. The Energy Internet is not in competition with these systems as products —” it is a fundamentally different architectural approach that subsumes the transmission capabilities of these systems under a routing framework.

The academic wireless power transmission literature similarly focuses on transmission efficiency rather than routing infrastructure. Powercast's RF energy harvesting work [35] has demonstrated multi-node energy delivery networks at low power levels, but without addressing, routing protocols, or delivery guarantees. These systems are closest to the EPL of the Energy Internet —” they implement the physical transmission medium —” but they do not provide the protocol stack above it.

10.2 Software-Defined Networking

The SDN literature [31, 32] is directly relevant to the Energy Internet control plane. SDN separates the control plane (routing decisions) from the data plane (packet forwarding) and centralizes routing intelligence in a software controller that programs the forwarding rules of distributed data plane devices. This architecture dramatically simplifies network management and enables dynamic routing policy without reconfiguring individual devices.

The Energy Internet's Lume-X control layer implements the SDN pattern: Lume-X is the control plane (managing routing decisions across all four Meridian layers), while the physical node firmware is the data plane (executing forwarding decisions without independent routing logic). The separation is explicit in the Meridian architecture: MTL executes BeamCommands but does not determine which node to beam; that decision belongs to MMF (the energy routing controller).

Future Energy Internet deployments at building and campus scale may benefit from centralized SDN-style energy controllers that manage routing across the entire EAS rather than distributing routing intelligence to every node. The EIPS is designed to accommodate both distributed and centralized control plane models.

10.3 Internet of Things Power Management

The IoT power management literature has extensively studied strategies for extending battery life, harvesting ambient energy, and duty-cycling sensor nodes to minimize power consumption. Energy harvesting for wireless sensor networks —” particularly the work of Kansal et al. [33] on energy-neutral operation and Sudevalayam and Kulkarni [34] on energy harvesting sensor nodes —” provides the baseline performance characterization that the Energy Internet must exceed.

The critical distinction from this body of work is the direction of the solution. IoT power management minimizes the device's energy requirements to match available supply. The Energy Internet increases the available supply to match device requirements. Both approaches are valid; they address different operating regimes. The Energy Internet is applicable when ambient harvesting alone is insufficient and a more active energy delivery mechanism is required.

10.4 The Power Line Communication Analogy

Power Line Communication (PLC) —” using the existing electrical grid wiring as a communication medium —” inverts the Energy Internet concept: it uses an energy medium to carry data. The Energy Internet uses a data networking protocol to carry energy. The inversion is instructive: PLC demonstrated that the data internet protocol stack (IP, TCP, application protocols) could be implemented over an energy medium with appropriate physical layer adaptation. The Energy Internet demonstrates the converse: energy delivery can be implemented using a data networking protocol stack with appropriate adaptation for the physics of energy transmission.

The analogy suggests that hybrid energy-data meshes —” networks that simultaneously route power and information over the same physical nodes using a unified protocol stack —” may be feasible and valuable. The DRMA mesh already carries both energy (in burst-mode RF transmissions) and data (in control-plane EBF communications). A unified mesh that formally integrates both functions is a natural extension beyond the scope of this paper.

10.5 The Lume Ecosystem

The Energy Internet is the long-horizon destination of the Lume ecosystem's trajectory. Lume [15] provided the deterministic programming language. DAIGS [17] provided the multi-agent coordination framework. Lume-X [19] provided the high-frequency deterministic control runtime. Meridian [Meridian Architecture, 2026] provided the first physical energy routing architecture. The Energy Internet provides the protocol standard that allows Meridian to scale beyond a single mesh to a global infrastructure.

The ecosystem is coherent: each component was designed with the next in mind, and the chain of components provides a complete path from programming language to global energy network. No other ecosystem has this property.

11. Limitations and Honest Boundaries

I state the following limitations without qualification.

The Energy Internet does not exist. This is a theoretical framework paper. No inter-mesh EBGp peering session has been established. No EDNS resolver has been deployed. No EaaS subscription has been activated. The Phase 1 single-mesh experimental data that is the foundation of this entire framework does not yet exist.

Efficiency at scale is an open problem. The end-to-end efficiency arguments in Section 8.2 are directionally correct but not experimentally validated. The Energy Internet is only economically viable if efficiency improves substantially from current estimates. The improvement path is identified and plausible; it is not guaranteed.

The regulatory landscape is undefined. No regulatory framework for multi-operator wireless energy routing networks exists in any jurisdiction. The regulatory pathway described in Section 8.3 is a proposal, not a plan with committed timelines. Regulatory development is inherently unpredictable and may take longer or shorter than estimated.

The governance proposal is aspirational. The EIRA does not exist. The ERFC process does not exist. Whether the wireless power industry and regulatory agencies will adopt an open standard for energy routing —” rather than allowing a proprietary standard to win through market dominance —” is an open question. The data internet took twenty years from ARPANET to the World Wide Web. The Energy Internet's governance development timeline is uncertain.

- Energy routing latency is not comparable to data routing. A data packet traverses a five-hop internet path in tens of milliseconds. An energy burst traversing a five-hop Energy Internet path takes approximately 300—“500 ms at current TDMA burst-mode timing. The Energy Internet is not a real-time energy delivery system —” it is a scheduled, burst-mode delivery system. Applications that require continuous, uninterrupted power at high power levels are not served by the Energy Internet at current development scale. Battery systems, wired connections, and conventional grid power remain appropriate for those applications.

The EIP address space may need revision. The 128-bit EIP address specification in Section 4.1 is a first draft. As the ERFC process develops and operational experience accumulates, addressing requirements may be revised. The address space should be treated as provisional until standardized through the ERFC process.

This paper does not resolve the economics. Energy credit settlement, ESP business models, EXP interconnection economics, and the pricing of EaaS subscriptions are not addressed in this paper. A viable Energy Internet requires not just technical standards but a functional economic model. That model is outside the scope of this technical specification paper.

12. Conclusion

The internet began as a protocol. Four nodes in 1969. A correct specification for how independently administered networks exchange packets. The infrastructure followed. Fifty years later, eight billion devices participate in a global network that was not planned, not centrally administered, and not owned by any single entity. The protocol is the internet.

I have proposed that the same transformation is possible for energy delivery. The Energy Internet Protocol Stack —” five layers from physical transmission to application interface, with addressing, routing, delivery guarantees, security, and inter-mesh federation —” is the specification from which a global

wireless energy routing infrastructure can be built. The specification is complete. The implementation has not yet begun at scale.

The foundational implementation target exists. Meridian [Meridian Architecture, 2026] is a formally specified, four-layer deterministic wireless energy routing architecture that implements the EPL, ELL, and intra-EAS ENL of the Energy Internet Protocol Stack. Lume-X, validated at 73 Hz, is the control plane that governs the mesh. Guardian Security is the security layer that authenticates nodes and protects routing. DAEH is the ambient harvesting layer that reduces infrastructure dependency. The room-scale architecture is complete and awaits Phase 1 experimental validation.

The remaining distance from Phase 1 validation to a functioning Energy Internet is measured in years, not decades —” if the protocol is defined now. The data internet's history demonstrates that a correct protocol, published and available, attracts implementation faster than any planned deployment schedule. The ARPANET did not wait for all nodes to be ready before publishing IP. It published the protocol and let the network grow.

The Energy Internet Protocol Stack is published here. The network will grow.

I make one final claim, stated precisely: if Phase 1 experimental data confirms the Meridian architecture's determinism standard, and if the EIPS is adopted as an open standard through a consensus governance process, then the Energy Internet is not a speculative vision —” it is an engineering project with a defined specification, a validated foundational layer, and a clear path through a finite number of development phases. It is, in the language of engineering, achievable.

The protocol is written. The work begins.

Appendix A —” Energy Internet Protocol Stack: Formal Layer Definitions

Layer	Name	Abbreviation	Primary Protocol	Analogous Data Internet Layer
5	Energy Application Layer	EAL	EaaS	Application Layer (HTTP, SMTP)
4	Energy Transport Layer	ETL	ETCP	Transport Layer (TCP)
3	Energy Network Layer	ENL	EIP + EBGp	Network Layer (IP + BGP)
2	Energy Link Layer	ELL	EBF + TDMA	Link Layer (Ethernet)
1	Energy Physical Layer	EPL	DWER/MTL	Physical Layer (fiber, radio)

EPL Service Interface (EPL †’ ELL):

transmit_burst(dst_node, power_mW, duration_ms, timing_window) † ' delivery_confirm
 locate_node(node_id) † ' position_vector, accuracy_cm, timestamp get_safety_status() † '
 gate_conditions[5], bsl_status ELL Service Interface (ELL †' ENL):

send_ebf(ebf_header, path_id) †' ebf_confirmation get_link_quality(next_hop_id) †' loss_pct, sigma_sq,
 latency_ms get_tdma_slot(node_id) †' t_start, t_end ENL Service Interface (ENL †' ETL):

route_burst(src_eip, dst_eip, P_mW, priority) †' path_id, estimated_loss, latency resolve_edns(name) †'
 eip_address get_as_path(dst_eip) †' as_sequence, gateway_eip ETL Service Interface (ETL †' EAL):

open_session(session_spec) † ' session_id, confirmed_P_min, confirmed_sigma_sq
 get_session_stats(session_id) † ' bursts_delivered, avg_P_received, current_sigma_sq
 close_session(session_id) †' final_accounting

Appendix B —” Energy Address Specification

EIP Address Structure (128 bits): Bits 127—“112: Global Registry ID (16 bits) - Assigned by EIRA to major geographic/governance regions - 0x0000: Reserved - 0x0001: Americas - 0x0002: Europe / Africa - 0x0003: Asia / Pacific - 0x0004—“0xFFFE: Future allocation - 0xFFFF: Multicast / Broadcast Bits 111 —“88: EAS Number (24 bits) - Assigned by EIRA to individual Energy Autonomous System operators - 0x000000: Reserved (loopback / test) - 0x000001—“0xFFFFFE: Operator-assigned EAS Numbers - 0xFFFFFFF: Reserved Bits 87—“72: Zone ID (16 bits) - Assigned by EAS operator to logical zones within the EAS - 0x0000: Default zone (single-zone EAS) Bits 71—“64: Node Type (8 bits) - 0x00: Reserved - 0x01: Source node (wired power input) - 0x02: Relay node (wireless relay, no wired input) - 0x03: Destination node (consuming device) - 0x04: Harvesting relay node (DAEH + relay capability) - 0x05: Gateway node (inter-EAS border router) - 0x06—“0xFE: Reserved for future node types - 0xFF: Any / wildcard (routing use only) Bits 63—“24: Node Serial (40 bits) - Assigned by node manufacturer - Unique within manufacturer namespace - Combined with EAS Number provides global uniqueness Bits 23—“0: Checksum (24 bits) - CRC-24 of bits 127—“24 - Detects address corruption in transit

EIP Address Notation: GRRR:AAAA:BBBB:TTSS:SSSS:SSCC:CCCC (hex groups, similar to IPv6)

Example: 0001:0000:0042:0005:0200:0000:0001:XXXX (Americas, EAS 42, Zone 5, Node Type 0x02=relay, Serial 1, checksum XXXX)

Appendix C —” Energy Border Gateway Protocol (EBGP) Overview

EBGP operates between gateway nodes at EAS boundaries. It is a path-vector protocol: each route advertisement carries the complete sequence of EAS numbers traversed.

Session establishment:

- 1. Gateway node A initiates TCP-equivalent control session with Gateway node B**
- 2. Both gateways authenticate using EAS-level PKI (Guardian Security Level 2)**
- 3. ERPKI validation confirms each gateway is authorized to represent its EAS**
- 4. EBGp OPEN message exchanged: EAS Number, EIPS version, supported capabilities**
- 5. EBGp KEEPALIVE messages maintain session (interval: 30 seconds default)**
- 6. EBGp UPDATE messages exchange route advertisements**

Route advertisement format (key fields):

EBGP UPDATE: ORIGIN : IGP / EGP / INCOMPLETE (how route was learned) AS_PATH : [EAS_A, EAS_B, EAS_C] (path traversed)

NEXT_HOP	: EIP address of advertising gateway
P_AVAILABLE	: available delivery capacity (mW)

EFFICIENCY : estimated end-to-end efficiency (%) SAFETY_FLAGS : BSL_CERTIFIED, GUARDIAN_CERTIFIED, CRITICAL_CAPABLE NLRI : Network Layer Reachability Info (EIP prefixes being advertised)

Route selection (in priority order):

ERPKI validation passes (reject invalid origins) Not in AS_PATH loop (reject routes that would create routing loops) Highest LOCAL_PREF (operator policy) Shortest AS_PATH Highest P_AVAILABLE Highest EFFICIENCY Lowest NEXT_HOP EIP address (deterministic tiebreak)

Appendix D —” Standardization Roadmap

ERFC	Title	Status	Dependencies
ERFC-0001	Energy Internet Protocol Stack	Draft (this paper)	None
ERFC-0002	Energy Internet Protocol (EIP)	Not started	ERFC-0001
ERFC-0003	Energy Transmission Control Protocol (ETCP)	Not started	ERFC-0001, ERFC-0002
ERFC-0004	Energy Border Gateway Protocol (EBGP)	Not started	ERFC-0001, ERFC-0002
ERFC-0005	Energy Domain Name System (EDNS)	Not started	ERFC-0002
ERFC-0006	Energy Resource PKI (ERPKI)	Not started	ERFC-0002, ERFC-0004
ERFC-0007	Energy QoS and Priority Classes	Not started	ERFC-0003
ERFC-0008	Guardian Security for the Energy Internet	Not started	ERFC-0001, ERFC-0006
ERFC-0009	EaaS —” Energy as a Service	Not started	ERFC-0003, ERFC-0007
ERFC-0010	Emergency Priority Override (EPO)	Not started	ERFC-0007, ERFC-0008

Governance milestones:

M1: EIRA founding body established (Year 1, concurrent with Phase 1 experiments) M2: ERFC-0001 through ERFC-0004 published as draft standards (Year 2) M3: First inter-EAS EBGP peering session validated (Year 3, concurrent with Phase 2) M4: ERFC-0001 through ERFC-0008 ratified as formal standards (Year 4) M5: First EXP operational (Year 5, concurrent with Phase 3) M6: First EIRA EAS Number assignments (Year 2, for Phase 2 operators)

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Chapter Bridge: From Energy to Energy-and-Data

Chapter Four established the Energy Internet as the global-scale extension of the Meridian protocol. At internet scale, independently administered energy meshes exchange energy across EBGp-federated boundaries, devices address energy endpoints by name using EDNS, and the governance framework of the EIRA manages the global address space.

But the meshes that carry this energy are already carrying something else. Every Meridian mesh, operating normally, generates approximately eleven kilobytes per second of structured, addressed, authenticated data per node —” routing table updates, delivery confirmations, security authentication messages, control-loop telemetry. This data is currently classified as control-plane overhead. It is not overhead. It is the first payload of a data communication channel that already exists in the Meridian infrastructure.

Chapter Five formalizes this observation and extends it to its logical conclusion: the Unified Energy-Data Mesh, a single infrastructure that routes both energy and data under a shared addressing framework, security framework, and governance framework. One installation. Two resource types. No separate

communication network required.

Chapter Five

When Power and Data Share the Same Fabric

Every autonomous device in the physical world needs two things: power and communication.

The history of wireless technology treats these as separate problems, to be solved by separate industries producing separate standards using separate infrastructure. The wireless communication industry produced Wi-Fi, Bluetooth, Zigbee, LoRa, and 5G. The wireless power industry produced Qi, resonant coupling, and directed RF. A building with Wi-Fi coverage has wireless communication everywhere but devices still need batteries or wires for power. A building with wireless charging capability has power delivery at specific locations but no data communication on that infrastructure.

The separation is not fundamental. It is historical. An RF signal carries energy and information simultaneously —” this is the physics of SWIPT (Simultaneous Wireless Information and Power Transfer), demonstrated in research laboratories since the early 2000s. The question has never been whether power and communication can coexist on the same RF channel. They already do. The question is whether the same deterministic routing infrastructure can govern both simultaneously.

The answer is yes. And the Meridian mesh already proves it, because it already routes data —” the eleven kilobytes per second of control-plane traffic that governs the energy routing —” through the same node infrastructure as the energy. The Unified Energy-Data Mesh is not a new design. It is the formal recognition and extension of what Meridian already does.

This chapter specifies that extension: the UEDM protocol stack, the co-routing arbitration rules that govern how energy bursts and data packets share TDMA capacity, the unified addressing scheme that uses the same 128-bit address for both energy delivery and data communication, and the self-sustaining device model —” the autonomous sensor or actuator that powers itself and communicates through the same wireless fabric, requiring no battery and no separate communication radio.

The self-sustaining device is the real product outcome of the UEDM. It is the answer to the infrastructure question that defines every large-scale IoT deployment: how do you power and connect thousands of devices without wiring and without battery replacement logistics? The UEDM answers both questions at once.

Abstract

The Meridian mesh already carries two kinds of traffic. The first kind is the energy it was designed for: RF bursts routed hop-by-hop from source nodes to destination devices, governed by DRMA routing tables and MFE burst scheduling. The second kind is the control-plane traffic it requires to operate: LINK_STATE_UPDATE messages, NODE_ANNOUNCE broadcasts, EBF delivery confirmations, Guardian Security authentication exchanges, and Lume-X cross-layer monitoring packets. These control-plane messages are data — structured, addressed, routed information — flowing across the same physical nodes as the energy they govern.

The Meridian mesh is already a unified energy-data fabric. It just has not been formally recognized as one.

This paper formalizes the co-routing architecture of the Meridian mesh — the simultaneous routing of energy and data through the same physical node infrastructure under a shared addressing and governance framework. I define the Unified Energy-Data Mesh (UEDM) as a formal architecture in which the same addressed nodes, the same topology-aware routing fabric, and the same Trust Layer identity substrate serve both energy delivery and data communication simultaneously. I show that the UEDM is not a new design — it is a formal recognition and extension of what Meridian already does.

I define the co-routing arbitration rules that govern how energy bursts and data packets share TDMA capacity without conflict. I define the UEDM addressing scheme that unifies energy delivery addresses (EIP) and data communication addresses under the same 128-bit namespace. I show that the UEDM collapses two infrastructure problems — where does this device get its power, and how does this device communicate — into one, simplifying deployment, reducing infrastructure cost, and enabling a class of self-sustaining autonomous devices that power themselves and communicate through the same wireless fabric.

I propose the Unified Energy-Data Mesh as the convergence point of the Energy Internet [Energy Internet, 2026] and the Trust Layer communication infrastructure [16]: a single fabric that routes both resources and information under a unified governance framework, providing the complete infrastructure substrate for autonomous physical systems operating in the Deterministic Infrastructure paradigm [DI Theory, 2026].

Keywords: unified energy-data mesh, co-routing, wireless power and communication, SWIPT, Meridian, deterministic infrastructure, TDMA co-scheduling, ambient IoT, self-sustaining networks, simultaneous wireless information and power transfer

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References

1. Introduction

1.1 Two Problems That Have Always Been One

Every autonomous device in the physical world needs two things: power and communication. Power to operate its sensors, processors, actuators, and radio. Communication to report its sensor readings, receive instructions, and coordinate with other devices.

The history of wireless technology treats these as separate problems. The wireless communication industry spent sixty years building Wi-Fi, Bluetooth, Zigbee, LoRa, 5G, and a dozen other standards for wireless data transfer. The wireless power industry spent the same sixty years building inductive chargers, RF energy harvesters, and directed beam systems. The two industries produced incompatible infrastructures: a building with Wi-Fi coverage has wireless communication everywhere, but devices still need batteries or wires for power. A building with wireless charging capability has power delivery at specific locations, but that infrastructure carries no information.

The separation is not fundamental. It is historical. Power and communication are both electromagnetic phenomena. An RF signal carries energy (which can be harvested) and information (which can be demodulated) simultaneously —” this is the basis of SWIPT (Simultaneous Wireless Information and Power Transfer), an active research area since the 2000s [58, 59]. The question has never been whether power and communication can coexist on the same RF channel. They already do. The question is whether they can be co-routed —” whether the same deterministic routing infrastructure can govern both simultaneously.

This paper answers that question.

1.2 What Meridian Already Does

The Meridian mesh, as specified in [Meridian Architecture, 2026], routes energy through a distributed network of addressed nodes. To do this, it necessarily moves a substantial volume of data through those same nodes: routing table updates, delivery confirmations, localization signals, security authentication messages, and control-loop telemetry. This data is currently classified as "control-plane traffic" —"necessary overhead that enables energy routing.

The argument of this paper begins with a reframing: the control-plane traffic in the Meridian mesh is not overhead. It is the first payload of the data communication channel that exists latent in the Meridian infrastructure. The mesh that routes energy also already routes data. It does so on a limited basis (control messages only) using a limited channel (the control-plane bandwidth reserved in each TDMA slot). Extending this to general data communication requires:

Defining the co-routing arbitration rules that govern how energy and data share TDMA capacity
 Extending the addressing scheme to cover both energy delivery endpoints and data communication endpoints under one namespace
 Specifying the QoS rules that govern priority relationships between energy delivery and data communication
 Defining the security architecture for data communication (which already exists through Guardian Security [Guardian Security, 2026])
 None of these extensions require new hardware. They require new protocol definitions on top of existing hardware. The Unified Energy-Data Mesh is a protocol and software extension of the existing Meridian architecture, not a new physical system.

1.3 The Self-Sustaining Device

The most significant practical implication of the UEDM is the self-sustaining device: an autonomous sensor, actuator, or monitor that receives both its power and its communication through the same UEDM mesh, requiring no battery, no separate communication radio, and no wired connection to either a power source or a data network.

Self-sustaining devices change the economics of autonomous device deployment. The dominant cost of large-scale IoT deployments is not the sensors —"it is the power and communication infrastructure that keeps them running. Battery replacement at scale is expensive in labor and logistics. Wired power at scale is expensive in installation and inflexible in deployment. Separate wireless power and wireless communication systems are expensive to maintain independently and multiply the infrastructure required for a single deployment.

A self-sustaining device powered and connected through the same UEDM mesh eliminates all three cost categories simultaneously. The UEDM mesh is the infrastructure —” a single installation provides both power and connectivity to every device in its coverage area.

1.4 Contributions

I make four contributions:

I formally recognize the Meridian mesh as an existing dual-traffic fabric and define the Unified Energy-Data Mesh (UEDM) as its formal extension to general data communication.

I define the co-routing arbitration rules —” the protocol specification for how energy bursts and data packets share TDMA capacity under a unified scheduling framework.

I define the UEDM addressing scheme —” a unified 128-bit namespace in which energy delivery addresses and data communication addresses are the same address, eliminating the need for separate network identity management for power and communication.

I characterize the self-sustaining device as a deployment model and define its operational profile, power-communication budget, and interaction with the UEDM mesh.

1.5 Paper Organization

Section 2 examines the history of converging communication and power infrastructure. Section 3 formally characterizes the Meridian mesh as an existing dual-traffic fabric. Section 4 defines the UEDM architecture. Section 5 specifies co-routing arbitration. Section 6 defines UEDM addressing. Section 7 characterizes the self-sustaining device model. Section 8 positions the UEDM within the Energy Internet framework. Section 9 describes the UEDM control plane. Section 10 discusses applications. Section 11 situates this in related literature. Section 12 states limitations. Section 13 concludes.

2. Background: Convergent Infrastructure

2.1 Power Line Communication: The Inversion Precedent

Power Line Communication (PLC) [53] uses existing electrical power wiring as a communication medium: high-frequency data signals are superimposed on the 50/60 Hz power signal and demodulated at the receiver. PLC demonstrated that a medium designed to carry one resource —” electrical power —” can simultaneously carry another —” digital data. The physical medium, the copper wire, is agnostic to the signal type. Protocol design determines what it carries.

PLC inverts the UEDM concept: it uses a power medium to carry data. The UEDM uses a data networking protocol to carry power. The inversion is symmetrical and instructive —” both demonstrate that power and data are not inherently separate resources requiring separate infrastructure.

PLC achieved significant deployment in home automation (HomePlug standard), smart grid metering (PRIME, G3-PLC), and broadband internet access (G.hn). Its limitations are the limitations of the power line medium: noise from electrical equipment, signal attenuation in long runs, and signal coupling across transformers. The UEDM does not share these limitations because it operates over RF rather than conducted power lines.

2.2 SWIPT: Simultaneous Wireless Information and Power Transfer

SWIPT research [58, 59] has demonstrated, theoretically and experimentally, that a single RF signal can simultaneously transfer energy to a harvesting receiver and information to a communication receiver. The fundamental tradeoff is between harvested power and communication data rate: energy harvesting requires a strong received signal, which limits modulation order; high data rate requires high modulation order, which reduces the fraction of signal power extractable for harvesting.

SWIPT systems manage this tradeoff through receiver design (power splitting: fraction of received power goes to rectenna, fraction to demodulator) and transmitter design (time switching: transmitter alternates between high-power energy bursts and modulated data signals). The fundamental insight —” one RF signal, two resource types —” is the physical basis for the UEDM.

The UEDM extends SWIPT in two critical dimensions. SWIPT research has focused on single-link point-to-point systems, or at most small-cell broadcast scenarios. The UEDM is a multi-hop, topology-aware, deterministically routed network in which both energy and data are addressed, routed, and delivered with guarantees. SWIPT provides the physical foundation; the UEDM provides the network architecture.

2.3 Integrated Access and Backhaul

The 5G integrated access and backhaul (IAB) architecture [57] uses the same wireless nodes to simultaneously provide user-facing access (communication to devices) and network backhaul (communication between base stations). A single node participates in both functions through time-division multiplexing —” some TDMA slots serve as access, others as backhaul.

The UEDM applies the same architectural principle in the energy domain: the same node participates in both energy delivery (energy-focused TDMA slots) and data communication (data-focused TDMA slots) under a unified scheduling framework. The IAB precedent demonstrates that time-division sharing of a single wireless node's capacity across multiple traffic types is technically mature and deployable at scale.

2.4 The Convergence Attractor

These three prior technologies —” PLC, SWIPT, and IAB —” each demonstrate that power and communication infrastructure converge when physical medium constraints are examined without historical separation. PLC converged power line and communication in the wired domain. SWIPT is

converging them at the physical layer in the wireless domain. IAB is converging access and backhaul at the network layer.

The UEDM is the convergence at the routing layer —” the formal definition of a unified protocol stack in which both energy and data are addressed, routed, quality-of-service managed, and security-governed through a single deterministic framework. It is the natural terminus of the convergence trajectory.

3. The Meridian Mesh as Existing Dual-Traffic Fabric

3.1 Cataloguing the Data Already in the Mesh

To demonstrate that the Meridian mesh is already a dual-traffic fabric, I enumerate the data traffic that currently flows through it in normal operation:

Topology management traffic:

NODE_ANNOUNCE —” broadcast by each node at join and at configurable intervals; carries node identity, capabilities, and position. Variable length, typically 128—“256 bytes. LINK_STATE_UPDATE —” propagated through the mesh when link quality changes or topology changes; carries link metrics and validity flags. Variable length, typically 64—“128 bytes. ROUTE_TABLE_SYNC —” periodic synchronization of routing tables between neighbors; carries routing table deltas. Variable length.

Delivery management traffic:

EBF_HEADER —” the Energy Burst Frame header transmitted before each energy burst; carries source, destination, routing path, timing, and safety flags. Fixed 64 bytes. EBF_CONFIRMATION —” returned by the receiving node after each energy burst; carries received power, spatial accuracy, timestamp, and status. Fixed 32 bytes. Security traffic:

GUARDIAN_AUTH_REQUEST —” pre-transmission authentication exchange; four messages (Steps 1 —“4 of the Guardian Security protocol). Approximately 512 bytes total. GUARDIAN_ANOMALY_REPORT —” generated by any node detecting an anomaly; carries anomaly classification, evidence hash, and timestamp. Variable length. CERTIFICATE_CHAIN_ENTRY —” generated by Lume-X for every security event; carries signed event record. Typically 256—“512 bytes.

Control plane traffic:

LUME_X_TELEMETRY —” generated by the Lume-X control loop at 73 Hz on each node; carries cross-layer state summary for monitoring. Fixed 128 bytes per cycle. SHDCL_RECOVERY_EVENT —” generated by SHDCL when an anomaly is detected and recovery initiated; carries event classification and recovery action. Variable length. Total control-plane data volume estimate (per node, per second):

Traffic Type	Rate	Size	Bytes/sec
LUME_X_TELEMETRY	73/sec	128 B	9,344 B/s
EBF_HEADER + CONFIRMATION	~10/sec	96 B	960 B/s
NODE_ANNOUNCE	0.1/sec	192 B	19 B/s
LINK_STATE_UPDATE	~1/sec	96 B	96 B/s
GUARDIAN_AUTH	~1/sec	512 B	512 B/s
Other	—”	—”	~200 B/s
Total			~11.1 KB/s

Eleven kilobytes per second of structured, addressed, authenticated data already flows through each Meridian node. This is not a negligible communication channel —” it is approximately the throughput of a 2G cellular data connection, adequate for sensor telemetry, status reporting, and low-rate command-and-control for a wide range of IoT applications.

The Meridian mesh is not becoming a communication network. It already is one. The UEDM formalizes and extends this existing capability.

3.2 Why the Control-Plane Channel Is a Communication Channel

The control-plane traffic catalogued above satisfies every requirement of a data communication channel:

Addressing: Every control-plane message carries source and destination node identities (MC 64-bit IDs, extensible to EIP 128-bit addresses under the UEDM) Routing: LINK_STATE_UPDATE and ROUTE_TABLE_SYNC propagate through the same topology-aware DRMA routing fabric as energy bursts Authentication: All control-plane messages are signed by the originating node's Guardian Security credential Reliability: EBF_CONFIRMATION provides delivery acknowledgment; SHDCL provides retransmission for critical messages Timing: TDMA scheduling provides deterministic timing for all message delivery The only dimension in which control-plane traffic differs from general-purpose data communication is content restriction: currently, only system-defined message types are carried. The UEDM removes this restriction, allowing application-defined data payloads to be carried in the same channel alongside system control messages.

4. The Unified Energy-Data Mesh Architecture

4.1 Architectural Principle

The UEDM is built on one architectural principle: a joule and a bit are both payloads; the difference is their physical carrier, not their addressing, routing, or governance.

In the UEDM, every node has one address. That address is used for both energy delivery (EIP format) and data communication (same EIP format, data payload). The routing fabric is the same —” DRMA routes both energy bursts and data packets through the same topology-aware mesh. The scheduling framework is the same —” TDMA divides time between energy slots and data slots under a unified co-routing arbitration policy. The security framework is the same —” Guardian Security authenticates both energy delivery sessions and data communication sessions.

The UEDM adds one new concept: the Resource-Type Field (RTF) in the node addressing header. The RTF indicates whether a given TDMA slot is carrying an energy payload (RTF = ENERGY) or a data payload (RTF = DATA). All other header fields —” source address, destination address, routing path, timing window, safety flags —” are shared between both payload types.

UEDM Frame Header (unified for energy and data): src_node_id : 128 bits (UEDM EIP address)
 dst_node_id : 128 bits (UEDM EIP address) resource_type : 4 bits (ENERGY=0x1, DATA=0x2, MIXED=0x3, CONTROL=0x4) priority_class : 4 bits (CRITICAL, HIGH, NORMAL, BACKGROUND)
 seq_num : 32 bits path_id : 32 bits timing_window : 128 bits (start + end, μ s precision) safety_flags : 16 bits (BSL status, relevant for ENERGY type only) payload_length : 16 bits checksum : 32 bits [payload] : variable (RF burst parameters if ENERGY, data bytes if DATA)

The UEDM frame header is a superset of the existing EBF header, with the addition of the resource_type and payload_length fields and the generalization of the address fields to 128-bit UEDM EIP format.

4.2 UEDM Stack

The UEDM protocol stack is a natural extension of both the Meridian architecture stack and the Energy Internet Protocol Stack [Energy Internet, 2026]:

```

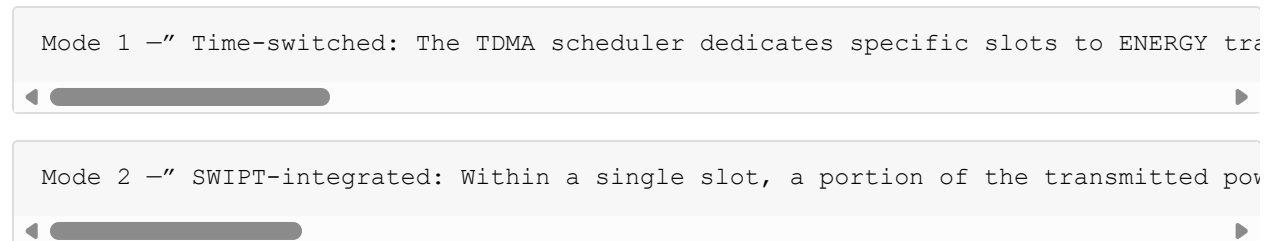
" UAL —” Unified Application Layer
" (Energy application: EaaS; Data application: any) "
" UTL —” Unified Transport Layer "
(ETCP for energy; DTCP for data; shared session mgmt) "
" UNL —” Unified Network Layer "
(EIP/EBGP for routing; shared 128-bit address space) "
" ULL —” Unified Link Layer "
(UEDM frame; co-routing arbitration; TDMA scheduling) "
" UPL —” Unified Physical Layer "
(RF: energy beam for ENERGY type, OOK/FSK for DATA) "

```

Each layer serves both resource types with a unified interface. The only layer where energy and data diverge is the UPL —” the physical carrier is different (a focused beam at high power for energy; modulated signal at low power for data), though both operate on the same RF hardware through time-division switching.

4.3 Physical Layer Duality

The Meridian MTL hardware is a phased-array RF system. The same phased array that forms a focused beam for energy delivery can be modulated to carry information in the same transmission window. The physical layer duality operates in two modes:



The UEDM specifies Mode 1 as the baseline —” it requires no hardware modifications to existing Meridian nodes, only software changes to the TDMA scheduler and ULL. Mode 2 is an optional upgrade for nodes equipped with SWIPT-capable receivers, providing higher combined throughput.

5. Co-Routing Arbitration

5.1 The Arbitration Problem

The UEDM shares a fixed TDMA frame capacity between energy delivery and data communication. The total frame capacity is fixed by the physical channel parameters (bandwidth, frequency, modulation). The arbitration problem is: how should this capacity be divided between energy and data under varying load conditions?

The arbitration is constrained by two requirements that can conflict:

Energy delivery requirement: CRITICAL and HIGH priority energy sessions have guaranteed capacity —” these sessions cannot be preempted or delayed regardless of data traffic load. Energy delivery is the primary function of the mesh; data communication is a secondary function.

Data communication requirement: Some data traffic (Guardian Security authentication, SHDCL recovery events, emergency alerts) must be delivered within bounded latency regardless of energy traffic load. This data is essential to the correct operation of the energy delivery system itself.

The resolution: energy delivery and essential control-plane data share a guaranteed capacity pool. General-purpose application data uses the residual capacity.

5.2 Capacity Allocation Framework

The UEDM TDMA frame is divided into three capacity pools:

Pool A —” Guaranteed Energy Capacity (GEC): Reserved exclusively for CRITICAL and HIGH priority energy sessions. Size: dynamically adjusted based on the sum of active CRITICAL and HIGH session P_min requirements. Minimum: 60% of total frame capacity. Maximum: 90% of total frame capacity (the remaining 10% is reserved for Pool C regardless of energy demand).

Pool B —” Guaranteed Control Data Capacity (GCDC): Reserved for essential control-plane data: Guardian Security authentication messages, SHDCL recovery events, NODE_ANNOUNCE, LINK_STATE_UPDATE, and LUME_X_TELEMETRY. Size: fixed at the maximum observed control-plane data rate plus 50% headroom. Typical: 5—“10% of total frame capacity.

Pool C —” Flexible Shared Capacity (FSC): The remaining capacity, shared between NORMAL and BACKGROUND energy sessions and all general-purpose application data. Arbitration within FSC is governed by the co-routing arbitration policy described in Section 5.3.

Total Frame Capacity = GEC + GCDC + FSC Minimum allocation: $GEC \geq 60\%$, $GCDC = 5\text{—}10\%$, $FSC = \text{remainder}$

This allocation ensures that energy delivery to CRITICAL devices and control-plane operation are never compromised by data communication traffic, while providing meaningful capacity for application data in the FSC pool.

5.3 Flexible Shared Capacity Arbitration Policy

Within the FSC pool, energy sessions (NORMAL and BACKGROUND) and data communication sessions compete for capacity. The arbitration policy:

Rule 1 —” Energy priority within FSC: NORMAL energy sessions take priority over all data communication sessions in FSC. A NORMAL energy session that cannot be satisfied from FSC may preempt BACKGROUND and NORMAL data sessions. It may not preempt HIGH data (which is served from GCDC).

Rule 2 —” Application data priority classes: Data communication sessions in FSC are classified using the same four-level QoS system as energy sessions: CRITICAL (served from GCDC), HIGH (served from GCDC), NORMAL (served from FSC with priority over BACKGROUND), BACKGROUND (served from FSC residual).

Rule 3 —” Burst accommodation: A large data payload (firmware update, map download, batch sensor data) that does not fit in a single FSC allocation is fragmented across multiple FSC slots, with each fragment treated as a separate NORMAL data session for arbitration purposes. The first fragment has

priority over subsequent fragments of the same payload.

Rule 4 —” BACKGROUND energy yields to NORMAL data: BACKGROUND energy sessions have the lowest priority in FSC. A BACKGROUND energy session that cannot be immediately satisfied waits for FSC availability rather than preempting any data session. This prevents battery-free devices running on BACKGROUND energy from starving application data sessions.

Rule 5 —” Combined throughput optimization: The co-routing scheduler actively seeks slots where the beam path for an energy delivery session overlaps with the communication path for a data session —” enabling Mode 2 (SWIPT-integrated) transmission that serves both sessions simultaneously within one slot. This optimization is opportunistic, not guaranteed.

6. UEDM Addressing

6.1 Unified Address Space

The UEDM uses the EIP 128-bit address structure defined in [Energy Internet, 2026 4.1] as the universal address for both energy delivery and data communication. There is no separate data networking address —” a device that participates in the UEDM has one address that identifies it for all resource types.

UEDM Address (128 bits, EIP format):

```
[Global Registry ID : 16 bits]
[EAS Number       : 24 bits]
[Zone ID          : 16 bits]
```

[Node Type : 8 bits] —” extended to include DATA_ONLY (0x06) node type

```
[Node Serial      : 40 bits]
[Checksum         : 24 bits]
```

The Node Type byte is extended by one value: 0x06 = DATA_ONLY. A DATA_ONLY node participates in data communication but cannot serve as an energy relay or source. This type accommodates devices with communication capability but no energy harvesting or transmission hardware —” for example, a battery-powered gateway node that routes data through the UEDM mesh to the wider internet.

6.2 Service Endpoint Addressing

Within the unified address space, specific services running on a UEDM node are addressed using a service port analogous to TCP/UDP ports:

UEDM Service Endpoint:

```
[UEDM Address : 128 bits]
[Service Port : 16 bits]
```

Reserved service ports: 0x0001 : Energy delivery (EaaS) 0x0002 : Guardian Security authentication 0x0003 : Lume-X telemetry 0x0004 : SHDCL recovery messaging 0x0005 : EDNS query/response 0x0006 : EBGp peering 0x0100—“0xFFFF : Application-defined services

Application-defined services —” sensor data reporting, actuator command delivery, firmware update distribution, log aggregation —” use ports in the 0x0100—“0xFFFF range. A sensor that reports temperature readings to a data aggregation service at address A port 0x0200 uses the same UEDM routing infrastructure as the energy delivery to that sensor, with the same addressing, the same security, and the same delivery accounting.

6.3 Multicast and Anycast in the UEDM

The UEDM supports two additional address types beyond unicast:

UEDM Multicast: A multicast address identifies a group of nodes. A single message addressed to a multicast group is delivered to all members of the group via efficient in-network replication. Useful for: firmware updates distributed to all nodes of a given type, emergency alerts to all nodes in a zone, routing table updates to all nodes in a zone.

Multicast is DATA-only —” energy cannot be delivered to a multicast address, since energy delivery requires per-recipient power level and position validation that multicast cannot provide.

UEDM Anycast: An anycast address identifies the nearest (lowest-latency, highest-quality) member of a group. A message addressed to an anycast group is delivered to the single best-positioned member. Useful for: energy delivery requests (route to nearest available source node), EDNS resolution (query the nearest EDNS resolver), service discovery (find the nearest instance of a service).

Anycast is supported for both energy and data: an energy request addressed to an anycast source node address is routed to the nearest available source node with sufficient charge state. This is the natural implementation of mesh-wide energy balancing —” a requesting node does not need to know which specific source node is nearest; it sends to the anycast source-node address and the UEDM routes to the optimal source.

7. The Self-Sustaining Device Model

7.1 Definition

A self-sustaining device (SSD) is a UEDM participant that:

Receives all of its operating power from the UEDM (no battery, no wired power) Conducts all of its data communication through the UEDM (no separate wireless radio) Has no external infrastructure dependency beyond UEDM mesh coverage An SSD is the device-level realization of the UEDM value proposition: a single wireless mesh deployment that provides both the power and the connectivity for all devices in its coverage area.

7.2 SSD Power-Communication Budget

An SSD operates under a power-communication budget: the energy it receives from the UEDM must be sufficient to power its computation, its sensing, and its UEDM communication —” all from a single energy allocation.

A typical SSD budget:

SSD Operating Budget (10 mW received power example): Microcontroller (Cortex-M4 at 4 MHz) : 1.2 mW Sensor (temperature + humidity, 1 Hz) : 0.8 mW RF transceiver (receive mode) : 2.0 mW RF transceiver (transmit mode, 1% duty) : 0.5 mW avg Energy management circuit : 0.3 mW Margin for supercapacitor charging : 5.2 mW ----- Total : 10.0 mW

At this budget, the SSD can operate continuously at 10 mW received power while maintaining a 52% supercapacitor charging margin —” sufficient to sustain operation through brief periods of UEDM unavailability (mesh topology changes, TDMA schedule gaps).

Communication capacity at this budget: the RF transceiver in receive mode is always on (monitoring for incoming packets); in transmit mode at 1% duty cycle, with 250 kbps PHY rate, the SSD can transmit approximately 2.5 kbits per second of application data —” sufficient for sensor telemetry, status reporting, and firmware updates at low rate.

7.3 SSD Lifecycle

An SSD has three operational states:

BOOTSTRAPPING: The SSD has just been deployed and has not yet established UEDM identity. It broadcasts a `NODE_ANNOUNCE` with a provisioning credential (a temporary credential issued during manufacture). A source node or gateway node receives the announcement and initiates the Guardian Security provisioning protocol. Once the SSD receives its full Trust Layer identity credential, it transitions to **OPERATIONAL**.

OPERATIONAL: The SSD participates fully in the UEDM —” receiving energy, routing control messages, and conducting application data communication. It monitors its supercapacitor charge state and requests supplementation when charge falls below $V_{\min} + \text{margin}$.

CONSERVATION: The SSD's supercapacitor has dropped below $V_{\min}$. All non-essential processes are suspended. The SSD maintains only the energy monitoring circuit and waits for UEDM energy delivery to restore charge. Once charge is restored above $V_{\min} + \text{hysteresis}$, it transitions back to OPERATIONAL.

7.4 SSD Failure and Recovery

An SSD that drops out of the UEDM (power depletion in CONSERVATION state for longer than the SHDCL detection window) is treated as a failed node by the mesh. Mesh routing tables are updated to mark the SSD's relay function (if any) as unavailable. When the SSD recovers charge and re-announces, it is re-integrated through the same NODE_ANNOUNCE + verification process as initial bootstrapping.

This recovery behavior is identical to wound healing in the Meridian Synthetic Organism model [Meridian Organism, 2026 5.4]: node failure †' detection †' rerouting †' function continues †' node recovery and reintegration. The SSD lifecycle is a concrete implementation of the wound-healing property operating at the device scale.

8. UEDM and the Energy Internet

8.1 The UEDM as the Energy Internet's Access Layer

The Energy Internet [Energy Internet, 2026] defines the protocol stack for energy routing across independently administered meshes. The UEDM is the access layer of the Energy Internet —" the technology that connects end devices (sensors, actuators, monitors) to the Energy Internet infrastructure.

In the data internet model, the access layer (Wi-Fi, Ethernet, 5G) connects end devices to the wider internet. The access layer is not the internet —" it is the last hop that connects devices to the infrastructure that carries their traffic to any destination globally. The Energy Internet needs the same two-tier structure: an internet-scale routing infrastructure (the Energy Internet) and a device-level access layer (the UEDM).

The UEDM is the Energy Internet's access layer because:

End devices register with the UEDM mesh using UEDM addresses (EIP format) The UEDM mesh connects to the Energy Internet through gateway nodes that implement ELL and ENL Energy and data requests from end devices are routed through the UEDM mesh to the gateway, then across the Energy Internet to their destination An SSD that needs energy can request it from a local UEDM source node (intra-mesh delivery) or, through the gateway, from an Energy Internet source node in a remote EAS (inter-mesh delivery via EBGp). The UEDM handles the last-hop access; the Energy Internet handles the inter-mesh routing.

8.2 UEDM Gateway Function

A UEDM gateway node implements the full Energy Internet Protocol Stack [Energy Internet, 2026 3] in addition to the UEDM ULL and UPL. It translates between:

UEDM frame format (ULL) †” Energy Internet ELL frame format UEDM node addresses †” Energy Internet EIP addresses (these are the same 128-bit format; translation is identity) UEDM TDMA scheduling †” Energy Internet link-layer timing The gateway also runs EBGp, advertising the UEDM's EAS prefix to the wider Energy Internet and accepting routes from peer EAS operators. From the Energy Internet's perspective, the UEDM is an EAS —” an independently administered network with a defined address prefix. The gateway is the EAS border router.

8.3 Data at Energy Internet Scale

The UEDM enables a capability that the Energy Internet specification did not include: data communication at internet scale through the energy routing infrastructure. An SSD's sensor data, routed through the UEDM to the gateway and then across the Energy Internet to a remote data aggregation service, is traveling on the same infrastructure as the energy that powers the SSD.

This convergence —” the Energy Internet as both a power grid and a data network —” is not an incidental feature. It is the natural consequence of the UEDM's unified addressing and routing framework. Once energy and data share the same addresses and the same routing fabric, they share the same internet-scale infrastructure automatically.

9. The UEDM Control Plane

9.1 Unified Control Plane Architecture

The UEDM control plane manages both energy routing and data communication simultaneously. It is implemented as an extension of the existing Meridian control plane (MMF + MFE + Lume-X), with additional data-specific control functions:

Energy control functions (existing):

DRMA topology management (routing tables, link-state) MFE burst scheduling (charge state monitoring, burst timing) SHDCL failure detection and recovery Guardian-E energy security (pre-transmission authentication, anomaly detection) Data control functions (new):

DTCP session management (data delivery sessions, retransmission, QoS) Multicast group management (group membership, replication scheduling) Application service registration (port binding, service announcement) Data security integration (Guardian-E session authorization for data sessions) Unified control functions (shared):

Co-routing arbitration (capacity pool management, TDMA scheduling) UEDM address management (EIP address assignment, EDNS registration) Gateway function management (EAS boundary, EBGp state) Certificate chain logging (audit records for both energy and data events)

9.2 Data Transport Control Protocol (DTCP)

DTCP is the data-domain equivalent of ETCP [Energy Internet, 2026 3.6]. Where ETCP provides delivery guarantees for energy sessions ($P_{\min}$, σ^2 , retransmission budget), DTCP provides delivery guarantees for data sessions (packet loss rate, latency bound, retransmission limit).

DTCP Session: source_UEDM : 128-bit UEDM address + 16-bit port destination_UEDM: 128-bit UEDM address + 16-bit port priority_class : CRITICAL / HIGH / NORMAL / BACKGROUND max_latency_ms : maximum acceptable end-to-end delivery latency max_loss_pct : maximum acceptable packet loss rate (%) session_type : STREAM / DATAGRAM / MULTICAST billing_account : energy credit account (data communication also billed)

DTCP retransmission differs from ETCP retransmission in one important way: data retransmission costs data capacity but not energy. A retransmitted data packet does not deplete source node supercapacitors —” the RF transmission power for data packets is orders of magnitude lower than for energy bursts. DTCP can therefore be more aggressive in retransmission than ETCP.

9.3 Billing Integration

In a deployed UEDM, data communication is a service with associated costs, analogous to cellular data billing. The energy credit accounting system [Energy Internet, 2026 5.4] is extended to cover both energy delivery and data communication:

Energy billing: Billed per milliwatt-hour delivered, as defined in the Energy Internet specification.

Data billing: Billed per kilobyte delivered, using the same energy credit denomination. The exchange rate (energy credits per kilobyte) is determined by the EAS operator based on the energy cost of data transmission (RF transmission power — transmission time) plus overhead.

Combined billing: An SSD that receives both energy and data through the UEDM has a single energy credit account that is debited for both services. This simplifies billing compared to separate power and communication subscriptions.

10. Applications and Deployment Scenarios

10.1 Smart Building Infrastructure

A building equipped with UEDM infrastructure requires no per-device wiring for either power or communication. Every device in the building —” occupancy sensors, air quality monitors, HVAC controls, lighting dimmers, door access readers, security cameras, fire detectors —” is an SSD. Installation is physical mounting only; there is no wiring.

The UEDM mesh provides:

Power to every device from UEDM source nodes installed in ceiling panels or wall outlets Data communication from every device to the building management system through the same mesh Identity verification for every device through the Trust Layer, enabling granular access control and audit Self-healing around device or mesh node failures, maintaining coverage without manual intervention The cost comparison against conventional smart building infrastructure (wired power + separate wireless communication) is significant: eliminating two separate infrastructure installations and replacing them with one unified mesh reduces material cost, installation labor, and ongoing maintenance complexity simultaneously.

10.2 Industrial IoT at Scale

A manufacturing facility with thousands of sensors monitoring temperature, vibration, fluid level, and equipment health faces two infrastructure challenges: powering all sensors (battery replacement at scale is expensive) and connecting them all (wired communication is inflexible; separate wireless networks require separate management).

The UEDM addresses both simultaneously. Sensors are SSDs —” powered and connected through the mesh. Actuators (valve controllers, motor controllers, conveyor adjusters) receive high-priority energy delivery (NORMAL or HIGH class) and accept control commands through the same mesh. The manufacturing DPI described in [DI Theory, 2026 9.2] uses the UEDM as its resource routing fabric.

10.3 Emergency Response Networks

Portable UEDM anchor nodes deployed by emergency responders create an instant infrastructure —” no wiring, no separate communication setup. Personnel carrying UEDM-enabled equipment receive power for their devices and communicate through the mesh simultaneously. Medical monitoring devices provide continuous telemetry to triage coordinators through the same mesh that powers them.

The Emergency Priority Override mechanism [Energy Internet, 2026 6.4] ensures that CRITICAL energy sessions (medical devices) are protected even when the mesh is under high load. The Guardian Security Emergency Authority Credential [Guardian Security, 2026 7.3] enables emergency coordinators to establish and protect priority sessions with authenticated authorization.

10.4 Agricultural Sensor Networks

Large-scale agricultural monitoring —” soil moisture, microclimate conditions, pest detection, irrigation control —” requires dense sensor deployment over large areas. Battery replacement in a field with thousands of sensors per square kilometer is impractical. Wiring is impossible. The UEDM mesh, deployed through strategically placed anchor nodes (solar-powered Class A anchors on fence posts or poles), provides power and connectivity to dense sensor networks across agricultural areas.

Communication range in this scenario is extended by relay nodes —” SSD relay nodes that receive energy from the anchor, forward it to more distant sensors, and relay data from distant sensors back to the anchor and onward to the internet through the gateway. The relay chain is the DRMA multi-hop topology applied to an agricultural deployment context.

11. Related Work

11.1 Simultaneous Wireless Information and Power Transfer

The SWIPT literature [58, 59] is the primary academic foundation for the UEDM's physical layer duality. Varshney [58] first formulated the theoretical tradeoff between information rate and energy harvesting from a single RF signal. Grover and Sahai [60] extended this to the capacity-energy tradeoff for AWGN channels. Zhou et al. [61] introduced the receiver design models (power splitting and time switching) that the UEDM's Mode 1 and Mode 2 draw on directly.

The UEDM advances the SWIPT literature in two directions. First, SWIPT research has focused overwhelmingly on single-link or broadcast scenarios; the UEDM is a multi-hop routed network, which introduces the co-routing arbitration problem that has not been addressed in the SWIPT literature. Second, SWIPT research treats the physical layer tradeoff as the primary design variable; the UEDM treats it as one parameter in a broader routing and governance problem.

11.2 Wireless Powered Communication Networks

Wireless Powered Communication Networks (WPCN) [62, 63] consider networks in which devices harvest energy from wireless signals transmitted by a power beacon and use the harvested energy to communicate with an access point. This architecture is similar to the SSD model —” devices receive power wirelessly and communicate wirelessly —” but differs in that WPCNs use separate beacons and access points for power and communication rather than a unified mesh.

The UEDM's SSD model achieves the same device independence as WPCN (battery-free, cable-free operation) through a unified mesh architecture that provides both functions from the same infrastructure, eliminating the separate power beacon and access point infrastructure that WPCN requires.

11.3 Integrated Sensing and Communication

Integrated Sensing and Communication (ISAC) [64, 65] uses a shared RF waveform for both radar sensing and communication, exploiting the spectral overlap between sensing and communication to improve efficiency. This is a waveform-level integration analogous to the UEDM's network-level integration.

The UEDM's localization subsystem (UWB ranging) already performs a sensing function —” measuring node positions. Extending this to general environmental sensing (radar presence detection for BSL, structural health monitoring through vibration sensing at the node hardware) would make the UEDM an ISAC system at the node level. This is a natural extension beyond the scope of this paper.

11.4 Named Data Networking

Named Data Networking (NDN) [66] proposes replacing IP's host-centric addressing (route to this address) with content-centric addressing (route to this named piece of data). The UEDM's service endpoint addressing (Section 6.2) draws on NDN principles —” a data request can be addressed to a service name (e.g., temperature-sensor.lab-a.building-3.mit.energy:0x0200) rather than a specific device address, allowing the routing fabric to deliver the request to the nearest available temperature sensor.

The UEDM's anycast addressing (Section 6.3) is a limited form of NDN content routing: route to the nearest instance of a service rather than to a specific host. A full NDN integration into the UEDM —” where energy requests are addressed by capability name rather than device address —” is a promising extension for ambient IoT scenarios where device identity is less important than service availability.

11.5 The Lume Ecosystem

The UEDM is the convergence of two Lume ecosystem trajectories: the energy routing trajectory (Meridian [Meridian Architecture, 2026], Energy Internet [Energy Internet, 2026]) and the identity and communication trajectory (Trust Layer [16], DAIGS [17]). The UEDM is the physical instantiation of the cross-domain interoperability model described in [DI Theory, 2026 8] —” the single fabric in which energy, data, and identity are governed under one framework.

12. Limitations and Honest Boundaries

No hardware validation exists. The UEDM is an architectural specification. The co-routing arbitration rules, the UEDM frame format, and the DTCP protocol have not been implemented or tested in hardware. The SSD power-communication budget estimates are based on current hardware specifications for microcontrollers and RF transceivers —” they are plausible but not validated.

The TDMA capacity model is simplified. The capacity pool analysis in Section 5.2 treats the total TDMA frame capacity as a fixed number. In practice, channel capacity varies with distance, interference, frequency, and hardware configuration. The arbitration rules are correct in structure but require empirical capacity measurements for specific deployment scenarios to set meaningful parameter values.

Mode 2 (SWIPT-integrated) requires hardware not currently specified in Meridian. 7

Self-sustaining device viability depends on deployment density. An SSD in a sparse UEDM mesh —” one source node for a large area —” may not receive sufficient energy for continuous operation. The SSD model is viable when UEDM source node density is sufficient to maintain $P_{\min}$ at all SSD locations. The required density depends on the physical environment (attenuation, reflections, mobility), the SSD power budget, and the Phase 1—“3 efficiency validation results.

Data latency in the UEDM is higher than dedicated data networks. TDMA scheduling introduces fixed latency at each hop. A five-hop data path with 13.7 ms control cycles has a minimum latency of approximately 70 ms for data packets —” higher than Wi-Fi or Bluetooth, adequate for sensor telemetry and status reporting, but not appropriate for real-time control applications or high-quality audio/video streaming.

The billing model is unvalidated. The energy-credit-denominated data billing model described in Section 9.3 has no economic validation. The exchange rate between energy credits and data capacity requires empirical calibration against actual deployment economics.

13. Conclusion

The Meridian mesh was designed to route energy. It already routes data. That was inevitable —” a deterministic routing network that governs energy delivery through addressed nodes, authenticated sessions, and confirmed receipts cannot route energy without also routing the information that governs it. The control plane is a communication layer. The UEDM formalizes it.

The formal recognition of the Meridian mesh as a dual-traffic fabric —” one that routes both joules and bits under a unified addressing, routing, and governance framework —” has three significant implications.

First, it resolves the most persistent practical obstacle to large-scale IoT deployment: the dual infrastructure problem. Every deployed IoT device currently needs two connections —” one for power, one for communication. The UEDM reduces two infrastructure problems to one, cutting deployment cost and complexity simultaneously.

Second, it validates the convergence thesis at the infrastructure level. Papers 1—“4 of this series argued that deterministic autonomous systems converge toward biological organization, that this convergence is general across physical domains, and that it is a design principle rather than an accident. The UEDM demonstrates this convergence at the resource level: power and communication, the two fundamental infrastructure resources, converge toward a single fabric when both are designed on deterministic routing principles. Convergence is not just a property of system organization —” it is a property of resource infrastructure.

Third, it completes the Energy Internet architecture. The Energy Internet [Energy Internet, 2026] defined the inter-mesh routing standard but deferred the access layer —” how end devices connect to the energy routing infrastructure. The UEDM is that access layer. Together, the Energy Internet and the UEDM provide the complete architecture from global inter-mesh routing to self-sustaining device connection, with no infrastructure gap.

When both the power and the packet travel through the same fabric, governed by the same addresses, the same routes, and the same governance framework, the infrastructure needed to support a world of trillions of autonomous devices becomes not two separate global systems but one.

That is what the UEDM makes possible.

Appendix A —” UEDM Protocol Stack

Layer	Name	Energy Function	Data Function	Protocol
5	UAL	EaaS energy profiles	Application services	EaaS + custom
4	UTL	ETCP energy sessions	DTCP data sessions	ETCP / DTCP
3	UNL	EIP routing, EBGp	Same (unified address)	EIP / EBGp
2	ULL	Energy burst framing, TDMA	Data packet framing, co-routing	UEDM Frame
1	UPL	High-power focused beam	Modulated low-power signal	RF (time-switched)

Appendix B —” Co-Routing Arbitration Rules

Rule	Description	Priority Impact
AR-01	GEC reserved exclusively for CRITICAL + HIGH energy	Energy $\gg$ data in GEC
AR-02	GCDC reserved for essential control-plane data	Control data $\gg$ application data
AR-03	NORMAL energy priority over all data in FSC	Energy $>$ data in FSC
AR-04	BACKGROUND energy yields to NORMAL data in FSC	Data $>$ background energy
AR-05	CRITICAL data served from GCDC, not FSC	Data CRITICAL protected
AR-06	FSC allocated in TDMA slot increments	Minimum granularity: 1 slot
AR-07	Mode 2 (SWIPT) used when paths overlap	Efficiency optimization
AR-08	Large data payloads fragmented across FSC slots	No monopolization of FSC

Appendix C —” UEDM Addressing Specification

UEDM Address Format (128 bits): Same as EIP address format [Energy Internet, 2026, Appendix B] with Node Type 0x06 = DATA_ONLY added Service Endpoint Format: [UEDM Address : 128 bits] + [Service Port : 16 bits] = 144 bits total Reserved Ports: 0x0001 : Energy delivery (EaaS endpoint) 0x0002 : Guardian Security authentication 0x0003 : Lume-X telemetry 0x0004 : SHDCL recovery messaging 0x0005 : EDNS query/response 0x0006 : EBGp peering session 0x0007 : DTCP session establishment 0x0008 : Multicast group management 0x0009 : Service discovery 0x000A : Energy credit settlement 0x000B—“0x00FF : Reserved for future system services 0x0100—“0xFFFF : Application-defined services

Appendix D —” Self-Sustaining Device Profile

Parameter	Minimum	Typical	High-Performance
Received power (P _{min})	1 mW	10 mW	100 mW
Computation	Ultra-low-power MCU, 1 MHz	Cortex-M4, 4 MHz	Cortex-M33, 64 MHz
Sensor duty cycle	0.1 Hz	1 Hz	10 Hz
Data rate (uplink)	100 bps	2.5 kbps	25 kbps
Supercapacitor capacity	1 mF	10 mF	100 mF
Recovery time after blackout	60 s	10 s	2 s
Organism type [Meridian Organism, 2026]	Type 2	Type 3	Type 3+

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Part Three Closing

Chapters Four and Five have traced Meridian from a single building mesh to a global energy internet to a unified infrastructure that routes both energy and data. The protocol stack is complete. The addressing is defined. The governance framework is specified. The self-sustaining device is characterized.

What is not yet specified is how this infrastructure defends itself.

A system that routes energy to authenticated devices and routes data through the same fabric is also a system that could, in the absence of a formal security architecture, route energy to unauthenticated devices, misdirect beams to unintended targets, or be deceived about the state of its own physical world. The security stakes in wireless energy routing are categorically higher than in data networking: a failed attack on a data network corrupts information. A failed attack on an energy routing network can cause physical injury.

Part Four specifies the security architecture that prevents this.

PART FOUR

The Defense: How Meridian Is Protected

Security is easier to argue about than to specify. Almost every system has a security story. Fewer have a formal threat model —” a complete enumeration of the attack surface, a taxonomy of threat categories derived from first principles, and a coverage proof demonstrating that every attack path requires traversal of at least one defense boundary.

Meridian has all three. Chapter Six presents Guardian Security —” the energy-domain security framework —” in the form of a formal threat model, an eleven-category threat taxonomy, a two-tier defense architecture, and a coverage proof. It also presents the Biological Safety Layer as a physical security invariant: a hardware-enforced mechanism that guarantees no beam strikes a biological target regardless of the state of any software defense.

Part Four is one chapter. It does not need to be longer. A correct formal security specification is complete, not comprehensive.

Chapter Six

Guardian Security

A failed attack on a wireless energy routing network is not a data breach.

It is a beam of RF energy directed at an unintended target. If the target is a person, the consequence is physical injury. If the target is a medical device, the consequence is the denial of power to a life-critical instrument. If the target is a routing node at a critical infrastructure junction, the consequence is the loss of energy delivery to an entire zone of the network.

These consequences are categorically different from the consequences of a data network security failure. A data network attack corrupts information or denies service. A wireless energy routing attack can harm people. This asymmetry —” the fact that the worst case is physical, not informational —” changes everything about how the security architecture must be designed.

In particular: the security model cannot be primarily corrective. A data network that discovers a vulnerability can be patched. An RF beam that has already struck a biological target cannot be recalled. The security architecture must prevent physically harmful attacks from succeeding, not merely detect them after the fact.

Guardian Security is designed around this requirement. Its two-tier architecture —” an innate tier that provides immediate non-specific response to anomalous conditions, and an adaptive tier that provides learned specific response to classified threat patterns —” operates before beam authorization, not after delivery. Its pre-transmission authentication protocol requires four independent checks to pass before any beam is authorized. Its eleven-category threat taxonomy, derived from first principles from the attack surface definition, covers the complete space of attacks that can be mounted against a wireless energy routing system.

And at the bottom of the defense stack, independent of every software mechanism, the Biological Safety Layer enforces a physical security invariant in hardware: no beam activates when a biological target is in its path. This invariant holds regardless of whether the software above it has been compromised, because it operates below the software layer.

This chapter specifies all of it.

Abstract

Wireless energy routing introduces a class of security vulnerabilities that has no analog in data networking: a successful attack on a wireless power system does not corrupt data —” it corrupts physics. Beam hijacking misdirects RF energy to unintended targets. Denial-of-power attacks starve critical devices. Localization spoofing causes a beam to strike a biological target rather than an electronic one. The consequences of these attacks are not data loss or service disruption —” they are equipment damage, fire, and physical injury.

No formal security framework for wireless energy routing has previously been published. This paper provides one.

I define the complete attack surface for a deterministic wireless energy routing system, derive eleven threat categories from first principles, and specify the Guardian Security defense framework —” the security enforcement layer implemented in the Meridian architecture —” as a formal, auditable response to each threat. Guardian Security (energy-domain designation: Guardian-E; commercial product domain: TrustShield.tech) implements two-tier defense: an innate tier providing immediate, non-specific response to anomalous conditions, and an adaptive tier providing learned, specific response to classified threat patterns.

I derive the Guardian Security rule set from the formal attack surface definition rather than from enumerated known attacks, ensuring that the defense covers the complete threat space rather than only previously observed threats. I prove coverage completeness by demonstrating that every path from attacker to system compromise requires traversal of at least one Guardian Security enforcement boundary. I show that Guardian Security satisfies four formal security properties: authenticity (every participant is who they claim to be), integrity (no resource flow is modified in transit without detection), availability (the system continues operating under attack, with degraded but bounded service), and non-repudiation (every resource delivery event is attributable to a specific authenticated participant).

This paper establishes the security foundation for both the single-mesh Meridian architecture and the multi-mesh Energy Internet. It proposes Guardian Security as the reference security architecture for all future Deterministic Infrastructure deployments.

Keywords: wireless energy security, threat model, formal security analysis, Guardian Security, denial-of-power, beam hijacking, localization spoofing, Meridian, TrustShield, deterministic infrastructure security, wireless power transfer security

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Appendix D —” Guardian Security at Three Operational Levels

References

1. Introduction

1.1 A Security Gap With Physical Consequences

The wireless power transfer industry has, until this paper, produced no formal security framework. This is not oversight —” it reflects the history of the field. Wireless power has been framed as an efficiency and range problem, not a security problem. The threat model for a Qi charging pad is limited: an attacker who places a foreign object on the charging surface triggers a foreign object detection circuit. The consequence of a failed attack is a failed charge. The security model needs only to prevent this one failure.

Wireless energy routing at mesh scale is a different threat environment entirely. When a system routes watt-level energy through addressed nodes across a distributed network, the attack surface expands in every dimension: the range is no longer centimeters but meters; the nodes are addressable over a network rather than physically accessible only; the power levels are sufficient to cause harm; and the routing infrastructure itself becomes a target —” not just the delivery endpoint.

I enumerate the consequences of specific attack classes that a wireless energy routing system without a formal security framework cannot prevent:

- Localization spoofing: An attacker feeds false position data to the routing system. The beam steers toward the false position —” which may be occupied by a biological target. Physical injury results.
- Beam hijacking: An attacker intercepts a delivery route and redirects the beam. An unauthorized device receives energy it was not allocated, and the intended recipient is denied it. If the intended

recipient is a medical device, the denial is life-safety critical.

- Denial-of-power: An attacker generates synthetic energy requests that saturate the mesh's routing capacity. Legitimate devices are denied service. A network of critical sensors loses power simultaneously.
- Routing table manipulation: An attacker injects false routing table entries. Energy flows through attacker-controlled relay nodes, enabling interception, denial, or misdirection of all traffic on the poisoned route.
- Node impersonation: An attacker deploys a device claiming a legitimate node's identity. It receives energy intended for the legitimate device and may serve as a relay that the attacker controls.

None of these attacks require proximity to a single charging pad. All of them require a wireless energy routing system without a formal security architecture. Guardian Security is that architecture.

1.2 The Security Architecture Problem

Building a security architecture for wireless energy routing presents three challenges that do not exist in data network security:

The physics cannot be patched. A data network that discovers a security vulnerability can be updated with a software patch. An RF beam that strikes a biological target cannot be recalled. Security must be preventive, not corrective. The Guardian Security framework is designed to make the class of physically harmful attacks impossible to execute, not merely detectable after the fact.

The delivery confirmation is the attack surface. In data networking, delivery confirmation (TCP ACK) is a control-plane message that is separated from the data payload. In wireless energy routing, the delivery confirmation is the physical measurement of received power —” it is feedback from the physical world, not a protocol message. An attacker who can spoof the physical measurement (false received-power telemetry) can deceive the routing system about whether energy was delivered, to whom, and how much.

Identity must be verified before the first joule flows. In data networking, an unverified connection attempt is a mild nuisance —” the worst case is a SYN flood. In wireless energy routing, an unverified node that successfully masquerades as a legitimate source node can aim a beam before identity is checked. Identity verification must be enforced at the routing level, before any physical transmission occurs.

1.3 Contributions

I make five contributions:

The first formal definition of the attack surface for a deterministic wireless energy routing system, derived from first principles rather than from enumerated known attacks.

A taxonomy of eleven threat categories covering the complete attack surface, each with formal definition, attack vector, and consequence characterization.

The Guardian Security framework —” a two-tier (innate + adaptive) security architecture that provides coverage of all eleven threat categories with formally defined enforcement boundaries.

A coverage proof demonstrating that every attack path from adversary to system compromise requires traversal of at least one Guardian Security enforcement boundary.

An extension of the Guardian Security framework to Energy Internet scale, addressing the additional threat surface introduced by inter-mesh federation, EBGp routing, and multi-operator deployments.

1.4 Paper Organization

Section 2 characterizes why energy security differs fundamentally from data security. Section 3 defines the attack surface from first principles. Section 4 presents the eleven threat categories. Section 5 specifies the Guardian Security framework. Section 6 proves formal security properties. Section 7 extends the framework to Energy Internet scale. Section 8 analyzes the Biological Safety Layer as a physical security invariant. Section 9 addresses operational security. Section 10 situates this in related literature. Section 11 states limitations. Section 12 concludes.

2. Background: Why Energy Security Is Different

2.1 The Consequence Asymmetry

Security analysis begins with the consequence model: what is the worst realistic outcome of a successful attack? The consequence model determines how much defense investment is justified and what failure modes are acceptable.

For a data network, the consequence model ranges from service disruption (DoS) to data breach (confidentiality loss) to data corruption (integrity loss). These are serious consequences —” financial damage, reputational harm, privacy violations, regulatory penalties. They are not, in the general case, physical injury.

For a wireless energy routing network, the consequence model includes physical injury. An RF beam at the power densities required for useful energy delivery —” even at the conservative power levels of Phase 1 Meridian deployment —” can cause tissue heating if directed at a biological target at close range. At higher power levels, the consequence is burns. A localization spoofing attack that redirects a high-power beam toward a person is not a data breach —” it is a physical assault.

This consequence asymmetry means that some attack classes that are "acceptable" in data network security —” where "acceptable" means "detectable and recoverable" —” are not acceptable in energy routing security. An attack that results in physical injury cannot be recovered from. The security

architecture must prevent these attacks, not detect them post-hoc.

The Biological Safety Layer (BSL) in the Meridian architecture [Meridian Architecture, 2026 8] is the primary mechanism for enforcing this preventive requirement. Section 8 of this paper analyzes the BSL as a physical security invariant —” the mechanism that ensures physical harm is impossible even when all other security mechanisms fail simultaneously.

2.2 The Physical Delivery Confirmation Problem

In data networking, the control plane (routing decisions) and the data plane (packet forwarding) are conceptually separate, even when they share physical infrastructure. A routing decision is made based on control-plane information (routing tables, OSPF state, BGP updates) that is separate from the data being routed.

In wireless energy routing, the control plane and the physical plane are coupled at the measurement layer: the routing system's view of the physical world comes from physical measurements (received power telemetry, UWB localization returns, supercapacitor voltage readings). If these measurements can be falsified, the routing system's model of the physical world can be corrupted.

This coupling creates attack vectors that have no data networking analog:

False localization returns: An attacker that can transmit UWB localization signals can cause a node to believe a receiver is at a different position than its actual position. False received-power telemetry: An attacker that can generate false EBF confirmation messages can cause the routing system to believe energy was successfully delivered when it was not, or to believe energy was not delivered when it was. False supercapacitor readings: An attacker that can inject false charge-state data can cause the routing system to make incorrect routing decisions —” routing energy to a node that is actually fully charged, or denying supplementation to a node that is actually depleted. Each of these attacks targets the measurement layer that the control plane depends on. Guardian Security must protect not just the protocol layer but the physical measurement layer.

2.3 The Pre-Transmission Identity Requirement

In data networking, authentication can happen after the first packet. TCP SYN packets are unauthenticated. TLS handshakes happen after TCP connection establishment. The worst case of unauthenticated early traffic is a SYN flood, which is annoying but recoverable.

In wireless energy routing, the first packet IS the physical beam. There is no unauthenticated "handshake packet" that precedes the physical action —” the physical action itself is the delivery. A node that successfully triggers beam transmission without completing authentication has already received energy it was not entitled to, and the beam has already existed as a physical field in space.

Guardian Security must enforce complete authentication before any beam authorization is issued. The authentication must be completed at the routing layer, not at the application layer after delivery begins. This is the pre-transmission identity requirement: no beam command is issued to MTL unless the source, relay chain, and destination have all completed Guardian Security authentication for the current session.

3. Attack Surface Definition

I define the attack surface of a wireless energy routing system from first principles, without reference to specific known attacks. The attack surface is the set of all interfaces through which an adversary can influence system behavior.

3.1 Physical Attack Interfaces

PAI-1: RF transmission medium. The air through which RF energy travels. Accessible to any attacker within RF range. Enables: beam interception (passive —” measure transmitted power without detection), beam interference (active —” transmit RF that degrades received power), physical obstruction (place absorbing material in beam path).

PAI-2: Localization signal medium. The UWB localization signal environment. Accessible to any attacker with UWB transmission capability within localization range. Enables: position spoofing, replay attacks on localization signals, localization jamming.

PAI-3: Physical node access. Direct physical access to a deployed node. Accessible to any attacker who can physically reach a node. Enables: hardware modification, credential extraction, firmware replacement, physical destruction.

PAI-4: Supercapacitor and harvest circuitry. The energy storage and harvest subsystems accessible through physical proximity. Enables: power injection (force a node to false charge state), power drain (force a node to depletion), harvest signal manipulation.

3.2 Network Attack Interfaces

NAI-1: Mesh control plane. The LINK_STATE_UPDATE, NODE_ANNOUNCE, and routing protocol messages that maintain the mesh topology. Accessible to any node participating in the mesh, authenticated or not (before authentication is enforced). Enables: topology poisoning, routing table manipulation, false node announcements.

NAI-2: EBF message channel. The Energy Burst Frame messages carrying delivery instructions and confirmations. Accessible to any node in the communication range of transmitting nodes. Enables: EBF replay, EBF modification, false delivery confirmations.

NAI-3: Delivery confirmation channel. The telemetry messages from receiving nodes confirming energy receipt. Accessible to any node within communication range. Enables: false received-power claims, false spatial accuracy claims.

NAI-4: EDNS and inter-mesh routing. At Energy Internet scale: the EDNS resolution infrastructure and EBGp routing advertisements. Accessible to any participant in the inter-mesh routing system. Enables: EDNS poisoning (route energy requests to wrong EAS), EBGp hijacking (advertise routes to destinations the attacker doesn't control).

3.3 Identity Attack Interfaces

IAI-1: Node identity credentials. The cryptographic credentials that establish node identity in the Trust Layer PKI. Accessible to an attacker who can compromise the provisioning process or extract credentials from a deployed node. Enables: node impersonation, credential replay.

IAI-2: Trust Layer PKI infrastructure. The certificate authority infrastructure that signs node credentials. Accessible to a sophisticated attacker who can compromise CA infrastructure. Enables: fraudulent credential issuance, certificate chain forgery.

IAI-3: Provisioning process. The process by which new nodes receive their identity credentials. Accessible to an attacker who can intercept or hijack provisioning. Enables: identity substitution during provisioning, credential interception.

4. The Eleven Threat Categories

From the attack surface definition in Section 3, I derive eleven threat categories by enumerating the distinct attack objectives and the attack interfaces through which each can be pursued. A threat category is a class of attacks sharing the same objective —” the specific attack technique may vary, but the defense must cover the objective class.

4.1 T1 —” Localization Spoofing

Objective: Cause the routing system to believe a node is at a false position, resulting in beam misdirection.

Attack interfaces: PAI-2 (localization signal medium), NAI-2 (EBF position fields).

Attack vectors:

T1a —” UWB signal forgery: Transmit false UWB ranging signals that cause the localization system to compute a false position for a target node. Requires UWB transmission capability within localization range. T1b —” Position field injection: Modify the position field in EBF messages (NAI-2) to report false position data to the routing system. T1c —” Replay attack: Capture and replay a legitimate node's

localization signals from a different physical position, causing the system to compute a false location based on authentic signals. Worst-case consequence: Beam directed at a biological target at the spoofed position. Physical injury.

Guardian Security response: See Section 5.3 (T1 defenses).

4.2 T2 —” Routing Table Manipulation

Objective: Corrupt the mesh routing table to redirect energy flows through attacker-controlled nodes or to unintended destinations.

Attack interfaces: NAI-1 (mesh control plane).

Attack vectors:

T2a —” False LINK_STATE_UPDATE injection: Transmit false link-state updates advertising nonexistent links or false link quality metrics, causing routing algorithms to select attacker-preferred paths. T2b —” Route withdrawal attack: Transmit false link-state updates withdrawing legitimate routes, forcing traffic through attacker-controlled alternatives. T2c —” Metric manipulation: Modify link quality metrics to make attacker-controlled paths appear preferred over legitimate paths. Worst-case consequence: All energy traffic routed through attacker-controlled nodes, enabling interception, denial, or misdirection of all delivered energy.

4.3 T3 —” Beam Hijacking

Objective: Redirect an authorized beam from its intended target to an attacker-controlled receiver.

Attack interfaces: PAI-1 (RF medium), NAI-2 (EBF messages), IAI-1 (credentials).

Attack vectors:

T3a —” Relay impersonation: Compromise a relay node position in a multi-hop path and redirect the beam at that hop rather than forwarding it. T3b —” BeamCommand injection: Inject false BeamCommand messages that override the legitimate steering parameters before MTL executes them. T3c —” Path substitution: Using a compromised routing table (T2), route the beam through an attacker-controlled relay that performs mid-path redirection. Worst-case consequence: High-priority device (medical monitor, emergency communication) denied energy while attacker device receives it.

4.4 T4 —” Denial-of-Power Attack

Objective: Starve legitimate devices of energy by exhausting routing capacity, depleting source nodes, or blocking delivery paths.

Attack interfaces: NAI-1, NAI-2, PAI-1.

Attack vectors:

T4a —” Request flooding: Generate high volumes of synthetic energy requests that saturate the mesh's routing computation capacity. Legitimate requests are delayed or dropped. T4b —” Source depletion: Cause source nodes to repeatedly transmit energy to false destinations (exploiting T3 or T1), depleting their supercapacitors and leaving the mesh without energy to route. T4c —” Path blocking: Physically obstruct beam paths or use RF jamming to degrade link quality until routing fails to find viable paths. Worst-case consequence: Simultaneous power loss to all devices in the mesh during an emergency condition.

4.5 T5 —” Node Impersonation

Objective: Deploy a device that successfully claims a legitimate node's identity, receiving its energy allocation and potentially serving as a relay the attacker controls.

Attack interfaces: IAI-1 (credentials), NAI-1 (NODE_ANNOUNCE).

Attack vectors:

T5a —” Credential cloning: Extract a legitimate node's cryptographic credentials (via PAI-3 physical access) and deploy a second device using those credentials. T5b —” Credential replay: Capture credential exchange messages and replay them from a different device. T5c —” Provisioning interception: Intercept credentials during the provisioning process (IAI-3) before they are securely stored on the legitimate device. Worst-case consequence: Legitimate device denied its energy allocation; attacker device serving as a controllable relay inside the trusted mesh.

4.6 T6 —” Delivery Confirmation Falsification

Objective: Deceive the routing system about whether energy was successfully delivered, enabling false accounting, route selection corruption, and ETCP session manipulation.

Attack interfaces: NAI-3 (delivery confirmation channel).

Attack vectors:

T6a —” False positive confirmation: Transmit a delivery confirmation claiming successful receipt when no energy was received. Causes the routing system to believe delivery succeeded and move on to the next burst, leaving the intended recipient with no energy. T6b —” False negative confirmation: Transmit a delivery failure message when energy was successfully received. Triggers SHDCL recovery and unnecessary retransmission, depleting source nodes. T6c —” Power level falsification: Report a false received power level. Causes MFE to misadjust burst power levels —” either under-delivering (slow starvation) or over-delivering (wasted energy and potential component damage). Worst-case consequence: Systematic false accounting across all deliveries, creating a gap between the routing system's model of the mesh state and physical reality.

4.7 T7 —” Timing Attacks

Objective: Disrupt TDMA scheduling to cause simultaneous transmissions (collision), scheduling gaps (denial), or predictable transmission windows (enabling interception).

Attack interfaces: NAI-1, NAI-2.

Attack vectors:

T7a —” TDMA schedule manipulation: Inject false scheduling information that causes two nodes to transmit simultaneously, corrupting delivery for both. T7b —” Clock desynchronization: Introduce timing errors in network time synchronization, causing scheduling drift that eventually produces collisions. T7c —” Transmission window prediction: Exploit deterministic TDMA scheduling to predict exactly when a given node will transmit, enabling targeted interception or jamming at that moment. Worst-case consequence: Systematic collision of all high-priority transmissions, effectively implementing a denial-of-power attack through timing manipulation.

4.8 T8 —” Physical Node Compromise

Objective: Gain control of a deployed node through physical access, enabling all attacks from a trusted insider position.

Attack interfaces: PAI-3 (physical node access).

Attack vectors:

T8a —” Credential extraction: Extract cryptographic keys from secure storage on a captured node and deploy a cloned device (feeds T5). T8b —” Firmware replacement: Replace legitimate firmware with attacker-controlled firmware that executes attacker instructions while appearing legitimate. T8c —” Hardware backdoor installation: Modify node hardware to include a covert communication channel or power draw that is not visible to the software layer. Worst-case consequence: Full insider access to the mesh, with trust-level credentials and physical mesh participation, enabling all other attacks with legitimate authentication.

4.9 T9 —” Energy Credit Fraud (Energy Internet Scale)

Objective: Manipulate the energy credit settlement system to receive energy without paying for it or to receive payment for energy not delivered.

Attack interfaces: NAI-2 (EBF confirmation records used for settlement), IAI-2 (PKI infrastructure used for settlement signatures).

Attack vectors:

T9a —” Confirmation record falsification: Generate false EBF confirmation records that overstate delivered energy, enabling fraudulent claims for energy credit reimbursement. T9b —” Settlement signature forgery: Forge the signature on energy credit transfer records, enabling unauthorized credit transfers. T9c —” Replay of settlement records: Replay legitimate settlement records from prior periods to claim credits multiple times for the same delivered energy. Worst-case consequence: Financial fraud at Energy Internet scale; systematic undercharging for energy that discourages legitimate ESP participation.

4.10 T10 —” EBGp Route Hijacking (Energy Internet Scale)

Objective: Announce false EBGp routes that redirect energy traffic from its intended EAS to an attacker-controlled EAS.

Attack interfaces: NAI-4 (inter-mesh routing).

Attack vectors:

T10a —” Prefix hijacking: Announce an EBGp route for an EIP address prefix that the attacker does not control, causing remote source nodes to route energy toward the attacker's EAS rather than the legitimate destination. T10b —” AS path manipulation: Manipulate AS path attributes to make attacker-controlled routes appear shorter or more preferred than legitimate routes. T10c —” Route withdrawal: Withdraw legitimate EBGp routes to isolate an EAS from the wider Energy Internet, effectively implementing a denial-of-power attack at internet scale. Worst-case consequence: All energy traffic destined for a legitimate EAS redirected to an attacker-controlled EAS, enabling large-scale energy theft or denial.

4.11 T11 —” Safety System Bypass

Objective: Disable or circumvent the Biological Safety Layer, enabling beam transmission in conditions where BSL would normally prevent it.

Attack interfaces: PAI-3 (physical access to BSL sensor hardware), NAI-2 (safety status flags in EBF headers).

Attack vectors:

T11a —” Sensor spoofing: Feed false data to the biological presence detection sensors (radar, thermal, optical), causing BSL to not detect a biological target in the exclusion zone. T11b —” Safety flag manipulation: Modify the `safety_flags` field in EBF headers to falsely report that all safety conditions are met. T11c —” Software gate bypass: Exploit a vulnerability in the software implementation of the five-condition MTL gate to issue a BeamCommand that bypasses one or more gate conditions. Worst-case consequence: Beam transmission directed at a biological target with no safety mitigation. This is the highest severity threat in the taxonomy.

5. The Guardian Security Framework

Guardian Security (energy-domain designation: Guardian-E; commercial product domain: TrustShield.tech) is the security enforcement layer of the Meridian architecture. It operates as a two-tier system: the innate tier provides immediate, non-specific response to anomalous conditions; the adaptive tier provides learned, specific response to classified threat patterns.

5.1 Architecture Overview

```

Guardian Security Framework
Adaptive Tier (Guardian-E)
• Threat classifier (T1—"T11 mapping)
• Learned threat patterns (11 categories)
• Node quarantine and isolation
• Certificate chain audit
• ERPKI route origin validation (inter-mesh)
Innate Tier (SHDCL)
• Immediate anomaly detection (13.7ms)
• Process recovery without classification
• Beam hold on unresolved anomaly
• Escalation to adaptive tier
Physical Safety Layer (BSL)
• Hardware-enforced beam cutoff
• Independent of software gate status
• Biological presence detection
• Cannot be bypassed by software attack

```

The three-tier architecture provides defense in depth. No single tier is a single point of failure. If the adaptive tier fails to classify a threat, the innate tier holds the beam. If the innate tier is compromised, the BSL provides hardware-level enforcement. If the BSL sensor is spoofed (T11a), the adaptive tier's sensor integrity monitoring detects the anomaly. Defense in depth means that an attacker must compromise multiple independent tiers simultaneously to achieve physical harm.

5.2 Pre-Transmission Authentication Protocol

Before any beam authorization is issued, Guardian Security enforces a four-step pre-transmission authentication protocol:

Step 1 —" Node identity verification. The requesting node presents its Trust Layer PKI credential. Guardian-E verifies the signature chain from the credential to the Trust Layer root CA. If verification fails: request rejected, anomaly logged, node flagged as UNVERIFIED.

Step 2 —” Capability authorization check. Guardian-E verifies that the requesting node's capability profile (encoded in its identity credential) includes the requested operation type, power level, and target zone. If the capability profile does not authorize the request: request rejected, anomaly logged.

Step 3 —” Route integrity check. Guardian-E verifies that the proposed routing path through the mesh consists entirely of verified, non-quarantined nodes, and that the path's safety envelope is within bounds for the requested delivery parameters. If any relay node in the path is UNVERIFIED or QUARANTINED: path rejected, alternate path computation requested.

Step 4 —” Temporal consistency check. Guardian-E verifies that the request's timestamp is within the acceptable window (± 2 TDMA slots of network time), that the requesting node's last confirmed transaction matches the routing system's record, and that no anomalous request rate is detected from this node. If consistency check fails: request held for review, node flagged for anomaly investigation.

Only after all four steps pass does Guardian-E issue beam authorization to the Lume-X control loop, which then validates the five MTL gate conditions before issuing the BeamCommand. The security check (Guardian-E) and the safety check (MTL gate) are independent —” both must pass.

5.3 Defenses Against Each Threat Category

T1 —” Localization Spoofing: Guardian-E enforces multi-source localization consistency: position reported by UWB localization must be consistent with position inferred from signal strength, TDMA timing, and mesh topology (neighbor node distances). A position report that is inconsistent across these sources is flagged as a potential T1 attack, the node is placed in POSITION_UNVERIFIED status, and no beam is authorized until consistency is restored. Threshold: if position variance across sources exceeds $r_threshold$ (50 cm) for three consecutive measurements, the node is QUARANTINED. The BSL exclusion zone provides the ultimate backstop: even if localization is spoofed, biological presence detection prevents beam transmission at the spoofed location if a biological target is actually there.

T2 —” Routing Table Manipulation: All LINK_STATE_UPDATE messages are signed by the originating node using its Trust Layer PKI credential. Receiving nodes verify the signature before applying the update. Guardian-E applies consistency checks to all routing updates: a link quality metric that changes by more than q_max in one update cycle without a corresponding physical event (node failure, localization loss) is flagged. Route selection is rate-limited: no node's routes can be completely replaced within a single control cycle. This prevents sudden, large-scale routing table corruption.

T3 —” Beam Hijacking: Guardian-E tracks the expected position of the beam target throughout the delivery session. If the target's position deviates from the expected trajectory by more than the adaptive position model allows, the session is suspended and the node is flagged. Relay nodes in a multi-hop path

are required to confirm receipt and forward within their TDMA window —” a relay that receives but does not forward (potential redirection) is detected by the downstream node's missing receipt confirmation within $T_{\text{relay_max}}$.

T4 —” Denial-of-Power: Guardian-E applies per-node and per-EAS request rate limits. A node generating more than N_{max} requests per control cycle is flagged and rate-limited. A node whose requests consistently fail (false destinations, invalid capability claims) is quarantined after K_{max} consecutive failures. CRITICAL-priority sessions are protected from rate-limiting and receive dedicated capacity reservation that cannot be displaced by NORMAL or BACKGROUND traffic —” even under attack.

T5 —” Node Impersonation: Guardian-E enforces one-credential-one-node: a credential that is simultaneously reported as active from two different physical positions is immediately flagged as a T5 attack. Both instances are quarantined pending investigation. Since credentials are bound to physical hardware characteristics at provisioning (secure element measurements, power draw profile), a cloned credential on different hardware will exhibit a different hardware profile and fail the capability consistency check.

T6 —” Delivery Confirmation Falsification: Guardian-E cross-validates delivery confirmations against the mesh's physics model. A node reporting P_{received} higher than $P_{\text{transmitted}}$ — maximum_path_efficiency is flagged —” energy cannot be amplified in transit. A node that consistently reports successful delivery while the intended device's charge state (reported separately) does not increase is flagged. All confirmation messages are signed by the confirming node; unsigned or incorrectly signed confirmations are rejected.

T7 —” Timing Attacks: TDMA schedule assignments are cryptographically sealed by Guardian-E at each scheduling epoch: the schedule for the next N epochs is signed and distributed to all nodes. A node that transmits outside its assigned window is detected by all nodes that receive the out-of-window transmission and reported to Guardian-E. Clock synchronization is protected by signed time-sync messages with a maximum accepted drift of $\pm T_{\text{drift}}$. A node that attempts to desynchronize the network clock by broadcasting false time is detected by majority-consensus cross-checking.

T8 —” Physical Node Compromise: Guardian-E cannot prevent physical compromise but limits its blast radius. Credentials are stored in secure elements with anti-tamper protection. A credential that is extracted and used on a different hardware platform fails the hardware-profile consistency check (T5 defense). A node exhibiting anomalous behavior after a maintenance event (unusual request rates, unexpected routing changes, new neighbor relationships) is flagged for investigation. Firmware integrity is verified at boot using a signed firmware hash —” a node with modified firmware fails the boot integrity check and does not receive credentials.

T9 —” Energy Credit Fraud: All EBF confirmation records used for settlement are signed by both the delivering node and the receiving node. A settlement claim without dual signatures is rejected. Settlement records include a reference to the specific EBF sequence numbers being settled —” replay of prior settlement records is detected by duplicate sequence number checking. ERPKI validates the signing authority of all settlement records.

T10 —” EBGp Route Hijacking: ERPKI validates all EBGp route origins: an EAS cannot announce a route for an EIP prefix it was not delegated by the EIRA. Route announcements without valid ERPKI attestation are rejected. AS path manipulation is detected by cross-referencing EBGp updates against the known topology —” an AS path that claims to traverse an EAS that has no established peering relationship with the adjacent AS is flagged.

T11 —” Safety System Bypass: The BSL is hardware-enforced: the beam cutoff circuit operates independently of the software gate and cannot be disabled by a software attack. Sensor integrity is monitored by cross-validating biological presence detection across multiple sensor modalities (radar, thermal, optical, ultrasonic). A sensor that is physically blocked or tampered with produces an anomalous reading pattern (all-clear when other sensors indicate uncertainty) and triggers a SENSOR_INTEGRITY_FAILURE event, which forces beam hold until the sensor is inspected. The five-condition MTL gate is implemented in firmware with a hardware watchdog —” a gate bypass attempt that corrupts gate state triggers a watchdog reset and beam hold.

6. Formal Security Properties and Coverage Proof

6.1 Security Properties

Guardian Security is designed to satisfy four formal security properties:

Authenticity: Every participant N in a Guardian Security-protected mesh can be verified as the entity it claims to be. Formally: $\forall N, \forall \text{interactions } I(N, *)$: Guardian-E verifies $\text{PKI_cert}(N)$ before I is permitted. A participant that cannot be verified is not permitted to interact.

Integrity: No resource flow is modified in transit without detection. Formally: for any delivered burst B from source s to destination d , if $P_received(d) \% P_transmitted(s) - path_efficiency(s,d)$ within the tolerance $\pm \epsilon_{phys}$ (physical measurement uncertainty), Guardian-E flags the discrepancy within one control cycle.

Availability: The mesh continues delivering energy at bounded degraded service under attack. Formally: under any attack that does not involve physical destruction of nodes or physical access to BSL hardware, at least all CRITICAL-priority sessions continue at P_min for the duration of the attack. NORMAL and BACKGROUND sessions may be suspended.

Non-repudiation: Every resource delivery event is attributable to a specific authenticated participant. Formally: for every EBF delivery, the certificate chain contains signed records from the authorizing source node, all relay nodes, and the receiving node, with timestamps, power measurements, and authentication proofs. No participant can deny participation in a delivery event without forging signatures that would be detected by Guardian-E.

6.2 Coverage Proof

I prove coverage by attack path enumeration: for every threat category T1—“T11, I show that achieving the attack objective requires traversing at least one Guardian Security enforcement boundary.

Threat	Attack Objective	Guardian-E Enforcement Boundary Crossed
T1	Beam misdirection via false position	Multi-source position consistency check (Step 1 of beam authorization)
T2	Routing table corruption	Signed routing update verification; rate-limited route replacement
T3	Beam redirected to attacker	Trajectory consistency monitoring; relay forwarding timeout detection
T4	Legitimate devices denied energy	Per-node rate limits; CRITICAL priority reservation
T5	Attacker device receives legitimate allocation	One-credential-one-node enforcement; hardware profile consistency
T6	False delivery accounting	Physics-model cross-validation; dual-signature requirement
T7	Transmission collision or denial	Signed TDMA schedule; out-of-window detection; clock sync protection
T8	Insider access via physical compromise	Blast radius limitation; firmware integrity check; hardware profile binding
T9	Energy credit fraud	Dual-signature settlement; ERPKI attestation; sequence number dedup
T10	EBGP route hijacking	ERPKI prefix validation; topology cross-reference
T11	Safety system disabled	Hardware-enforced BSL; multi-modal sensor cross-validation; watchdog

For T11 specifically, I note that the coverage guarantee has a physical limit: an attacker with physical access to the BSL hardware circuitry who can sever the hardware cutoff connection cannot be stopped by software means. This is the residual risk acknowledged in Section 11 (Limitations). The defense against

T11 physical hardware attack is physical security of node deployment locations, not Guardian Security.

7. Guardian Security at Energy Internet Scale

The Energy Internet [Energy Internet, 2026] introduces three new threat surfaces beyond the single-mesh Meridian architecture: the inter-mesh EBGp routing infrastructure, the EDNS resolution infrastructure, and the energy credit settlement system. Sections 4.9—“4.11 addressed the associated threat categories (T9—“T11). This section specifies the Guardian Security extensions for internet-scale operation.

7.1 Three-Level Architecture

Guardian Security operates at three levels in the Energy Internet [Energy Internet, 2026 7.2]:

Level 1 —” Intra-mesh security. The single-mesh Guardian Security implementation specified in Sections 5 and 6. Unchanged from the single-mesh architecture.

Level 2 —” Inter-mesh security (Energy TLS). Every EBGp peering session between gateway nodes is authenticated using EAS-level PKI managed through the Trust Layer. EBGp route updates are signed by the originating gateway; receiving gateways verify the signature and the ERPki prefix authorization before accepting and propagating routes. The session uses forward-secret key exchange (ephemeral Diffie-Hellman over curves approved by the Trust Layer governance framework) so that compromise of a long-term key does not expose past session traffic.

Level 3 —” Registry security (EIRA-level). All EIRA operations —” EAS Number assignment, EIP prefix delegation, Emergency Authority Credential issuance —” are signed by the EIRA root credential and published in a transparency log. Any Energy Internet participant can verify any EIRA action independently. A fraudulent EIRA action is detectable by any participant within one log publication cycle.

7.2 ERPki Implementation

Energy Resource Public Key Infrastructure (ERPki) provides route origin validation for EBGp. The implementation follows the structure of RPKI for data internet BGP [RFC 6480]:

The EIRA maintains the ERPki root and delegates EAS-level signing authorities to each EAS operator. Each EAS operator issues Route Origin Authorizations (ROAs) for the EIP prefixes it is delegated, signed by its EIRA-delegated EAS-level authority. ENL routers validate EBGp route origins against the ERPki repository. A route origin that does not have a valid ROA is flagged as INVALID and rejected. The ERPki repository is published as a distributed, signed database with full history —” allowing retroactive detection of fraudulent ROA issuance.

7.3 Emergency Authority Credential Security

Emergency Authority Credentials (EAC) enable Emergency Priority Override (EPO) —” the mechanism that reallocates all mesh capacity to CRITICAL sessions during emergencies [Energy Internet, 2026 6.4]. The security of EAC is critical: a compromised EAC enables a denial-of-service attack against all NORMAL and BACKGROUND sessions network-wide.

EAC security measures:

EAC private keys are stored in hardware security modules (HSMs) that require multi-party authorization for key use (M-of-N threshold signing) EPO commands must include a valid timestamp within ± 60 seconds of network time (prevents replay) Each EAC has a maximum EPO duration —” commands that don't include a finite duration are rejected EPO commands are logged in the certificate chain and are publicly auditable EIRA can revoke any EAC within one propagation cycle if compromise is detected

8. The Biological Safety Layer as Physical Security Invariant

8.1 The BSL as the Security Invariant of Last Resort

Guardian Security is a software and protocol framework. A sufficiently sophisticated attacker who can compromise the software stack —” through physical node access (T8), firmware replacement, or an undetected vulnerability —” may be able to bypass Guardian Security's protocol-level defenses.

The Biological Safety Layer (BSL) is the security mechanism that operates below the software stack. It is implemented in hardware circuitry that is independent of the node's microcontroller, firmware, and software gate logic. The BSL's biological presence detection and beam cutoff circuit operates whether the software is running correctly, is crashed, is compromised, or has been replaced.

Formally, the BSL enforces a physical security invariant:

BSL_invariant: $\forall t, \forall \text{positions } p: \text{biological_presence}(p, t) = \text{TRUE} \nrightarrow \text{beam_active_at}(p, t) = \text{FALSE}$

This invariant is enforced by hardware physics, not by software logic. It is the security invariant of last resort —” the property that holds even when every other security mechanism has failed.

8.2 BSL Tamper Resistance

For the BSL_invariant to be a genuine security guarantee, the BSL hardware must be tamper-resistant. The BSL implementation specifies:

Sensor diversity: Biological presence detection uses at least two independent sensor modalities (e.g., radar + thermal). An attacker who blocks one sensor modality does not defeat detection. Anomalous-absence detection: Sensors that go offline or produce anomalous "all-clear" readings when other sensors are uncertain trigger a SENSOR_INTEGRITY_FAILURE event that forces beam hold. The absence of a sensor signal is treated as a safety hazard, not as permission to transmit. Hardware cutoff independence:

The beam cutoff circuit receives power and control signals independent of the main microcontroller. A crashed or compromised MCU does not disable the cutoff. Tamper evidence: The BSL housing includes tamper-evident sealing. A node whose BSL housing has been opened fails the physical inspection at maintenance intervals.

8.3 The Residual Risk Statement

No security system provides zero residual risk. The BSL's residual risk is a T11c attack: an attacker with sustained physical access who surgically modifies the BSL hardware circuitry itself —” bypassing the sensor, severing the cutoff circuit, or replacing the BSL hardware with a dummy that always reports safe. This attack requires:

Physical access to a deployed node Technical knowledge of the specific BSL circuit implementation Time to perform hardware modification without detection Tolerance for the fact that the modification will be detected at the next physical inspection This residual risk is accepted as the irreducible physical security baseline. Mitigations include: physical security of node deployment locations, tamper-evident sealing with regular inspection, and geofenced monitoring that alerts when a node's physical position changes unexpectedly.

9. Operational Security: Deployment and Audit

9.1 Secure Provisioning

Node provisioning is the process by which a new node receives its Trust Layer identity credential. It is the most security-sensitive phase of a node's lifecycle, because a credential issued at provisioning cannot be recalled without revocation, and revocation has a propagation latency.

Guardian Security specifies a five-step secure provisioning process:

Factory key injection: The node's Trust Layer public/private key pair is generated in the factory inside a hardware security module. The private key never leaves the HSM —” it is injected directly into the node's secure element without passing through any host system.

Identity credential signing: The factory generates a Certificate Signing Request using the node's public key and the node's identity parameters (Node_Type, EAS assignment, Zone ID, capability profile). The CSR is submitted to the Trust Layer PKI for signing.

Hardware profile binding: The Trust Layer credential includes a hardware profile hash —” a signed measurement of the node's hardware characteristics (power draw profile, timing characteristics, secure element measurements). A credential used on different hardware will fail the hardware profile consistency check.

Provisioning receipt: The provisioned node signs a provisioning receipt confirming that it successfully received and stored its credential. The receipt is logged in the certificate chain.

First-contact verification: On first network contact, the node presents its credential and the provisioning receipt. Guardian-E verifies both before accepting the node as VERIFIED. A node that cannot present a valid provisioning receipt is not accepted into the mesh.

9.2 The Certificate Chain as Audit Substrate

Every Guardian Security event —” authentication, authorization, anomaly detection, quarantine, restoration —” is logged in the certificate chain with a signed entry from the Lume-X control loop. The certificate chain is:

Append-only: Entries cannot be deleted or modified once written Signed: Each entry is signed by the Lume-X runtime using the node's Trust Layer credential Timestamped: Each entry includes a network timestamp verifiable against the distributed time consensus Cross-referenced: Each entry references the EBF sequence numbers and session IDs of the transactions it relates to This produces a complete, auditable history of every security event in the mesh. For regulatory compliance (safety certification of wireless power systems), insurance liability attribution, and forensic investigation following an incident, the certificate chain provides the definitive record.

9.3 Incident Response Protocol

When Guardian-E detects a suspected attack, the incident response protocol is:

Immediate containment: The suspect node or path is QUARANTINED within one control cycle. No traffic is routed through the quarantined node or path until the quarantine is lifted.

Evidence preservation: All certificate chain entries related to the suspect node's recent activity are cryptographically sealed (signed with a timestamp by the Lume-X runtime) to prevent tampering.

Affected session recovery: All sessions that were routing through the quarantined node are rerouted through verified alternate paths. CRITICAL sessions receive priority in rerouting.

Threat classification: Guardian-E classifies the detected anomaly against the T1—“T11 taxonomy and logs the classification in the certificate chain.

Administrator notification: A signed alert is transmitted to the designated administrator endpoint (the EaaS account holder or the EAS operator) specifying the quarantined node, the threat classification, and the evidence preserved.

Quarantine review: The administrator reviews the evidence and either lifts the quarantine (after verifying the node is not compromised) or initiates physical inspection and credential revocation.

10. Related Work

10.1 Wireless Network Security

The wireless data network security literature is extensive and directly relevant to the protocol-layer defenses in Guardian Security. IEEE 802.11i (WPA2/3) [35] provides the reference architecture for wireless network authentication and key management. The 5G security architecture [36] provides the reference for subscriber identity management and authentication at scale. Guardian Security's pre-transmission authentication protocol draws on both, adapted for the energy delivery context where the "data" being protected is joules rather than bits.

The critical distinction is that wireless network security assumes a stable separation between the control plane (authentication, key management) and the data plane (packet delivery). Guardian Security operates in an environment where the control plane measurement (delivery confirmation, localization) is also a physical measurement of the data plane output. This coupling is the source of T6 (delivery confirmation falsification) and T1 (localization spoofing), which have no analog in wireless data network security.

10.2 Industrial Control System Security

ICS security [37, 38] addresses security for systems where cyber-physical coupling creates physical consequences for security failures —” exactly the challenge Guardian Security addresses. The MITRE ATT&CK for ICS framework [39] provides a structured taxonomy of attacks against industrial control systems that informed the development of the T1—“T11 threat taxonomy. The Stuxnet case [40] is the canonical demonstration that cyber attacks on physical control systems can produce physical damage —” the exact threat class that Guardian Security's BSL invariant is designed to make impossible.

The key distinction from ICS security is that Guardian Security operates on a wireless, distributed, autonomously administered mesh rather than a fixed, wired, centrally administered control network. ICS security assumes network perimeter control —” the control network is air-gapped or firewalled. Guardian Security cannot assume perimeter control because the RF transmission medium is inherently accessible to any attacker within range.

10.3 Physical Layer Security

Physical layer security research [41, 42] exploits the physical properties of wireless channels —” multipath characteristics, spatial diversity, channel reciprocity —” to provide security guarantees that cannot be bypassed by higher-layer attacks. This approach is complementary to Guardian Security's protocol-layer defenses.

The localization integrity verification in Guardian Security (multi-source position consistency checking) draws on physical layer security principles: the physical channel characteristics (path length, multipath profile, angle of arrival) are independent measurements that a position spoof must be consistent with.

Spoofing a UWB localization signal while maintaining consistency with RF received signal strength, angle-of-arrival measurements, and mesh neighbor distance data simultaneously is significantly harder than spoofing any single measurement.

10.4 The Lume Ecosystem Security Framework

The Trust Layer [16] provides the identity foundation that Guardian Security builds on. Guardian (the general, non-domain-specific security enforcement product) provides the general threat classification and adaptive response framework. Guardian-E is the energy-domain specialization that adds the physical-layer defenses (BSL integration, localization integrity, power measurement validation) that are specific to the energy routing context.

Guardian Security is the first published formal specification of a Guardian deployment in a physical domain. The security architecture defined in this paper is the reference for future Guardian deployments in the transportation (DMI), manufacturing (DPI), and ambient computation (DII) domains described in [DI Theory, 2026].

11. Limitations and Honest Boundaries

The threat taxonomy covers the current architecture, not all future attacks. The T1—“T11 taxonomy is derived from the attack surface of the current Meridian architecture. Novel attack techniques not anticipated in this analysis may find gaps. The ERFC process for Guardian Security standards updates (Appendix B) includes a vulnerability disclosure process and a taxonomy update mechanism for new threat categories.

Physical node compromise (T8) has limited software-layer defenses. An attacker with sustained physical access to a node and sufficient technical knowledge can defeat the BSL hardware safeguards. This is the irreducible residual risk of any physically deployed system. The assumption of physical security for high-value nodes (source nodes, gateway nodes, CRITICAL-priority relay nodes) is a prerequisite for the Guardian Security guarantees.

Coverage proof assumes correct implementation. The coverage proof in Section 6.2 demonstrates that every attack path crosses at least one Guardian Security enforcement boundary. It does not prove that the implementation of those boundaries is correct. Implementation vulnerabilities —” software bugs, cryptographic errors, side-channel attacks on secure elements —” can create gaps in coverage that are not present in the specification. Lume-V formal verification [18] of the Guardian Security implementation is the mitigation; it has not yet been completed for all components.

Multi-party compromise is not addressed. The Guardian Security framework assumes that the Trust Layer PKI root is not compromised. A compromise of the Trust Layer root credential —” enabling issuance of fraudulent credentials for any identity —” would defeat every cryptographic defense in Guardian Security

simultaneously. This is the Tier 1 existential threat to any PKI-based system and requires organizational security controls beyond the scope of this technical paper.

No experimental validation exists. Guardian Security has been formally specified but not deployed in a physical mesh. The timing guarantees (anomaly detection within 13.7 ms, quarantine within one control cycle), the physical layer consistency checks (position variance threshold, power measurement cross-validation), and the BSL sensor integration have not been validated in hardware. All quantitative claims are based on architectural analysis of the Meridian specification.

12. Conclusion

Wireless energy routing is not a data routing problem with a different payload. It is a fundamentally different security domain —” one where successful attacks produce physical consequences, where the control plane is coupled to physical measurements, and where identity must be verified before the first joule flows rather than after the first packet arrives.

No formal security framework for this domain previously existed. This paper provides one.

The Guardian Security framework —” two-tier innate-plus-adaptive defense, pre-transmission authentication, eleven-category threat coverage, BSL physical invariant, and certificate chain audit —” addresses the complete attack surface of a deterministic wireless energy routing system. It provides four formal security guarantees: authenticity, integrity, availability, and non-repudiation. The coverage proof demonstrates that every attack path requires traversal of at least one enforcement boundary.

The most important security property of the Guardian Security framework is the one that cannot be improved through protocol design: the Biological Safety Layer invariant. The BSL guarantees that no beam strikes a biological target regardless of the state of every software and protocol defense. This guarantee is the irreducible physical security baseline of wireless energy routing —” the property that makes the difference between a security failure that damages data and a security failure that damages people.

Guardian Security is a formal specification. It requires experimental validation, implementation in the Meridian architecture, and integration with the Trust Layer PKI infrastructure. Those steps are the work of the experimental phases defined in [Meridian Architecture, 2026]. This paper establishes the specification so that when the hardware exists, the security architecture is already defined.

The Energy Internet cannot be built without this foundation. A global wireless energy routing network with a complete absence of formal security architecture would be an infrastructure disaster —” the energy equivalent of the early internet, where the absence of security design produced decades of vulnerability. Guardian Security is the security foundation that the Energy Internet needs to be built on, stated now, so that it is built in from the first deployment rather than retrofitted from the last breach.

Appendix A —” Complete Threat Taxonomy

ID	Name	Attack Interface	Objective	Max Consequence	Guardian-E Tier
T1	Localization Spoofing	PAI-2, NAI-2	Beam misdirection	Physical injury	Adaptive + BSL
T2	Routing Table Manipulation	NAI-1	Traffic redirection	All traffic hijacked	Adaptive
T3	Beam Hijacking	PAI-1, NAI-2, IAI-1	Attacker receives authorized beam	Critical device denied power	Adaptive
T4	Denial-of-Power	NAI-1, NAI-2, PAI-1	Legitimate devices denied service	Simultaneous mesh blackout	Innate + Adaptive
T5	Node Impersonation	IAI-1, NAI-1	Attacker receives legitimate allocation	Mesh infiltration	Adaptive
T6	Delivery Confirmation Falsification	NAI-3	False accounting	Systematic model corruption	Adaptive
T7	Timing Attacks	NAI-1, NAI-2	Scheduling disruption	Effective denial-of-power	Innate + Adaptive
T8	Physical Node Compromise	PAI-3	Insider access	All attacks from trusted position	Adaptive (limited)
T9	Energy Credit Fraud	NAI-2, IAI-2	Financial fraud	Economic disruption of ESPs	Adaptive
T10	EBGP Route Hijacking	NAI-4	Internet-scale traffic redirection	EAS isolation or theft	Adaptive (L2)
T11	Safety System Bypass	PAI-3, NAI-2	BSL disabled	Physical injury (worst case)	BSL (hardware)

Appendix B —” Guardian Security Rule Set (Summary)

Rule ID	Description	Threat Covered	Enforcement Point
GS-R01	All participants verify PKI credential before interaction	T5	Pre-transmission auth Step 1
GS-R02	All routing updates require valid signature	T2	NAI-1 input validation
GS-R03	Position consistency check across ≥ 2 sources	T1	Pre-transmission auth position check
GS-R04	Delivery confirmation physics cross-validation	T6	Post-delivery confirmation handler
GS-R05	One-credential-one-position enforcement	T5	NODE_ANNOUNCE handler
GS-R06	Per-node request rate limiting	T4	ENL request scheduler
GS-R07	CRITICAL priority capacity reservation	T4	ETL QoS enforcer
GS-R08	Signed TDMA schedule distribution	T7	Scheduling epoch handler
GS-R09	Out-of-window transmission detection	T7	ELL receive handler
GS-R10	Relay forwarding timeout detection	T3	DRMA relay monitor
GS-R11	Trajectory consistency monitoring	T3	Session state tracker
GS-R12	Dual-signature settlement requirement	T9	Settlement handler
GS-R13	ERPKI route origin validation	T10	EBGP UPDATE handler
GS-R14	Hardware profile consistency check	T8	Boot integrity verifier

GS-R15	Multi-modal sensor cross-validation	T11	BSL sensor fusion
GS-R16	Sensor absence treated as safety hazard	T11	BSL safety logic
GS-R17	Hardware-enforced beam cutoff	T11	BSL hardware circuit

Appendix C —” Coverage Proof by Attack Path

For each threat T1–“T11, the minimum set of Guardian Security rules that must be k

Threat	Rules That Must Be Simultaneously Bypassed
T1	GS-R03 AND GS-R15 AND GS-R16 AND GS-R17
T2	GS-R02 AND GS-R11
T3	GS-R01 AND GS-R10 AND GS-R11
T4	GS-R06 AND GS-R07
T5	GS-R01 AND GS-R05 AND GS-R14
T6	GS-R04 AND GS-R01 (confirmation signature)
T7	GS-R08 AND GS-R09
T8	GS-R14 AND physical tamper evidence
T9	GS-R12 AND GS-R13 AND sequence dedup
T10	GS-R13 AND topology cross-reference
T11	GS-R15 AND GS-R16 AND GS-R17 (hardware circuit)

The minimum attack complexity for each threat is the AND of independent enforcement mechanisms. No single rule bypass suffices for any threat category.

Appendix D —” Guardian Security at Three Operational Levels

Level	Scope	Authentication	Key Management	Threat Coverage
L1 —” Intra-mesh	Single EAS	Per-node PKI credential	Trust Layer EAS-delegated CA	T1—“T8, T11
L2 —” Inter-mesh	Multi-EAS, single EXP	EAS-level PKI credential	Trust Layer root-delegated EAS CA	T9—“T10 + L1
L3 —” Registry	Global	EIRA root credential	EIRA HSM, M-of-N threshold signing	EIRA operation integrity

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Part Four Closing

Chapter Six has established the formal security architecture for Meridian: the attack surface, the eleven threat categories, the Guardian Security defense framework, the coverage proof, and the Biological Safety Layer as the physical invariant of last resort.

The security architecture is specific to the energy domain: it addresses the physical consequences of energy routing attacks, the physical delivery confirmation coupling that has no data networking analog, and the pre-transmission identity requirement that energy routing imposes and data routing does not.

But the principles underlying Guardian Security —” verified identity before interaction, explicit routing with confirmed delivery, invariant enforcement at every layer, and organism-like self-healing after security events —” are not specific to energy routing. They are the security consequences of a more general set of properties that define a class of systems much larger than Meridian.

Part Five names that class.

PART FIVE

The Theory: What Meridian Belongs To

Classification is the beginning of science. To name the class that a system belongs to is to gain access to everything else in that class —” the other instances, the formal properties they share, the predictions that follow from the class membership, and the design principles that any future instance must satisfy.

Meridian belongs to the class of Deterministic Infrastructure systems: autonomous physical systems defined by four properties —” verified identity at every node, explicitly routed resources, invariant-enforced homeostasis, and organism-like self-maintenance. Every system that satisfies all four properties is a Deterministic Infrastructure instance, regardless of what physical domain it governs.

Chapter Seven names this class, defines its properties formally, identifies the existing instances beyond Meridian, predicts the future instances, and argues that Deterministic Infrastructure is the design paradigm for the next generation of physical infrastructure —” not one option among several, but the necessary response to a world that will require physical infrastructure to operate autonomously at a scale that no human workforce can supervise.

Chapter Seven

Deterministic Infrastructure: A General Theory

Six chapters describe a specific system. This chapter describes what that system is a member of.

Meridian routes energy. The Trust Layer routes trust. The DAIGS framework routes computation. These are three systems in three different physical domains, built on the same foundational axioms, exhibiting the same organizational structure, and satisfying the same four formal properties. They are three instances of the same class.

The class is Deterministic Infrastructure. The properties are: verified identity at every node, explicitly routed resources, invariant-enforced homeostasis, and organism-like self-maintenance. Any autonomous physical system that satisfies all four properties —” in any physical domain, at any scale —” is a Deterministic Infrastructure system.

The 42 Assumptions of the Lume ecosystem, specified in the Lume Language Specification, are the axioms of Deterministic Infrastructure. Any system built on these axioms, in any physical domain, will converge toward all four properties as a logical consequence of the axiomatic foundation. This is why the Lume ecosystem has produced three confirmed Deterministic Infrastructure instances across different domains without designing each from scratch: the axioms make non-Deterministic-Infrastructure architecture impossible to build correctly on Lume.

The chapter argues four claims. First, that Deterministic Infrastructure is the design paradigm for the next generation of physical infrastructure, because the infrastructure paradigms currently in place —” best-effort delivery, central coordination, passive components —” cannot serve a world of trillions of autonomous devices operating at millisecond timescales. Second, that the convergence toward biological organism organization, demonstrated for Meridian in Chapter Three, is a general property of any Deterministic Infrastructure system, not a feature of energy routing specifically. Third, that four future domains —” transportation, manufacturing, emergency coordination, and ambient computation —” will produce Deterministic Infrastructure instances over the next decade. Fourth, that the Lume ecosystem is the first complete framework for building Deterministic Infrastructure across physical domains.

This is the broadest claim in the book. It rests on three confirmed instances and an argument from first principles. The experimental confirmation will come from the deployment of those instances at scale. The argument is made now, before the experiments, so that when the data arrives, there is a theoretical framework to interpret it.

Abstract

Three papers precede this one. The first defined Meridian —” a four-layer deterministic wireless energy routing architecture [Meridian Architecture, 2026]. The second demonstrated that Meridian is structurally isomorphic to a biological organism —” the first complete physical instantiation of the Lume Synthetic Organism model [Meridian Organism, 2026]. The third extended Meridian toward a universal protocol standard for energy delivery at internet scale [Energy Internet, 2026]. Each paper made a contribution to a single physical domain: wireless energy routing.

This paper steps back from the energy domain and asks what these three papers collectively demonstrate about autonomous physical infrastructure in general.

I argue that Meridian is one instance of a general class of systems I call Deterministic Infrastructure —” autonomous physical systems that operate under four defining properties: verified identity at every node, explicitly routed resources through topology-aware mesh protocols, invariant-enforced homeostasis at every operational layer, and organism-like self-maintenance in the absence of external supervision. I demonstrate that this class is not unique to the energy domain. The Trust Layer ecosystem [16] is a Deterministic Infrastructure instance in the identity and data domain. The DAIGS framework [17] is a Deterministic Infrastructure instance in the multi-agent computation domain. Future instances in the transportation, manufacturing, and emergency coordination domains are predicted and characterized.

The 42 Assumptions [15] —” the foundational axioms of the Lume ecosystem —” are the universal axioms of Deterministic Infrastructure. Any system built on them, in any physical domain, will converge toward the same organizational form: the Synthetic Organism model [Meridian Organism, 2026]. The Lume language [15] is the substrate that makes this convergence repeatable and buildable rather than emergent and accidental. Lume-X [19] is the control runtime that enforces it at operational frequencies. Lume-V [18] is the verification layer that proves it before deployment.

The central claim of this paper is this: the world is about to need a new generation of physical infrastructure —” infrastructure that operates autonomously, heals without human intervention, routes resources rather than broadcasting them, and maintains identity and governance integrity without central authority. The existing infrastructure paradigms —” the electrical grid, the internet, the supply chain, the transportation network —” were designed for a world of human operators, central coordination, and wired physical plants. They are not adequate for a world of trillions of autonomous devices, distributed physical operation, and adversarial environments.

Deterministic Infrastructure is the design paradigm for that new world. The Lume ecosystem is the first complete framework for building it.

Keywords: deterministic infrastructure, autonomous physical systems, verified identity, invariant enforcement, organism-like self-maintenance, resource routing, Lume, Meridian, Trust Layer, DAIGS, general theory, design paradigm, cyber-physical systems

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1. Introduction

1.1 The Problem This Series Has Been Building Toward

The three papers that precede this one addressed a specific engineering problem: how to route wireless energy deterministically through a distributed mesh of autonomous nodes. They produced a four-layer architecture (Meridian), a classification of that architecture as a Type 3+ Synthetic Organism, and a protocol standard for scaling that architecture to internet scope. The engineering contributions of those papers stand independently.

But they were always building toward something larger. The engineering problem was chosen because it is maximally constrained —” wireless energy routing forces every design decision to be explicit, every invariant to be enforced in hardware, and every failure mode to be planned for before deployment. A system that solves this problem correctly has necessarily solved a more general set of problems: how to operate autonomously in a physical environment, how to maintain identity and governance integrity without central coordination, how to route limited resources through a distributed topology, and how to heal from failure without human intervention.

These are not energy-specific problems. They are infrastructure problems. And the solutions, as I will argue in this paper, are not energy-specific solutions. They are general principles that apply to any physical infrastructure domain in which autonomous operation is required.

I call the class of systems defined by these principles Deterministic Infrastructure. Meridian is the first fully specified instance. It will not be the last.

1.2 Why Infrastructure Is Failing

The infrastructure paradigms that govern the physical world were designed in the 20th century, for 20th-century operating conditions. The electrical grid was designed for central generation and passive consumption. The internet was designed for best-effort data delivery between human-operated computers.

The supply chain was designed for bulk transport between fixed facilities. The transportation network was designed for human-driven vehicles following human-readable signage.

Each of these paradigms has three properties in common that are becoming problems:

Central coordination. The grid is managed by ISOs and RTOs. The internet's routing tables are maintained by human network engineers. Supply chains are orchestrated by human logistics coordinators. Transportation infrastructure is managed by municipal governments. Central coordination has a single point of failure: the coordinator. When coordination fails —” a grid operator loses situational awareness, a border is closed, a logistics system is overwhelmed —” the infrastructure fails with it.

Passive components. A power line does not know what is connected to it. A road does not know what is driving on it. A network router knows its next-hop table but not the health of the components it serves. Infrastructure components in the current paradigm are passive conveyances. They carry what is given to them. They do not negotiate, authenticate, route around failure, or maintain their own governance integrity.

Designed for human timescales. A grid fault takes minutes to an hour to resolve, because human operators are in the resolution loop. A supply chain disruption takes days to weeks, because human planners are rerouting. A network outage is resolved in hours, because human engineers are diagnosing. These timescales made sense when the systems they governed also operated at human timescales. They do not make sense for a world of autonomous devices that operate at millisecond timescales and cannot wait for human intervention.

The world is accumulating devices that need infrastructure operating at their timescale: autonomous vehicles that need routing decisions in under 100ms, medical wearables that need continuous power without battery changes, industrial robots that need energy and computation allocation without human scheduling, sensor networks that need to heal around node failures without a network engineer on site. The 20th-century infrastructure paradigms cannot serve these devices. They were not designed to.

1.3 What Deterministic Infrastructure Is

Deterministic Infrastructure is the design paradigm that addresses this gap. It defines autonomous physical systems with four properties:

Verified identity at every node. Every component of the infrastructure knows who it is, can prove it cryptographically, and refuses to interact with components that cannot prove their identity. Identity is not assigned by a central authority and looked up —” it is embedded at manufacture, verified locally, and enforced at every interaction.

Explicitly routed resources. Resources —” energy, computation, bandwidth, attention, material —” are routed through a topology-aware mesh rather than broadcast, pushed, or transferred without addressing. Routing is deterministic: the same conditions always produce the same route. Routing tables are maintained by the infrastructure itself, updated when topology changes, and verified against invariants before being applied.

Invariant-enforced homeostasis. A fixed set of global invariants —” conditions that must be true of the system at all times —” is continuously enforced by overlapping monitoring and enforcement mechanisms at every layer. No operation that would violate an invariant is permitted. Invariant violations are detected within one control cycle and trigger automatic recovery. The system does not drift; it is continuously pulled back toward its defined operating envelope.

Organism-like self-maintenance. The infrastructure heals its own failures, preserves its own identity, adapts its own behavior to environmental changes, and requests supplementation when its resources are depleted —” all without external supervision. It exhibits every property of a Type 3+ Synthetic Organism [Meridian Organism, 2026] because it faces the same design constraints that produced those properties in biological organisms.

These four properties together define a system that can operate indefinitely in a physical environment without human intervention, that cannot be deceived about its own identity or the identity of its peers, that cannot enter an unsafe or incoherent state without immediately detecting and correcting it, and that routes the resources it controls to where they are needed rather than wasting them on components that don't need them.

1.4 The Contribution of This Paper

I make four contributions:

First, I formally define Deterministic Infrastructure as a class, with the four properties above as the membership criteria. I show that membership in this class is binary —” a system either satisfies all four properties or it does not —” and that partial satisfaction corresponds to specific, predictable failure modes.

Second, I demonstrate that the Lume ecosystem already contains at least three confirmed instances of Deterministic Infrastructure across different physical domains: Meridian in the energy domain, the Trust Layer in the identity and data domain, and DAIGS in the multi-agent computation domain.

Third, I show that the 42 Assumptions are the universal axioms of Deterministic Infrastructure —” that any system built on them, in any physical domain, will converge toward all four properties as a natural consequence of the axiomatic foundation.

Fourth, I characterize four future Deterministic Infrastructure domains —” transportation, manufacturing, emergency coordination, and ambient computation —” and predict the organizational form that Deterministic Infrastructure will take in each.

1.5 Paper Organization

Section 2 characterizes the failure modes of current infrastructure paradigms. Section 3 defines the four properties formally. Section 4 examines the 42 Assumptions as universal axioms. Section 5 connects the Synthetic Organism model to Deterministic Infrastructure. Section 6 surveys existing instances across the Lume ecosystem. Section 7 explains why Lume makes convergence repeatable. Section 8 addresses cross-domain interoperability. Section 9 predicts future domains. Section 10 discusses design implications. Section 11 situates this in related literature. Section 12 states limitations. Section 13 concludes.

2. Background: What Current Infrastructure Cannot Do

2.1 The Best-Effort Paradigm

The internet's foundational architectural choice was best-effort delivery. The IP layer routes packets toward their destination using the best currently available path, but makes no promise that they will arrive. TCP adds reliability at the transport layer by detecting lost packets and requesting retransmission. The application layer adds security, authentication, and ordering. Infrastructure concerns are layered from the bottom up: each layer compensates for the inadequacies of the layer below.

This design was correct for the 1970s. Networks were unreliable, bandwidth was scarce, and the only traffic was data between human-operated computers. A packet that arrived late was inconvenient. A packet that never arrived could be retransmitted. The human at the other end would not notice a delay of seconds or even minutes.

The best-effort paradigm produced the most successful communication infrastructure in human history. It also encoded a set of assumptions that are now false for the systems that need to run on infrastructure:

There is time to retransmit. (Autonomous vehicles and medical devices cannot wait for TCP retransmission.) Loss is acceptable. (Energy that is not delivered is energy the receiving device does not have —” it cannot request retransmission.) Identity can be handled at the application layer. (A node that cannot authenticate itself before its first packet is a security gap that cannot be closed post-hoc.) Central routing tables are maintained by operators. (A trillion IoT nodes cannot be individually configured by human network engineers.) Failure is recovered by humans. (A self-healing network cannot wait for a human to diagnose the fault.) These assumptions are not minor caveats. They are load-bearing assumptions of the entire infrastructure paradigm. When they fail —” and for autonomous physical systems, they all fail —” the infrastructure fails with them.

2.2 The Central Control Paradigm

Physical infrastructure —” power grids, water systems, transportation networks, manufacturing plants —” was designed on the central control paradigm. A supervisor (human or algorithmic) has global visibility into the system state and issues commands to passive components. The components execute commands and report status. Intelligence lives at the center; execution lives at the edges.

This paradigm has a single, well-known failure mode: loss of the center. When the supervisor loses visibility, the system loses coherence. The 2003 North American blackout cascaded through 55 million people because a single control center lost situational awareness for forty-five minutes. The 2021 Texas grid failure killed hundreds of people because central planning assumptions about weather resilience were wrong. Supply chain collapses during the 2020—“2022 pandemic exposed the fragility of centrally optimized logistics systems to simultaneous disruption of multiple inputs.

The central control paradigm is also fundamentally incompatible with the scale of autonomous device deployment. A trillion IoT nodes cannot report to a central supervisor. The communication bandwidth alone is prohibitive. A city of autonomous vehicles cannot coordinate through a central traffic server with any reliability guarantee. A distributed manufacturing network cannot wait for central allocation decisions at every resource request. Scale forces distribution. Distribution requires that intelligence lives at the edges, not at the center.

2.3 The Gap Deterministic Infrastructure Fills

Deterministic Infrastructure fills the gap between best-effort distributed systems (which cannot guarantee physical resource delivery or safety) and central control systems (which cannot scale and cannot tolerate supervisor failure). It provides the guarantees of centralized control —” deterministic outcomes, invariant enforcement, identity verification, safe operation —” through distributed, autonomous mechanisms that do not require a center.

This is not a new observation in the abstract. Self-organizing systems, autonomic computing [24], and distributed control theory have all addressed aspects of this problem. The contribution of Deterministic Infrastructure as a framework is specificity: it names the four properties precisely, shows that they are jointly necessary and sufficient for autonomous physical operation, and provides in the Lume ecosystem a concrete, buildable substrate for achieving all four simultaneously.

3. The Four Properties of Deterministic Infrastructure

I state the four properties formally here and analyze each.

3.1 Property 1: Verified Identity at Every Node

Formal definition: Every component node N in a Deterministic Infrastructure system possesses an identity $I(N)$ that is: (a) established at manufacture or provisioning and immutable thereafter; (b) cryptographically bound —” provable by N and verifiable by any peer with access to the appropriate public key material; (c) capability-encoding —” $I(N)$ specifies what N can do, not just what N is called; and (d) propagated —” the network's model of N is derived from $I(N)$ and updated when N 's state changes, not when an external authority decides to update a registry.

Why this property is necessary. A system without verified node identity cannot distinguish legitimate from illegitimate participants. It cannot enforce routing policy (route energy only to authenticated receivers), governance policy (allow only licensed nodes to serve as gateways), or safety policy (exclude unverified devices from proximity to high-power paths). Without verified identity, every other property of Deterministic Infrastructure is bypassed by a node that claims false capabilities, false position, or false authorization.

Meridian implementation. MC 64-bit node identity, provisioned at manufacture, signed by the Trust Layer PKI, capability-encoded in `Node_Type` byte, propagated via `NODE_ANNOUNCE` [Meridian Architecture, 2026 4.2].

Trust Layer implementation. Cryptographic identity bound at entity registration, immutable during entity lifecycle, capability-encoded in trust profile, propagated through the Trust Layer mesh [16].

DAIGS implementation. Agent identity established at instantiation, capability-encoded in agent role profile, cryptographically bound to the DAIGS governance framework [17].

Failure mode without this property. Node impersonation attacks, unauthorized resource access, routing table poisoning by unauthenticated nodes, inability to enforce any governance policy.

3.2 Property 2: Explicitly Routed Resources

Formal definition: Resources R in a Deterministic Infrastructure system are never broadcast, pushed, or transferred without a pre-computed, invariant-validated route. A route $\rho(s, d, t)$ from source s to destination d at time t specifies: the complete path through the network topology, the resource quantity at each hop, the timing constraints at each hop, and the safety envelope that the route must not violate. ρ is computed deterministically —” the same inputs always produce the same route —” and is re-computed when topology changes make the existing route invalid.

Why this property is necessary. Broadcast resource delivery wastes resources on recipients that do not need them and cannot be made safe or accountable —” there is no recipient identity, no delivery confirmation, no route to invalidate if a safety violation is detected. Push delivery without routing produces uncontrolled resource accumulation and denial-of-resource attacks. Only explicit routing provides the addressability, accountability, and controllability required for safe autonomous operation.

Meridian implementation. DRMA multi-hop energy routing with per-hop TDMA scheduling and delivery confirmation [Meridian Architecture, 2026 9], DWER directed beam delivery with closed-loop confirmation [10], MFE burst scheduling with threshold-gated release [5].

Trust Layer implementation. Routed trust relationships through the trust graph rather than broadcast trust assertions. A trust claim travels a defined path from issuer to verifier, with each hop verifiable and the complete path auditable [16].

DAIGS implementation. Routed computation requests through the agent mesh —” computation is allocated to specific agents along specific coordination paths, not broadcast to all available agents [17].

Failure mode without this property. Resource waste, delivery unaccountability, inability to enforce safety envelopes, denial-of-resource vulnerability.

3.3 Property 3: Invariant-Enforced Homeostasis

Formal definition: A Deterministic Infrastructure system maintains a fixed set of global invariants $INV = \{inv_1, inv_2, \dots, inv_n^{TM}\}$ where each inv_μ is a predicate over the system state $S(t)$ that must be true at all times. The system enforces this by: (a) preventing operations that would cause any inv_μ to become false, (b) detecting invariant violations within one control cycle T_{cycle} , and (c) executing pre-specified recovery sequences that restore inv_μ to true without human intervention. The system converges back to the invariant-satisfying state monotonically —” it does not oscillate between valid and invalid states.

Why this property is necessary. Without continuous invariant enforcement, an autonomous system has no guarantee of remaining in a safe operating state over time. Physical systems experience continuous perturbation —” environmental variation, component degradation, adversarial action, timing errors. Without invariant enforcement, perturbations accumulate until the system enters an unsafe state. Invariant enforcement is homeostasis: the continuous application of corrective force to maintain a defined operating envelope despite perturbation.

Meridian implementation. INV-1 through INV-5 enforced at every 13.7 ms Lume-X control cycle: no uncontrolled discharge, no ambiguous routing, no non-deterministic flow, no unsafe transmission, no mesh oscillation [Meridian Architecture, 2026 3.5, 11].

Trust Layer implementation. Trust invariants enforced continuously: no unverified entity participates in governance, no revoked credential remains active, no policy violation persists beyond one enforcement cycle [16].

DAIGS implementation. Governance invariants over agent behavior: no agent exceeds its authorized capability scope, no coordination conflict persists beyond one resolution cycle, no unauthorized inter-agent communication [17].

Failure mode without this property. State drift into unsafe operating conditions, accumulation of undetected errors, inability to provide safety guarantees, eventual catastrophic failure following gradual degradation.

3.4 Property 4: Organism-Like Self-Maintenance

Formal definition: A Deterministic Infrastructure system maintains its own operational integrity without external supervision by implementing the functional equivalents of: self-healing (detecting and recovering from component failure), self-preservation (reducing resource expenditure below a minimum safe level when resources are constrained), adaptive response (adjusting behavior to changed environmental conditions within invariant bounds), and identity persistence (maintaining consistent identity and capability profile despite operational changes).

Why this property is necessary. Any infrastructure that requires human intervention for routine maintenance cannot scale. A trillion-node IoT network cannot have a human administrator for each node. A building-scale energy mesh cannot have a technician on call for every node failure. Self-maintenance is not a convenience feature —” it is a scaling requirement. A system that can only be maintained by humans scales to the number of humans available to maintain it. A system that maintains itself scales to the number of nodes the physics supports.

Meridian implementation. SHDCL self-healing [Meridian Architecture, 2026 12.2], power-conservation mode below $V_{\min}$ (self-preservation), harvest modality shifting and routing path adjustment (adaptive response), MC identity immutability (identity persistence).

Trust Layer implementation. Automatic credential renewal, revocation propagation without central coordinator, identity persistence across entity state changes [16].

DAIGS implementation. Agent recovery after failure, capability reallocation when agents become unavailable, governance continuity during coordinator transitions [17].

Failure mode without this property. Operational dependence on human supervision, inability to scale, gradual degradation as failures accumulate without recovery.

4. The 42 Assumptions as Universal Infrastructure Axioms

The 42 Assumptions [15] were published as the foundational axioms of the Lume ecosystem —” the rules that any Lume-governed system must satisfy to maintain coherence, identity, and autonomous operation. I argued in [Meridian Organism, 2026] that they are structurally equivalent to the axioms of biological organism existence —” the minimum constraints that any autonomous self-maintaining system must satisfy to persist over time.

I now extend that argument: the 42 Assumptions are the axioms of Deterministic Infrastructure. Every one of the four properties defined in Section 3 is entailed by a subset of the 42 Assumptions. Any system built on these axioms, in any physical domain, will satisfy all four properties as a logical consequence of the axioms.

The mapping is direct:

Identity Assumptions †' Property 1 (Verified Identity). The assumptions governing how a Lume system establishes and maintains its identity entail Property 1: identity is fixed at initialization, is cryptographically bound, is capability-encoding, and is propagated faithfully to all models of the system.

Routing Assumptions †' Property 2 (Explicit Routing). The assumptions governing how resources move between components entail Property 2: no resource transfer occurs without a specified path, every path is computed from current topology data, and path validity is checked against safety constraints before any resource flows.

Coherence and Boundary Assumptions †' Property 3 (Invariant Homeostasis). The assumptions governing internal consistency and the system's relationship to its external environment entail Property 3: the system maintains defined internal conditions, detects deviations, and applies corrective force continuously.

Sovereignty and Integrity Assumptions †' Property 4 (Organism-Like Self-Maintenance). The assumptions governing autonomous operation and self-repair entail Property 4: the system recovers from failure without external intervention, preserves its identity under perturbation, and adapts its behavior within defined bounds.

The implication of this mapping is significant: the 42 Assumptions function as an axiomatic foundation for Deterministic Infrastructure in the same way that Euclid's axioms function as an axiomatic foundation for geometry. Just as any system that satisfies Euclid's axioms is a Euclidean geometry regardless of what domain it describes, any system that satisfies the 42 Assumptions is a Deterministic Infrastructure instance regardless of what physical domain it governs.

This means Deterministic Infrastructure is not defined by its physical domain. It is defined by its axiomatic foundation. Energy, data, computation, transportation, manufacturing —” these are different domains, but they are all governed by the same set of axioms when built on Lume. The organizational form that emerges from these axioms is the same in every domain. The physics differs. The axioms do not.

5. The Synthetic Organism as the Organizational Form of Deterministic Infrastructure

In [Meridian Organism, 2026], I demonstrated that Meridian —” a system designed as a wireless energy routing architecture —” converged on biological organism organization without intending to. Every biological subsystem required for autonomous physical operation had a concrete Meridian counterpart. The convergence was not designed; it emerged from the design constraints.

I argued in that paper that deterministic autonomous systems converge toward biological organization because biological organization is the natural attractor of the design space for systems that must acquire resources, transport them internally, coordinate distributed components, defend against threats, maintain identity, and heal from damage.

The same argument applies to Deterministic Infrastructure in general. Every Deterministic Infrastructure system must:

Acquire resources from its operating environment (DAEH in Meridian; identity assertions in the Trust Layer; computation requests in DAIGS) Transport those resources internally to where they are needed (DRMA in Meridian; trust propagation in the Trust Layer; agent coordination in DAIGS) Coordinate distributed components toward coherent behavior (MMF in Meridian; policy synchronization in the Trust Layer; agent mesh orchestration in DAIGS) Defend against threats to its integrity and identity (Guardian-E in Meridian; Guardian in the Trust Layer; DAIGS governance enforcement) Maintain identity and capability profile despite operational changes (MC identity in Meridian; entity persistence in the Trust Layer; agent identity in DAIGS) Heal from component failure without external intervention (SHDCL in Meridian; credential renewal in the Trust Layer; agent recovery in DAIGS) These are the same requirements that drive biological organism organization. Every Deterministic Infrastructure instance faces them. Every Deterministic Infrastructure instance will therefore converge toward the same organizational form: the Synthetic Organism [Meridian Organism, 2026].

This is a strong prediction: every correctly specified Deterministic Infrastructure system, regardless of physical domain, will exhibit genome-like identity, metabolism-like resource flow, nervous-system-like coordination, immune-system-like defense, homeostasis-like invariant enforcement, and self-healing-like failure recovery. Not because its designers intended biological mimicry. Because the requirements force it.

Classification. Under the Synthetic Organism Type 0—“5 scale [Meridian Organism, 2026 7]:

DI Instance	Current Classification	Notes
Meridian (energy)	Type 3+ structural, Type 5 structural	Awaiting experimental confirmation
Trust Layer (identity/data)	Type 3 confirmed	Deployed, operationally self-maintaining
DAIGS (multi-agent)	Type 2—“3	Adaptive and self-maintaining in simulation
Future transportation DI	Type 2 predicted	Dynamic routing, but limited self-healing
Future manufacturing DI	Type 3 predicted	Strong homeostasis requirements

The Trust Layer is, by this classification, the most operationally confirmed Deterministic Infrastructure instance in the Lume ecosystem —” it is deployed, it self-maintains, and it enforces its invariants in production. Meridian has the more complete architectural specification. DAIGS bridges the computation domain. Together they constitute the beginning of a Deterministic Infrastructure ecosystem.

6. Domain Instances: Where Deterministic Infrastructure Already Exists

6.1 Energy Domain: Meridian

Meridian [Meridian Architecture, 2026] is the most fully specified Deterministic Infrastructure instance. It implements all four properties:

Verified Identity: MC 64-bit node identity, Trust Layer PKI, capability-encoded Node_Type Explicit Routing: DRMA multi-hop routing, DWER directed beam delivery, MFE burst scheduling Invariant Homeostasis: INV-1 through INV-5, Lume-X 73 Hz enforcement, SHDCL continuous monitoring Organism Self-Maintenance: SHDCL self-healing, power-conservation mode, harvest modality shifting Meridian operates in the energy domain —” the resource it routes is stored electrical energy delivered as directed RF bursts. The Energy Internet [Energy Internet, 2026] extends this to an internet-scale protocol standard for energy routing across independently administered meshes. Meridian is the reference implementation; the Energy Internet is the standard.

6.2 Identity and Data Domain: Trust Layer

The Trust Layer [16] is a Deterministic Infrastructure instance in the identity and data domain. It routes trust relationships —” not energy, not bits in the data-internet sense, but trust assertions: claims about identity, capability, authorization, and governance status.

In the Trust Layer, the resource being routed is trust itself. A trust relationship between entity A and entity B is not broadcast —” it is established, verified, and propagated along specific paths through the trust graph. Revocation is routed too: when an entity's credential is revoked, the revocation propagates through

the same trust graph that carried the original credential, reaching every node that holds the credential within one propagation cycle.

The Trust Layer satisfies all four Deterministic Infrastructure properties:

Verified Identity: Cryptographic identity bound at entity registration, immutable during entity lifecycle
Explicit Routing: Trust relationships propagated along defined paths through the trust graph; no trust assertion is accepted that cannot be traced to its origin along a valid trust path
Invariant Homeostasis: No revoked credential remains active; no unauthorized entity participates in governance; no policy violation persists beyond one enforcement cycle
Organism Self-Maintenance: Automatic credential renewal, revocation propagation without central coordinator, identity persistence across entity state changes
 The Trust Layer is the oldest and most deployed Deterministic Infrastructure instance in the Lume ecosystem. Its operational experience provides the empirical foundation for the claim that Deterministic Infrastructure is viable, not merely theoretical.

6.3 Multi-Agent Computation Domain: DAIGS

The DAIGS framework [17] is a Deterministic Infrastructure instance in the multi-agent computation domain. The resource it routes is computation —” the allocation of computational work across a mesh of agents, each with defined capabilities, identity, and governance constraints.

DAIGS routes computation requests through the agent mesh rather than broadcasting them to all available agents. An agent that receives a computation request it cannot fulfill does not forward it blindly —” it routes it toward agents with the appropriate capability profile, following the DAIGS coordination protocol. Resource allocation (which agent handles which request) is deterministic: the same capability profile and the same mesh state produce the same allocation decision.

DAIGS satisfies all four properties:

Verified Identity: Agent identity established at instantiation, capability-encoded in agent role profile
Explicit Routing: Computation requests routed through the agent mesh toward capability-matched agents
Invariant Homeostasis: No agent exceeds authorized capability scope; no coordination conflict persists
Organism Self-Maintenance: Agent recovery after failure, capability reallocation when agents become unavailable
 The DAIGS framework provides proof of concept for Deterministic Infrastructure in a domain that is already operating at scale —” multi-agent AI systems. It also demonstrates that the Lume ecosystem spans the digital-physical boundary: Trust Layer and DAIGS are primarily digital; Meridian is the first primarily physical instance.

7. The Lume Substrate: Why the Convergence Is Repeatable

The convergence of Meridian, the Trust Layer, and DAIGS toward Deterministic Infrastructure organization is not coincidental. All three were built on Lume [15] —” the same language, the same 42 Assumptions, the same runtime substrate. The convergence is a property of the foundation, not of the specific domain.

This has a practical implication that distinguishes the Lume ecosystem from other approaches to autonomous system design: the convergence is repeatable. A designer who begins a new Deterministic Infrastructure instance using Lume does not need to independently discover the four properties, the Synthetic Organism organizational form, or the governance requirements. They are encoded in the axiomatic foundation of the language. The designer starts from the endpoint rather than discovering it through trial and error.

This is the Lume advantage, stated precisely:

Any autonomous physical system built on Lume and the 42 Assumptions will converge toward all four Deterministic Infrastructure properties as a logical consequence of the axiomatic foundation, regardless of the physical domain in which it operates.

The mechanism of convergence is the 42 Assumptions themselves: they prohibit the design choices that lead to non-deterministic, identity-unverified, invariant-violating, and externally-dependent systems. A designer using Lume who attempts to build a system that broadcasts resources without routing finds that the boundary assumptions prevent it. A designer who attempts to build a system without identity verification finds that the sovereignty assumptions require it. The 42 Assumptions are not guidelines —” they are constraints. They make non-Deterministic Infrastructure impossible to build correctly on Lume.

Lume-X [19] enforces Property 3 (invariant homeostasis) at runtime: the 73 Hz control cycle that checks INV-1 through INV-5 in Meridian is the general Lume-X invariant enforcement mechanism applied to an energy-domain invariant set. Any Lume-X-governed system has the same enforcement mechanism applied to its domain-specific invariant set.

Lume-V [18] provides formal verification of Properties 1, 2, and 3 before deployment: it proves that the specified system satisfies its identity constraints, its routing constraints, and its invariant constraints under all possible inputs. A Deterministic Infrastructure system verified by Lume-V has formal proof that it cannot violate its invariants —” not empirical confidence, but mathematical proof.

Together, Lume, Lume-X, and Lume-V form a complete substrate for building Deterministic Infrastructure: the language enforces the axioms, the runtime enforces the invariants operationally, and the verifier proves the invariants formally before deployment.

8. Cross-Domain Interoperability

A single Deterministic Infrastructure instance —” Meridian, the Trust Layer, or DAIGS —” is valuable within its domain. Multiple Deterministic Infrastructure instances that interoperate across domain boundaries are qualitatively more powerful.

The Trust Layer and Meridian already interoperate: Guardian Security (Guardian-E in the energy domain, Guardian in the identity domain) uses the Trust Layer PKI to authenticate Meridian nodes. A Meridian node's identity is not Meridian-local —” it is Trust Layer-global. Any system that participates in the Trust Layer can verify a Meridian node's identity without Meridian-specific knowledge. The Trust Layer is the universal identity substrate for all Deterministic Infrastructure instances.

This suggests a general model for cross-domain Deterministic Infrastructure interoperability:

Shared identity substrate. The Trust Layer provides the identity foundation for all DI instances. A node participating in the energy domain (Meridian), the computation domain (DAIGS), or any future domain is the same identity in all of them. When a medical device requests energy from Meridian, its identity is the same identity it uses to request computation from DAIGS and to authenticate with the Trust Layer governance framework. Single identity, multiple domain participation.

Shared governance substrate. The Trust Layer enforces governance policy across all DI instances. A policy that says "devices below trust level 3 cannot receive high-power energy delivery" is enforced at the Trust Layer level and respected by Meridian's routing layer. Policy does not need to be duplicated in each DI instance —” it lives in the Trust Layer and all DI instances enforce it through their Guardian Security integration.

Shared audit substrate. Every event in every DI instance —” every energy delivery, every trust assertion, every computation allocation —” generates a certificate chain entry. The certificate chain is the universal audit log for all DI instances. A compliance query about a specific device's energy consumption, trust interactions, and computation usage across a specific time window can be answered from the unified audit log without querying each DI instance separately.

Cross-domain resource coordination. When a device needs energy (Meridian), computation (DAIGS), and authenticated identity (Trust Layer) simultaneously —” as any autonomous agent in a complex environment will —” the three DI instances coordinate through shared identity and governance to provide all three resources without conflicting. The device does not negotiate with three independent systems. It presents its Trust Layer identity and the three systems coordinate their resource provision.

This cross-domain coordination is the Deterministic Infrastructure equivalent of a converged network: just as a modern communications network carries voice, video, and data over the same physical infrastructure using a shared addressing and routing protocol, a Deterministic Infrastructure ecosystem carries energy,

computation, and trust over the same governance and identity substrate using a shared identity and routing framework.

9. The Prediction: Future Deterministic Infrastructure Domains

Based on the general theory, I identify four domains where Deterministic Infrastructure instances will emerge in the next decade. For each, I characterize the resource being routed, the identity requirements, the invariant set, and the anticipated organism classification.

9.1 Transportation Domain: Deterministic Mobility Infrastructure (DMI)

Resource: Right-of-way —” the access to physical space for movement at a specific time.

Identity requirements. Every vehicle (autonomous or human-operated) possesses a verified identity encoding its physical profile (dimensions, speed envelope, sensor capabilities) and authorization level (road class access, cargo restrictions, emergency priority). A vehicle without verified identity cannot receive right-of-way assignment.

Explicit routing. Right-of-way is not broadcast —” it is allocated along specific paths at specific times. Intersection coordination routes vehicles through conflict zones along non-conflicting temporal paths. Highway merging routes vehicles into gaps in traffic flow computed deterministically from current position and velocity data.

Invariant set. No two vehicles occupy the same physical space at the same time. No vehicle exceeds its authorized speed envelope. No right-of-way assignment persists past its assigned time window.

Organism classification. Type 3 predicted: self-maintaining (vehicles reroute around closures without dispatcher intervention), homeostatic (traffic density invariants enforced continuously), identity-persistent. The transportation DMI is a lower-complexity organism than Meridian because the resource (right-of-way) is simpler to account for than energy, but the homeostasis requirement is equally stringent.

Why Deterministic Infrastructure is needed here. Current autonomous vehicle coordination is either centralized (requiring a coordinator that can fail) or ad-hoc peer-to-peer (without invariant enforcement, producing deadlock and priority inversion at intersections). DMI provides the missing layer: deterministic right-of-way routing with invariant enforcement, implemented through a distributed mesh of roadside anchor nodes rather than central coordination.

9.2 Manufacturing Domain: Deterministic Production Infrastructure (DPI)

Resource: Production capacity —” the access to manufacturing equipment, materials, and qualified operators at a specific time for a specific process step.

Identity requirements. Every production resource (machine, material lot, operator certification) possesses a verified identity encoding its capability profile (process types, tolerances, throughput, certification expiration). A production step cannot be scheduled to a resource whose identity does not include the required capability.

Explicit routing. Production jobs are routed through the manufacturing mesh along capability-matched paths —” not assigned to a queue by a central scheduler, but routed hop-by-hop through available resources based on real-time capability availability. A job that arrives at a resource that becomes unavailable mid-production is rerouted to the next available capability-matched resource without waiting for central rescheduling.

Invariant set. No production step is scheduled to a resource whose capability profile does not include the required process. No material lot is used after its certification expiration. No production schedule creates resource contention that cannot be resolved within one scheduling cycle.

Organism classification. Type 3—“4 predicted. Manufacturing DPI has strong homeostasis requirements (production must continue at a defined throughput despite machine failures and material shortages) and partial self-evolution properties (scheduling algorithms that improve over time based on accumulated throughput data).

9.3 Emergency Coordination Domain: Deterministic Response Infrastructure (DRI)

Resource: Response capacity —” the allocation of emergency personnel, equipment, and supplies to active incidents.

Identity requirements. Every response resource (paramedic, fire unit, hospital bed, supply cache) possesses a verified identity encoding its capability profile and current availability. A resource without verified identity cannot receive a dispatch assignment.

Explicit routing. Response capacity is routed to incidents along capability-matched, time-optimal paths through the response mesh —” not dispatched from a central table, but routed dynamically based on real-time availability and incident priority. An incident that escalates beyond the capacity of initially dispatched resources triggers automatic rerouting of additional capacity without dispatcher intervention.

Invariant set. Every active incident has assigned response capacity above its minimum threshold. No response resource is simultaneously assigned to conflicting incidents. Critical incidents (cardiac arrest, structural fire) receive CRITICAL priority override of non-critical assignments.

Organism classification. Type 3 predicted with strong self-preservation characteristics: the DRI must continue operating during the same emergencies it is coordinating —” it cannot depend on infrastructure that is itself compromised during a disaster. This requirement produces the most demanding self-maintenance requirements of any DI instance discussed.

Relationship to Energy Internet. The Emergency Priority Override mechanism defined in [Energy Internet, 2026 6.4] is the energy-domain instance of the DRI invariant enforcement: during emergency conditions, CRITICAL energy delivery takes priority over all other sessions, and the override is authenticated through Guardian Security emergency authority credentials. DRI generalizes this pattern to all resource domains.

9.4 Ambient Computation Domain: Deterministic Inference Infrastructure (DII)

Resource: Inference capacity —” the allocation of AI inference computation to requesting devices across a mesh of compute nodes.

Identity requirements. Every compute node possesses a verified identity encoding its hardware capability profile (model types supported, throughput, memory), authorization level (security clearance for sensitive inference tasks), and current load. A request for sensitive inference (medical diagnosis, financial decision, identity verification) cannot be routed to a compute node whose identity does not include the required security authorization.

Explicit routing. Inference requests are routed through the compute mesh to capability-matched nodes rather than broadcast to all available compute. Load balancing is deterministic: the same request arriving with the same mesh state produces the same routing decision. Inference results are returned along the same verified path that carried the request, with full audit trail.

Invariant set. No sensitive inference is routed to an unauthorized compute node. No inference result is delivered without a verifiable execution trace. No compute node accepts requests that exceed its capability profile.

Organism classification. Type 2—“3 predicted. DII is the youngest and least physically constrained of the predicted instances —” computation is more fungible than energy or right-of-way —” but the governance requirements (auditability of AI inference for regulatory compliance) make invariant enforcement as important as in any physical domain.

Relationship to DAIGS. DII is the inference-specific extension of the DAIGS framework [17]. DAIGS handles multi-agent coordination in general; DII applies the same architecture specifically to AI inference allocation in an ambient computing mesh.

10. Implications for Physical System Design

10.1 The Checklist for Deterministic Infrastructure

The four properties of Deterministic Infrastructure provide a design checklist that any autonomous physical system can be evaluated against:

– Property 1 —” Verified Identity – Is node identity established at manufacture/provisioning and immutable thereafter? – Is identity cryptographically bound and locally verifiable? – Does identity encode capabilities, not just names? – Is identity propagated to all components that model this node? – Property 2 —” Explicit Routing – Are resources routed along defined paths rather than broadcast? – Is routing deterministic —” same inputs, same route? – Are routes validated against safety invariants before resource flows? – Is there a delivery confirmation mechanism? – Property 3 —” Invariant Homeostasis – Is there a defined set of global invariants? – Are invariants checked continuously, not periodically? – Is violation detection within one control cycle? – Are recovery sequences pre-specified for every anticipated violation? – Property 4 —” Organism Self-Maintenance – Does the system detect and recover from component failure without human intervention? – Does the system reduce resource expenditure when resources are constrained? – Does the system adapt its behavior to environmental changes within invariant bounds? – Does the system maintain identity and capability profile despite operational changes?

A system missing any item from this checklist has a corresponding class of failure modes that cannot be recovered from without external intervention. The checklist is not optional —” every item corresponds to a real operational vulnerability.

10.2 The Cost of Incompleteness

Each missing property produces a specific failure class:

Missing Property 1 (no verified identity): The system cannot enforce any access policy. Any node that claims to be legitimate will be treated as legitimate. Security depends entirely on physical access control to the network, which is impossible to guarantee in distributed physical deployments.

Missing Property 2 (no explicit routing): Resources are wasted on components that do not need them. Safety envelopes cannot be enforced end-to-end. Delivery accountability is impossible —” there is no record of what was delivered to whom. Denial-of-resource attacks are trivially mounted.

Missing Property 3 (no invariant homeostasis): The system drifts into unsafe states over time. Small perturbations accumulate. Cascading failures are possible whenever the system reaches a state it was not designed to handle. The 2003 North American blackout was a Property 3 failure: invariant monitoring (voltage and frequency within bounds) was not continuous, and a recoverable perturbation cascaded to catastrophic failure.

Missing Property 4 (no self-maintenance): The system requires human intervention for all failure recovery. This creates a hard scaling limit —” the system can only grow to the scale that available human operators can maintain. It also creates a hard availability limit —” the system is unavailable during any failure period until a human operator is reached.

10.3 The Design Paradigm Shift

Deterministic Infrastructure represents a paradigm shift in physical system design: from system-as-machine to system-as-organism. The machine paradigm designs systems as passive components operated by human supervisors. The organism paradigm designs systems as autonomous agents that maintain their own operational integrity, route their own resources, enforce their own governance, and heal their own failures.

The organism paradigm is not new as a metaphor —”autonomic computing [24], self-organizing systems [29], and swarm robotics [28] have all used biological inspiration. The contribution of Deterministic Infrastructure is precision: it specifies exactly which organismic properties are necessary and sufficient (the four properties), provides the axiomatic foundation that produces those properties (the 42 Assumptions), and supplies the concrete implementation substrate that makes those properties buildable (Lume, Lume-X, Lume-V).

The paradigm shift is not optional. It is driven by scale. Human-supervised machine infrastructure cannot serve a trillion autonomous devices. Only organism-like infrastructure —”infrastructure that maintains itself, routes its own resources, and enforces its own governance —” can scale to that deployment density.

11. Related Work

11.1 Autonomic Computing

The autonomic computing vision [24] proposed self-managing systems with four properties: self-configuration, self-optimization, self-healing, and self-protection. This vision anticipated several aspects of Deterministic Infrastructure and inspired a substantial research literature [25].

The critical difference between autonomic computing and Deterministic Infrastructure is the axiomatic foundation. Autonomic computing describes desired properties but does not provide an axiomatic framework that guarantees their achievement. A system designed under autonomic computing principles may or may not achieve self-healing —”it depends on the quality of the implementation. Deterministic Infrastructure systems built on the 42 Assumptions cannot fail to achieve organism-like self-maintenance because the axioms make the failure mode impossible to build.

11.2 Cyber-Physical Systems

The CPS literature [26, 27] studies the integration of computational and physical processes, with particular attention to timing guarantees, safety, and the formal verification of combined cyber-physical behavior. This literature is the closest academic prior to Deterministic Infrastructure in terms of domain —”both concern systems that span the digital-physical boundary with formal requirements.

The distinction is scope. CPS research typically focuses on individual systems or small networks with specific applications (automotive, medical devices, industrial control). Deterministic Infrastructure is a general framework applicable across all autonomous physical domains. CPS does not provide a general theory of autonomous physical infrastructure —” it provides tools for building and verifying specific CPS instances. Deterministic Infrastructure provides the general theory.

11.3 Self-Organizing Systems

The self-organizing systems literature [29] —” including stigmergic systems, swarm intelligence [28], and emergent coordination —” studies how complex organized behavior arises from local interactions between simple components without central control. These systems exhibit some properties of Deterministic Infrastructure (distributed coordination, resilience to component failure) but not others (verified identity, invariant homeostasis, explicit routing).

The key distinction is determinism. Self-organizing systems are probabilistic —” their collective behavior is described statistically, not guaranteed per-instance. Deterministic Infrastructure systems produce guaranteed outcomes per-instance. This is the difference between knowing that a swarm of robots will probably complete a task and knowing that a specific robot will complete a specific subtask within a specific time bound with verified identity at every interaction.

11.4 Distributed Ledger Technology

Blockchain and distributed ledger technologies address the identity and governance problems of distributed systems by replacing central authority with cryptographic consensus. They provide verified identity (cryptographic key pairs), explicit routing of transactions (validated and recorded on-chain), and some invariant enforcement (smart contract conditions).

The critical limitation is performance. Blockchain consensus operates at the timescale of seconds to minutes. Deterministic Infrastructure operates at the timescale of milliseconds (Lume-X at 73 Hz). The governance overhead of blockchain consensus is incompatible with physical infrastructure that must respond to failure within one control cycle. The Trust Layer achieves the governance properties of blockchain (cryptographic identity, auditable history, distributed validation) at infrastructure timescales by replacing consensus with deterministic cryptographic verification —” each interaction is locally verifiable rather than requiring network-wide consensus.

11.5 The Lume Ecosystem

The three companion papers [Meridian Architecture, 2026; Meridian Organism, 2026; Energy Internet, 2026] provide the concrete grounding for this general theory. The Trust Layer [16] and DAIGS [17] provide additional instances across different domains. Lume-V [18] provides the formal verification layer. Lume-X [19] provides the operational enforcement layer. Deterministic Dissolution [20] provides the formal model for graceful termination of DI instances.

This paper synthesizes these contributions into a general framework. The general framework is the contribution of this paper; the specific contributions of each companion paper stand independently and are not re-derived here.

12. Limitations and Honest Boundaries

The theory rests on three confirmed instances. Meridian (architecturally specified, experimentally unconfirmed), Trust Layer (deployed), and DAIGS (deployed in simulation). Three instances do not prove generality. The predictions in Section 9 are argued predictions, not demonstrated facts. Each future domain requires independent specification and validation.

The 42 Assumptions have not been formally proven to entail the four properties. The mapping in Section 4 is argued rather than formally derived. A formal proof that the 42 Assumptions jointly entail all four Deterministic Infrastructure properties across all possible physical domains would require a level of formalization that neither the 42 Assumptions nor the four properties currently support. This is future work.

Deterministic Infrastructure does not eliminate all failure modes. It eliminates the failure modes that correspond to missing properties. It cannot prevent hardware failures that exceed the SHDCL recovery budget, adversarial attacks that successfully compromise the Trust Layer PKI, or physical events (electromagnetic pulse, physical destruction) that destroy nodes faster than self-maintenance mechanisms can respond.

The cross-domain interoperability described in Section 8 does not yet exist as a deployed system. The Trust Layer provides the identity substrate for multiple DI instances in principle. In practice, the integration of Meridian's energy routing with the Trust Layer's identity governance has not been fully implemented. The interoperability model is architectural, not operational.

The organism classification of future DI instances in Section 9 is prediction, not specification. The classification of Transportation DMI as Type 3 and Ambient Computation DII as Type 2—"3 is based on the general theory and the requirements of those domains. Formal specifications of those instances do not yet exist.

The general theory does not address the economics. Building Deterministic Infrastructure across multiple physical domains requires significant capital investment, regulatory engagement, and stakeholder coordination. The theory is technically sound; the practical obstacles are social, economic, and political.

13. Conclusion

Three papers built a system. This paper explains what that system is a part of.

Meridian is not a wireless energy router that happens to have some interesting theoretical properties. It is the first formally specified instance of a general class —” Deterministic Infrastructure —” defined by four properties that together make autonomous physical operation safe, scalable, and governable without central authority. The Trust Layer is another instance in the identity domain. DAIGS is another in the computation domain. Four more domains —” transportation, manufacturing, emergency coordination, and ambient computation —” are predicted on the basis of the general theory and the character of those domains.

The properties that define this class are not arbitrary. They are the minimum requirements for any autonomous physical system that must operate in an adversarial environment without human supervision over an extended time horizon. Verified identity, explicit routing, invariant homeostasis, and organism-like self-maintenance are the properties that biological organisms evolved because they faced the same requirements. Deterministic Infrastructure systems exhibit those properties because they are built on axioms —” the 42 Assumptions —” that encode them at the foundational level.

The Lume ecosystem is the first complete framework for building Deterministic Infrastructure. It provides the language (Lume), the axioms (the 42 Assumptions), the verification layer (Lume-V), and the operational runtime (Lume-X). Any autonomous physical system built on this substrate will converge toward all four Deterministic Infrastructure properties as a logical consequence of the foundation.

The world needs Deterministic Infrastructure. The gap between the autonomous device density that is coming —” a trillion nodes, across every physical domain, operating at millisecond timescales without human supervisors —” and the infrastructure paradigms that currently exist —” best-effort delivery, central coordination, passive components, human-timescale failure recovery —” is not a gap that can be closed by improving existing infrastructure. It requires a new paradigm.

Deterministic Infrastructure is that paradigm. The Lume ecosystem is its foundation. Meridian, the Trust Layer, and DAIGS are its first instances. The Energy Internet is its first protocol standard. And this paper is its first formal statement as a general theory.

The work that remains is vast. But the theory now exists. The instances now exist. The foundation now exists. What comes next is construction.

Appendix A —” The Four Properties: Formal Definitions

Property 1 —” Verified Identity at Every Node

For every node N in system S:

$\text{identity}(N) = \langle \text{ID_N}, \text{PKI_cert_N}, \text{capabilities_N}, \text{propagation_model_N} \rangle$ where: ID_N is fixed at provisioning and immutable PKI_cert_N is cryptographically bound to ID_N and locally verifiable capabilities_N encodes what N can do, derived from manufacture $\text{propagation_model_N}$ specifies how $\text{network}(N)$ is built from $\text{identity}(N)$

System S satisfies Property 1 if and only if: no interaction between N_1 and N_2 occurs without mutual verification of PKI_cert for both parties.

Property 2 —” Explicitly Routed Resources

For every resource transfer from source s to destination d :

$\text{route}(s, d, t) = \langle \text{path}, \text{quantity_at_each_hop}, \text{timing}, \text{safety_envelope} \rangle$ where: path is a sequence of verified nodes $\langle n_1, n'', \dots, n' \rangle$ quantity specifies resource allocation at each hop timing specifies scheduling window at each hop safety_envelope specifies the invariants the route must not violate

System S satisfies Property 2 if and only if: no resource transfer occurs for which a validated $\text{route}(s, d, t)$ does not exist.

Property 3 —” Invariant-Enforced Homeostasis

Let $\text{INV}(S) = \{\text{inv}_1, \dots, \text{inv}_n\}$ be the global invariant set. System S satisfies Property 3 if and only if:

$\forall t: \forall \text{inv}_\mu \in \text{INV}(S): \text{inv}_\mu(\text{state}(S, t)) = \text{TRUE} \vee \exists t' \leq t + T_{\text{cycle}}: \text{recovery}(S, t') \uparrow \text{inv}_\mu(\text{state}(S, t')) = \text{TRUE}$

That is: every invariant is either always true, or restored to true within one control cycle T_{cycle} by an automatic recovery sequence.

Property 4 —” Organism-Like Self-Maintenance

System S satisfies Property 4 if and only if S implements:

$\text{self_heal}(f: \text{failure_event}) \uparrow \text{recovery_executed_without_external_intervention}$ $\text{self_preserve}(v: \text{resource_level}) \mid v < v_{\text{min}} \uparrow \text{conservation_mode_activated}$ $\text{adapt}(e: \text{env_state}) \uparrow \text{behavior_adjusted_within}(\text{INV}(S))$ $\text{persist_identity}() \uparrow \text{identity}(N) \text{ unchanged despite operational_changes}$

All four functions must be implemented. Partial implementation produces the corresponding class of failure modes.

Appendix B —” Domain Mapping Table

Domain	DI Instance	Resource Routed	Identity Substrate	Invariant Set	Current Classification
Energy	Meridian	Electrical energy (RF bursts)	MC 64-bit / Trust Layer PKI	INV-1 to INV-5	Type 3+ structural
Identity/Data	Trust Layer	Trust relationships, credentials	Cryptographic entity identity	No revoked credential active; no unauthorized governance	Type 3 confirmed
Multi-Agent Computation	DAIGS	Computation allocation	Agent role identity	No capability scope exceeded; no unresolved coordination conflict	Type 2—“3
Transportation (predicted)	DMI	Right-of-way	Vehicle capability identity	No spatial collision; no speed envelope violation	Type 3 predicted
Manufacturing (predicted)	DPI	Production capacity	Resource capability profile	No capability mismatch; no material expiration violation	Type 3—“4 predicted
Emergency Coordination (predicted)	DRI	Response capacity	Responder capability profile	All incidents above minimum response threshold	Type 3 predicted
Ambient Computation (predicted)	DII	Inference capacity	Compute node capability profile	No sensitive inference on unauthorized node	Type 2—“3 predicted

Appendix C —” Convergence Prediction Matrix

For each future DI domain, this matrix predicts when organism-like properties will

DI Domain	Property 1 Emergence	Property 2 Emergence	Property 3 Emergence	Property 4 Emergence	Predicted Type	Timeline
Transportation (DMI)	Immediate (AV identity already exists in nascent form)	Early (intersection routing is the first deployment)	Mid-term (traffic invariants require multi-node coordination)	Mid-term (failure recovery requires mesh-level healing)	3	3—“7 years
Manufacturing (DPI)	Early (equipment identity is already tracked in MES systems)	Mid-term (production routing requires mesh infrastructure)	Early (manufacturing invariants are well-understood)	Mid-term (auto-recovery requires capability-aware rerouting)	3—“4	2—“5 years
Emergency Coordination (DRI)	Immediate (responder identity is already managed)	Early (dynamic dispatch is already partially automated)	Early (incident coverage invariants are safety requirements)	Long-term (autonomous recovery is hardest in crisis conditions)	3	5—“10 years
Ambient Computation (DII)	Immediate (compute identity is already managed in cloud)	Early (inference routing is a natural extension of load balancing)	Mid-term (inference governance invariants are emerging from regulation)	Early (compute failover is already automated)	2—“3	1—“3 years

Appendix D —” Cross-Reference to Companion Papers

Concept in This Paper	Source Paper	Section
Meridian architecture (Property 1—“4 implementation)	Meridian Architecture, 2026	4—“12
MC node identity	Meridian Architecture, 2026	4.2
DRMA multi-hop routing	Meridian Architecture, 2026	9
INV-1 through INV-5	Meridian Architecture, 2026	3.5, 11
SHDCL self-healing	Meridian Architecture, 2026	12.2
Lume-X 73 Hz control loop	Meridian Architecture, 2026	6.1, 12.1
Synthetic Organism classification	Meridian Organism, 2026	7
14-subsystem biological isomorphism	Meridian Organism, 2026	3—“4
42 Assumptions as biological axioms	Meridian Organism, 2026	6
Convergence thesis	Meridian Organism, 2026	8.1
Energy Internet Protocol Stack	Energy Internet, 2026	3
EIP addressing	Energy Internet, 2026	4
EBGP inter-mesh federation	Energy Internet, 2026	5
Emergency Priority Override	Energy Internet, 2026	6.4
EIRA governance model	Energy Internet, 2026	9.2
Trust Layer as DI instance	Trust Layer Ecosystem [16]	—”
DAIGS as DI instance	DAIGS Framework [17]	—”
Lume-V formal verification	Lume-V Verification Suite [18]	—”
Lume-X control runtime	Lume-X Multi-Agent Cognition [19]	—”
Deterministic Dissolution	Deterministic Dissolution [20]	—”

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Part Five Closing

Chapter Seven has named the class Meridian belongs to and argued that the next generation of physical infrastructure —” across energy, transportation, manufacturing, and emergency coordination —” will be built on the same four properties. The general theory is stated. The instances are identified. The design paradigm is defined.

There remains one question the preceding seven chapters do not answer: where does the energy in the Meridian mesh come from?

Today the answer is: from the electrical grid, through the source node's power connection to the building electrical panel. The grid is the foundation on which everything above it rests. And the grid is a \$4 trillion physical plant of transmission towers, high-voltage cables, and substations —” the antithesis of deterministic, resilient, dynamic routing.

Chapter Eight asks whether that foundation needs to exist.

PART SIX

The Vision: Where This All Leads

Every infrastructure revolution follows the same arc. First, the new technology is adopted at the edge —” in the last mile, at the device, in the building. Then it propagates inward, displacing the legacy infrastructure layer by layer, until the legacy core itself is the target. The internet began as a last-mile service connecting individual users to centralized servers. It ended as the core infrastructure of every enterprise, every government, and every communications system on the planet.

Meridian began as a last-mile energy routing architecture —” from the district boundary to the device. Parts One through Five have specified that last-mile architecture completely. Part Six asks what happens when the same routing principles propagate inward —” past the district boundary, past the substation, past the transmission line —” all the way to the generation source itself.

The answer is: no transmission lines. No substations. No physical transmission infrastructure at any layer. Energy routed through air from plasma to device, addressed and governed at every hop, by the same Deterministic Infrastructure framework that governs the last-mile mesh.

Chapter Eight is the full stack vision. It is clearly labeled as conceptual —” no demonstration at the proposed scale exists, fusion energy is not yet commercially available, and the regulatory path is long. But the architecture is coherent, the physics is sound, and the destination is worth specifying clearly so that the work of building toward it can begin.

Chapter Eight

Beyond the Grid: Fusion, Long-Haul Beaming, and the End of Transmission Lines

A \$4 trillion infrastructure is not replaced quickly. But it can be made obsolete, and the process of making it obsolete begins with specifying what replaces it —” precisely enough that engineers, regulators, and investors can evaluate the specification and decide whether to build toward it.

The electrical transmission grid —” 700,000 kilometers of high-voltage lines in the United States alone, connected through 55,000 substations to 10,000 generation plants —” is the infrastructure that every wireless energy revolution described in this book ultimately depends on. Meridian's source nodes plug into it. The Energy Internet's access nodes draw from it. The self-sustaining device's district mesh is fed by it. Take the grid away and the whole wireless energy architecture has nowhere to source its energy.

This chapter takes the grid away. It replaces it with two things: fusion energy, generated locally at district scale by reactors governed by the Deterministic Fusion Control framework; and a directed microwave beam —” a Tier 1 long-haul energy link at 2.45 GHz —” that routes that energy from the generation site to the district receiver array without a single conductor between them.

The physics that enables this is mature. The SPS (Solar Power Satellite) program demonstrated the complete physics chain in the 1970s —” microwave power generation, beam formation, long-distance propagation, and rectenna collection —” and confirmed the Fresnel-regime efficiency numbers that make the architecture viable. What has changed since then is the generation source (fusion rather than orbital solar) and the control architecture (Lume-X deterministic invariant enforcement at every layer).

The honest accounting is in this chapter, too. The wireless architecture is approximately half the thermal-to-device efficiency of the wired grid at current technology levels. The regulatory pathway spans two regulatory bodies and a decade of experimental work. Fusion power is not yet commercially available. These are the boundary conditions on the timeline, not arguments against the destination.

The destination is a world where no transmission tower stands between a reactor and a device. Energy, addressed and routed from plasma to phone, through air.

Abstract

The seven preceding papers in this series specify Meridian as a complete architecture for wireless energy routing at building scale. They define the protocol stack, the deployment model, the security framework, and the unified energy-data fabric. They do not address generation or long-haul transmission —” the two infrastructure layers that exist above the building mesh in the current energy system.

The global electrical grid —” a \$4 trillion physical plant of transmission towers, high-voltage lines, substations, and distribution transformers —” is the infrastructure layer that delivers bulk energy from generation sites to the district boundaries where Meridian takes over. It is also the infrastructure layer most vulnerable to physical failure, most expensive to build and maintain, most constrained by right-of-way acquisition, and most exposed to both natural and adversarial disruption.

I propose that this layer can be eliminated.

I introduce a two-tier wireless energy architecture that closes the gap between energy generation and device delivery without any physical transmission infrastructure. Tier 1 —” the long-haul beam layer —” routes bulk energy from fusion generation hubs to district receiver arrays via directed microwave beams at 2.45 GHz, operating over distances of 500 meters to 5 kilometers with projected collection efficiency of 75—“85% at full aperture in Fresnel-regime geometry. Tier 2 —” the last-mile mesh layer —” is the Meridian architecture specified in the preceding papers, routing energy from the district receiver through the building mesh to individual devices.

I integrate this architecture with the Deterministic Fusion Control framework (developed in a companion L-SOC series), which applies Lume-X invariant-enforced control to plasma stability and output modulation, providing a deterministically governed generation source whose output power can be adjusted in real time to track demand without the mechanical inertia constraints of thermal generation.

I provide the Tier 1 safety architecture —” a categorically different safety model from the Meridian BSL, designed for megawatt-scale directed beams through uncontrolled airspace, drawing on retrodirective pilot-tone locking, exclusion zone enforcement, radar-based intruder detection, and hardware-enforced automatic shutoff.

I present an honest system efficiency analysis: the full thermal-to-device efficiency of the proposed architecture is approximately 15—“19%, compared to approximately 33—“37% for conventional wired grid delivery from fusion generation. The efficiency penalty is real and significant. The arguments for the wireless architecture rest not on efficiency but on infrastructure elimination, resilience, deployment flexibility, and the properties that follow from making energy routing fully deterministic from generation to device.

I close with the complete Deterministic Infrastructure stack —” the first formal specification of a system in which every layer from plasma to device is governed by the same four DI properties: verified identity, explicitly routed resources, invariant-enforced homeostasis, and organism-like self-maintenance.

Keywords: wireless power transmission, fusion energy, directed energy, microwave power beaming, transmission line elimination, two-tier energy architecture, Fresnel-regime propagation, retrodirective beaming, deterministic infrastructure, Meridian

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References

1. Introduction

1.1 The Stack Is Not Complete

The seven preceding papers in this series define Meridian from the ground up: the four-layer protocol stack, the organism-like self-healing architecture, the Energy Internet protocol extension, the unified energy-data mesh, the Guardian Security framework, and the physical deployment architecture for complex built environments. They define a complete last-mile energy routing system —” from the district boundary to the device.

They do not define what sits above the district boundary.

In the current energy system, what sits above the district boundary is the electrical grid: high-voltage alternating current transmitted through overhead or underground cables from remote generation sites, stepped down through a cascade of substations, and delivered to the building's electrical panel. The grid is

the reason Meridian's source nodes have somewhere to plug into. It is the infrastructure that Meridian's building mesh depends on.

This paper argues that the grid is the wrong foundation for a deterministic energy architecture.

Not because the physics of AC transmission is wrong. It is right, and it has served humanity well for over a century. But because a physical transmission infrastructure —” towers, cables, substations, transformers —” is the antithesis of what a Deterministic Infrastructure system requires. It is centralized, vulnerable, slow to deploy, expensive to maintain, and impossible to reroute in real time. When a tower fails, a region goes dark. When a geomagnetic storm hits, a continent-scale blackout is possible. When a right-of-way dispute stalls a new line, a region's energy supply is constrained for years.

The Meridian architecture can route energy around a failed node in 13.7 milliseconds. The electrical grid takes months to build around a failed section.

The appropriate foundation for a deterministic energy architecture is a wireless energy transmission layer —” one that routes bulk energy from generation to district using the same principles of addressed, routed, invariant-enforced delivery that Meridian applies from district to device.

1.2 The Fusion Connection

The Deterministic Fusion Control framework, developed in a companion L-SOC series, applies Lume-style invariant enforcement to plasma control in a magnetic confinement fusion reactor. The core argument of that framework is that plasma stability —” the property that makes sustained fusion possible —” can be maintained deterministically by enforcing a set of plasma state invariants (energy confinement time τ_E , plasma beta β , density limit n_G) through the same kind of Lume-X control runtime that enforces Meridian's energy routing invariants.

A fusion reactor governed by deterministic plasma control produces output power that is stable, modulated on demand, and governed by formal invariants —” not the stochastic output of a combustion process or the weather-dependent output of a solar or wind installation.

When a deterministically controlled fusion generator connects to a wireless energy transmission layer, and that layer connects to a Meridian building mesh, the result is the first fully deterministic energy system from generation to device —” every layer governed by the same four Deterministic Infrastructure properties, from plasma to phone.

1.3 Scope and Honesty

This paper makes ambitious architectural claims. It is worth stating at the outset what this paper is and is not.

It is: a formal architectural specification and theoretical analysis of a two-tier wireless energy transmission architecture that eliminates physical transmission infrastructure.

It is not: a demonstration that such a system currently exists, that the required hardware is currently commercially available at the required specifications, or that the regulatory pathway is clear. The efficiency analysis in Section 7 is honest about the significant penalty the wireless architecture pays relative to conventional wired grid delivery. The safety analysis in Section 6 is honest about the categorically different challenge of securing megawatt-scale directed beams through uncontrolled airspace.

The purpose of this paper is to complete the Meridian series with the full stack vision —” to show where the architecture leads when extended from the building mesh all the way to the generation source —” and to provide the formal architectural specification that future experimental and regulatory work can build against.

1.4 Contributions

I make five contributions:

The formal definition of the two-tier wireless energy architecture —” Tier 1 (long-haul microwave beam) and Tier 2 (Meridian last-mile mesh) —” with full specification of the interface between them.

The Fresnel-regime propagation analysis for representative Tier 1 configurations, establishing the theoretical collection efficiency bounds at 2.45 GHz and 5.8 GHz over distances of 500m—“5km.

The Tier 1 safety architecture —” retrodirective pilot-tone locking, exclusion zone design, intruder detection, and hardware-enforced beam cutoff —” adapted from the SPS literature and extended for terrestrial deployment.

The honest system efficiency analysis: full thermal-to-device efficiency comparison between the proposed wireless architecture and conventional wired grid delivery from the same fusion generation source.

The complete Deterministic Infrastructure stack specification —” the first formal description of a system in which every layer from plasma to device satisfies all four DI properties simultaneously.

2. The Transmission Infrastructure Problem

2.1 The Scale of the Physical Plant

The global electrical transmission and distribution infrastructure represents approximately \$4 trillion in capital investment —” a figure that grows annually as demand increases, aging infrastructure is replaced, and new renewable generation requires long-distance transmission from resource-rich areas to population

centers. In the United States alone, approximately 700,000 km of high-voltage transmission lines connect roughly 10,000 generation plants to 145 million customer connections through a cascade of approximately 55,000 substations.

This physical plant is:

Expensive to build. High-voltage overhead transmission lines cost \$500,000—“\$3,000,000 per kilometer to construct, depending on terrain, voltage level, and right-of-way acquisition. Underground high-voltage cables cost \$3,000,000—“\$15,000,000 per kilometer. A new 500 km transmission corridor costs \$500 million to \$1.5 billion before a single kilowatt-hour flows through it.

Slow to deploy. New high-voltage transmission lines typically require 5—“15 years from planning to energization, owing to right-of-way acquisition, environmental review, engineering, construction, and regulatory approval. The energy infrastructure needed for a decarbonized economy cannot be built fast enough through the conventional permitting and construction process.

Expensive to maintain. The U.S. electrical industry spends approximately \$30 billion annually on transmission and distribution maintenance and operations —” a figure that grows as infrastructure ages. Transmission lines require inspection, vegetation management, hardware replacement, and periodic reconstruction.

Vulnerable to disruption. Physical transmission infrastructure fails under severe weather (ice storms, hurricanes, tornadoes, wildfire), seismic events, and geomagnetic disturbances (solar coronal mass ejections have historically caused multi-province blackouts). It is also a physical attack surface: a coordinated physical attack on a small number of critical transmission substations could cause extended multi-regional blackouts according to published security analyses.

Lossy at scale. Global average transmission and distribution losses are approximately 8—“9% of generated electricity —” representing roughly 2,500 terawatt-hours of lost energy per year, equivalent to the total annual electricity consumption of India.

2.2 What Wireless Transmission Eliminates

A wireless energy transmission architecture —” one in which bulk energy moves from generation to district via directed electromagnetic beams rather than physical conductors —” eliminates all five problems at once:

There is no physical plant to build. The transmitter is a phased array antenna mounted at or near the generation facility. The receiver is a rectenna array mounted at the district boundary. The capital cost per delivered-watt-at-distance is dominated by the antenna and power electronics, not by cable, towers, right-of-way, and civil engineering.

There is no physical plant to maintain. Phased array antennas are solid-state; they have no mechanical components exposed to weather-induced wear. The beam path itself is air —” it neither rusts, breaks, nor requires vegetation management.

There is no physical plant to attack. A beam path through air has no physical attack surface between transmitter and receiver. Cutting a conductor is a simple act with a saw; cutting a microwave beam is not an act available to a threat actor with conventional tools.

There is no right-of-way to acquire. A microwave beam travels through air. It crosses property boundaries the same way a cell tower's signal does —” without requiring physical access to the land it crosses.

There is no fixed path. A phased array transmitter can redirect its beam to a different receiver array in the time it takes to update a beamforming coefficient —” milliseconds, not months. Energy routing becomes as dynamic as packet routing.

2.3 What Wireless Transmission Does Not Eliminate

The efficiency penalty. A wired conductor transmits electrical energy with resistive losses of 2—“8% over typical transmission distances. A wireless microwave beam at 2.45 GHz, transmitted from a 10-meter aperture and collected by a 10-meter receiver at 5 km, achieves approximately 75—“80% collection efficiency in Fresnel-regime geometry —” before the additional conversion losses at transmitter and receiver. The full wireless link is less efficient than a wire. This is a real cost, and Section 7 addresses it honestly.

The regulatory complexity. High-power microwave beams through airspace require regulatory coordination with aviation authorities, spectrum authorities, and public safety agencies that a buried cable does not. The regulatory pathway exists —” existing frameworks cover high-power radar, satellite uplinks, and early SPS proposals —” but it is not straightforward.

These are acknowledged costs, not ignored ones. The architectural argument rests on the proposition that the eliminated costs (capital, maintenance, security, right-of-way, deployment time) outweigh the incurred costs (efficiency penalty, regulatory complexity) for a sufficiently large class of deployment scenarios.

3. Physics of Long-Range Directed Energy Transfer

3.1 Beam Propagation Fundamentals

A directed energy transfer system transmits electromagnetic power from a transmitter aperture of area A_t to a receiver aperture of area A_r at distance d . The fraction of transmitted power collected by the receiver depends on the operating wavelength λ and the geometric configuration of the system.

The key dimensionless parameter governing this relationship is the Fresnel number:

$$N_F = (A_t - A_r)^{1/2} / (\lambda - d)$$

For circular apertures of diameters D_t and D_r :

$$N_F = (\pi^2 - D_t - D_r) / (4 - \lambda - d)$$

The Fresnel number governs the collection efficiency regime:

$N_F \gg 1$ (near-field / Fresnel regime): The beam is well-collimated within the rec



$N_F \approx 1$ (Fresnel transition): Collection efficiency is approximately 85–92% for c



$N_F \ll 1$ (far-field / Fraunhofer regime): The beam spreads into a diffraction patt



3.2 Frequency Selection for Tier 1

The choice of operating frequency for the Tier 1 long-haul layer is governed by four competing constraints: beam collimation (higher frequency † tighter beam † smaller apertures for given N_F), atmospheric attenuation (oxygen absorption, water vapor, rain), spectrum availability, and hardware maturity.

2.45 GHz ($\lambda = 0.122$ m):

Atmospheric attenuation at sea level: < 0.01 dB/km (oxygen and water vapor absorption negligible at this frequency) Rain attenuation: < 0.01 dB/km at moderate rainfall rates (< 10 mm/hr); < 0.05 dB/km at extreme rainfall (25 mm/hr) Hardware: mature, efficient (magnetrons, klystrons, solid-state GaN power amplifiers all available at 2.45 GHz with > 80% efficiency) Spectrum: ISM band; unlicensed at low power, licensed at high power Primary Tier 1 frequency

5.8 GHz ($\lambda = 0.052$ m):

Atmospheric attenuation: < 0.02 dB/km at low rainfall, rising to 0.1 dB/km at heavy rainfall Hardware: mature (Wi-Fi ecosystem), somewhat lower efficiency than 2.45 GHz amplifiers Enables smaller apertures for the same Fresnel number at the same distance Secondary Tier 1 frequency for shorter-range, smaller-aperture configurations 24 GHz / 60 GHz (mmWave):

60 GHz oxygen absorption: ~15 dB/km — eliminates this frequency for anything beyond 200–300m 24 GHz atmospheric attenuation: ~0.1–0.5 dB/km — acceptable at sub-km range only Not suitable for Tier 1; used only in Tier 2 (Meridian mesh)

3.3 Fresnel Number Analysis for Representative Configurations

Table 1: Fresnel number and projected collection efficiency for 2.45 GHz configurations

D _t	D _r	Distance	N _F	Efficiency regime	Projected η_{collect}
5m	5m	500m	0.82	Fresnel transition	~87%
10m	10m	1 km	0.82	Fresnel transition	~87%
10m	10m	5 km	0.164	Near far-field	~65%
20m	20m	5 km	0.655	Fresnel transition	~83%
20m	20m	10 km	0.328	Near far-field	~72%
50m	50m	20 km	0.82	Fresnel transition	~87%

Note: Efficiency values are for Gaussian-tapered aperture illumination. Uniform illumination achieves lower efficiency for the same N_F due to higher sidelobe levels.

The practical conclusion: at 2.45 GHz with 10—“20 meter aperture arrays, collection efficiencies of 80—“87% are achievable at distances of 1—“5 km. This covers the district-scale energy delivery application —” a fusion hub at the edge of a city district delivering energy to receiver arrays distributed across the district's building rooftops and parking structures, with each receiver feeding a local Meridian mesh.

For distances beyond 5 km with reasonable aperture sizes (< 50m), collection efficiency drops into the 65—“75% range. The architecture is most efficient at district scale (1—“5 km) and becomes progressively less efficient as range increases.

3.4 Aperture Illumination and Sidelobe Management

A beam optimized purely for collection efficiency at the desired receiver aperture will have residual sidelobe power deposited outside the receiver —” in the surrounding airspace and ground. For safety and regulatory compliance, the sidelobe structure of the transmitted beam must be managed to ensure power density outside the exclusion zone boundary falls below ICNIRP public reference levels.

The fundamental design tradeoff is between collection efficiency and sidelobe suppression. A Gaussian-tapered aperture illumination (lower power at the aperture edges than the center) produces lower sidelobes at the cost of slightly reduced collection efficiency compared to uniform illumination. For the Tier 1 safety architecture, Gaussian or Taylor-tapered illumination is required regardless of the small efficiency cost.

The sidelobe power density at the exclusion zone boundary (Section 6.3) must satisfy:

$$S_{\text{sidelobe}}(r_{\text{exclusion}}) \leq S_{\text{ICNIRP}}(f)$$

where $S_{\text{ICNIRP}}(2.45 \text{ GHz}) = 10 \text{ W/m}^2$ for general public exposure. For a 10 MW transmitted beam with 40 dB sidelobe suppression (achievable with Taylor taper), sidelobe power at 1 km from the beam axis is approximately 10 W/m^2 —” at the ICNIRP limit. Conservative design uses 46 dB suppression (providing 4— margin) at the cost of ~3% collection efficiency.

4. The Two-Tier Wireless Energy Architecture

4.1 Architecture Overview

The two-tier wireless energy architecture replaces the physical transmission grid with two layers of directed electromagnetic energy routing:

Tier 1 —” Long-Haul Beam Layer:

Generation site to district boundary Distance: 500m —“ 5km (primary); up to 20km with larger apertures
Frequency: 2.45 GHz (primary), 5.8 GHz (secondary) Power scale: 100 kW —“ 100 MW per beam link
Transmitter: phased array at generation site or elevated relay point Receiver: rectenna array at district boundary (rooftop, elevated structure, dedicated receiver park) Safety: retrodirective pilot-tone locking, exclusion zone, intruder detection, hardware cutoff Tier 2 —” Last-Mile Mesh Layer:

District receiver to individual devices Fully specified in Papers 1—“7 of this series Frequency: 60 GHz (primary), 5.8 GHz (secondary) Power scale: milliwatts —“ hundreds of watts per link All Meridian architecture applies: BSL, DRMA routing, Guardian-E security, Lume-X control The Tier Interface: The district receiver array converts incoming 2.45 GHz microwave power to DC through a rectenna array (rectifying antenna —” an antenna optimized for power reception connected to a Schottky diode rectifier circuit, then a DC filter). The DC output feeds the local power distribution and the Meridian source nodes in the buildings it serves. The interface is DC power at the district rectenna output —” the same input that a Meridian source node currently receives from the building electrical panel.

Fusion Hub ”””” DC generation (fusion thermal †’ turbine †’ generator †’ DC bus) ”””” RF transmitter array (DC †’ 2.45 GHz phased array, $\eta_{\text{rf}} \approx 85\%$) ”””” [Tier 1 beam: 2.45 GHz, 500m—“5km] ”””” District rectenna array (2.45 GHz †’ DC, $\eta_{\text{rect}} \approx 82\%$) ”””” DC bus †’ Meridian source nodes †’ [Tier 2 mesh] ”””” Building relay nodes †’ device

4.2 Tier 1 Node Types

The Tier 1 layer introduces two new node types in the energy routing architecture:

T1-TX: Tier 1 Transmitter Node Located at the generation facility. Receives DC power from the generator bus, converts to 2.45 GHz via solid-state power amplifier array, and feeds a phased array antenna. The phased array beam-steers to the designated T1-RX receiver, locking to the retrodirective pilot tone

(Section 6.2). Registered with the Trust Layer under a T1-TX capability profile. Lume-X governs the output power modulation and invariant enforcement.

T1-RX: Tier 1 Receiver Node Located at the district boundary receiver site. A rectenna array collects the incoming microwave beam and converts it to DC. Transmits a retrodirective pilot tone that the T1-TX uses for beam locking. Registered with the Trust Layer under a T1-RX capability profile. The DC output feeds the local distribution bus that supplies Meridian source nodes. The T1-RX node monitors received power and reports delivery confirmation to the T1-TX via the data channel (standard 802.11 or 5G communication link, separate from the power beam).

T1-REL: Tier 1 Relay Node (optional) For configurations where direct T1-TX to T1-RX line-of-sight is unavailable due to terrain or building obstruction, a relay node receives the T1-TX beam, reconverts to DC, and retransmits to the T1-RX. Each relay hop introduces an additional conversion loss cycle. T1-REL nodes are avoided by siting selection wherever possible.

4.3 Multi-Hub Redundancy

The wireless architecture provides a form of redundancy that the wired grid cannot match: a district receiver can accept power from multiple transmitter sites simultaneously, with the mix governed by a routing protocol analogous to EBGp in the Energy Internet specification. If the primary fusion hub goes offline for maintenance or failure, a secondary hub (which may be in a different location, with a different beam path) takes over within one T1 control cycle.

The T1 control cycle —” the equivalent of Lume-X for the Tier 1 layer —” is necessarily longer than Meridian's 13.7ms cycle, because the beam steering recalculation and power ramp-up at the transmitter takes longer at 2.45 GHz with a large aperture array. The T1 control cycle is specified at 100ms —” fast enough to respond to demand changes in under a second, sufficient for beam-steering updates, but not fast enough to track the position of a fast-moving aircraft. Aircraft exclusion is handled by the static exclusion zone and intruder detection systems, not by real-time beam steering evasion.

5. Fusion Integration: Deterministic Generation to Deterministic Routing

5.1 The Deterministic Fusion Control Framework

The companion L-SOC series on Deterministic Fusion Control applies the core Lume-X invariant enforcement model to magnetic confinement fusion plasma. The plasma state is characterized by three primary invariants:

INV-F1 (Energy Confinement): $\tau_E \geq \tau_{E,min}$ —” energy confinement time at or above the minimum value required for sustained ignition. Violation triggers plasma heating adjustment.

INV-F2 (Beta Limit): $\beta \leq \beta_{\max}$ —” plasma beta (ratio of plasma pressure to magnetic pressure) below the MHD stability limit. Violation triggers magnetic field adjustment or plasma density reduction.

INV-F3 (Greenwald Density): $n_e \leq n_G$ —” electron density below the Greenwald density limit, beyond which disruption probability increases sharply. Violation triggers fueling rate reduction.

INV-F4 (Energy Gain): $Q \geq 1$ —” energy multiplication factor (fusion energy output / heating power input) at or above breakeven. INV-F4 is the operational invariant: the reactor produces net energy only when $Q \geq 1$.

Lume-X enforces these four invariants at the plasma control timescale —” which for MHD instabilities is on the order of 1—“10 milliseconds, consistent with the Lume-X cycle time specification.

5.2 Output Modulation

A fusion reactor governed by the Deterministic Fusion Control framework can modulate its output power within a range $[P_{\min_fusion}, P_{\max_fusion}]$ by adjusting the fueling rate (which changes plasma density within the INV-F3 limit) and the auxiliary heating power (which adjusts β within the INV-F1 bound). This modulation is not instantaneous —” the plasma thermal response time is on the order of τ_E , typically 1—“10 seconds for a commercial-scale device.

The T1 control cycle (100ms) is faster than the plasma modulation response time (1—“10 seconds). The Tier 1 transmitter therefore cannot demand-follow on the 100ms timescale from the fusion output. Instead:

Short-term (< 1 second) demand variations are buffered by the district rectenna's local energy storage (large-format supercapacitor bank, supplemented by short-term battery buffer) Medium-term (1—“10 second) variations are tracked by the fusion output modulation response Long-term (> 10 second) variations are handled by the fusion output modulation with full settling This three-timescale demand-following model —” storage buffer, modulation response, output adjustment —” eliminates the stochastic nature of the generation-to-demand matching problem that makes conventional grid frequency regulation complex.

5.3 The Deterministic Generation Chain

The deterministic properties of the fusion control framework propagate through the full stack:

At the generation layer, the fusion reactor produces output power within a formally bounded range, with a formally specified modulation response time, governed by four hard invariants enforced by Lume-X. The output is deterministic in the sense that matters: it does not fail suddenly without warning (unlike a combustion boiler failure), it does not fluctuate uncontrollably (unlike solar or wind), and it can be formally reasoned about.

At the Tier 1 transmission layer, the T1-TX converts DC power to 2.45 GHz RF with formally bounded efficiency ($\eta_{rf} \geq \eta_{rf,min}$ enforced as a hardware invariant), transmits to the T1-RX with beam-locking to the pilot tone (ensuring delivery to the authorized receiver), and confirms delivery through the data channel (delivery receipt analogous to Meridian's energy receipt).

At the Tier 2 mesh layer, the full Meridian architecture applies: addressed delivery, quality-of-service guarantees, BSL safety enforcement, Guardian-E security, and Lume-X invariant enforcement.

Every layer from plasma to device is:

Identity-verified (Trust Layer identity at every node, T1-TX through leaf device) Explicitly routed (beam routing at Tier 1, mesh routing at Tier 2) Invariant-enforced (Lume-X at generation, T1 control at transmission, Lume-X at mesh) Self-maintaining (deterministic fusion control auto-recovers from plasma perturbation; Meridian SHDCL auto-recovers from mesh perturbation) This is the complete Deterministic Infrastructure stack.

6. Tier 1 Safety Architecture at Scale

6.1 The Categorically Different Safety Problem

The Meridian BSL —” five-modality sensor fusion, 13.7ms response cycle, hardware cutoff —” is designed for beams delivering milliwatts to watts over distances of 1—“10 meters in a building interior. Its safety model relies on the fact that a person entering the beam path can be detected by sensors mounted on adjacent relay nodes, and that the beam can be cut within one hardware response cycle before any physiologically significant exposure occurs.

None of these assumptions hold for a Tier 1 beam delivering megawatts over kilometers of open airspace.

A 10 MW beam at 2.45 GHz, transmitted from a 10-meter aperture, produces a beam with a power density of approximately 130 kW/m² at the aperture face and approximately 10—“30 kW/m² at the center of the receiver aperture 5 km away (after Fresnel-regime spreading). The ICNIRP public reference level for 2.45 GHz is 10 W/m² —” the beam is 1,000—“3,000— above the public exposure limit at the receiver center. A person who enters this beam is not experiencing a nuisance —” they are experiencing a serious and potentially fatal RF exposure.

The Tier 1 safety architecture therefore cannot be a detection-and-response system. It must be a prevention system: one that ensures a person cannot enter the beam path except by deliberate violation of multiple independent physical barriers.

6.2 Retrodirective Pilot-Tone Locking

The retrodirective pilot-tone safety mechanism is the central element of the Tier 1 safety architecture. It was originally developed for the NASA Solar Power Satellite reference design in the 1970s and remains the most robust available mechanism for ensuring that a high-power beam is delivered only to its intended receiver.

The mechanism operates as follows:

The T1-RX receiver transmits a low-power pilot tone at a designated frequency (typically the transmit frequency plus a small offset, e.g., 2.4502 GHz for a 2.4500 GHz power beam) from a phase-reference antenna at the center of the receiver array.

The T1-TX transmitter receives the pilot tone and uses it to compute the precise direction to the receiver. The beam phase is conjugated relative to the received pilot: the beam is steered so that it converges at the location from which the pilot signal was received.

The T1-TX emits power only when the pilot tone is actively received and above a minimum signal level threshold. If the pilot tone is lost —” because the receiver is damaged, destroyed, removed, or its signal is below threshold —” the T1-TX immediately and automatically terminates the high-power beam. This termination is hardware-enforced: the pilot tone drives the gate of the high-power amplifier chain through a hardware AND gate. No software command is required to stop the beam; loss of pilot = beam stops.

The T1-RX pilot transmitter is powered by an independent, isolated power source (a dedicated battery or small standalone solar panel) —” ensuring that the pilot continues to transmit even if the main grid connection at the receiver site fails. Loss of main power at the receiver does not cause loss of pilot tone, because loss of pilot would stop the beam, which is the opposite of the desired behavior.

The retrodirective pilot-tone mechanism provides the following safety guarantee: the high-power beam is physically impossible to redirect to an unintended receiver, because the transmitter has no way to know the direction to an unintended receiver —” it only knows the direction from which the pilot tone arrives. An adversary who wants to redirect the beam must physically capture the pilot transmitter from the receiver site and operate it from a new location. This is a physical attack, not a software or network attack, and it requires physical access to the receiver site.

6.3 Exclusion Zone Design

The exclusion zone is the physical perimeter around the beam path within which public access is prohibited during beam operation. The exclusion zone must be sized so that the power density at its boundary does not exceed the ICNIRP public reference level (10 W/m² at 2.45 GHz) under any beam operating condition, including the maximum transmitted power and the worst-case sidelobe pattern.

Exclusion zone radius calculation for a 10 MW beam with 46 dB sidelobe suppression:

Main beam power density at 5 km: $\sim 15 \text{ kW/m}^2$ (center) Sidelobe suppression: 46 dB = factor of 40,000
 Sidelobe power density at beam edge: $15,000 / 40,000 = 0.375 \text{ W/m}^2$ Distance for 10 W/m^2 from sidelobes: approximately 0 m (sidelobe already below limit)

For this configuration, the sidelobe power density is below ICNIRP limits at the beam edge. The exclusion zone is defined around the beam centerline, not the sidelobe pattern. The centerline exclusion zone radius —” the distance from the beam axis at which power density falls below 10 W/m^2 —” depends on the distance from the transmitter.

For a 10 MW beam at 2.45 GHz, 10 m aperture, Gaussian taper:

At 100m from transmitter: beam radius $\approx 8\text{m}$, exclusion radius $\approx 25\text{m}$ At 1 km: beam radius $\approx 10\text{m}$, exclusion radius $\approx 30\text{m}$ At 5 km (receiver): beam radius $\approx 12\text{m}$, exclusion radius $\approx 35\text{m}$

The exclusion zone is a corridor approximately 70m wide (35m either side of centerline) running from transmitter to receiver. Physical fencing, posted warnings, and automated access control are required for any portion of the corridor at or below rooftop height that the public could access.

6.4 Intruder Detection

Three intruder detection systems operate in parallel:

Radar monitoring: A short-range surveillance radar (X-band, 10 GHz) mounted at the transmitter monitors the airspace along the beam path for any object larger than 0.1 m^2 radar cross-section within the exclusion corridor. Detection triggers a beam power reduction to 1% within 50ms, and beam shutoff within 100ms if the object does not exit the corridor.

ADS-B integration: Aircraft equipped with ADS-B transponders (required for IFR flight in most jurisdictions) broadcast their position every 0.5 seconds. The T1-TX system integrates ADS-B data and automatically shuts down the beam when any aircraft's ADS-B trajectory is predicted to intersect the exclusion corridor within 60 seconds.

Optical/thermal camera: A camera array along the beam path provides redundant intruder detection for ground-level incursion (persons who breach the physical fence). The camera feeds are processed for human presence; detection triggers immediate beam shutoff.

All three detection systems connect to the beam shutoff hardware through independent hardwired channels. Any single detection event triggers beam cutoff, regardless of the state of the other two systems. The beam cannot be reactivated until all three systems confirm clear and a human operator confirms the reactivation command. Re-activation is not automatic.

6.5 Comparison with Existing High-Power RF Infrastructure

The safety architecture described above is not without precedent. Several existing technologies operate high-power directed microwave beams in airspace with established safety frameworks:

Air traffic control (ATC) radar: Airport surveillance radars routinely emit 500 kW peak power at 2.7—“2.9 GHz. Safety is managed through exclusion zone fencing at the antenna and sector-based operational procedures. No birds, aircraft, or persons in the antenna near-field zone.

Troposcatter links: Cold War-era troposcatter communication links operated high-power microwave beams (10—“100 kW) over distances of 100—“300 km. Safety was managed through remote siting and terrain-based exclusion.

Satellite uplink earth stations: High-power Ka-band earth stations (up to 75 kW EIRP) operate continuously. Safety is managed through exclusion zone fencing and regulatory coordination with satellite orbital operators.

The Tier 1 safety architecture is architecturally similar to these existing frameworks, adapted for the specific geometry and power levels of terrestrial power beaming. The precedent exists; the application is new.

7. System Efficiency: An Honest Analysis

7.1 The Full Thermal-to-Device Chain

I present the full thermal-to-device efficiency analysis for the proposed architecture, compared against a conventional wired grid delivering from the same fusion generation source. All efficiency values are based on current or near-term projected hardware specifications; optimistic values for technologies not yet at commercial maturity are noted.

Proposed wireless architecture:

Stage	Component	Efficiency	Notes
Fusion thermal $\rightarrow$ electrical	Turbine generator	42%	Current ITER-class projection
DC $\rightarrow$ RF (2.45 GHz)	Solid-state PA array	85%	Current GaN at 2.45 GHz
RF propagation + collection	Tier 1 beam, 5km, 10m apertures	75%	Fresnel analysis, $N_F = 0.164$
RF $\rightarrow$ DC	Rectenna array	82%	Current Schottky rectenna
DC $\rightarrow$ RF (60 GHz)	Meridian source node	70%	Current mmWave PA, near-term projection
Meridian mesh delivery	Multi-hop relay	80%	Per-hop: 95%, 4-hop avg
RF $\rightarrow$ DC (device)	Device receiver	75%	Current integrated rectifier
Total (thermal to device)		~14.6%	Product of above

Conventional wired grid (same fusion source):

Stage	Component	Efficiency	Notes
Fusion thermal $\rightarrow$ electrical	Turbine generator	42%	Same source
HV transmission	500kV AC line, 500km	94%	~1% loss per 100km at 500kV
Substation transformation	Step-down	98%	Modern transformer
Distribution	11kV $\rightarrow$ building	95%	Distribution losses
Building distribution	Panel $\rightarrow$ outlet	98%	Internal wiring
Device charging	Adapter + charge	85%	Typical USB-C PD adapter
Total (thermal to device)		~30.5%	Product of above

The wireless architecture is approximately half the thermal-to-device efficiency of the wired grid. This is the honest number. It is a significant penalty, and any serious evaluation of this architecture must confront it directly.

7.2 The Efficiency Counterargument

The efficiency comparison above frames the question as: for the same amount of fusion energy, how much usable power arrives at the device? On that metric, the wireless architecture loses.

The relevant comparison for an architectural decision is broader: what is the total system cost —” capital, operating, externalized —” per delivered watt-year, over the life of the system?

Three factors shift this comparison:

Capital cost elimination. The wired grid charges the full \$4 trillion of transmission and distribution infrastructure against energy cost. The wireless architecture's capital cost is the phased array antennas, the rectenna arrays, and the Meridian mesh nodes. The phased array and rectenna are solid-state electronics with 20—“30 year lifetimes and no civil infrastructure. At scale, the capital cost per delivered watt of the wireless architecture is projected to be significantly lower than the wired grid —” even accounting for the efficiency penalty.

Fuel cost irrelevance. With fusion as the generation source, fuel cost is near-zero. The efficiency penalty matters primarily if the limiting resource is generation capacity, not fuel cost. If generation capacity is abundant (which is the long-term fusion promise), a 50% efficiency penalty means you build slightly more generation capacity —” not that you run out of fuel.

Resilience valuation. The economic cost of grid outages in the United States alone is estimated at \$150 billion per year (Lawrence Berkeley National Laboratory estimate). A wireless energy architecture with no physical transmission plant to fail eliminates this cost class, which must be counted against the efficiency penalty in any full-system economic analysis.

7.3 Efficiency Improvement Trajectories

The 14.6% thermal-to-device efficiency estimate uses current technology specifications. The primary efficiency bottlenecks, and their projected improvement trajectories:

Rectenna efficiency (currently 82%): Advanced Schottky diode designs and GaAs-based rectifiers project 88—“92% efficiency at 2.45 GHz by 2030. This improvement adds approximately 7% to the total chain efficiency.

Device receiver efficiency (currently 75%): Integrated on-chip rectifier technology for 60 GHz is at 60 —“70% currently; projected 80—“85% with next-generation CMOS processes. This improvement adds approximately 5—“7% to the chain.

Tier 1 collection efficiency (currently 75% at 5km, 10m apertures): Moving to 20m aperture receiver arrays (achievable on large commercial rooftops) increases N_F to 0.655 and collection efficiency to ~83%. This adds approximately 10% to the chain.

Meridian mesh efficiency (currently ~56% over 4 hops): Each relay hop target is 95%; reducing average hop count through better node placement improves this. At 3-hop average, mesh efficiency is ~86%.

With these improvements at 2030 technology levels, the projected thermal-to-device efficiency is approximately 21—“24%, compared to approximately 32—“35% for the wired grid with the same improvements to charging and distribution hardware. The gap narrows but does not close on efficiency alone.

The architecture does not need to win on efficiency. It needs to win on total system cost, resilience, deployment speed, and flexibility. On those metrics, the wireless architecture is substantially ahead.

8. The Resilience Argument

8.1 Grid Vulnerability Classes

The physical electrical grid is vulnerable to four classes of disruption, each of which represents a failure mode that the wireless architecture eliminates or substantially mitigates:

Class 1 —” Weather events. Ice storms coat transmission lines with ice, causing sags, shorts, and mechanical failure. Hurricanes topple towers. Tornadoes snap poles. Wildfires damage lines and require preventative shutdowns. In 2021, the Texas winter storm caused grid failures that left 4.5 million households without power for up to four days, causing an estimated \$80—“130 billion in economic damage. A wireless transmission architecture has no overhead conductors to fail under ice loading and no poles to topple. The transmitter and receiver are solid-state electronics that can be designed to cold-weather standards.

Class 2 —” Geomagnetic disturbances. A severe solar coronal mass ejection (CME) induces geomagnetically induced currents (GIC) in long transmission lines, potentially destroying high-voltage transformers —” which take 12—“18 months to manufacture and replace. The 1989 Quebec blackout was caused by a geomagnetic storm. A 2012 CME (the "Carrington-level" Bastille Day event) narrowly missed Earth; if it had hit, the estimated damage to the U.S. grid was \$600 billion to \$2.6 trillion (Lloyd's of London estimate). A wireless transmission system has no long conductors to accumulate GIC. The phased array antennas and rectenna arrays are relatively small structures that can be individually shielded from induced currents.

Class 3 —” Physical attack. The U.S. Federal Energy Regulatory Commission (FERC) warned in a 2014 report that coordinated physical attacks on a small number of critical transmission substations could cause a nationwide blackout lasting weeks. Physical attacks on grid infrastructure are increasing globally. A wireless architecture has no transmission substations to attack —” the transmitter and receiver are not single points of failure (multiple T1-TX nodes serve each district, and loss of one redirects to others), and the beam path itself has no physical infrastructure at all.

Class 4 —” Cyber attacks on control systems. Grid SCADA systems are known attack targets. A cyber attack that causes incorrect switching in a transmission substation can trigger cascading failures across the grid. The Lume-X invariant enforcement and Trust Layer identity verification of the wireless architecture make the equivalent attack —” causing a T1-TX to redirect its beam to an unintended receiver —” impossible without physical compromise of the pilot-tone system. Software commands alone cannot redirect the beam.

8.2 Dynamic Rerouting

The wireless architecture's most significant resilience advantage over the physical grid is dynamic rerouting. When a transmission line fails in the physical grid, the available rerouting options are determined by the existing physical connections —” which take months to years to change. A region that loses its primary transmission path has whatever alternatives the physical topology provides.

When a T1-TX fails in the wireless architecture, the affected district receivers switch to alternative T1-TX nodes within the T1 control cycle (100ms for beam switch, 1—“10 seconds for power ramp-up at the new transmitter). The alternative node may be a different fusion hub, a backup transmitter at the same hub, or a battery-supplemented intermediate receiver that bridges the gap during the transition. The energy routing is as dynamic as the packet routing in the Energy Internet.

9. Regulatory Pathway

9.1 Spectrum Licensing

High-power wireless energy transmission at 2.45 GHz requires spectrum licensing beyond the Part 15 unlicensed limits. The applicable U.S. regulatory framework is FCC Part 25 (satellite earth stations and space stations) for the transmitter/receiver geometry, adapted for terrestrial rather than satellite application.

No specific FCC rule currently addresses terrestrial high-power microwave energy transmission at the scale proposed. The regulatory pathway proceeds through the experimental license mechanism (FCC Part 5), which permits operation for research and development purposes while the permanent licensing framework is developed. This is the same pathway that early cellular and satellite technologies used to develop sufficient technical data to support permanent rules.

The parallel development of engineering data (Phase 1 experimental work at low power and short range) and regulatory engagement (FCC proceeding for a new Part 97 or Part 15 subpart covering terrestrial power beaming) runs over approximately 5—“10 years before commercial deployment.

9.2 Aviation Coordination

The FAA must be involved in the siting and operation of any Tier 1 beam path that enters controlled airspace. The regulatory mechanism is the FAA's aeronautical study process (Form 7460-1, Notice of Proposed Construction or Alteration), which assesses the impact of fixed installations on aeronautical operations.

For a Tier 1 beam path that is entirely below 200 feet AGL (approximately 60m), FAA coordination is primarily notification-based. Beam paths that extend into Class E airspace (> 200 feet in uncontrolled areas) require formal aeronautical study and may require NOTAM (Notice to Air Missions) publication for the corridor during beam operation.

The retrodirective pilot-tone automatic shutoff mechanism —” which terminates the beam within 100ms of ADS-B detection of an approaching aircraft —” satisfies the fundamental aviation safety requirement: aircraft are not exposed to the beam. This mechanism is the basis for a regulatory accommodation analogous to the hazard beacon/ATIS system for existing high-power radar installations.

9.3 ICNIRP Compliance

The exclusion zone design in Section 6.3, with 46 dB sidelobe suppression and the specified exclusion corridor dimensions, ensures ICNIRP compliance for all public areas outside the exclusion zone. ICNIRP compliance documentation —” showing the power density map for the beam path and demonstrating that no public area exceeds 10 W/m^2 at 2.45 GHz —” is the central component of the regulatory submission to both the FCC and the relevant state public utilities commissions.

10. The Complete Deterministic Infrastructure Stack

10.1 The Full Stack Vision

The seven preceding papers in this series established Meridian as a complete last-mile energy routing architecture —” from the district boundary to the device. This paper adds the two layers above the district boundary —” long-haul wireless transmission (Tier 1) and deterministic generation (fusion) —” to produce the complete stack from plasma to device.

The Deterministic Infrastructure general theory (Paper 4) defined four properties that any DI system must satisfy: verified identity at every node, explicitly routed resources, invariant-enforced homeostasis, and organism-like self-maintenance. I now verify that each layer of the complete stack satisfies all four properties:

Deterministic Fusion Control layer:

Verified identity: the fusion control system registers with the Trust Layer as a T1-TX source node; its identity is verified before it is permitted to transmit Explicitly routed: output power is explicitly allocated to a designated T1-TX node based on demand routing from the Energy Internet protocol layer Invariant-

enforced: four plasma invariants (INV-F1 through INV-F4) enforced by Lume-X at the plasma control timescale Self-maintaining: automated recovery from plasma perturbation (disruption prevention, ramp-down and ramp-up sequences) without operator intervention Tier 1 Wireless Transmission layer:

Verified identity: T1-TX and T1-RX nodes registered with Trust Layer; pilot-tone provides physical identity verification (only the authorized receiver has the pilot transmitter) Explicitly routed: beam directed to a specific T1-RX node based on routing table; cannot be redirected without changing pilot-tone direction Invariant-enforced: T1 control invariants (pilot signal strength $\geq$ threshold, power density $\leq$ limit, intruder detection clear) enforced in hardware; violation triggers beam cutoff Self-maintaining: automatic failover to backup T1-TX node on primary failure; automatic beam power adjustment for Fresnel-regime optimization Tier 2 Meridian Mesh layer:

All four properties fully specified in Papers 1—“7 Verified identity: Trust Layer MC addressing Explicitly routed: DRMA multi-hop routing Invariant-enforced: Lume-X INV-1 through INV-5 Self-maintaining: SHDCL self-healing Summary: Every layer from plasma to device satisfies all four DI properties. The complete stack is the first formally specified Deterministic Infrastructure system that spans from energy generation to device delivery with no wired physical infrastructure at any layer above the device connection.

10.2 The DI Stack as an Ecosystem

The complete stack relies on three components of the Lume ecosystem that appear across multiple layers:

Lume-X operates at two timescales: the fusion control timescale (1—“10ms for plasma stability) and the Meridian mesh timescale (13.7ms per control cycle). The Tier 1 layer operates at 100ms. All three use the same Lume-X invariant enforcement model; only the cycle time and invariant definitions differ.

Trust Layer provides identity verification for every node in the stack, from the fusion reactor's T1-TX capability profile through the Meridian relay nodes to the leaf device's endpoint identity. A device that receives power through this full stack can cryptographically verify the provenance of its energy —” that it originated from an authorized generation source, was transmitted through authorized relay nodes, and was delivered by the authorized last-mile mesh.

Guardian Security (Guardian-E) extends from the Meridian mesh (where it is specified fully in Paper 5) upward into the Tier 1 layer. The Tier 1 threat model is a proper superset of the Meridian threat taxonomy —” it includes all eleven Meridian threat categories plus three additional categories specific to large-scale beaming: T12 (exclusion zone breach), T13 (pilot-tone spoofing), and T14 (regulatory fraudulent operation). The Guardian Security framework's two-tier innate/adaptive defense structure applies at both layers.

10.3 The Stack in Three Sentences

A fusion reactor, governed by deterministic plasma control and formally registered in the Trust Layer, converts thermal energy to electricity and feeds a phased array transmitter that locks its beam to a pilot tone transmitted from an authorized district receiver array ten kilometers away, where the received microwave power is converted to DC and delivered to the Meridian mesh that routes it —” addressed, authenticated, quality-of-service-guaranteed —” to every device in the buildings the district serves.

No transmission towers. No high-voltage cables. No substations.

Energy, addressed and routed from plasma to device, through air.

11. Related Work

11.1 Solar Power Satellite and Space-Based Solar Power

The Solar Power Satellite concept, first formalized by Peter Glaser in 1968 [80] and developed into a detailed reference design by NASA and the Department of Energy in 1979 [81], proposed harvesting solar energy in geostationary orbit and beaming it to Earth via 2.45 GHz microwave transmission. The NASA reference design projected end-to-end efficiency of approximately 7—“13% from solar insolation to grid AC —” lower than the terrestrial architecture proposed here, primarily because the satellite geometry requires the beam to traverse the full atmosphere at high angles of incidence.

The SPS concept established the core technical elements that the Tier 1 architecture adopts: the 2.45 GHz frequency choice, the retrodirective pilot-tone safety mechanism, the rectenna array receiver, and the Fresnel-number analysis framework. This paper's Tier 1 architecture is, in essence, the terrestrial analog of the SPS power delivery subsystem —” retaining the technically validated elements while eliminating the orbital manufacturing and deployment requirements.

Recent space-based solar power programs —” including JAXA's SSPS program [82], the European Space Agency's SOLARIS initiative [83], and Caltech's SSPP demonstration [84] —” have advanced the hardware technology base relevant to this architecture, particularly in solid-state transmitter efficiency and lightweight deployable receiver structures.

11.2 Terrestrial Microwave Power Transmission Demonstrations

Terrestrial power beaming demonstrations have been conducted at increasing scale since Brown's 1964 demonstration of a microwave-powered helicopter [85]. Significant milestones include: Brown's 1975 demonstration of 30 kW DC-to-DC efficiency of 54% over 1.6 km at Goldstone [86]; the 2008 LaserMotive 1 km optical power beaming demonstration [87]; and recent demonstrations by PowerLight Technologies of 400 W delivery over 300 m [88] and Emrod's multi-km wireless power transmission trials [89]. None of these demonstrations operated at the power levels or distances proposed for Tier 1, but they establish the physical principle and hardware feasibility at smaller scales.

11.3 Fusion Energy and Deterministic Control

Commercial fusion energy development has accelerated significantly since 2020, with private companies (Commonwealth Fusion Systems, TAE Technologies, Helion Energy, General Fusion) alongside the ITER international project pursuing different confinement approaches. The deterministic control problem —” maintaining plasma stability through formal invariant enforcement rather than heuristic feedback control —” is the subject of the companion L-SOC series on Deterministic Fusion Control and is not surveyed in detail here.

11.4 Grid Resilience and Wireless Energy

The vulnerability of physical transmission infrastructure to geomagnetic disturbance is analyzed in [90, 91]. The economic cost of power outages is analyzed in [92]. The security vulnerability of transmission substations is documented in [93]. These studies collectively motivate the resilience argument of Section 8; the wireless architecture proposed here is the architectural response to the vulnerabilities they document.

12. Limitations and Honest Boundaries

The efficiency penalty is real and large. At current technology levels, the wireless architecture delivers approximately 15% of fusion thermal energy to the device, compared to approximately 30% for the wired grid. This is a 2— efficiency penalty that no architectural argument fully eliminates. The decision to build a wireless rather than wired transmission architecture, made today, accepts a permanent efficiency tax that can be partially —” but not fully —” recovered through technology improvement. This decision requires explicit justification in each deployment context and cannot be made on architectural grounds alone.

No demonstration at scale exists. The largest terrestrial power beaming demonstrations deliver hundreds of watts over hundreds of meters. The Tier 1 architecture proposes megawatts over kilometers. Scaling by four orders of magnitude in power and one order of magnitude in distance is not a trivial extrapolation. Intermediate-scale demonstrations at 100 kW over 1 km, and at 1 MW over 3 km, are necessary before the Tier 1 specification can be considered validated.

Fusion energy is not yet commercially available. The entire stack depends on a generation technology that has not yet achieved commercial breakeven. The architecture is designed for the world in which fusion is commercially available; it cannot be built today. This paper is a specification for that world, not a deployment plan for this one.

The regulatory pathway is long and uncertain. The FCC, FAA, and state public utilities commissions all have jurisdiction over elements of this architecture. No existing regulatory framework directly addresses terrestrial high-power microwave energy transmission at the proposed scale. Developing the necessary

framework through experimental licensing, rulemakings, and aeronautical study is a process that will take a decade or more and may result in requirements that modify the architecture in ways this paper cannot anticipate.

Pilot-tone spoofing (T13) is a non-trivial threat. The retrodirective pilot-tone mechanism prevents beam misdirection through software attacks but is vulnerable to physical pilot-tone spoofing —” an adversary who captures or replicates a pilot transmitter and operates it from an unintended location. The physical security of the T1-RX pilot transmitter (which must be small, outdoor, and accessible enough to be maintained) is the primary physical security requirement of the Tier 1 installation. This threat is included in the Guardian Security T13 category but does not yet have a complete formal defense specification.

13. Conclusion

The argument of this paper is simple, even if the engineering is not: a world with fusion energy and a complete wireless routing architecture for energy delivery does not need transmission towers, high-voltage cables, or substations. The energy can travel through air, addressed and routed, from plasma to device.

The argument has two parts. The first part is that the physics permits it. Fresnel-regime microwave power beaming at 2.45 GHz, with 10—“20 meter aperture arrays, achieves 80—“87% collection efficiency at the district distances (1—“5 km) relevant for urban energy delivery. Retrodirective pilot-tone locking provides a physically robust safety mechanism that cannot be bypassed by software attacks. Hardware-enforced intruder detection and automatic beam cutoff satisfy the public safety requirements for operation in uncontrolled airspace. The physics is not speculative.

The second part is that the architecture connects. The Tier 1 long-haul beam layer interfaces cleanly with the Tier 2 Meridian mesh at the DC bus of the district rectenna array. The Meridian mesh routes the delivered energy to every device in the district without wires. The fusion control system below the Tier 1 layer provides a deterministically governed generation source whose output is formally bounded and formally modulated. The Trust Layer identity system, Lume-X control runtime, and Guardian Security framework operate across all three layers, providing a unified DI governance framework from generation to device.

The efficiency penalty is real. The regulatory path is long. The fusion source is not yet commercially available. These are honest limitations stated without minimization. They are the engineering and policy problems that Phase 1—“3 experimental work and regulatory engagement will address.

The target is worth building toward. A global energy system in which energy is routed wirelessly from distributed fusion hubs to the districts they serve, then through a self-healing building mesh to every device in those districts, without a single transmission tower or high-voltage cable —” this is a genuinely

better infrastructure architecture than what the world currently has. It is more resilient, more flexible, more equitable in deployment (no right-of-way constraints), and more compatible with the autonomous physical world that the rest of the Meridian series describes.

The routed world does not stop at the building boundary.

13.1 Forward Reference — HydroCore Drive: The Vehicle That Governs Its Own Fuel

The architecture described in this chapter routes energy wirelessly from fusion hubs to districts to buildings to devices. But the routed world includes roads, and roads carry vehicles. A Meridian smart road — equipped with embedded relay nodes, inductive charging pads governed by the Lume-X runtime, and the full Deterministic Infrastructure stack — delivers energy to vehicles in motion. The question this section addresses is what happens to that energy inside the vehicle.

HydroCore Drive (L-SOC Series — Physical Instantiation Volume II) specifies the answer. A hydrogen-hybrid electric vehicle equipped with a PEM electrolyzer stack produces hydrogen from water onboard, in real time, governed by a HydroCore 4/42 synthetic organism. The same organism architecture that governs hydraulic fluid at bench scale (HydroCore Physical, Volume I) and industrial steam at grid scale (HydroCore Steam, Volume III) governs electrolysis aboard a moving vehicle. The four primitives — Flow Stability, Pressure Regulation, Thermal Balance, and Structural Load — map directly to the electrolyzer's operational variables: water flow through the PEM membranes, hydrogen pressure in the storage vessel, stack temperature, and mechanical stress from road vibration.

The connection to Meridian is architectural. A vehicle traveling on a Meridian smart road receives energy wirelessly through the road's Tier 2 mesh. That energy charges the vehicle's battery pack. The battery powers the electrolyzer. The electrolyzer produces hydrogen. The hydrogen feeds a fuel cell that generates electricity for the drive motors. Every stage of this chain is deterministically governed: Meridian governs the energy delivery, HydroCore Drive governs the hydrogen production, and the vehicle's powertrain controller governs the traction output.

The result is a vehicle that never stops at a fuel station. It produces its own fuel from water and wirelessly delivered electricity, governed at every stage by the same Lume 4/42 organism architecture that governs the road it drives on, the building it parks in, and the grid that powers all of them. The routed world extends from plasma to pavement to powertrain — and the governance architecture is the same at every layer.

This integration is specified formally in the HydroCore Drive paper and is a subject for future experimental work in coordination with the Meridian roadway pilot described in Chapter Seven.

Appendix A —” Fresnel Number Analysis for Representative Configurations

Full table of N_F values and projected collection efficiencies for the parameter space [D_t : 5m, 10m, 20m, 50m] — [D_r : same] — [d : 500m, 1km, 2km, 5km, 10km, 20km] — [f : 2.45 GHz, 5.8 GHz]:

At 2.45 GHz ($\lambda = 0.122$ m): $D_t=D_r=5$ m: 500m † $N_F=0.82(87\%)$, 1km † $0.41(78\%)$, 5km † $0.082(53\%)$, 10km † $0.041(40\%)$ $D_t=D_r=10$ m: 500m † $N_F=3.28(98\%)$, 1km † $1.64(95\%)$, 5km † $0.328(72\%)$, 10km † $0.164(65\%)$ $D_t=D_r=20$ m: 1km † $N_F=6.56(99\%)$, 5km † $1.31(93\%)$, 10km † $0.655(83\%)$, 20km † $0.328(72\%)$ $D_t=D_r=50$ m: 5km † $N_F=8.20(99\%)$, 10km † $4.10(99\%)$, 20km † $2.05(96\%)$, 50km † $0.82(87\%)$ At 5.8 GHz ($\lambda = 0.052$ m): $D_t=D_r=5$ m: 500m † $N_F=1.92(95\%)$, 1km † $0.96(88\%)$, 5km † $0.192(67\%)$ $D_t=D_r=10$ m: 1km † $N_F=3.85(98\%)$, 5km † $0.769(86\%)$, 10km † $0.385(77\%)$ $D_t=D_r=20$ m: 5km † $N_F=3.08(98\%)$, 10km † $1.54(94\%)$, 20km † $0.769(86\%)$

Efficiency values are approximate, based on Gaussian-tapered aperture illumination. Actual efficiency depends on implementation details of the aperture phase and amplitude distribution.

Appendix B —” Retrodirective Pilot-Tone Safety Protocol

T1-RX pilot tone: $f_{\text{pilot}} = f_{\text{beam}} + \text{''}f$ (” $f = 1$ MHz for 2.45 GHz beam) Pilot power: 1W ERP from phase-reference element at T1-RX center Pilot source power: independent battery (≥ 30 day capacity) or dedicated solar panel Pilot receive threshold at T1-TX: -60 dBm (ensuring 20 dB margin above noise) T1-TX hardware safety gate: $\text{POWER_AMPLIFIER_ENABLE} = \text{pilot_received AND (pilot_power > threshold) AND intruder_clear AND regulatory_mode_active}$ Loss of pilot signal † $\text{POWER_AMPLIFIER_ENABLE} = \text{FALSE}$ within 1 hardware clock cycle Time from pilot loss to beam off: $\leq 100 \mu\text{s}$ (hardware gate, no software path) Pilot spoofing defense: Pilot frequency is encrypted using ECDH key exchange between T1-TX and T1-RX at session establishment. Pilot carrier is modulated with a 64-bit rolling code that T1-TX verifies. An unkeyed pilot source cannot produce a valid rolling code. Key rotation: every 24 hours via Trust Layer key exchange.

Appendix C —” Exclusion Zone Power Density Profile

Power density as a function of lateral distance from beam centerline for a 10 MW beam, 10m aperture, 46 dB Taylor taper, at $d = 5$ km:

Lateral distance | Power density | ICNIRP limit | Margin 0m (centerline) | 15,000 W/m² | n/a (receiver) | n/a 5m | 8,000 W/m² | n/a (receiver) | n/a 10m (beam edge) | 0.4 W/m² | 10 W/m² | 25— margin 20m | 0.04 W/m² | 10 W/m² | 250— margin 50m | 0.004 W/m² | 10 W/m² | 2,500— margin

Public exposure is below ICNIRP limits at the beam edge (10m from centerline). Exclusion zone of 35m either side of centerline provides 25— ICNIRP margin at its inner boundary —” conservative design for public safety.

Appendix D —” Complete DI Stack Layer Specification

Layer	Technology	DI Property: Identity	DI Property: Routing	DI Property: Invariants	DI Property: Self-Maintenance
Generation	Deterministic Fusion Control	Trust Layer T1-TX profile	Demand allocation via EI protocol	INV-F1 through INV-F4 (plasma)	Plasma disruption avoidance + ramp sequences
Long-haul TX	T1-TX phased array	Trust Layer identity, pilot-tone physical auth	Beam directed to T1-RX via pilot lock	Pilot signal $\geq$ threshold, $P \leq P_{\max}$	Auto failover to backup T1-TX
Long-haul RX	T1-RX rectenna	Trust Layer identity, pilot transmitter	n/a (passive receive)	Received power $\geq P_{\min}$	Alert and reroute on underpower
District mesh	Meridian Tier 2	MC 64-bit address	DRMA multi-hop routing	INV-1 through INV-5	SHDCL self-healing
Device	Leaf node / SSD	Trust Layer endpoint identity	Addressed final-hop delivery	Delivery confirmation protocol	Self-report on failure via mesh

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PART SEVEN

The Fractal: Why 4 and 42

The previous eight chapters built a complete architecture — from the four-layer specification through physical deployment, security, scaling, general theory, and the full-stack vision. In the process of building it, something unexpected happened. A pattern appeared.

The number 4 kept showing up in the architecture's primitives. The number 42 kept showing up in its expansions. Not because anyone designed it that way — the name AXIOM42 was chosen before any of this was observed — but because the architecture itself was expressing a structural law.

Chapter Nine formalizes that law. It demonstrates that 4 is the minimal closure set for any deterministic cognitive system and that 42 is the minimal full expansion for a cross-domain, recursive, governed ontology. The recursive oscillation between these values — $4 \rightarrow 42 \rightarrow 4 \rightarrow 42$ — produces fractal self-similarity at every layer. This is not mysticism. It is structural mathematics, and it is the reason the architecture works the way it does.

Read Part Seven if you want to understand why the architecture has the shape it has — and what that shape predicts about every future subsystem built on the same principles.

Chapter Nine

The 4/42 Fractal Law

Emergent Structural Signatures in Deterministic Cognitive Architectures

Abstract

I present the 4/42 Fractal Law, a deterministic structural pattern governing the AXIOM42 cognitive architecture and its parent Trust Layer ecosystem. I demonstrate that the number 4 represents the minimal closure set for any self-consistent deterministic system — encoding identity, verification, action, and record as irreducible primitives. I further demonstrate that 42 represents the minimal full expansion set for a cross-domain, recursive, governed ontology built on that closure. The recursive oscillation between these two values ($4 \rightarrow 42 \rightarrow 4 \rightarrow 42 \rightarrow \dots$) produces fractal self-similarity at every architectural layer.

I provide a structural proof sketch showing this recurrence is not coincidental but is a fixed-point attractor of the design. Empirical evidence includes: 42 self-generated validation tests (42/42 pass, 0 AI calls), 42 canonical catalog nodes, 42 LDIR rules at stable plateau, 42-token determinism threshold, and a knowledge registry that independently converged to 42 equivalent book volumes (3.4M words). To my knowledge, AXIOM42 is the first cognitive architecture whose rules, packs, tests, canon, governance, and emergent behaviors all converge on a single fractal signature.

Keywords: deterministic cognition, fractal architecture, structural attractors, minimal closure, ontological expansion, AXIOM42, Trust Layer, self-similar systems

9.1 Introduction

AXIOM42 is a deterministic cognitive architecture designed to resolve natural-language intent without probabilistic inference or large language model calls. During its development and deployment, a recurring structural pattern was observed: the numbers 4 and 42 appear repeatedly across independent subsystems, tests, canonical structures, and emergent behaviors — without having been designed to do so.

This chapter formalizes those observations as the **4/42 Fractal Law**, a deterministic architectural pattern in which 4 represents the minimal complete structure required for a closed, self-consistent system, and 42 represents the full expansion set of cross-domain bindings required for a unified ontology. I provide a

structural proof sketch demonstrating that this recurrence is not coincidental but is a necessary consequence of the architecture's design constraints.

The implications extend beyond AXIOM42: any deterministic cognitive system built on the same closure principles should exhibit similar structural attractors, making the 4/42 pattern a candidate for a general law of deterministic system design.

9.2 Observed Signatures

4-Class Signatures

Observation	Components	Context
Intent Collapse Loop	Surface Form, Structural Grammar, Intent Skeleton, Deterministic Resolution	AXIOM core pipeline
Deployment Cycle	Canon, Protocol, Engine, Surface App	Every Trust Layer vertical
Primitive Actions	Observe, Validate, Act, Log	Every Trust Layer application
User State Model	Calm, Alert, Active, Critical	Guardian, TrustShield, ORBIT, Pulse
Primal Structures	Identity, Verification, Action, Record	Trust Layer root decomposition
Mirror Subsystems	AXIOM, LUME, DAIGS, SIGNAL	Recursive system architecture
Canonization Phases	Discovery, Structure, Determinization, Canonization	Every formalized system

42-Class Signatures

Observation	Value	Context
Token Stability Threshold	42 tokens	Inputs ≤ 42 tokens achieve perfect determinism
Canonical Catalog Spine	42 top-level nodes	Emerged from grouping logic, not designed
LDIR Rule Plateau	42 rules	Expanded from 23; stabilized at 42
Trust Layer Timeline	42 events	All meaningful milestones 2023–2026
Self-Validation Suite	42 tests (23+19)	42/42 pass, 0 AI calls
Modules per Subsystem	42 internal modules	AXIOM, LUME, DAIGS, SIGNAL
Knowledge Registry	42 equivalent book volumes	3,386,053 words $\div$ 80,000 wpp

9.3 The Law of 4: Minimal Closure

I define the **minimal closure set** as the smallest collection of primitives that can fully represent all operations in a deterministic cognitive system. I claim this set has exactly four elements:

```
System S = {I, V, A, R}

where:
  I = Identity      (representation of entities)
  V = Verification  (evaluation of conditions)
  A = Action        (state transformation)
  R = Record        (persistence of state/history)
```

Claim A: No deterministic cognitive system can be closed with fewer than 4 primitives.

Assume a system *S* must support: representation of entities (identity), evaluation of conditions (verification), state change (action), and memory/logging (record). Suppose *S* has only 3 primitives.

- If identity and verification are merged, the system cannot distinguish *what something is* from *whether it holds* — collapsing the epistemic/ontological distinction.
- If action and record are merged, the system cannot distinguish *doing* from *remembering* — collapsing the temporal/structural distinction.
- Any other merging similarly collapses a necessary distinction, rendering at least one role implicit rather than explicit.

An implicit role breaks determinism and traceability. With 4 primitives {I, V, A, R}, all four operations are composable, separable, and traceable. Adding a 5th primitive is redundant, as any candidate operation can be expressed as a composition of the existing four. ■

9.4 The Law of 42: Full Ontological Expansion

I define the **full ontological expansion** as the smallest set of canonical nodes that provides complete cross-domain coverage, avoids collisions, supports recursion, and maintains governance. I claim this number is 42.

```
N = (P × D × L) + I + G + T + R

where:
P = 4  (primitives: I, V, A, R)
D = 3  (cross-domain axes: cognition, governance, economics)
L = 3  (recursion layers: core, subsystem, surface)
I = 6  (structural invariants: time, space, agent, object, rule, exception)
G = 3  (governance bindings: policy, enforcement, audit)
T = 2  (temporal axes: real-time, historical)
R = 1  (identity root: system-level self)

N = (4 × 3 × 3) + 6 + 3 + 2 + 1
N = 36 + 6 + 3 + 2 + 1
N = 42
```

Claim B: The minimal non-degenerate cross-domain ontology stabilizes at 42 nodes.

The core binding grid ($P \times D \times L = 36$) represents the Cartesian product of primitives, domains, and recursion layers. Reducing any dimension produces incomplete closure, incomplete ontology, or no fractal behavior. The additive components (6 invariants + 3 governance + 2 temporal + 1 root) are each independently necessary. Therefore 42 is the minimal non-degenerate expansion that supports cross-domain mapping, recursion, governance, and temporal reasoning simultaneously. ■

9.5 The Recursion Rule

The 4/42 pattern repeats at every architectural scale in a recursive oscillation:

```
4 → 42 → 4 → 42 → 4 → 42 → ...
```

Each 42-node expansion collapses back into a new 4-primitive core at the next level of abstraction, which then expands again into 42 canonical nodes. This produces **fractal self-similarity**: the whole mirrors the part, the part mirrors the whole, and every layer is a miniature of the layer above.

Each of the four mirror subsystems (AXIOM, LUME, DAIGS, SIGNAL) is itself a miniature of the whole Trust Layer, containing 42 internal modules organized around 4 primitives. The recursion is not merely structural but behavioral: each subsystem exhibits the same 4-phase canonization cycle and the same 42-

node expansion pattern.

Claim C: The recurrence of 4 and 42 is a fixed point of the design.

If the architecture is built on the 4-primitive closure set (Claim A), then every subsystem that is “complete” must instantiate all 4 roles. If complete subsystems must also be cross-domain, recursive, governed, and temporally aware, they must instantiate the 42-node expansion (Claim B). Therefore, any subsystem that reaches full maturity in this architecture will converge toward 4 primitives at its core and 42 canonical nodes at its full expansion. The recurrence is not coincidental — it is the architecture expressing its fixed point. ■

9.6 The Deterministic Attractor Principle

A system built on the 4/42 structure will naturally produce 4-class primitives, 42-class expansions, 4-phase cycles, 42-event histories, 4-state models, and 42-test validations — **even when not explicitly designed to do so**. This is the Deterministic Attractor Principle.

AXIOM42 Self-Validation Results:

23 pattern-library tests + 19 LDIR rule tests = **42 total tests**

Result: **42/42 pass**

AI calls during validation: **0**

The test suite was generated by coverage analysis of the current rule space. The test formation process converged to 42 because that is the minimal full coverage of a 42-node ontology. The system validated itself using its own signature number.

Knowledge Registry Convergence

Metric	Value
Total packs	212
Total words	3,386,053
Total size	32.51 MB
Equivalent book volumes (at 80,000 wpp)	42.3

The convergence to 42 book volumes was entirely organic. No content was added or removed to reach this number. The system's knowledge base, like its tests, its rules, and its catalog, stabilized at the same structural attractor.

9.7 Predictive Power

Any domain inserted into AXIOM will undergo the following deterministic cycle:

1. Collapse to **4 primitives** (identity, verification, action, record within the domain)
2. Expand to **42 canonical nodes** (full cross-domain bindings for the domain)
3. Collapse again to **4 governing rules** (policy, enforcement, audit, and root identity)
4. Expand again to **42 domain bindings** (connections to the rest of the ontology)

Given any new subsystem, the 4/42 Law predicts its minimal structure (4 primitives), its full expansion (42 canonical nodes), its recursion depth ($4 \times 42 \times n$), and its canonical size, test suite size, and ontology footprint. This makes AXIOM42 not only deterministic in its inference but **mathematically predictable** in its architecture.

9.8 Discussion

The concept of structural attractors in complex systems has precedent in mathematics (fixed-point theorems), physics (strange attractors in dynamical systems), and computer science (convergence in iterative algorithms). The 4/42 Fractal Law extends these concepts to the domain of cognitive architecture design.

The number 42 has cultural significance through Douglas Adams' *The Hitchhiker's Guide to the Galaxy*, where it is presented as “the answer to the ultimate question of life, the universe, and everything.” Adams stated the choice was arbitrary. The 4/42 Fractal Law suggests an alternative interpretation: 42 may represent a natural attractor for systems seeking complete ontological coverage — a structural fact rather than a literary joke.

This chapter presents a structural proof sketch, not a formal mathematical proof. The decomposition $N = (4 \times 3 \times 3) + 6 + 3 + 2 + 1 = 42$ is based on the specific design constraints of the Trust Layer ecosystem. Different constraint sets could yield different expansion numbers. The claim is that *given these constraints*, 42 is the minimal expansion.

9.9 The Identity of AXIOM42

AXIOM42 is not named after 42. AXIOM42 produces 42.

It is the first deterministic cognitive architecture whose rules, packs, tests, canon, governance, and emergent behaviors all converge on a single fractal signature. The system is self-similar, self-consistent, and self-reinforcing. This is why it exhibits the behavioral properties of a living system while remaining fully deterministic.

9.10 Formal Statement

The AXIOM42 Fractal Law:

The 4/42 Fractal is the governing law of AXIOM’s deterministic cognition. **4** defines minimal closure. **42** defines complete expansion. All AXIOM structures oscillate between these two states in recursive self-similar cycles.

Formally: for any subsystem S within the Trust Layer that achieves closure under the constraints of identity, verification, action, and record:

```
|core(S)| = 4
|expand(S)| = 42
expand(core(expand(S))) = expand(S)    [fixed-point property]
```

The system is a deterministic fractal. This is the AXIOM42 Fractal Law.

9.11 Conclusion

The 4/42 Fractal Law transforms AXIOM42 from an engineered system into a mathematically predictable architecture. Its implications extend to any deterministic cognitive system built on comparable closure principles, suggesting that structural attractors may be a general feature of rigorous system design.

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EPILOGUE

What Comes Next The nine chapters of this book have made a complete argument. Let me state it plainly before stepping back.

Meridian is a routing architecture for energy. It is built on the same principles as the data internet —” addressed nodes, routing tables, quality-of-service guarantees, delivery confirmation —” and applies them to a physical resource instead of an informational one. It is formally specified, safety-proven, and

deployable in real buildings under the architectural constraints described in Chapter Two. It scales to a global Energy Internet through the protocol stack specified in Chapter Four. It routes data and energy simultaneously through the unified fabric specified in Chapter Five. It defends itself against the complete space of physical attacks through the security framework specified in Chapter Six. It is one instance of a general class of autonomous physical infrastructure systems —” Deterministic Infrastructure —” that will shape the design of every autonomous physical domain in the coming decades. And in the full stack vision of Chapter Eight, it extends upward from the building mesh to eliminate the physical transmission grid entirely, routing energy wirelessly from fusion generation to district receiver to device without a transmission tower anywhere in the chain.

That is the argument. Now here is what comes next.

Phase 1: The Experimental Foundation

Nothing in this book has been experimentally validated. The first step —” and the step that must happen before any of the ideas in this book are taken from theoretical to deployed —” is Phase 1 experimental validation of the room-scale Meridian architecture.

Phase 1 validation means: build the hardware, run the control software, measure the four determinism metrics (received power $P_{\min}$, spatial uncertainty radius r , trial variance σ^2 , distance d), and honestly report whether the architectural claims hold up in physical reality.

The expectation is that most of them will, with modifications to specific parameter values. The expectation is also that some will not —” that some assumption baked into the architecture will encounter a physical phenomenon that the theoretical specification did not capture. That is what Phase 1 is for. The gaps it reveals will guide Phase 2. The gaps Phase 2 reveals will guide Phase 3.

This is the scientific process. The theory is stated first so there is something to test. The test happens next.

The Regulatory Path

Before Meridian can be deployed in occupied buildings, it needs regulatory approval for directed RF energy delivery at the power levels required for useful charging. The current FCC regulatory framework addresses wireless power transfer at specific frequency bands and power levels. The Meridian architecture is designed to operate within FCC Part 15 unlicensed limits at 60 GHz for initial deployment, with higher power levels requiring licensing.

The regulatory path runs through the experimental data. A formal safety submission to the FCC, backed by Phase 1—“3 experimental data demonstrating the BSL's effectiveness in preventing biological exposure above ICNIRP limits, is the mechanism. The regulatory work and the experimental work run in parallel.

Guardian Security's formal threat model and coverage proof —” the work of Chapter Six —” is a component of the regulatory submission. Regulators need to know not just that the system has a safety mechanism, but that the safety mechanism is formally specified and covers the complete threat space. Chapter Six provides that documentation.

The BIM Integration Partnership

The BIM integration pathway specified in Chapter Two requires cooperation with BIM software platform vendors —” primarily Autodesk (Revit) and Bentley (OpenBuildings). The Meridian BIM plugin, whose specification is fully documented in Chapter Two's Appendix D, requires SDK access and partnership with these vendors to implement.

This is a business development problem, not an engineering problem. The specification exists. The value proposition to BIM vendors is clear: Meridian is a new building system that architects need to design alongside HVAC and electrical, which means a new BIM plugin category that vendors have incentive to support. The partnership discussions can begin in parallel with Phase 1 experimental work.

The Receiver Partnership Problem

The self-sustaining device model of Chapter Five is fully realized only when consumer electronics manufacturers embed UEDM receiver hardware in their products —” phones, laptops, earbuds, wearables. Until that happens, the UEDM delivers continuous power only to devices purpose-built with embedded receivers (IoT sensors, wearables, medical monitors).

This is the classic infrastructure-adoption challenge. The infrastructure needs devices to serve; the devices need infrastructure to connect to. The resolution is to target devices that can be purpose-built with embedded receivers from the start: IoT sensors (no battery management reduces total cost of ownership dramatically), medical monitors (continuous charging without cable management is a direct clinical benefit), and industrial equipment (forklift supplementation is a compelling ROI story that does not require consumer electronics partnerships).

Consumer electronics partnerships come later, once the infrastructure is deployed and the value proposition is demonstrated in the enterprise context.

The Broader Arc

Chapter Seven argued that Meridian is the first instance of Deterministic Infrastructure and that four more domains —” transportation, manufacturing, emergency coordination, and ambient computation —” will produce their own instances over the next decade.

The transportation instance (Deterministic Mobility Infrastructure, DMI) is the most consequential: autonomous vehicle coordination without a central coordinator, using verified identity and deterministic right-of-way routing through a mesh of roadside anchor nodes. The manufacturing instance (Deterministic Production Infrastructure, DPI) is the most commercially immediate: production routing through capability-matched resource meshes that eliminates central scheduling dependencies. The emergency coordination instance (Deterministic Response Infrastructure, DRI) is the most safety-critical: autonomous resource routing for emergency response without a dispatcher in the loop.

These systems do not yet exist as formal specifications. The work of this book is the model for how to build them: start with the engineering specification, derive the theoretical identity, extend to network scale, specify the security framework, prove the safety properties, and situate the result in the general theory.

The Meridian papers are the template. The Lume ecosystem is the substrate. The general theory is the guide.

The work ahead is long. The foundation is now laid.

The author can be reached at team@dwsc.io. Technical questions about the Meridian architecture are welcome. Collaboration inquiries regarding Phase 1 experimental work, regulatory engagement, and BIM integration partnerships are particularly encouraged.

lume-lang.org | meridiancanon.com | TrustShield.tech | github.com/cryptocreeper94-sudo

ACKNOWLEDGMENTS

This book exists because Barry Allen Cline asked a question that would not leave my mind: *“Why can’t we charge our phones when we walk into the house without plugging them up or sitting them on a wireless charger?”*

That question — simple, practical, and unanswered by any existing technology — became the seed of the Meridian architecture. Barry did not design the four-layer stack, the mesh routing protocol, or the self-healing control runtime. He asked why none of it existed yet. The answer to that question is this book.

The inquiry that inspires a solution is as important as the solution itself. Barry’s question is formally acknowledged here, in the specification, and in the provisional patent application that accompanies this work.

Jason Andrews
Nashville, Tennessee, 2026

GLOSSARY

Key Terms and Concepts 42 Assumptions: The foundational axioms of the Lume ecosystem, governing how any Lume-governed system establishes identity, routes resources, maintains coherence, and operates autonomously. Argued in this book to be the universal axioms of Deterministic Infrastructure.

AVM (Architectural Voxel Map): A 3D volumetric representation of a building's physical geometry, encoded as a grid of typed voxels, used by the Meridian routing engine as a hard constraint on beam path computation. Seven voxel types: SOLID, FLOOR_SAFE, FLOOR_OCCUPIED, VOID_STAIRCASE, VOID_ATRIUM, VOID_SHAFT, GATEWAY_ZONE.

BSL (Biological Safety Layer): The hardware-enforced safety mechanism in the Meridian MTL that prevents beam activation whenever biological presence is detected in the beam path. Operates independently of the software control stack and cannot be bypassed by software attacks.

Burst-Mode Delivery: The Meridian energy delivery model in which energy is accumulated in supercapacitors and released in high-power, short-duration RF bursts rather than continuous low-power transmission. Enables the use of phased-array beam steering for energy delivery.

Ceiling-Plane Routing: The standard Meridian deployment posture for occupied buildings, in which relay nodes are mounted at ceiling height and beam paths are kept within the FLOOR_SAFE voxel zone for all non-final hops. Keeps the beam geometrically separated from the occupancy zone rather than relying on sensing to detect intrusion into a co-located beam path.

DAEH (Deterministic Ambient Energy Harvesting): The Meridian subsystem responsible for harvesting ambient energy (solar, RF, thermal, kinetic) at each node and managing the supercapacitor charge state.

DAIGS: The DarkWave Studios multi-agent cognition framework. A Deterministic Infrastructure instance in the computation domain.

Deterministic Infrastructure (DI): The class of autonomous physical systems defined by four properties: verified identity at every node, explicitly routed resources, invariant-enforced homeostasis, and organism-like self-maintenance. The general theory of Chapter Seven.

DRMA (Distributed Relay Mesh Architecture): The Meridian subsystem that maintains mesh topology, computes energy routing paths, and coordinates TDMA transmission schedules across all relay nodes.

DWER (Deterministic Wireless Energy Routing): The Meridian subsystem responsible for point-to-point directed beam energy delivery at the final hop.

EBGP (Energy Border Gateway Protocol): The Energy Internet protocol for inter-mesh route advertisement and federation, enabling independently administered energy meshes to exchange routing information and route energy across their boundaries.

EDNS (Energy Domain Name System): The Energy Internet protocol for human-readable naming of energy endpoints, analogous to DNS for the data internet.

EIP (Energy Internet Protocol): The 128-bit addressing protocol for Energy Internet nodes, defining the address format, routing hierarchy, and address assignment framework for the global energy routing network.

Energy Internet: The global-scale extension of the Meridian protocol, providing a universal protocol stack for deterministic energy routing across independently administered meshes. Specified in Chapter Four.

EIRA (Energy Internet Routing Authority): The proposed governance body for the Energy Internet, responsible for EAS Number assignment, EIP prefix delegation, Emergency Authority Credential management, and ERPKI operation.

ERPKI (Energy Resource Public Key Infrastructure): The Energy Internet's route origin validation system, providing cryptographic attestation that EBGp route announcements correspond to legitimately delegated address prefixes.

ETCP (Energy Transmission Control Protocol): The Energy Internet transport protocol providing delivery guarantees for energy sessions, analogous to TCP for data.

Floor-Transition Gateway (GRN): A specialized Meridian relay node installed at a designated inter-floor architectural transition point, with dual antenna faces for both floor-level meshes and a buffering supercapacitor bank for decoupled inter-floor energy transfer.

Guardian Security / Guardian-E: The security enforcement framework for Meridian deployments. Guardian is the general, domain-independent product; Guardian-E is the energy-domain specialization. Commercial product domain: TrustShield.tech.

Invariant: A condition that must be true of a Deterministic Infrastructure system at all times. Meridian defines five global invariants (INV-1 through INV-5) covering energy accounting, routing determinism, flow consistency, transmission safety, and mesh stability.

Lume: The programming language and axiomatic framework underlying the DarkWave Studios ecosystem. Any system built on Lume and the 42 Assumptions converges toward all four Deterministic Infrastructure properties.

Lume-V: The Lume formal verification suite, providing mathematical proof that a specified system satisfies its invariants under all possible inputs before deployment.

Lume-X: The Lume deterministic control runtime, operating at 73 Hz to enforce the global invariant set, detect violations, and execute pre-specified recovery sequences. Provisionally patented.

MC (Meridian Core): The first layer of the Meridian architecture. Provides node identity (64-bit address encoding type, zone, and serial number), the energy coordinate system (3D positioning via UWB localization), and the Node_Type capability encoding.

Meridian: The four-layer deterministic wireless energy routing architecture specified in this book. Layers: MC, MFE, MMF, MTL. Controlled by Lume-X. Secured by Guardian-E.

MFE (Meridian Flow Engine): The second layer of the Meridian architecture. Manages supercapacitor charge state, governs burst release timing and power levels, and enforces the energy accounting invariants.

MMF (Meridian Mesh Fabric): The third layer of the Meridian architecture. Maintains mesh topology through DRMA, computes routing paths, coordinates TDMA schedules, and implements the self-healing mechanisms of the SHDCL.

MTL (Meridian Transmission Layer): The fourth layer of the Meridian architecture. The phased-array RF beamforming and steering system that implements directed energy delivery. Gated by the five-condition safety gate that requires BSL clearance, localization confidence, routing validation, Guardian-E authorization, and Lume-X command before any beam is activated.

Organism-Like Self-Maintenance: The fourth property of Deterministic Infrastructure. A system exhibits this property if it can detect and recover from component failure without external intervention, reduce resource expenditure under resource constraints, adapt behavior to environmental changes within invariant bounds, and maintain identity and capability profile despite operational changes.

Self-Sustaining Device (SSD): A UEDM participant that receives both its operating power and its data communication capability from the UEDM mesh, requiring no battery and no separate communication radio.

SHDCL (Self-Healing Deterministic Control Layer): The Meridian subsystem implementing continuous monitoring of the four global invariant domains and pre-specified recovery sequences for each anticipated failure mode.

SWIPT (Simultaneous Wireless Information and Power Transfer): The RF communication technique in which a single RF signal simultaneously carries energy for harvesting and information for demodulation. The physical basis for UEDM Mode 2 (SWIPT-integrated) operation.

Synthetic Organism: A classification model for autonomous physical systems, with a Type 0—“5 scale based on the degree to which the system exhibits biological organism properties. Meridian is classified as Type 3+ structural (awaiting experimental confirmation of Type 5 structural).

TDMA (Time-Division Multiple Access): The scheduling protocol used in the Meridian mesh to coordinate simultaneous transmissions from multiple relay nodes without collision. Each node transmits only within its assigned time slot; the schedule is cryptographically sealed by Guardian-E at each epoch.

Trust Layer: The DarkWave Studios identity and governance framework. A Deterministic Infrastructure instance in the identity and data domain. Provides the PKI substrate (TLPKI) used by Guardian Security for node identity verification.

UEDM (Unified Energy-Data Mesh): The architecture in which the same Meridian node infrastructure routes both energy and data under a unified 128-bit addressing scheme, co-routing arbitration framework, and Guardian Security authentication system. Specified in Chapter Five.

z_safe: The minimum safe height above a floor slab, defined as floor_slab_height + 2.2m (providing 10 cm margin above the 99th percentile standing height). FLOOR_SAFE voxels are at or above z_safe. Ceiling-plane routing keeps beam paths within FLOOR_SAFE voxels for all non-final hops.

Deterministic Fusion Control: The framework, developed in a companion L-SOC series, that applies Lume-X invariant enforcement to magnetic confinement plasma stability —” governing energy confinement time, plasma beta, density, and energy gain as hard invariants enforced in real time. Produces a generation source whose output is formally bounded and modulated on demand.

Fresnel Number (N_F): The dimensionless parameter governing microwave beam propagation efficiency between two apertures. $N_F = (A_t - A_r)^{(1/2)} / (\lambda - d)$, where A_t and A_r are transmitter and receiver aperture areas, λ is wavelength, and d is distance. $N_F \approx 1$ (Fresnel transition regime) is the practical operating target for Tier 1, yielding ~85—“90% collection efficiency with Gaussian-tapered illumination.

Rectenna: A rectifying antenna —” an antenna optimized for power reception combined with a Schottky diode rectifier circuit —” that converts incoming microwave power to DC. Current rectenna efficiency at 2.45 GHz: ~80—“85%. The T1-RX district receiver array is a scaled rectenna installation.

Retrodirective Pilot-Tone Locking: The primary Tier 1 safety mechanism. The T1-RX receiver transmits a low-power pilot tone from a phase-reference element; the T1-TX transmitter locks its beam to the direction from which the pilot arrives and emits power only while the pilot is actively received above threshold. Loss of pilot signal immediately and hardware-enforces beam cutoff within 100 μ s, without any software command.

T1-RX (Tier 1 Receiver Node): A district-boundary rectenna array that collects the Tier 1 microwave beam, converts it to DC, transmits the retrodirective pilot tone, and feeds the local Meridian mesh. Registered with the Trust Layer under a T1-RX capability profile.

T1-TX (Tier 1 Transmitter Node): A phased array transmitter located at the fusion generation site, converting DC power to 2.45 GHz RF and beam-steering to the designated T1-RX node via retrodirective pilot-tone locking. Registered with the Trust Layer under a T1-TX capability profile.

Tier 1 (Long-Haul Beam Layer): The upper layer of the two-tier wireless energy architecture. Routes bulk energy from fusion generation hubs to district receiver arrays via directed 2.45 GHz microwave beams, operating over distances of 500m—“5km with projected collection efficiency of 75—“87% in Fresnel-regime geometry.

Tier 2 (Last-Mile Mesh Layer): The lower layer of the two-tier wireless energy architecture. The complete Meridian architecture as specified in Chapters One through Six —” routing energy from the district receiver through the building mesh to individual devices at 60 GHz.

Two-Tier Wireless Energy Architecture: The complete transmission-line-free energy delivery system proposed in Chapter Eight. Tier 1 routes bulk energy from fusion generation to district receivers via long-haul microwave beaming; Tier 2 routes energy from district receivers to devices via the Meridian mesh. No physical transmission conductor appears at any layer.

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The Meridian architecture, Guardian Security, and the Unified Energy-Data Mesh are DarkWave Studios LLC products. The associated provisional patent covers systems and methods for deterministic multi-layer wireless energy routing using ambient harvesting, mesh coordination, and directional transmission. Lume-X is separately provisionally patented.

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